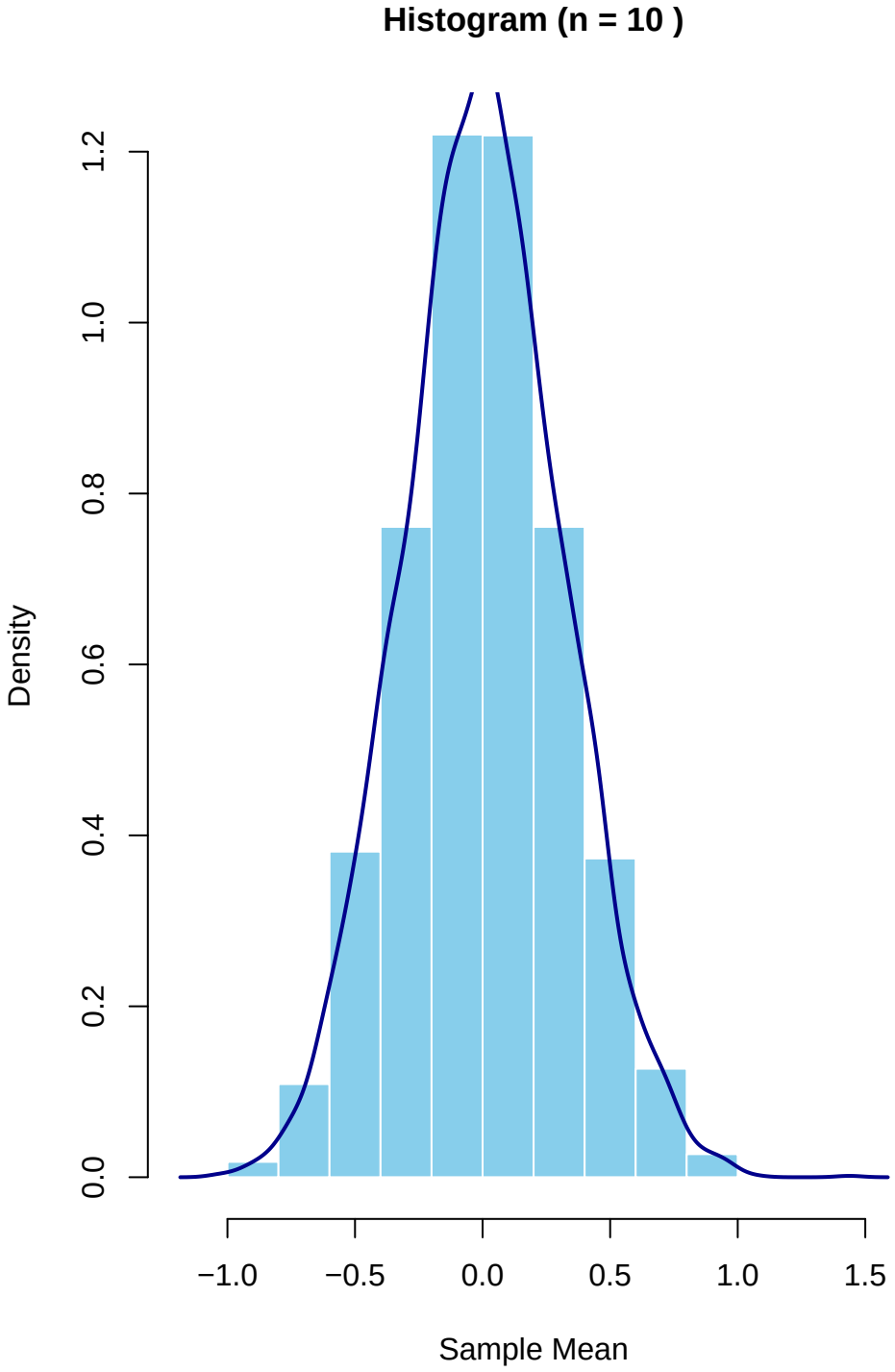
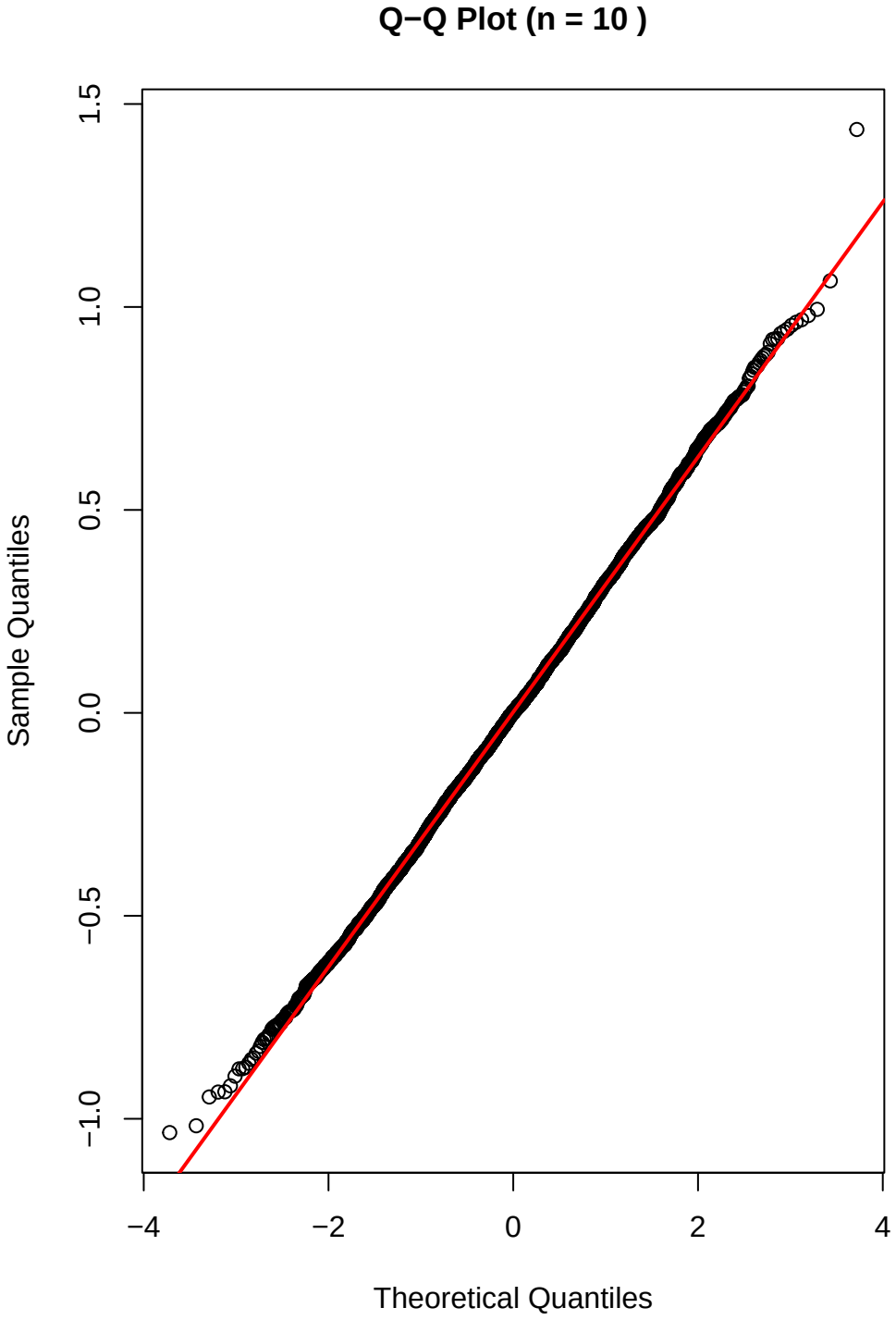
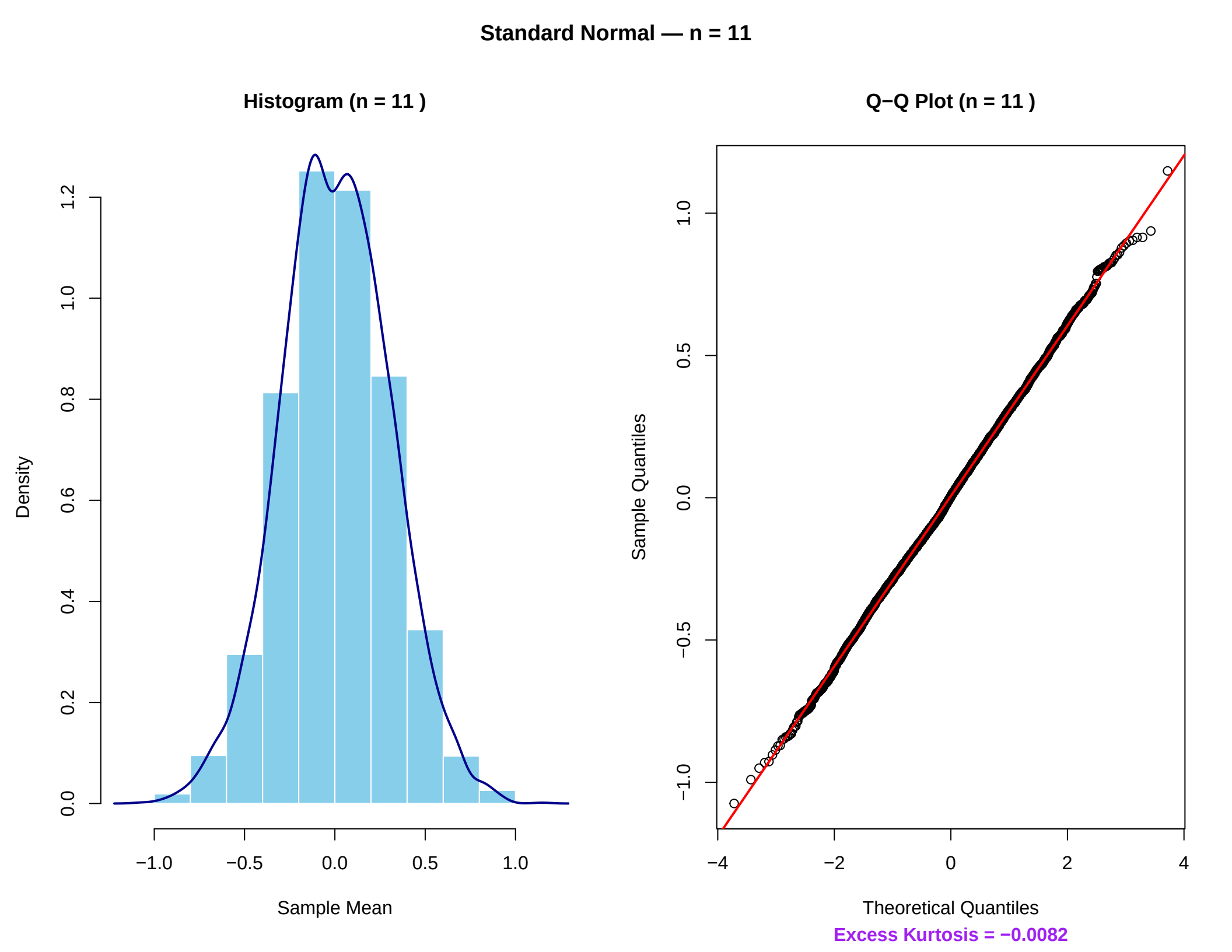
Standard Normal — n = 10

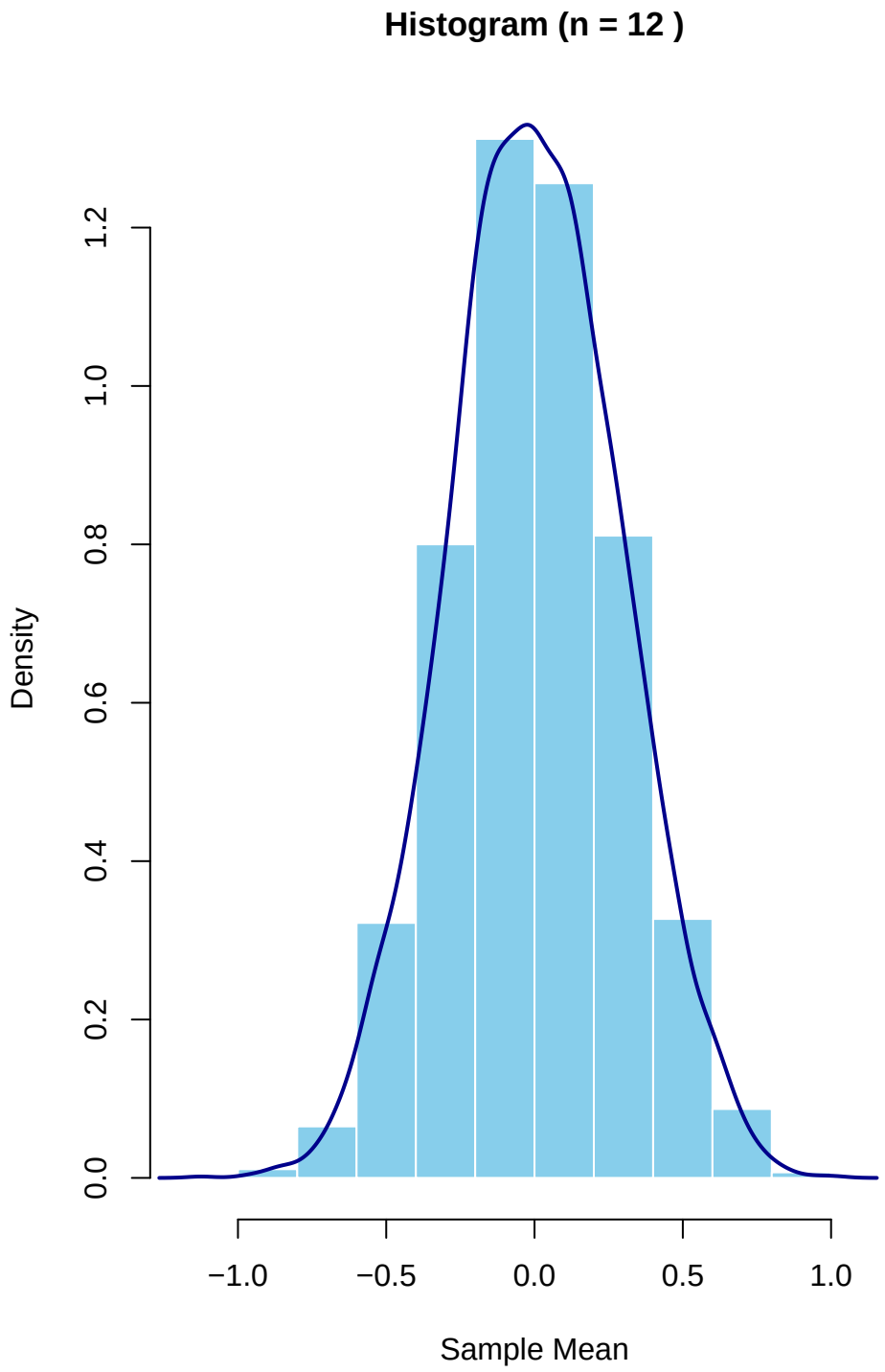


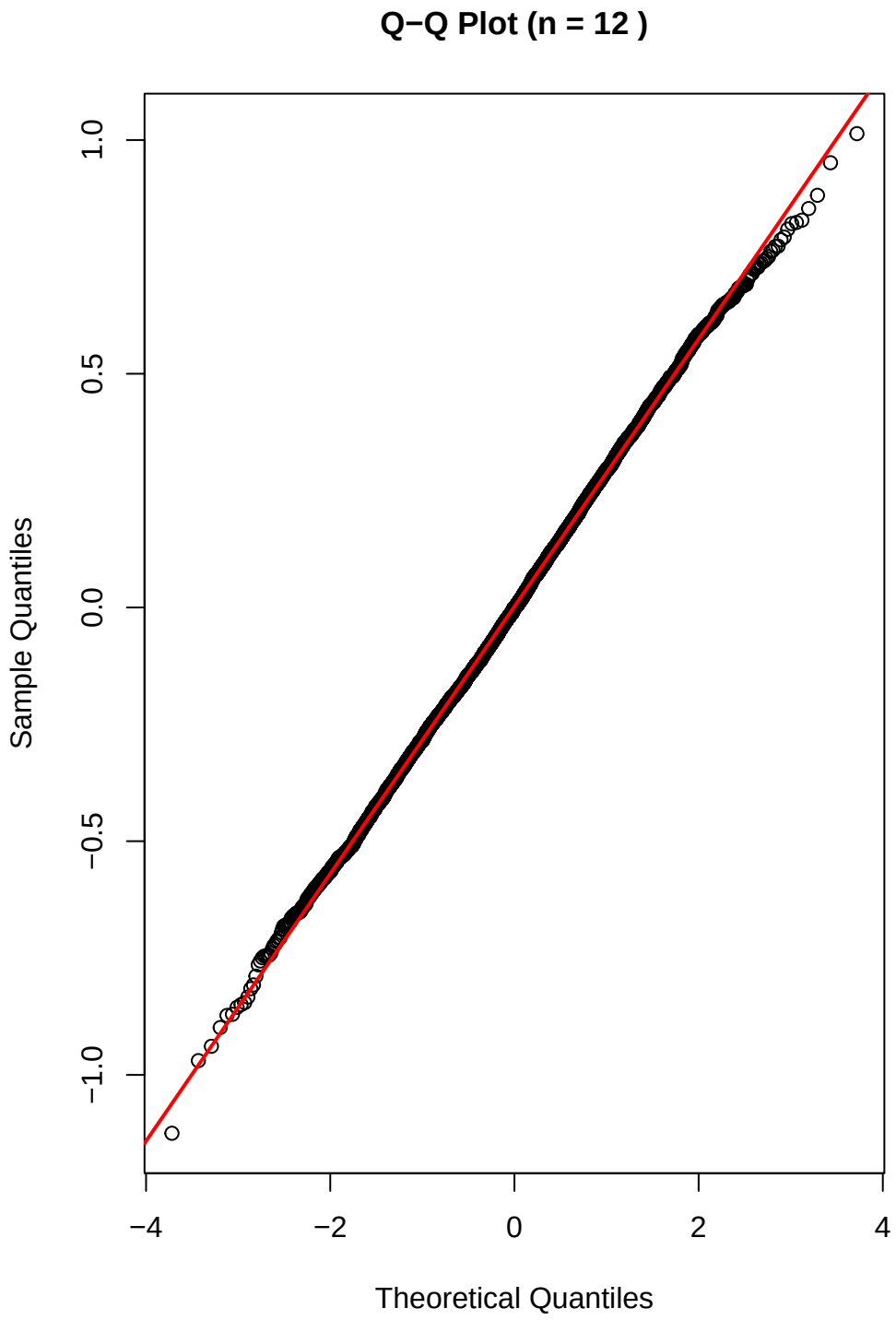


Excess Kurtosis = 0.0015

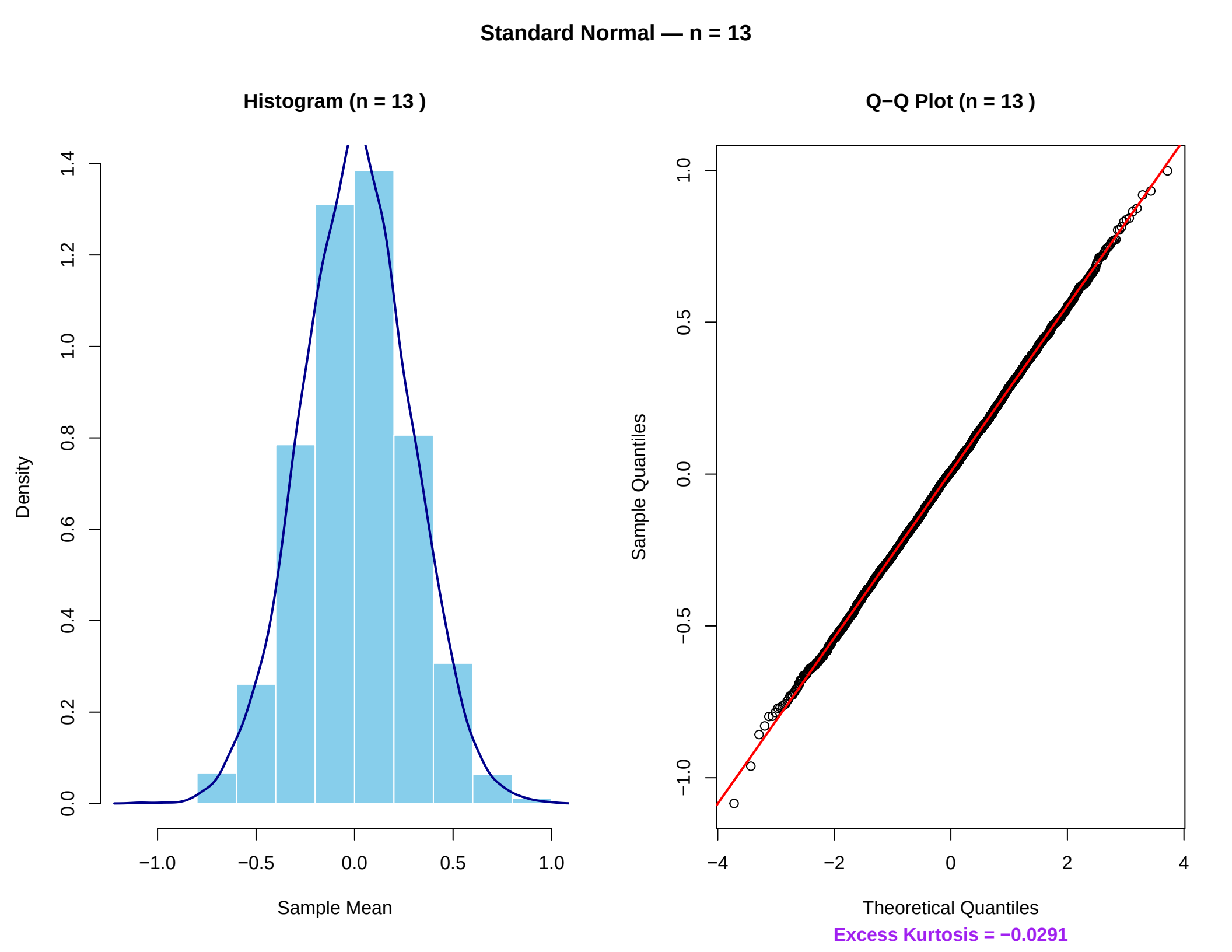

Standard Normal — n = 11
Histogram (n = 11)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
−1.0
−0.5
0.0
0.5
1.0
Sample Mean
Q−Q Plot (n = 11)
Sample Quantiles
−1.0
−0.5
0.0
0.5
1.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0082

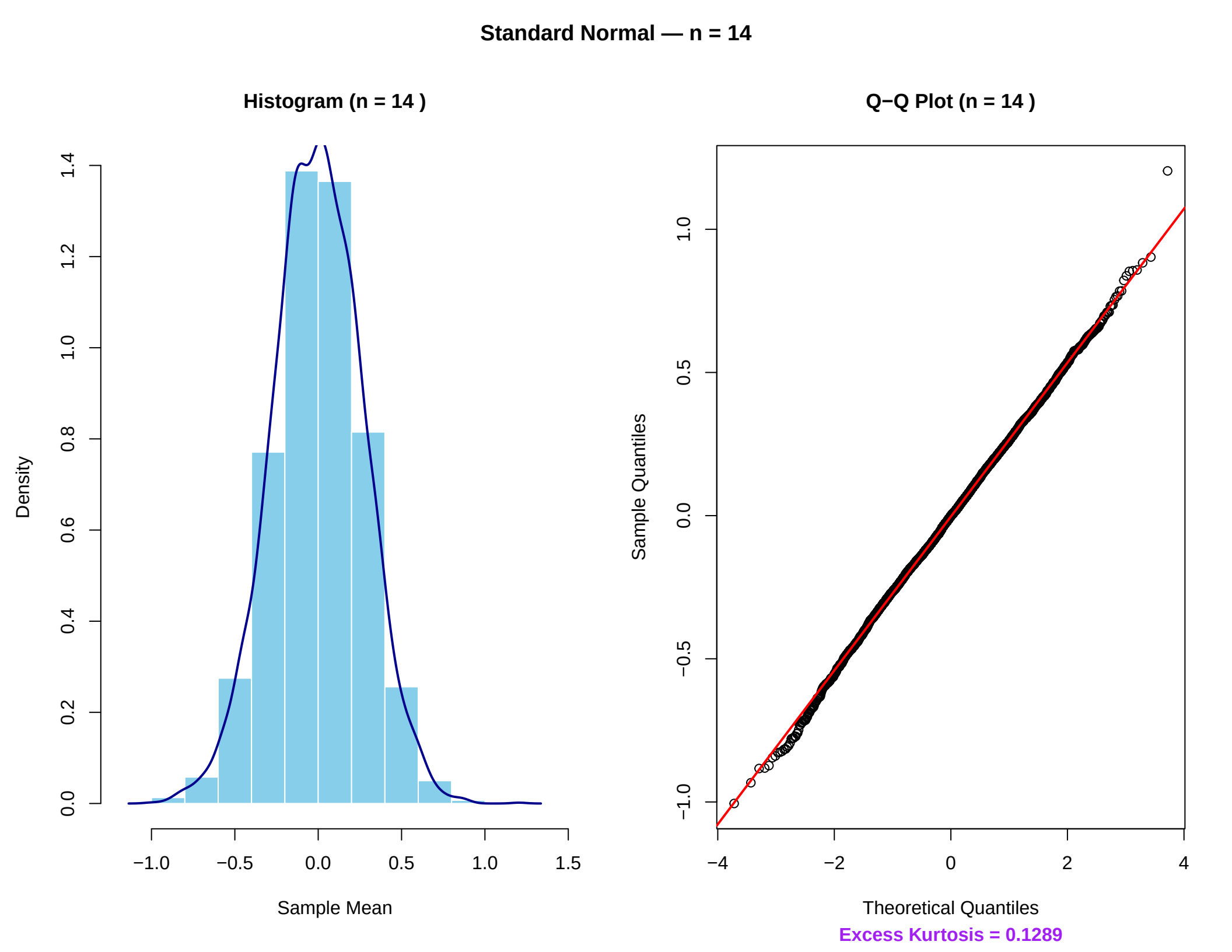
Standard Normal — n = 12

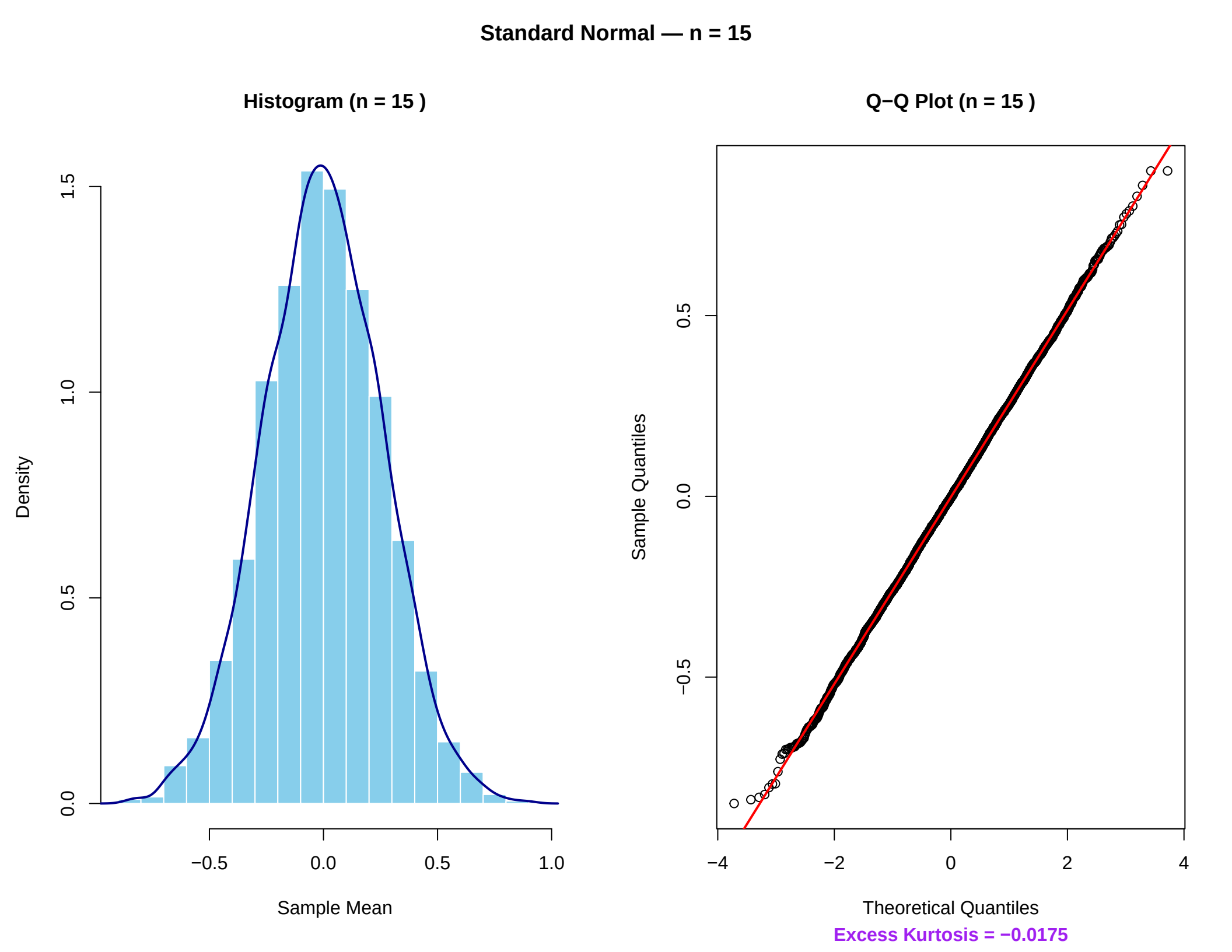




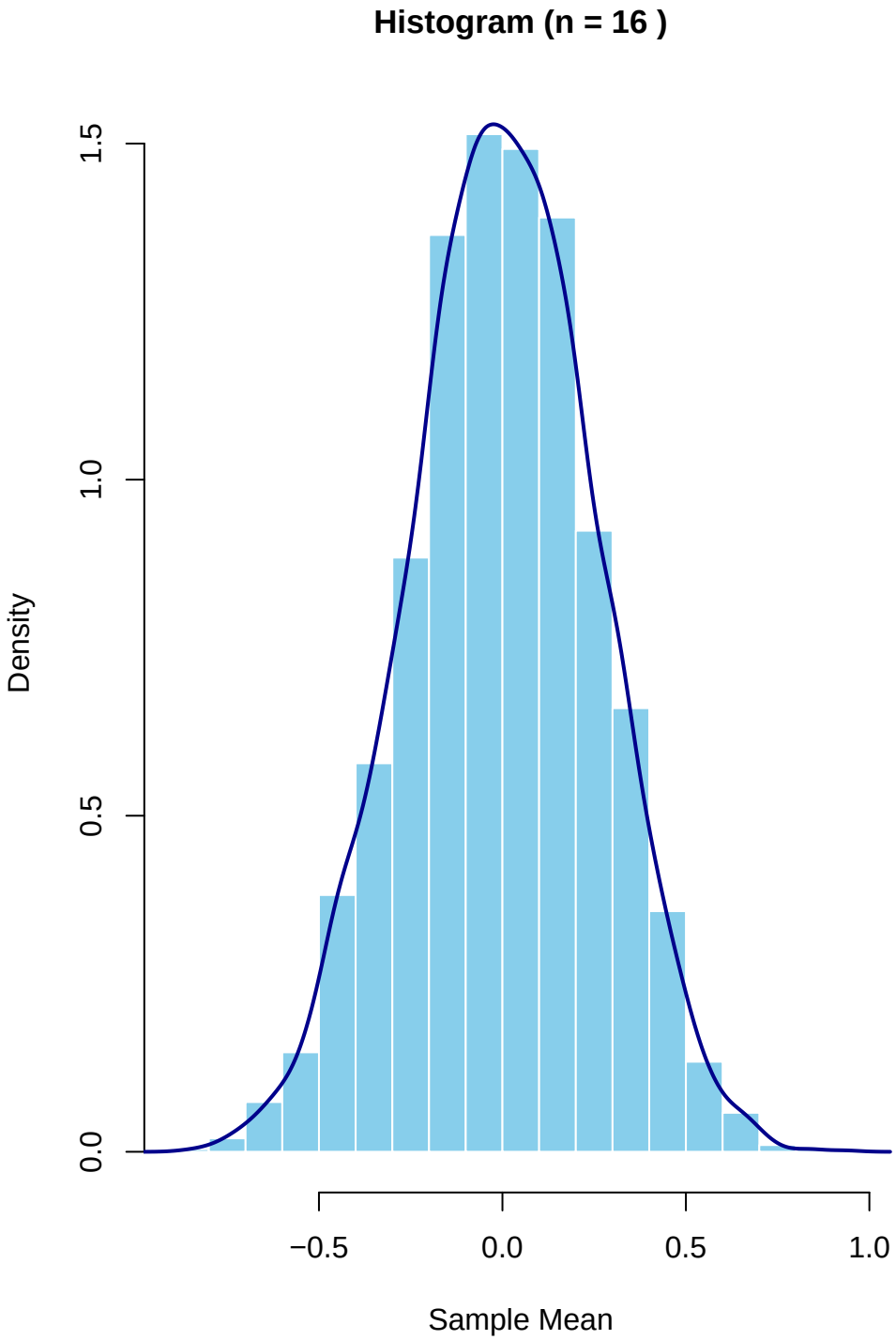
Excess Kurtosis = −0.1187

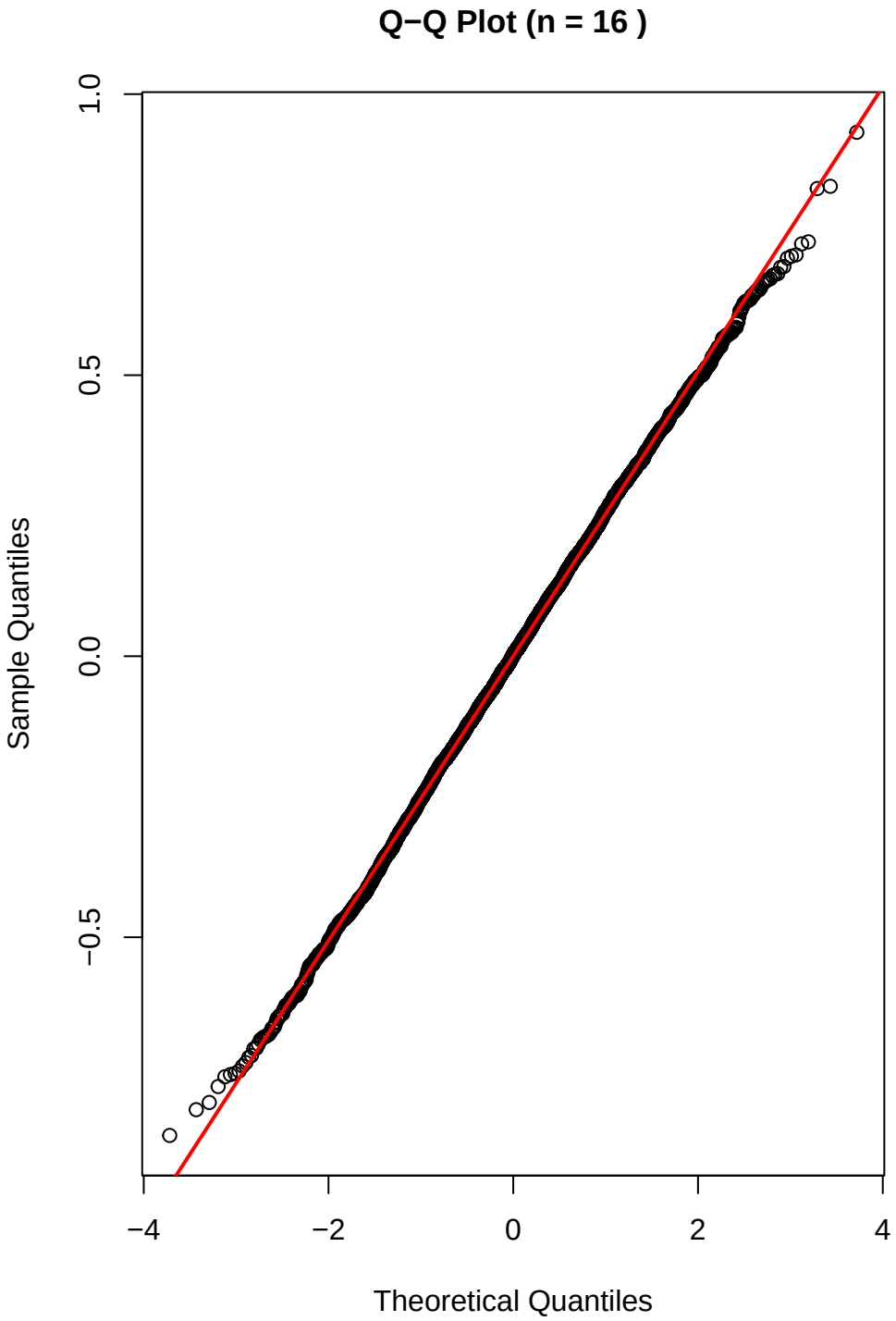

Standard Normal — n = 13
Histogram (n = 13)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
−1.0
−0.5
0.0
0.5
1.0
Sample Mean
Q−Q Plot (n = 13)
Sample Quantiles
−1.0
−0.5
0.0
0.5
1.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0291


Standard Normal — n = 14
Histogram (n = 14)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
−1.0
−0.5
0.0
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 14)
Sample Quantiles
−1.0
−0.5
0.0
0.5
1.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.1289

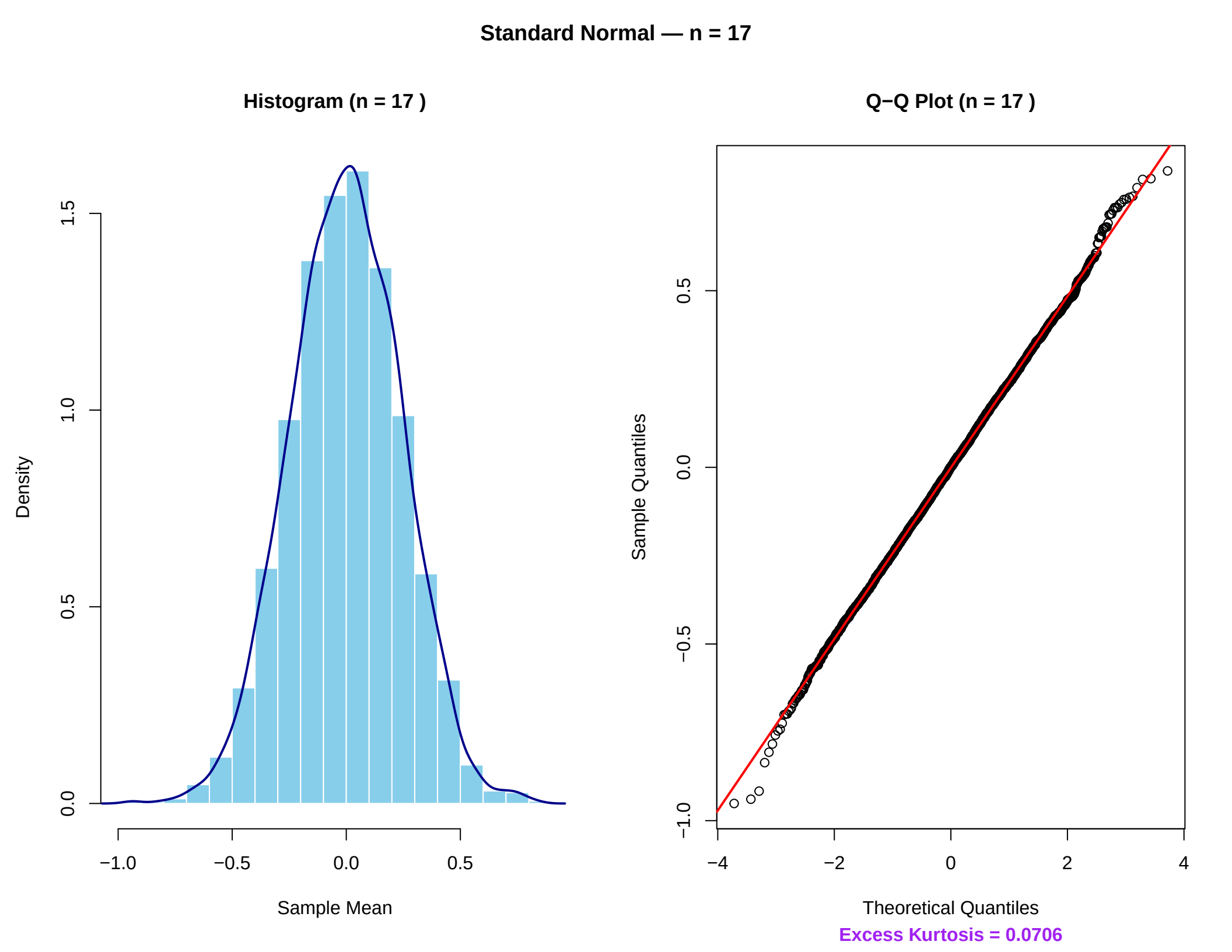

Standard Normal — n = 15
Histogram (n = 15)
Density
0.0
0.5
1.0
1.5
−0.5
0.0
0.5
1.0
Sample Mean
Q−Q Plot (n = 15)
Sample Quantiles
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0175

Standard Normal — n = 16

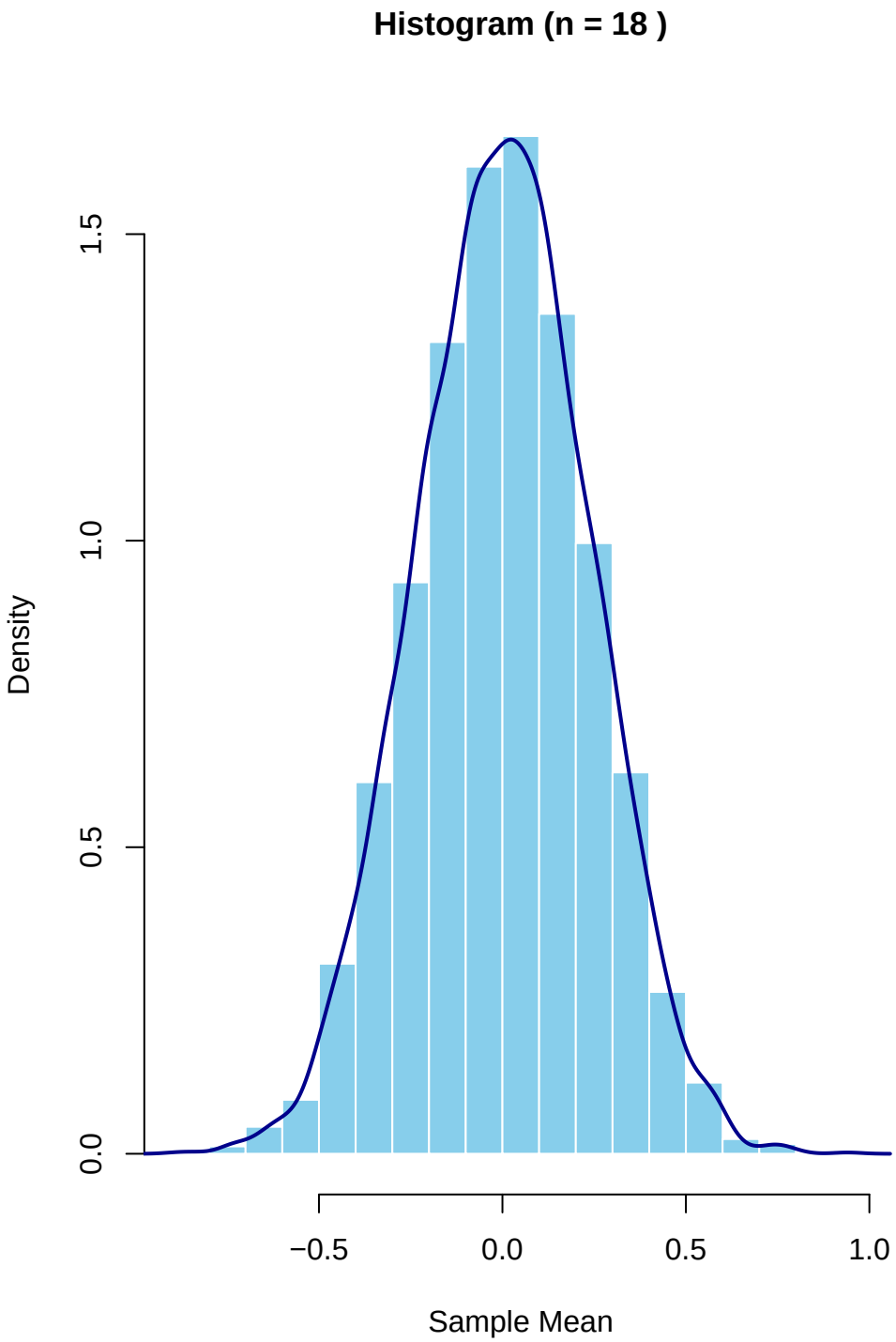


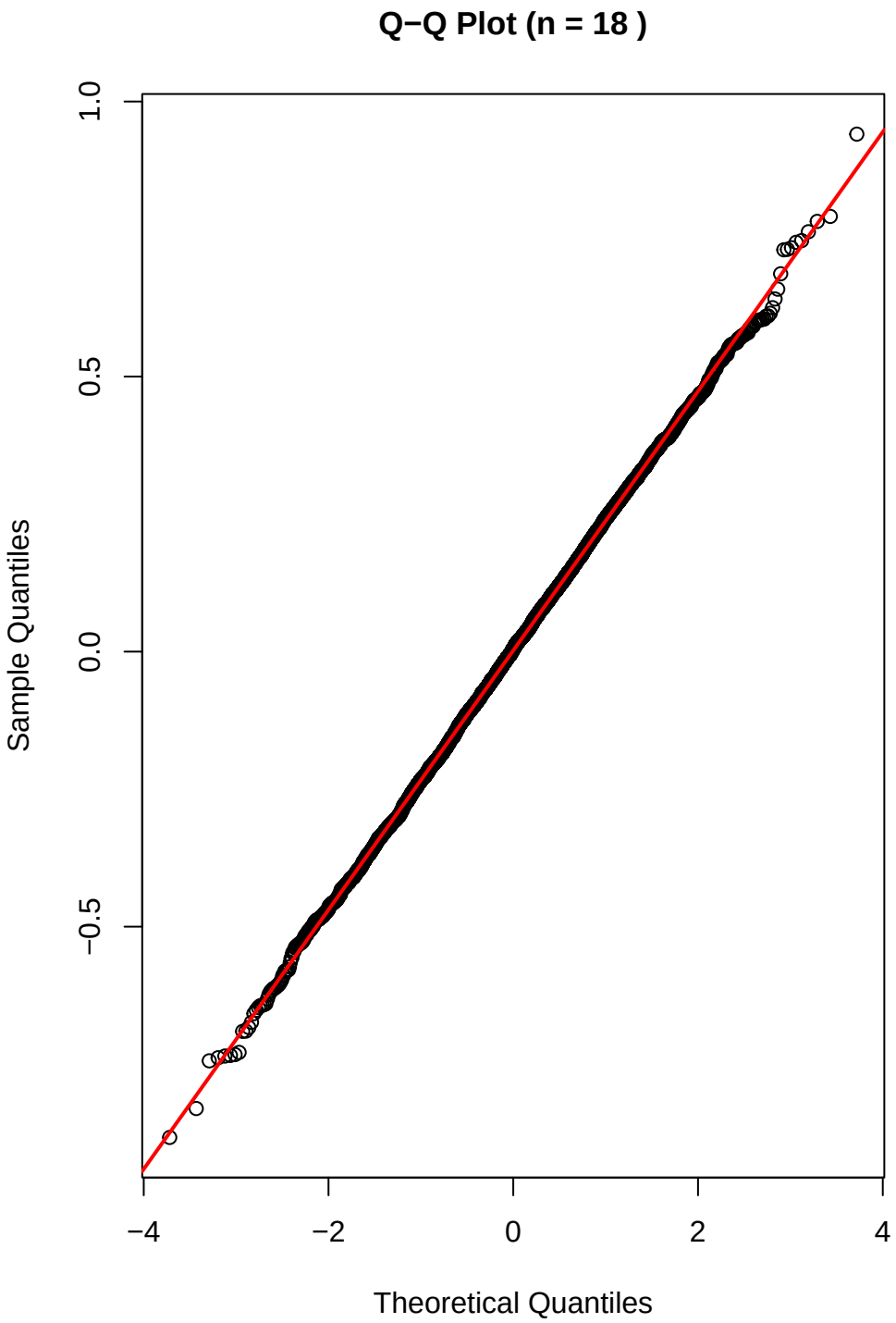


Excess Kurtosis = −0.119


Standard Normal — n = 17
Histogram (n = 17)
Density
0.0
0.5
1.0
1.5
−1.0
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 17)
Sample Quantiles
−1.0
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0706

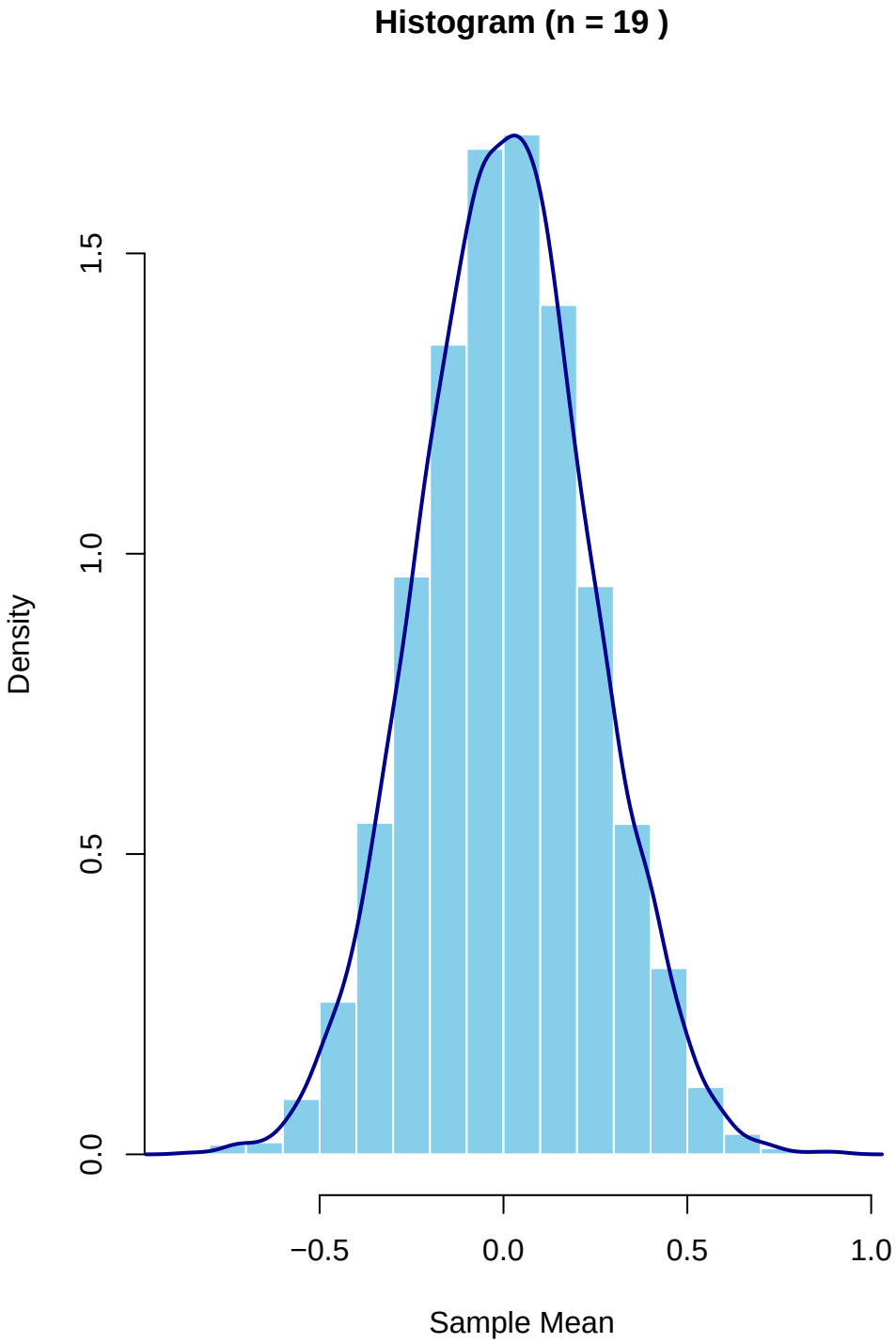
Standard Normal — n = 18

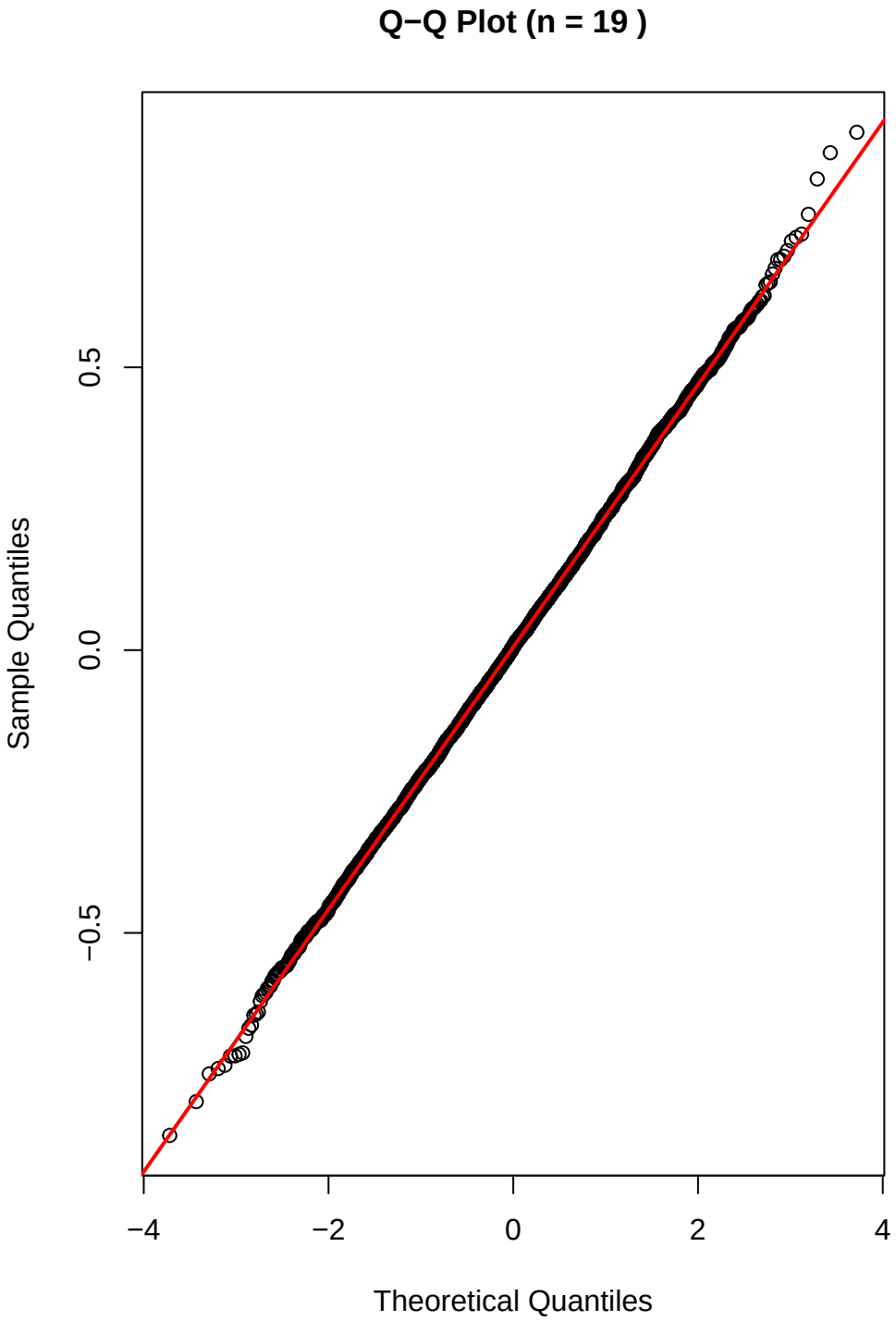




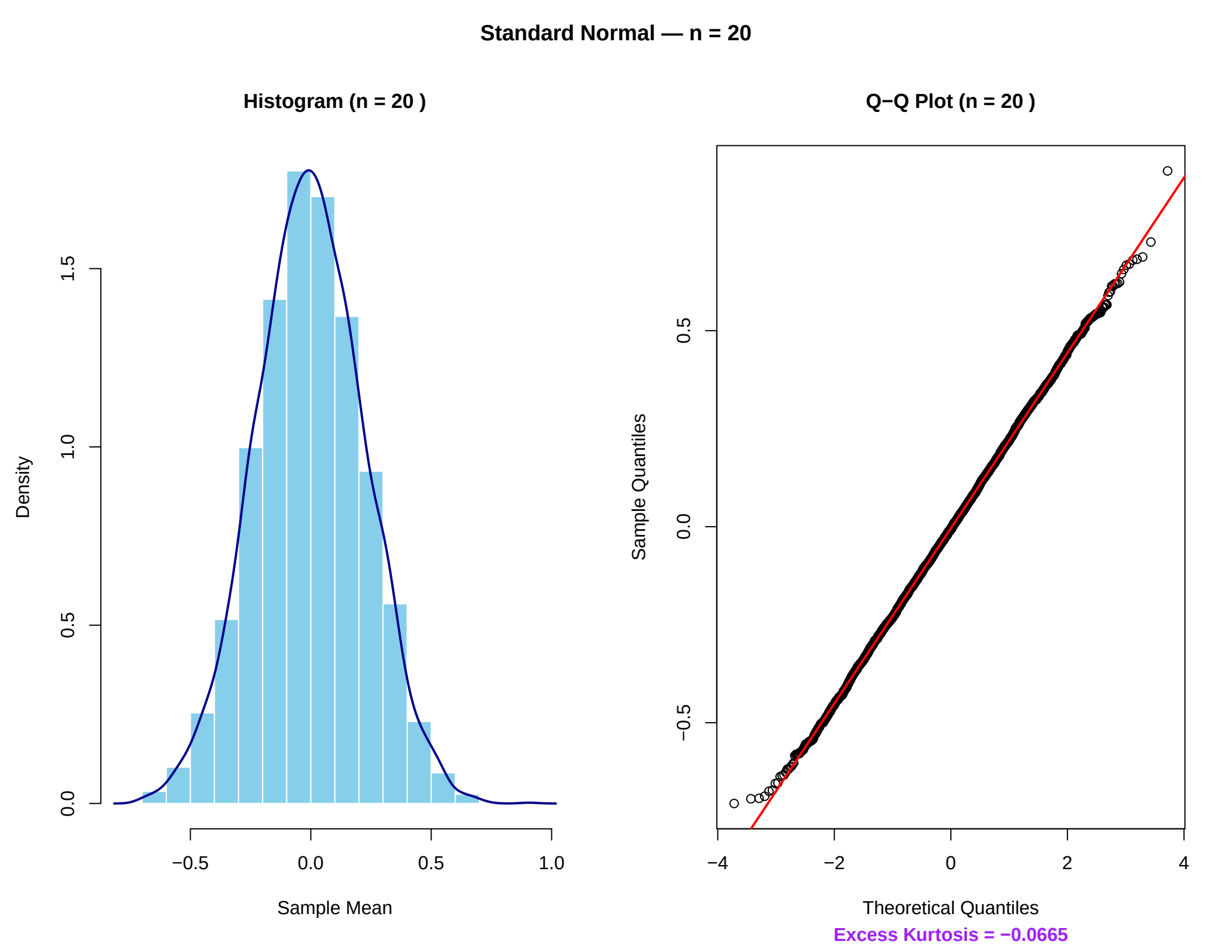
Excess Kurtosis = −0.0475

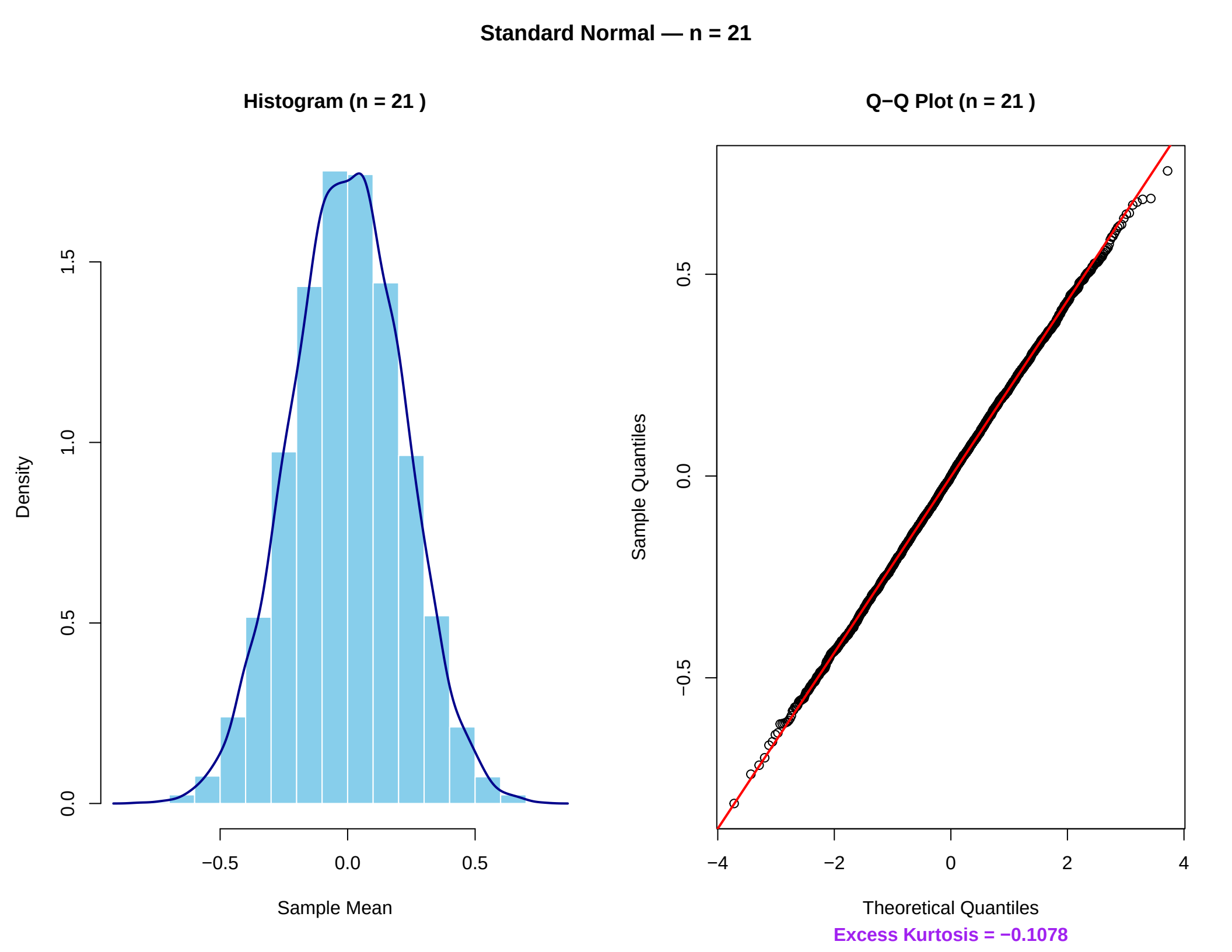
Standard Normal — n = 19



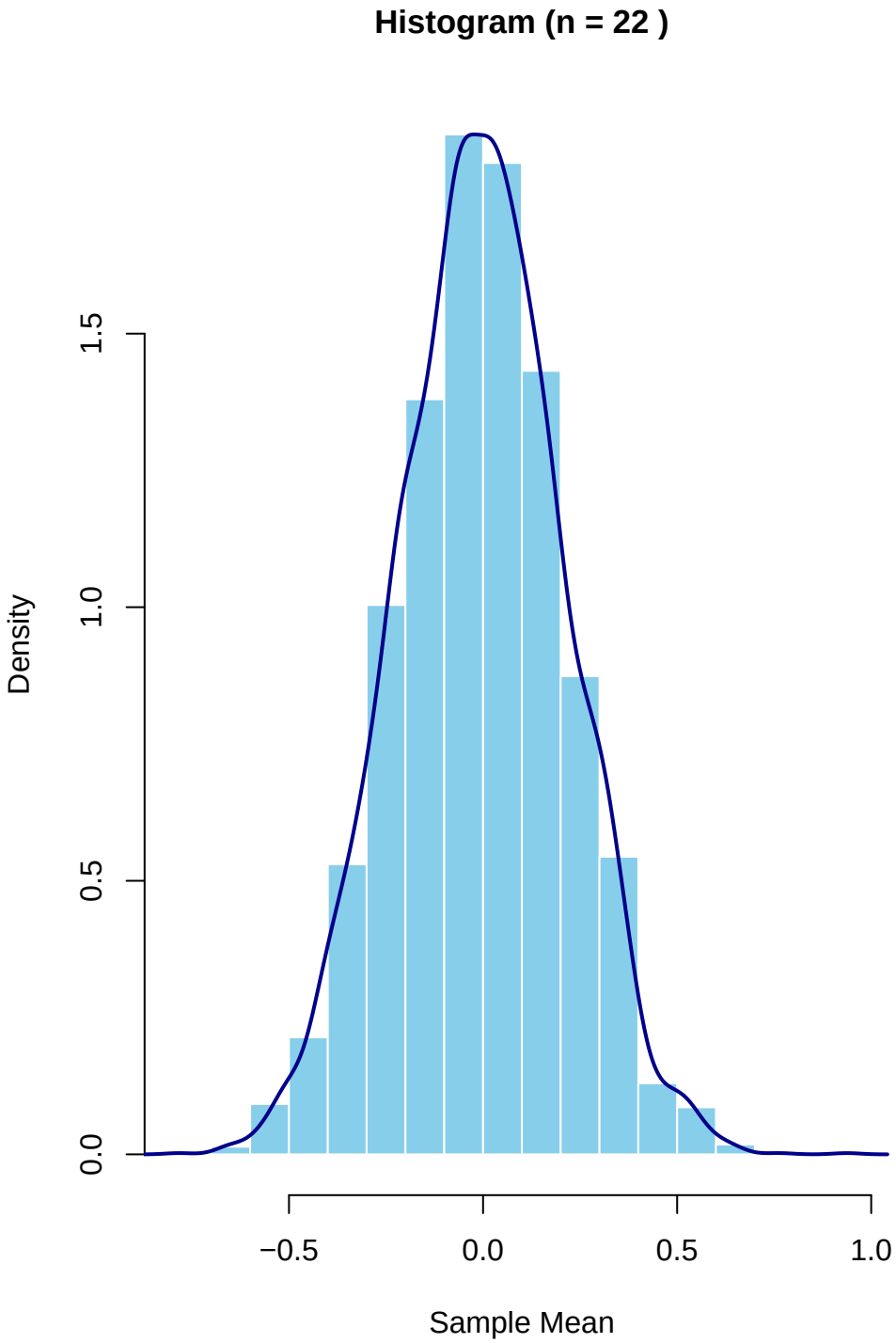


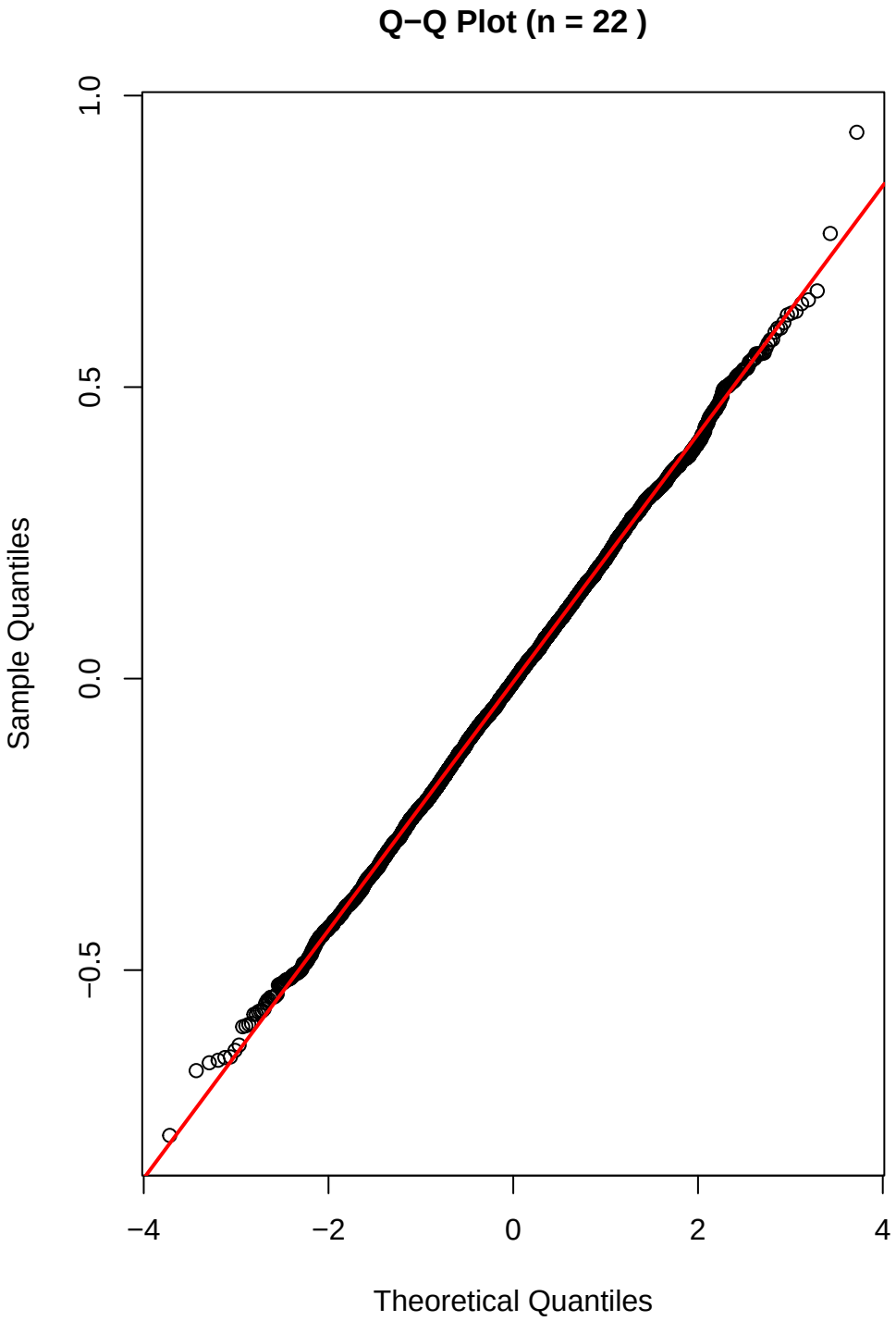
Excess Kurtosis = 0.0235


Standard Normal — n = 20
Histogram (n = 20)
Density
0.0
0.5
1.0
1.5
−0.5
0.0
0.5
1.0
Sample Mean
Q−Q Plot (n = 20)
Sample Quantiles
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0665

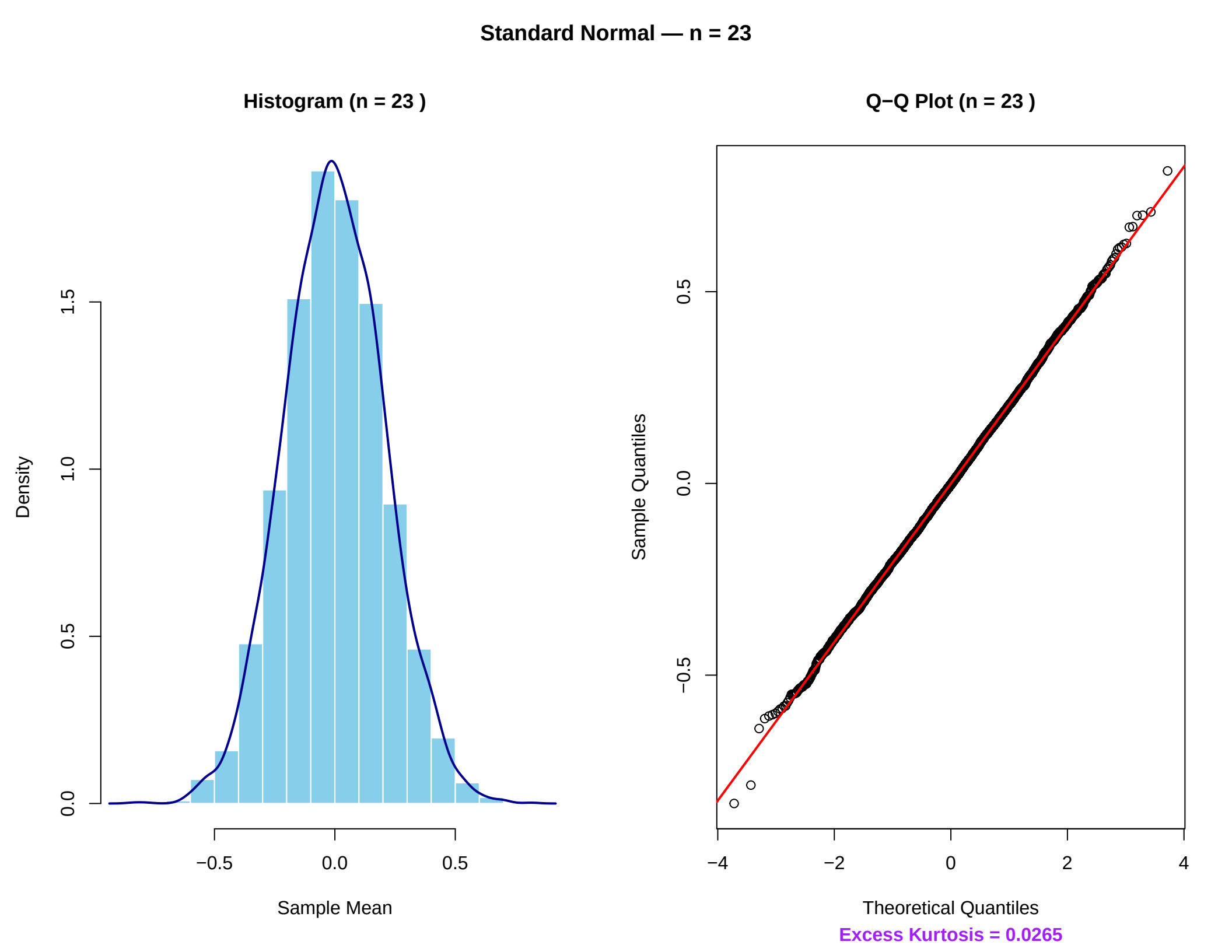

Standard Normal — n = 21
Histogram (n = 21)
Density
0.0
0.5
1.0
1.5
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 21)
Sample Quantiles
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.1078

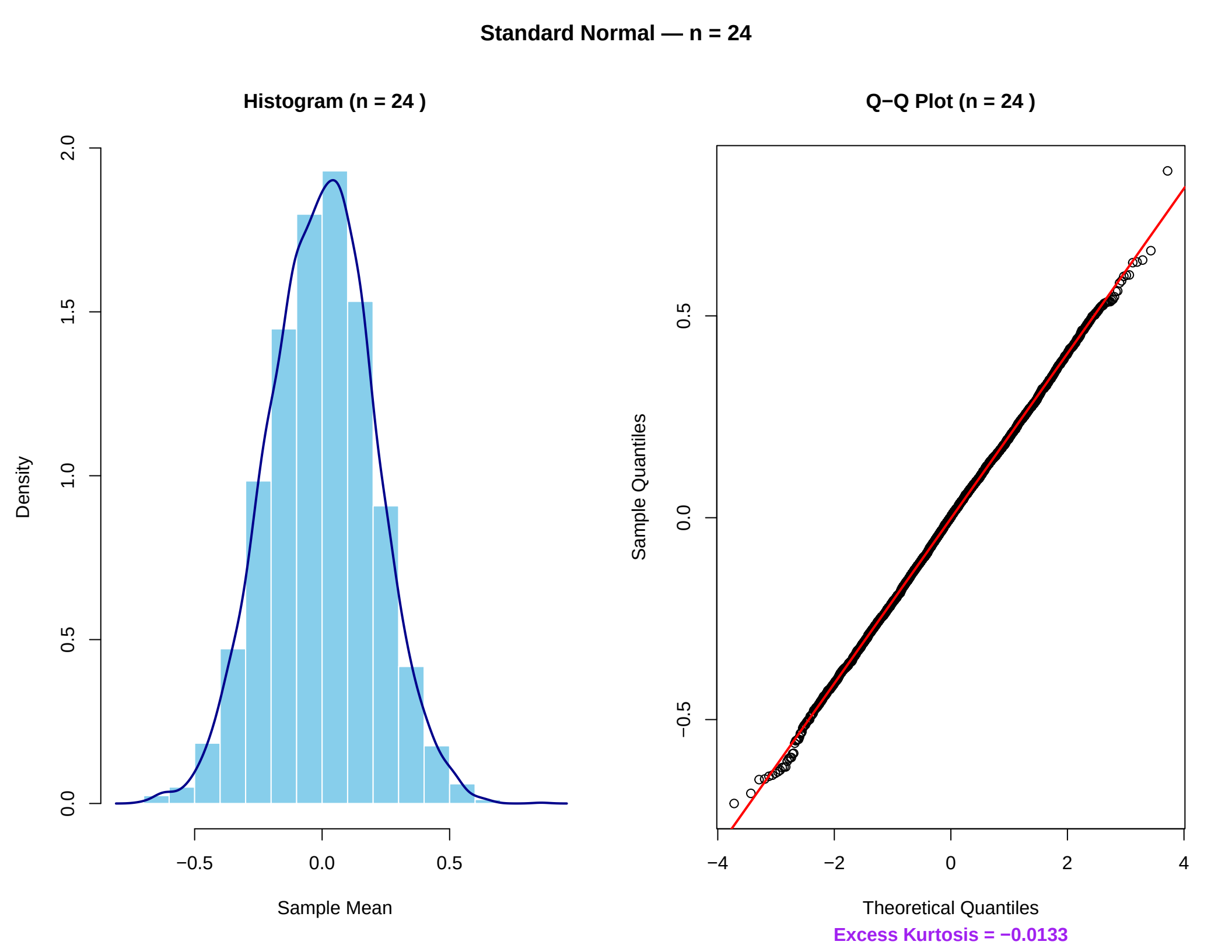
Standard Normal — n = 22

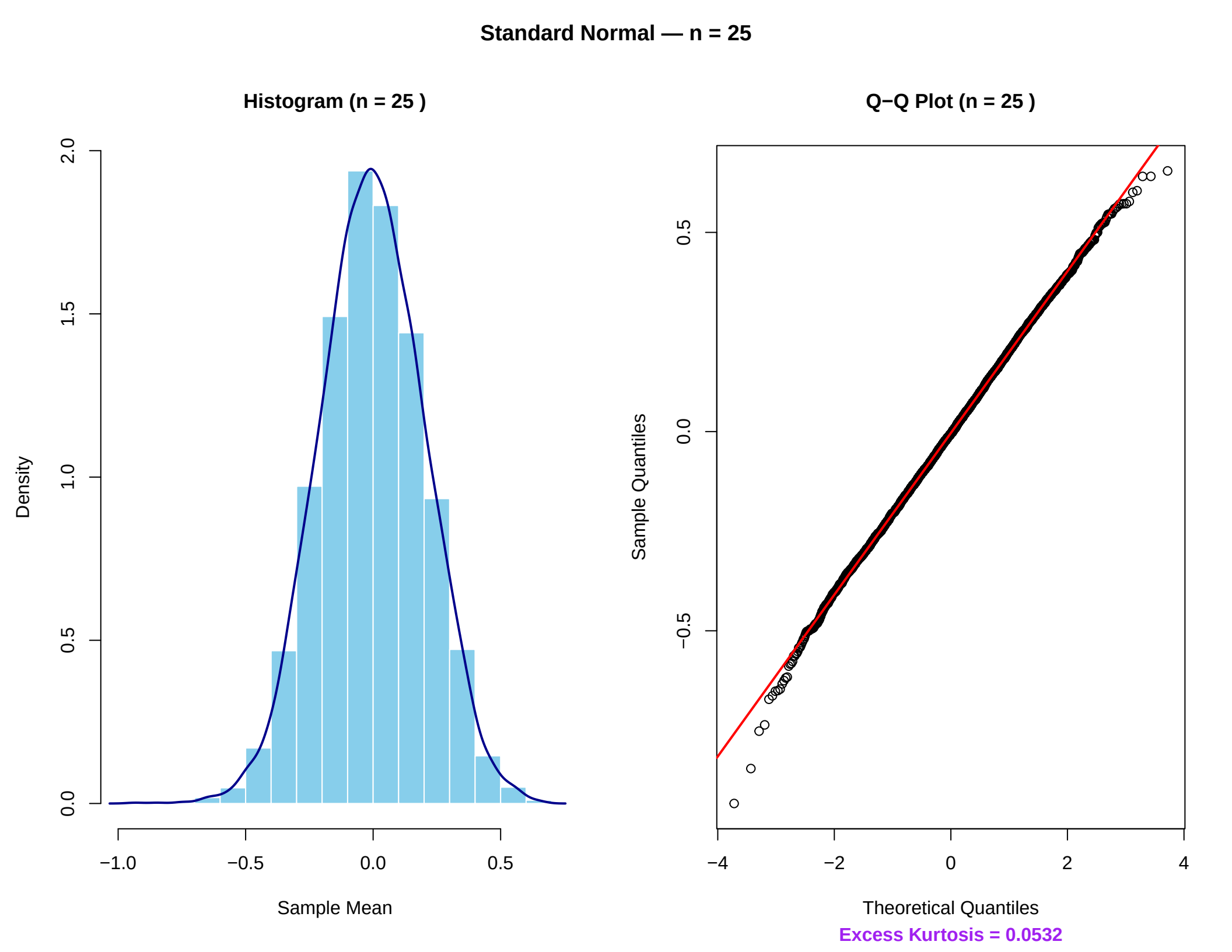


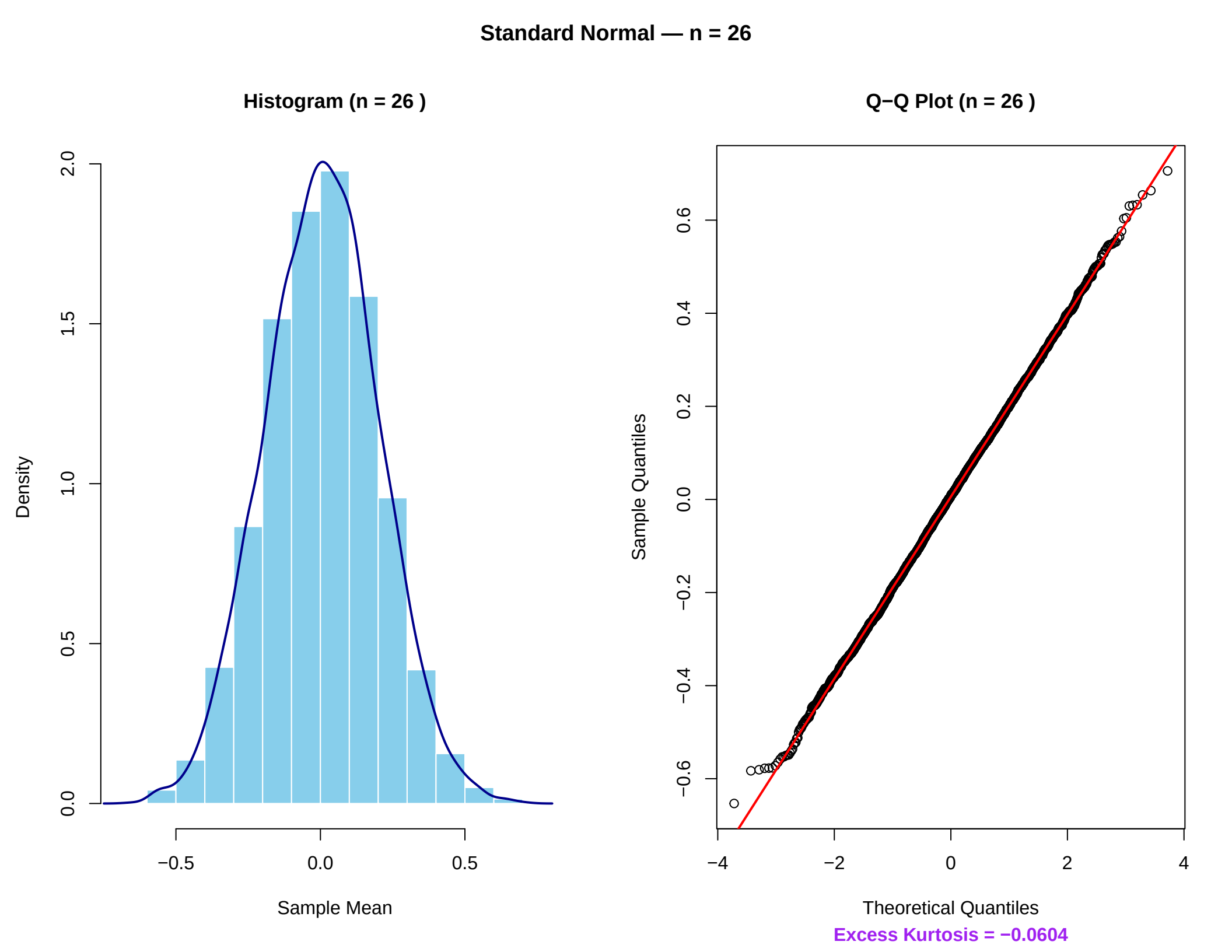


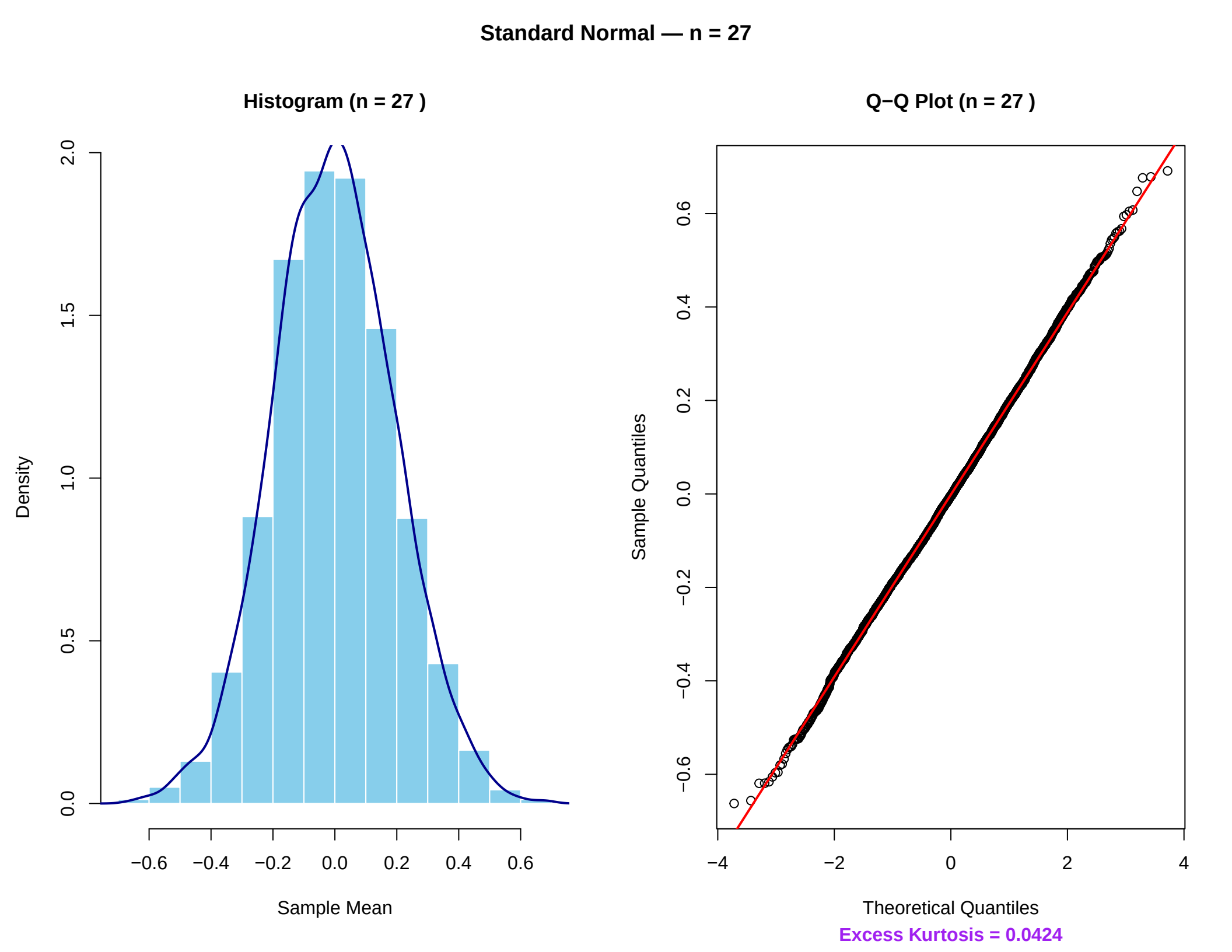
Excess Kurtosis = −0.0352

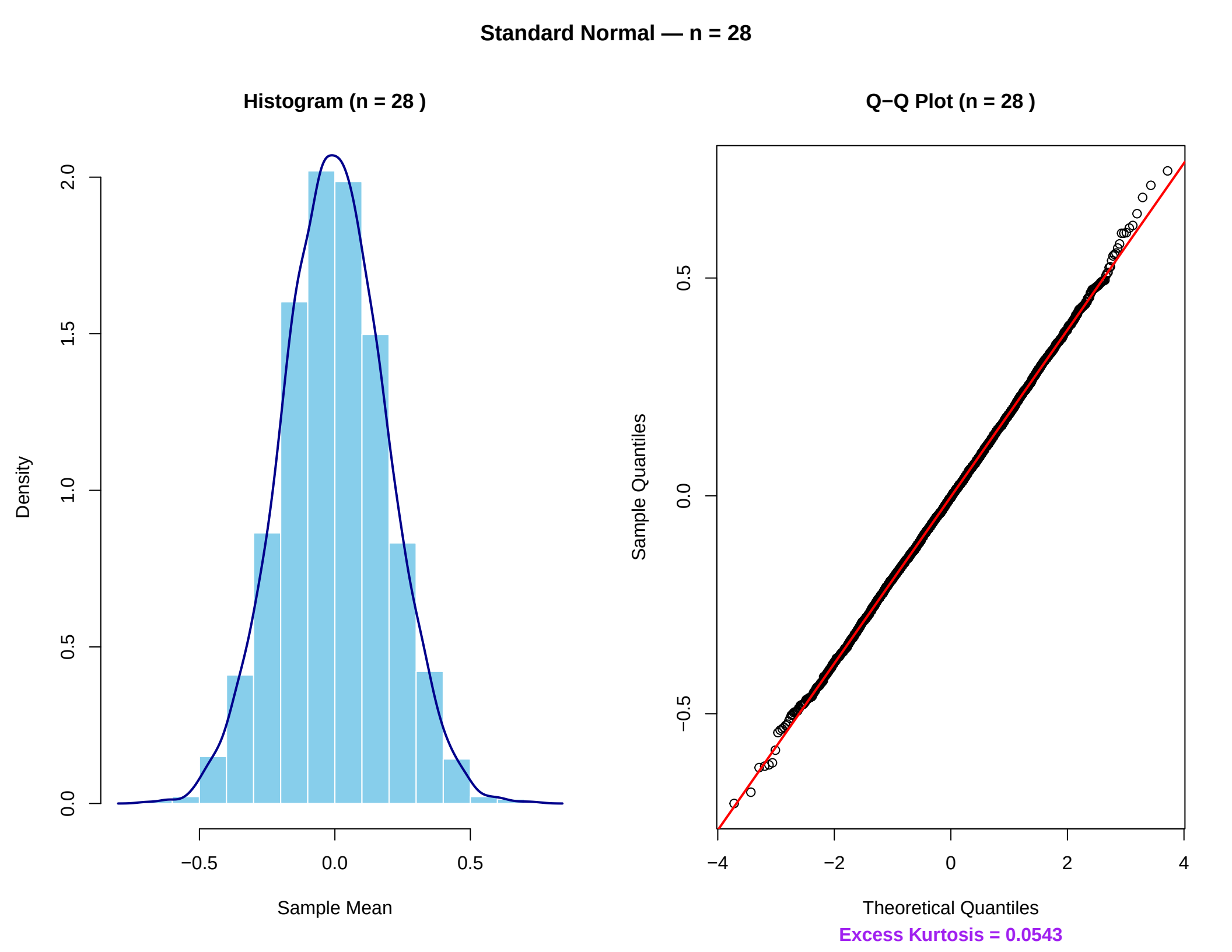

Standard Normal — n = 23
Histogram (n = 23)
Density
0.0
0.5
1.0
1.5
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 23)
Sample Quantiles
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0265

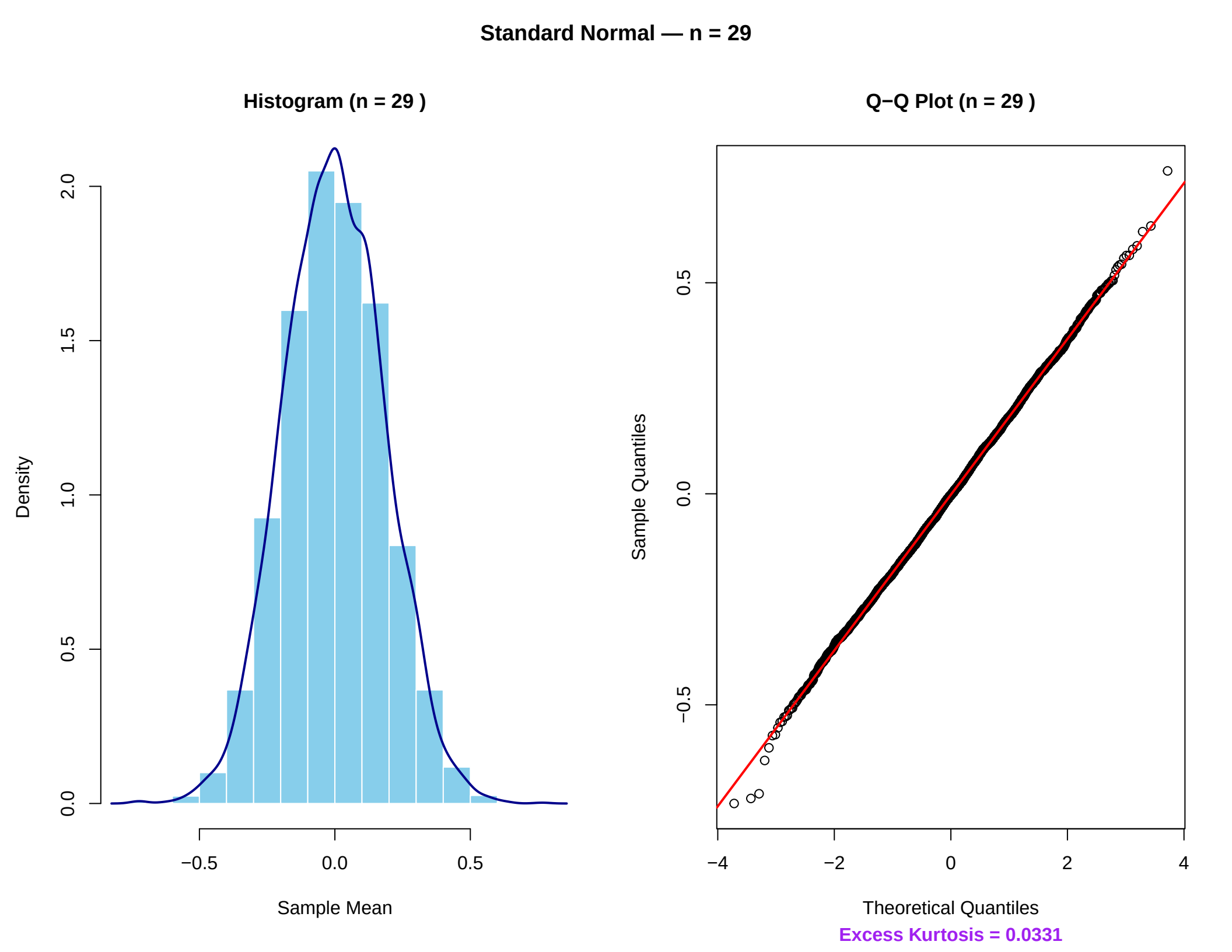

Standard Normal — n = 24
Histogram (n = 24)
Density
0.0
0.5
1.0
1.5
2.0
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 24)
Sample Quantiles
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0133


Standard Normal — n = 25
Histogram (n = 25)
Density
2.0
1.5
1.0
0.5
0.0
−1.0
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 25)
Sample Quantiles
0.5
0.0
−0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0532

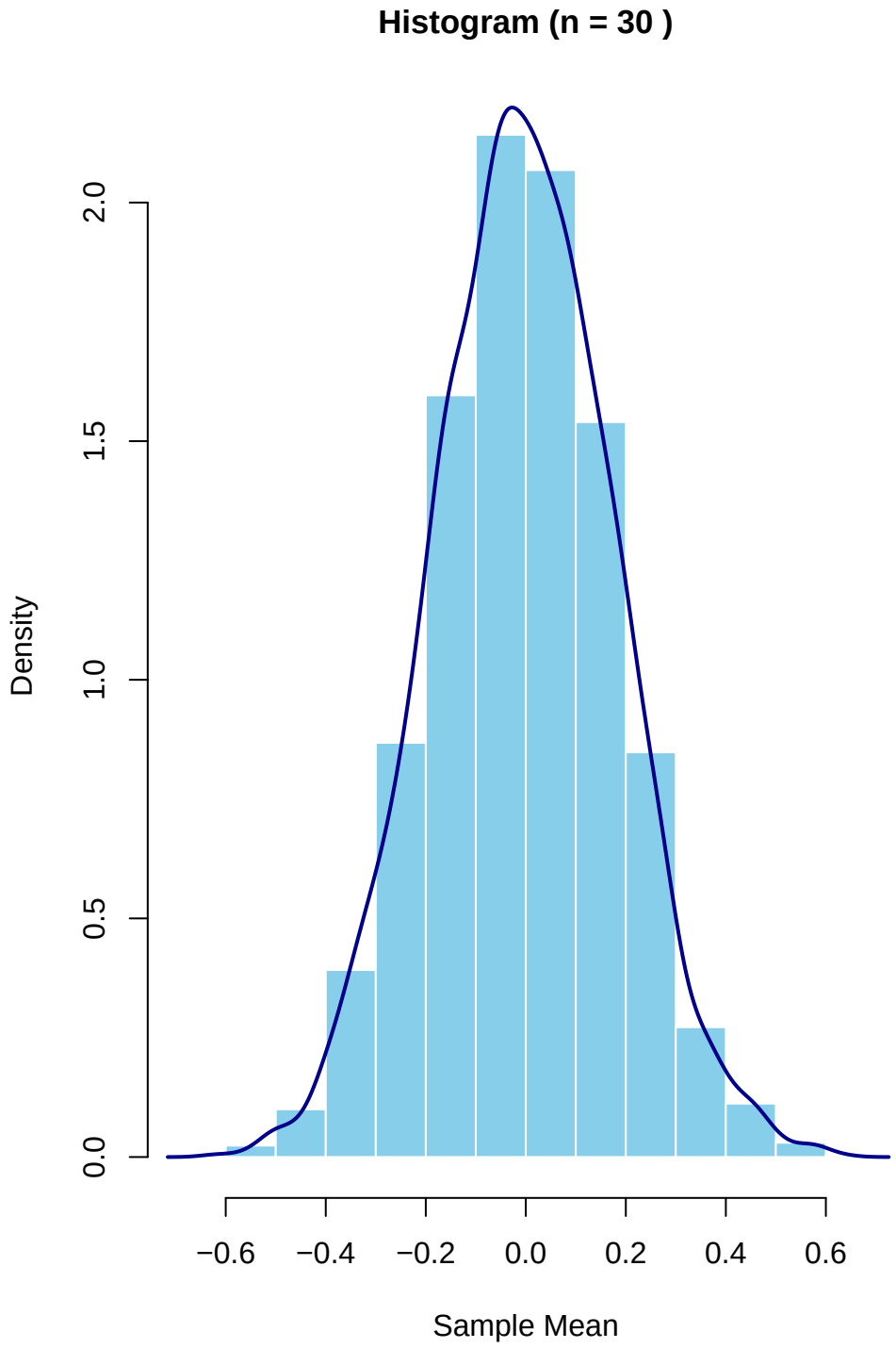

Standard Normal — n = 26
Histogram (n = 26)
Density
0.0
0.5
1.0
1.5
2.0
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 26)
Sample Quantiles
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0604

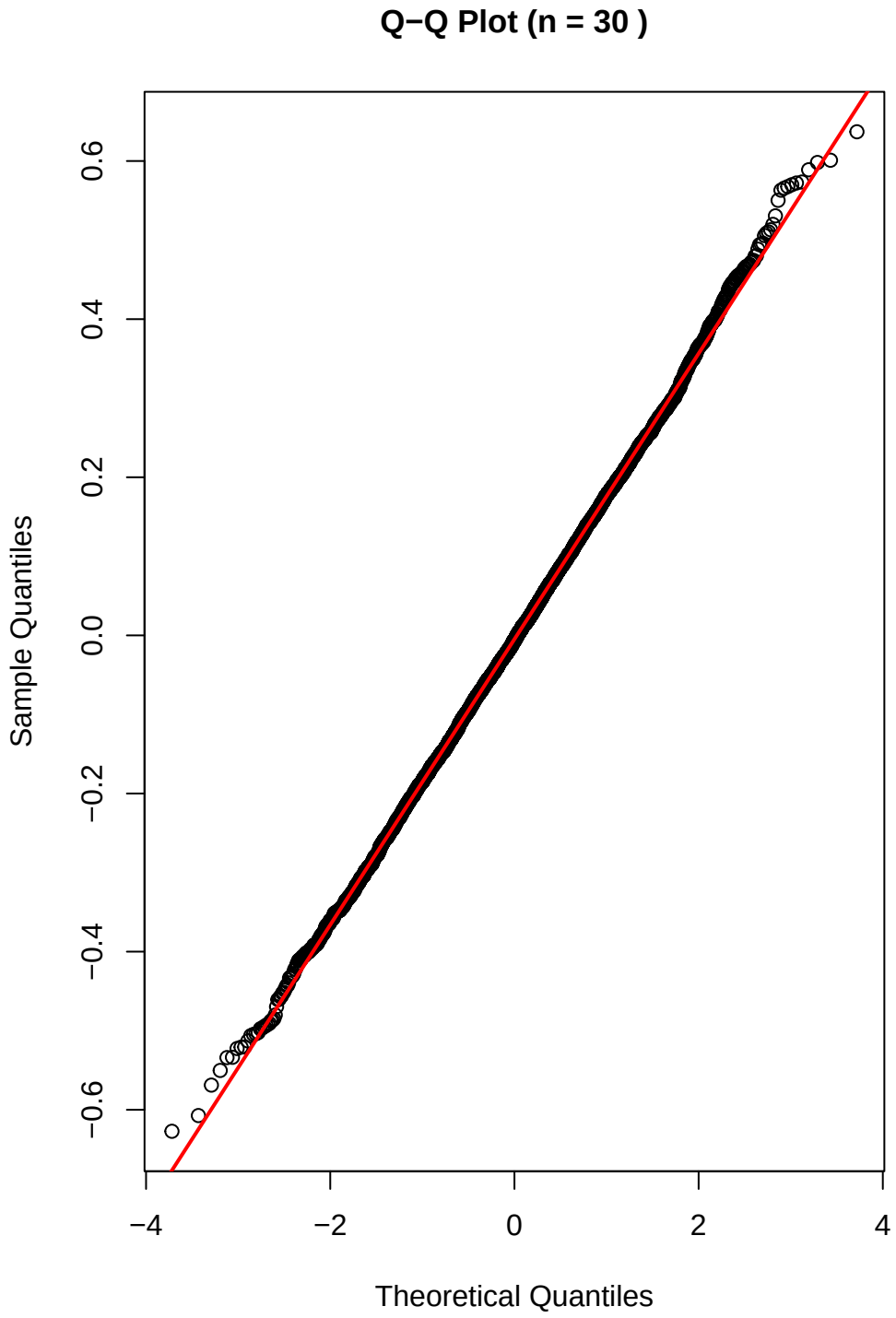

Standard Normal — n = 27
Histogram (n = 27)
Density
Sample Mean
Q−Q Plot (n = 27)
Sample Quantiles
Theoretical Quantiles
Excess Kurtosis = 0.0424


Standard Normal — n = 28
Histogram (n = 28)
Density
0.0
0.5
1.0
1.5
2.0
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 28)
Sample Quantiles
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0543

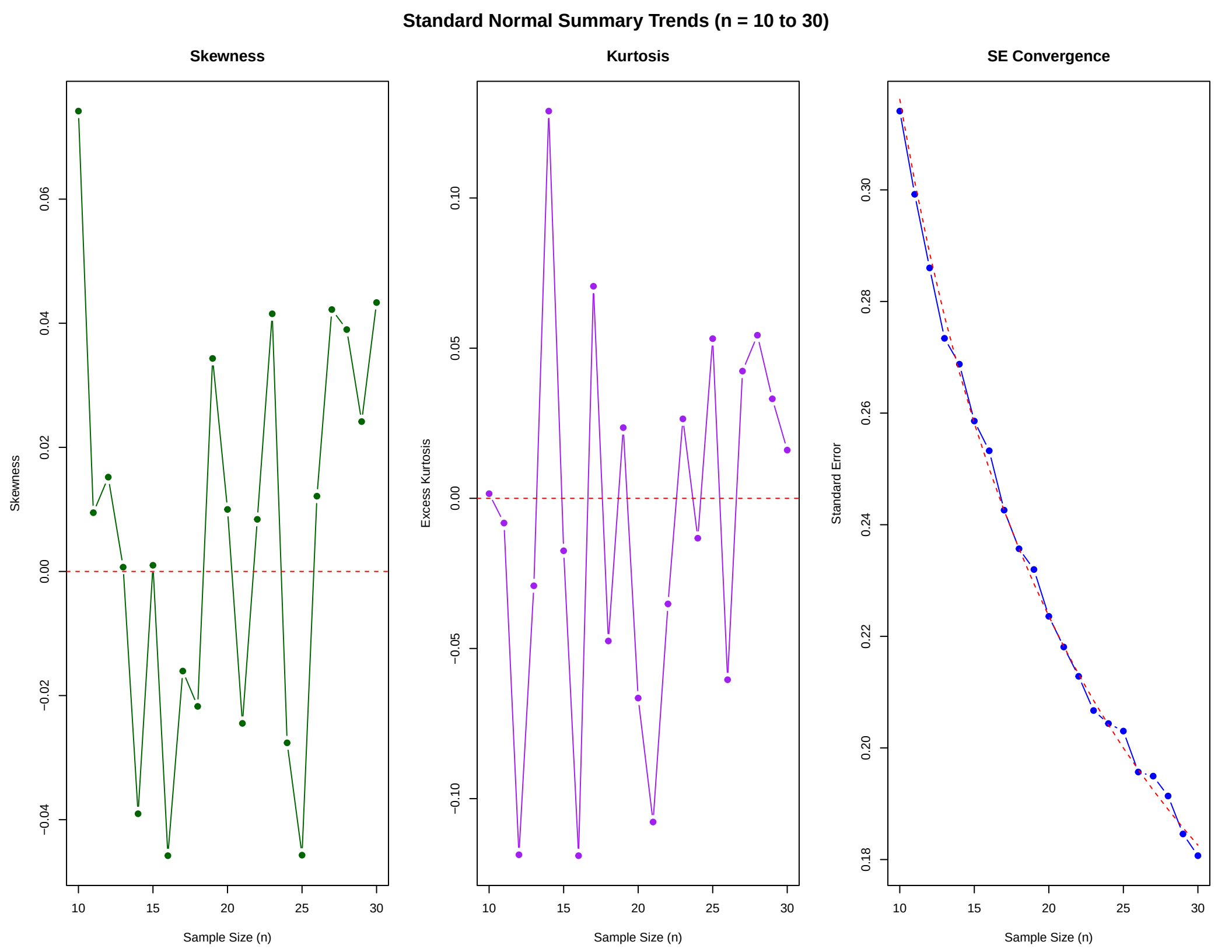

Standard Normal — n = 29
Histogram (n = 29)
Density
0.0
0.5
1.0
1.5
2.0
−0.5
0.0
0.5
Sample Mean
Q−Q Plot (n = 29)
Sample Quantiles
−0.5
0.0
0.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0331

Standard Normal — n = 30

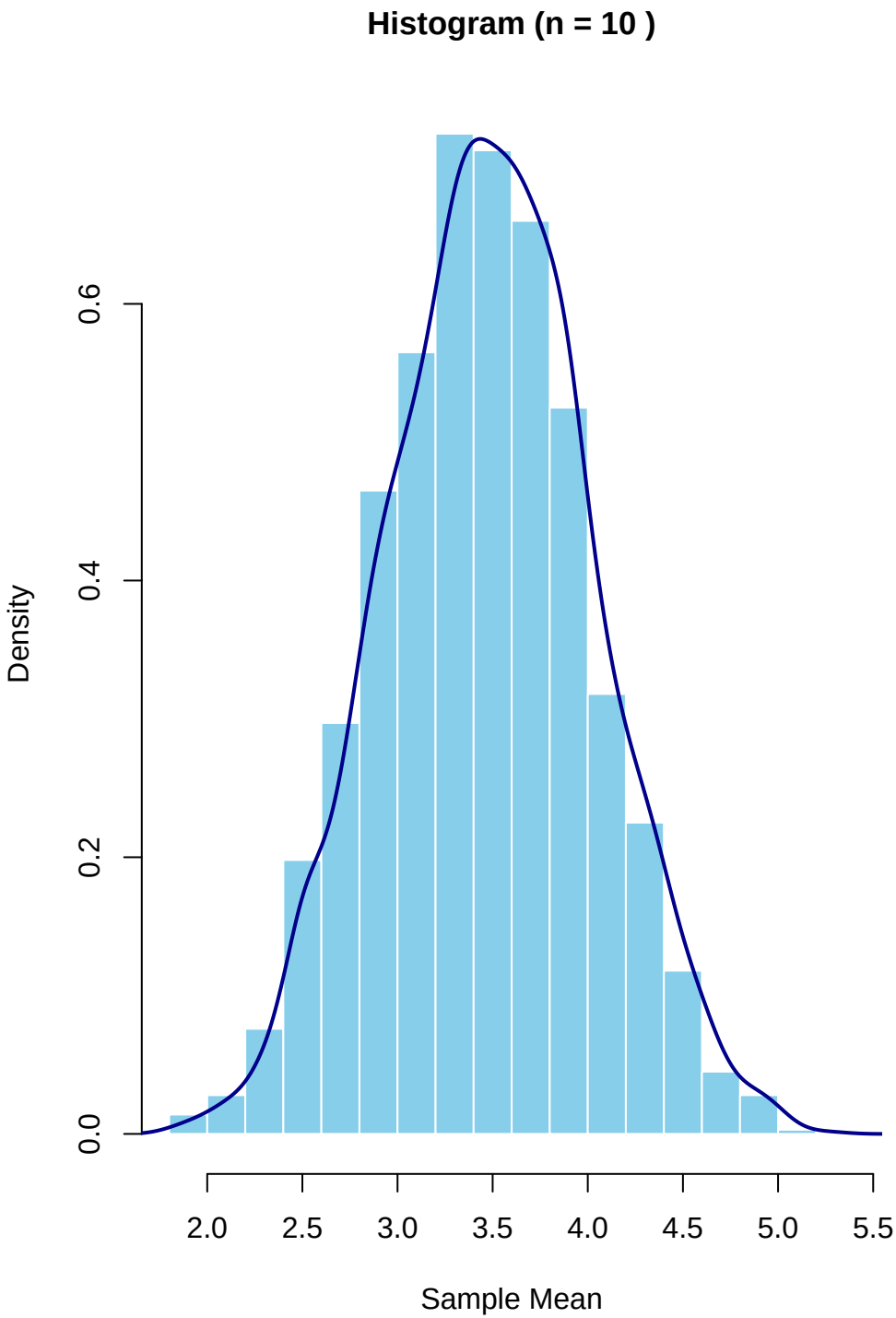


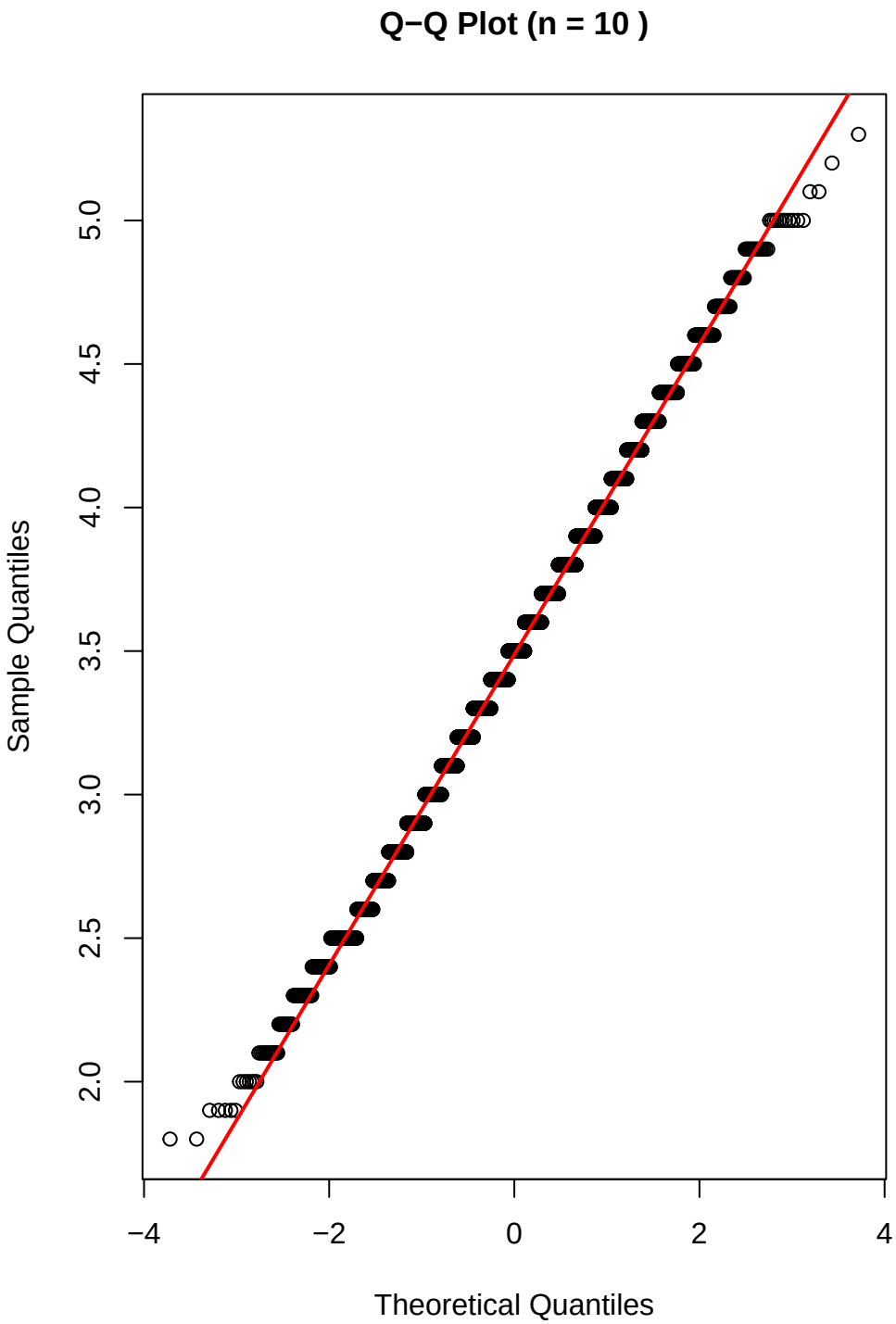


Excess Kurtosis = 0.0161

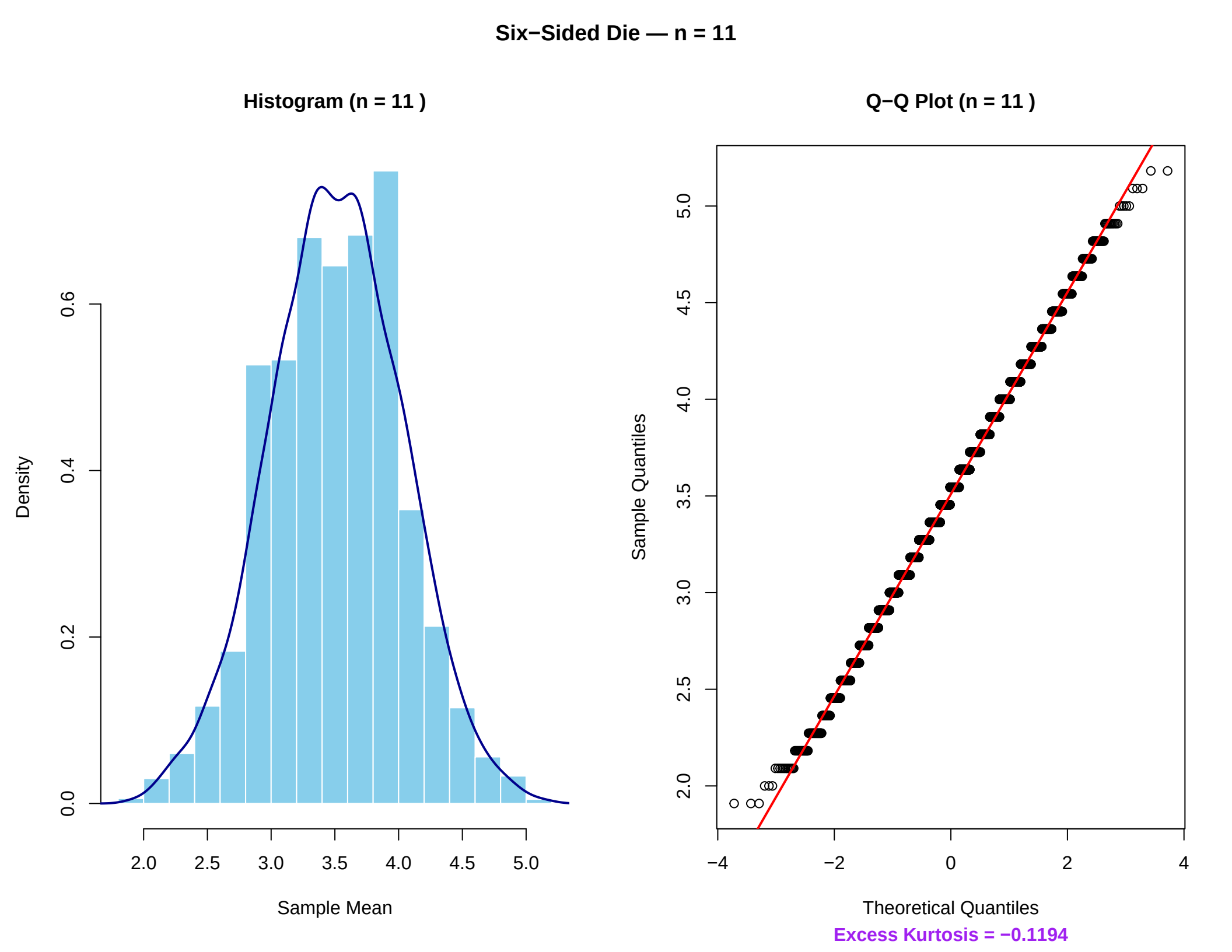

Standard Normal Summary Trends (n = 10 to 30)
Skewness
Skewness
0.06
0.04
0.02
0.00
−0.02
−0.04
10
15
20
25
30
Sample Size (n)
Kurtosis
Excess Kurtosis
0.10
0.05
0.00
−0.05
−0.10
10
15
20
25
30
Sample Size (n)
SE Convergence
Standard Error
0.30
0.28
0.26
0.24
0.22
0.20
0.18
10
15
20
25
30
Sample Size (n)

Six−Sided Die — n = 10

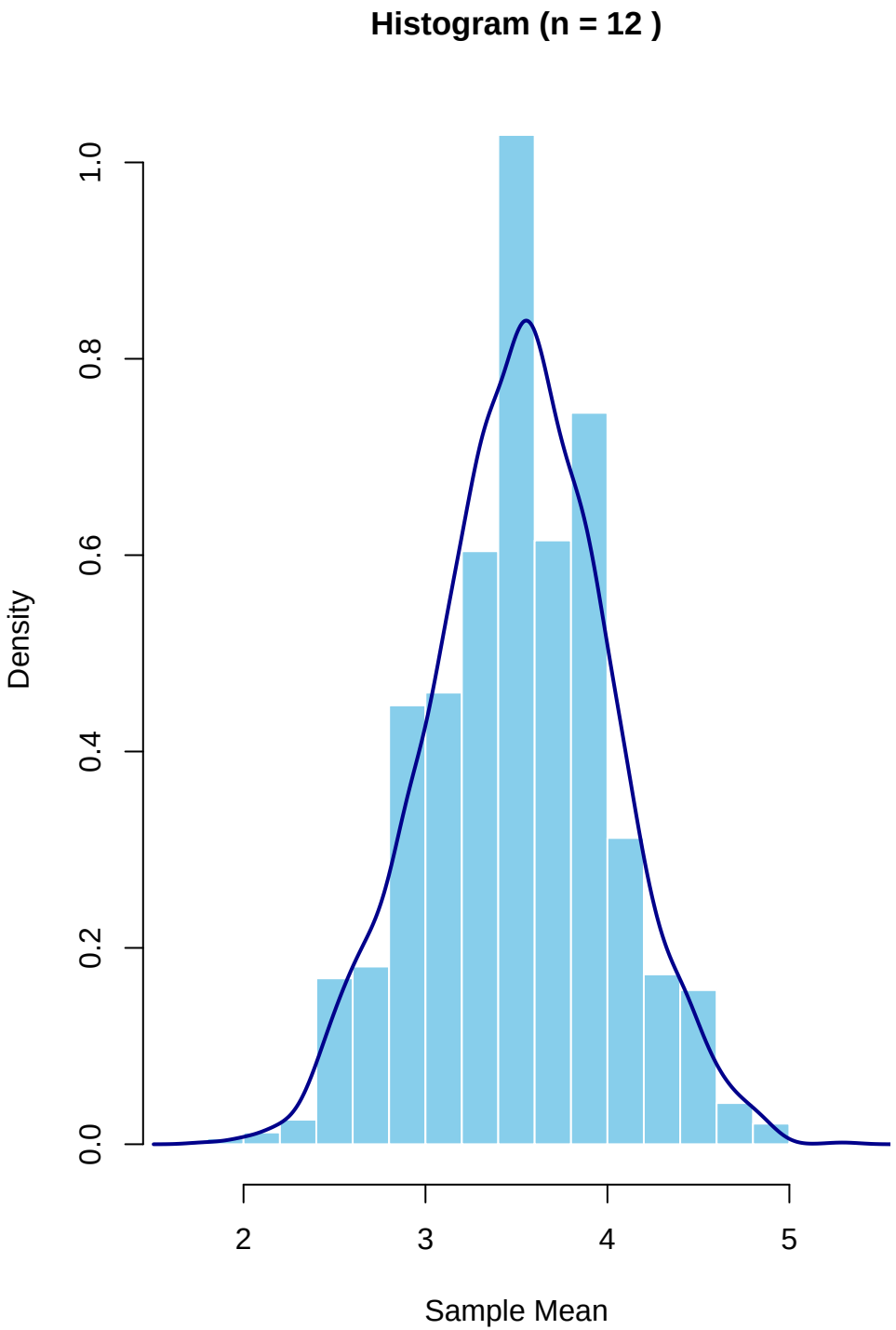


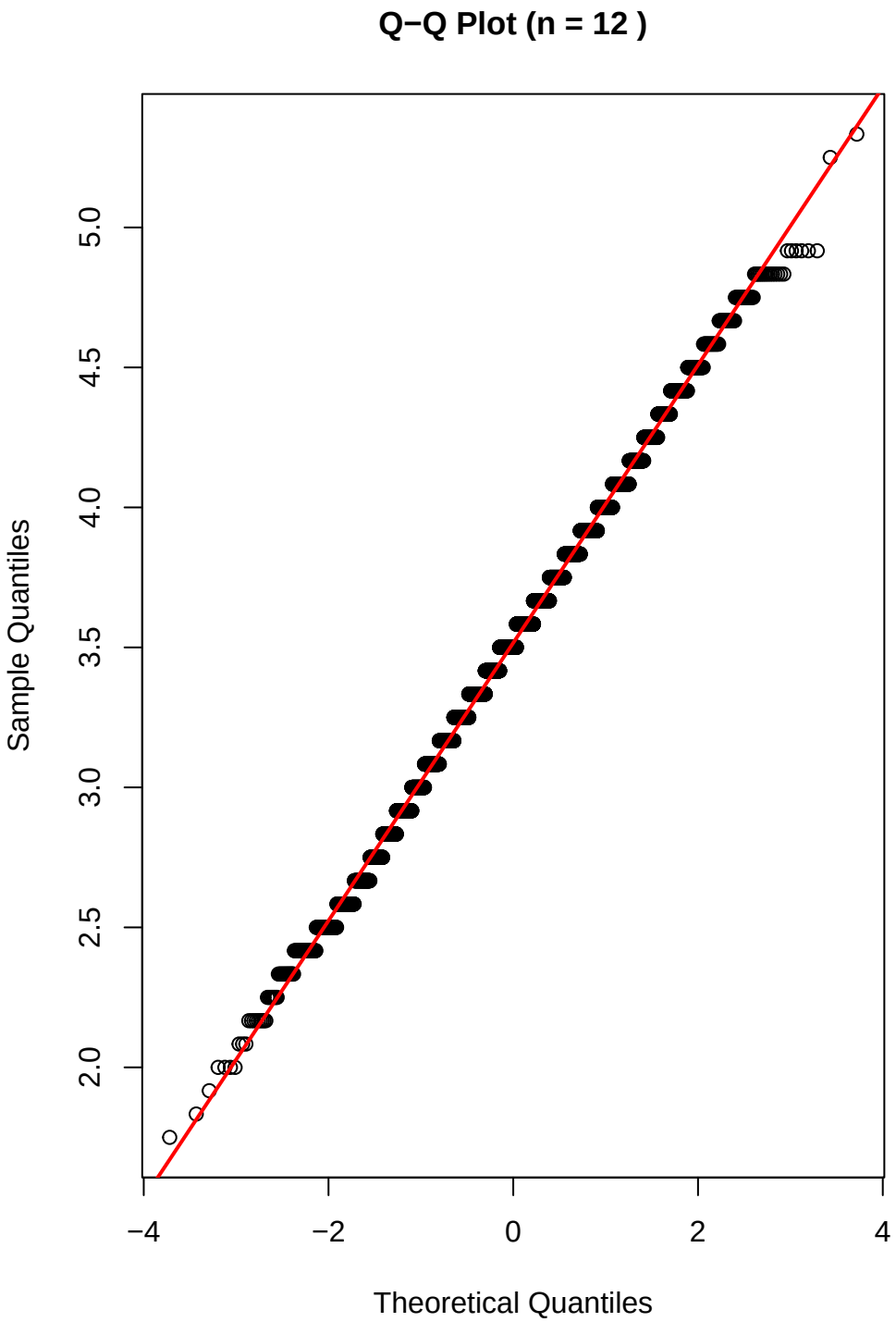


Excess Kurtosis = −0.1657


Six−Sided Die — n = 11
Histogram (n = 11)
Density
0.0
0.2
0.4
0.6
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample Mean
Q−Q Plot (n = 11)
Sample Quantiles
2.0
2.5
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.1194

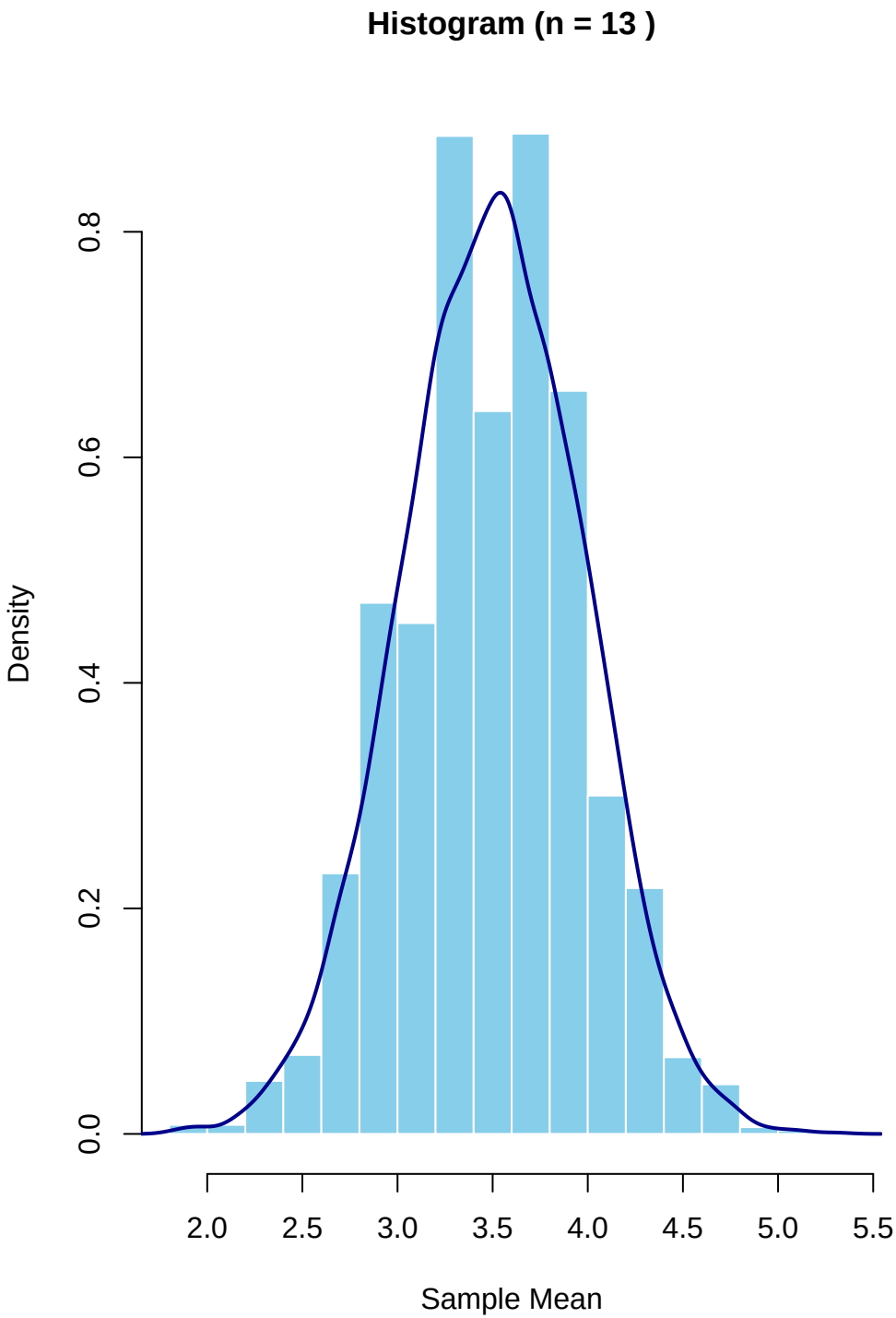
Six−Sided Die — n = 12

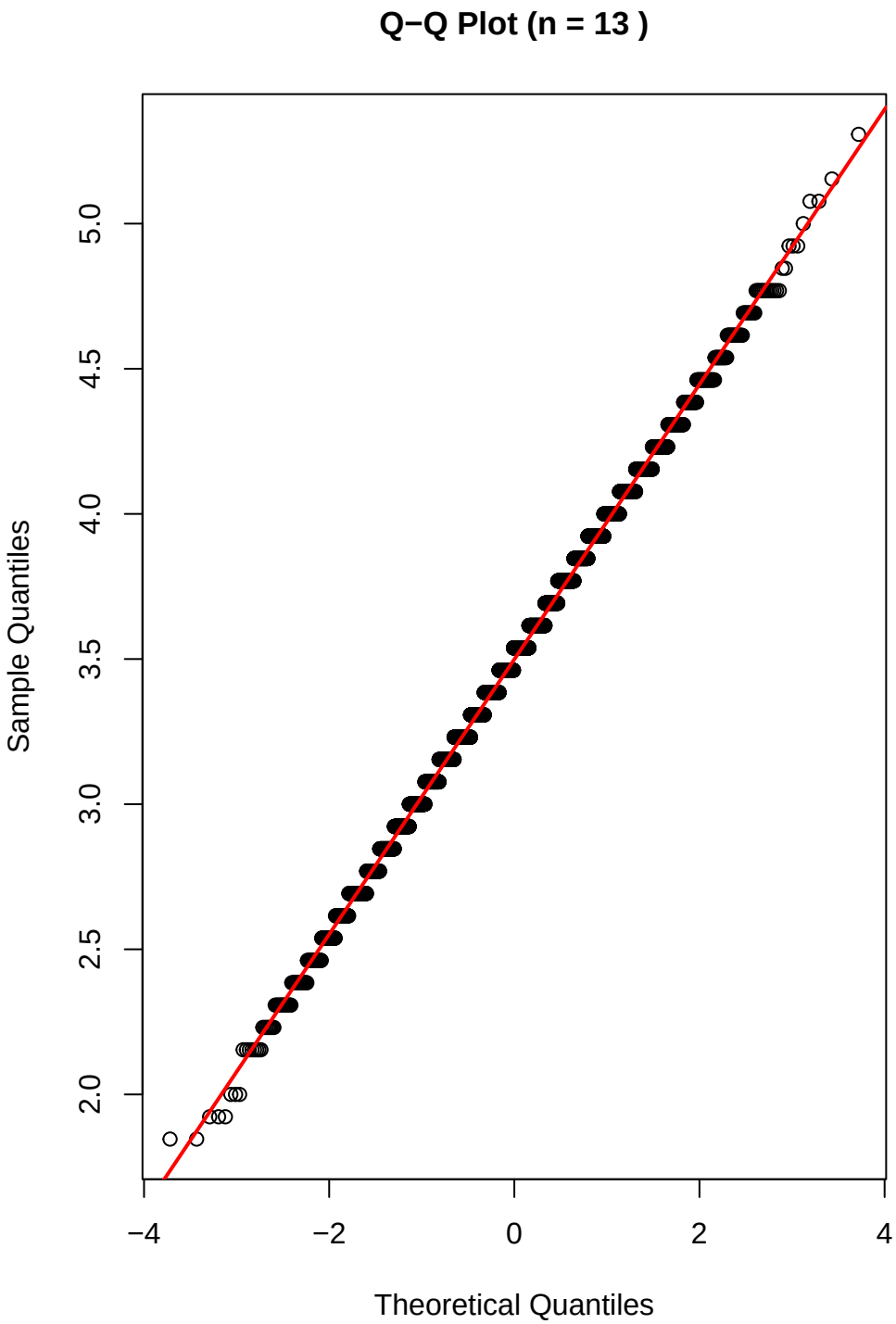




Excess Kurtosis = −0.048

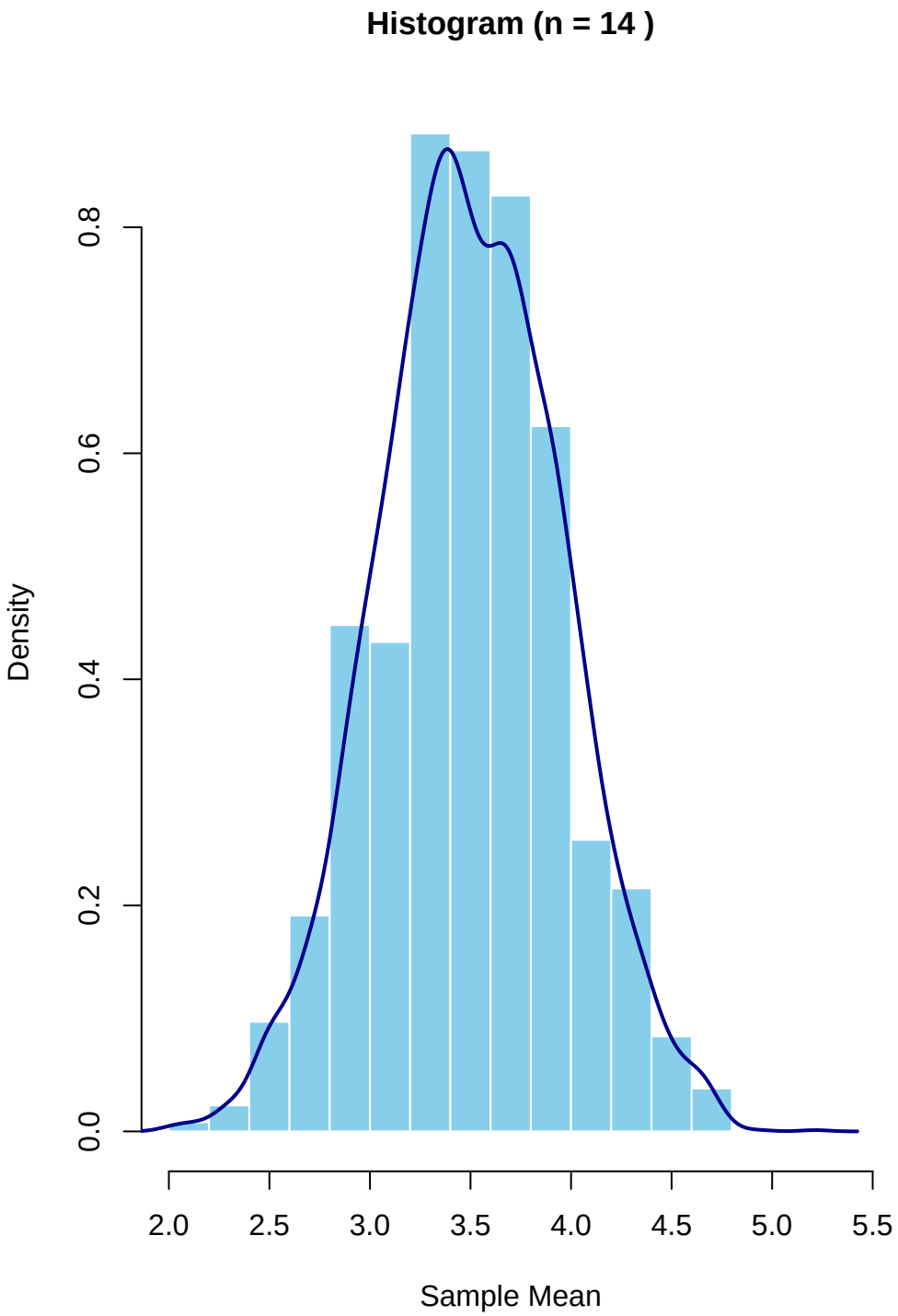
Six−Sided Die — n = 13

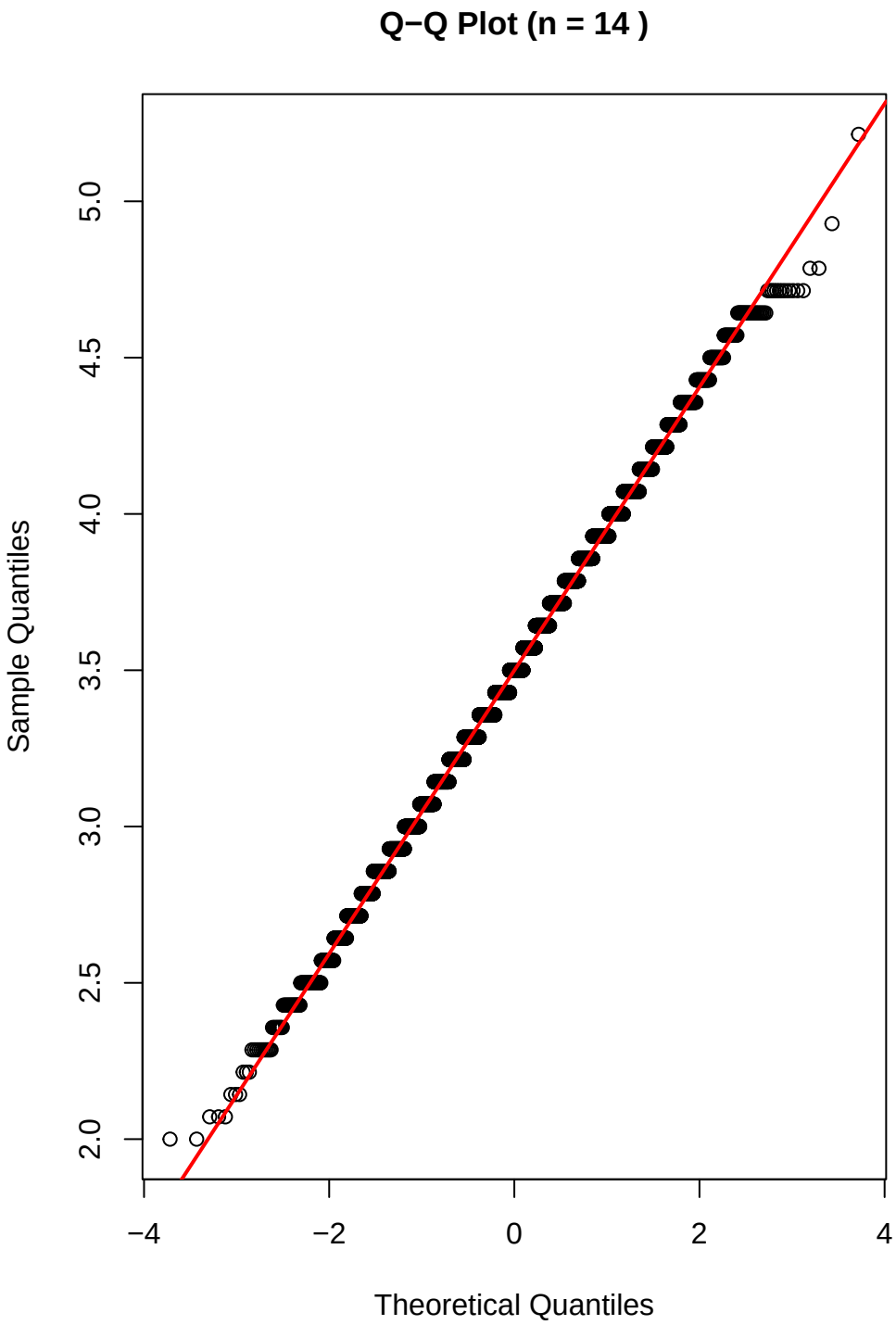




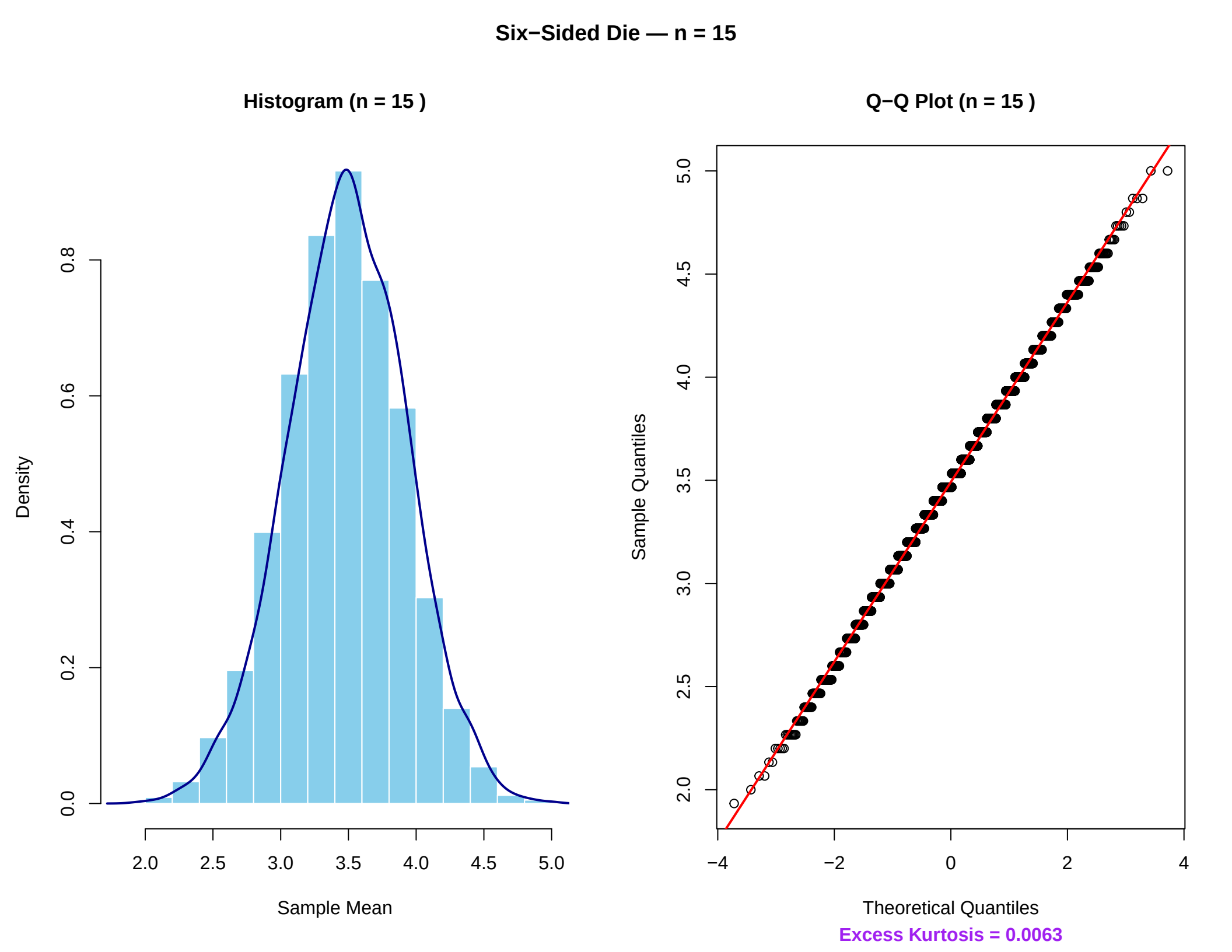
Excess Kurtosis = −0.0178

Six−Sided Die — n = 14

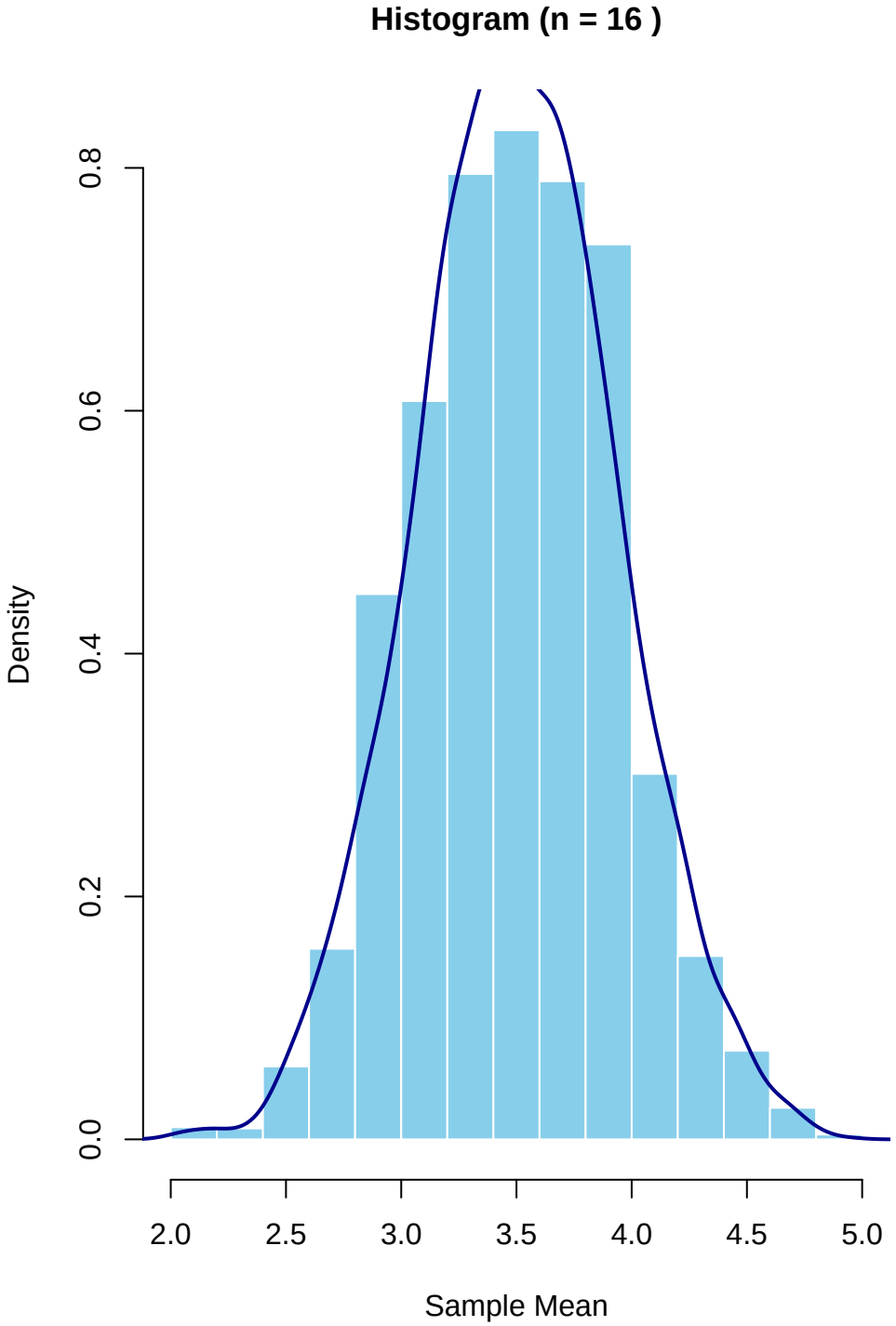


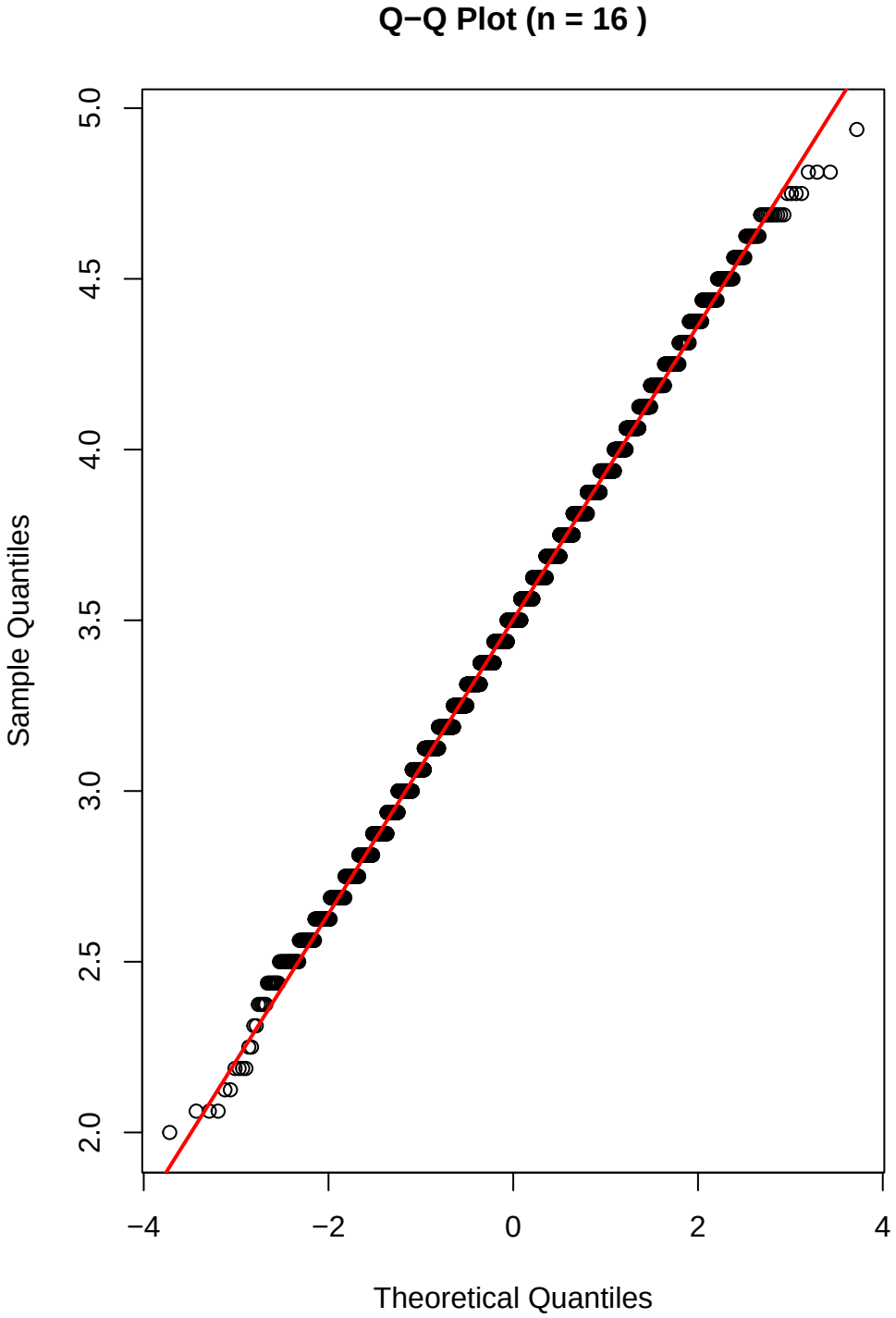


Excess Kurtosis = −0.1137


Six−Sided Die — n = 15
Histogram (n = 15)
Density
0.0
0.2
0.4
0.6
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample Mean
Q−Q Plot (n = 15)
Sample Quantiles
2.0
2.5
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0063

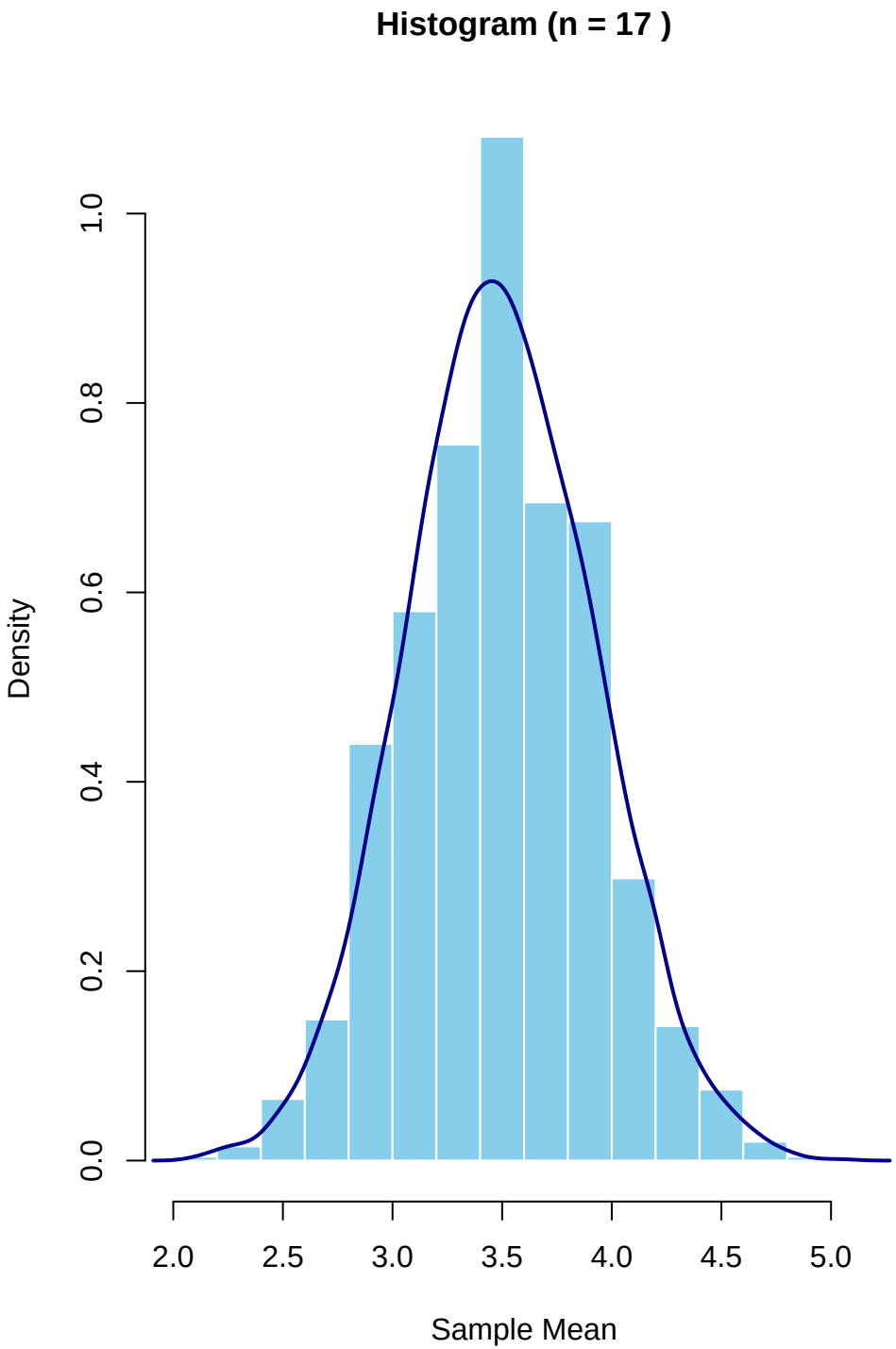
Six−Sided Die — n = 16

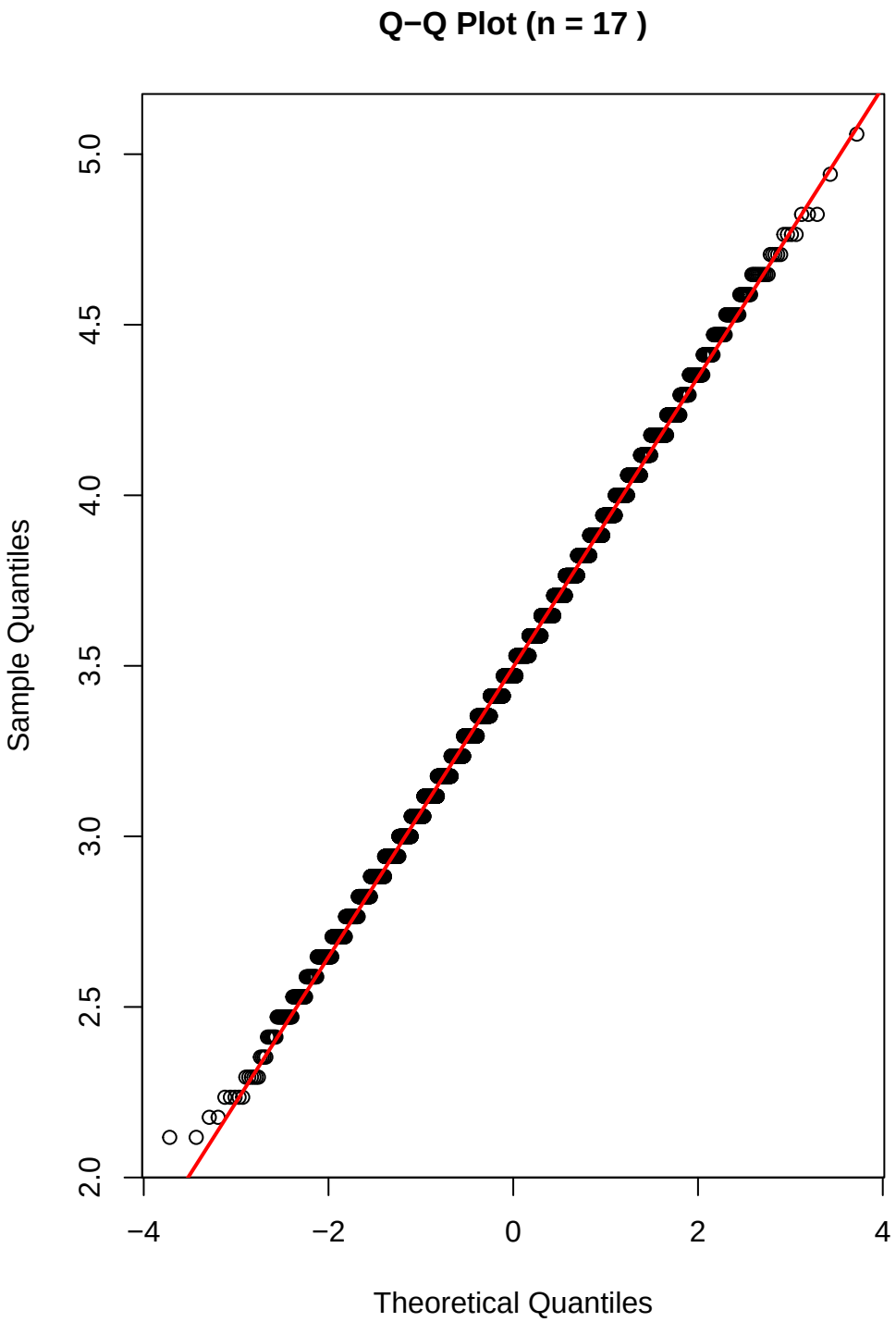




Excess Kurtosis = −0.0669

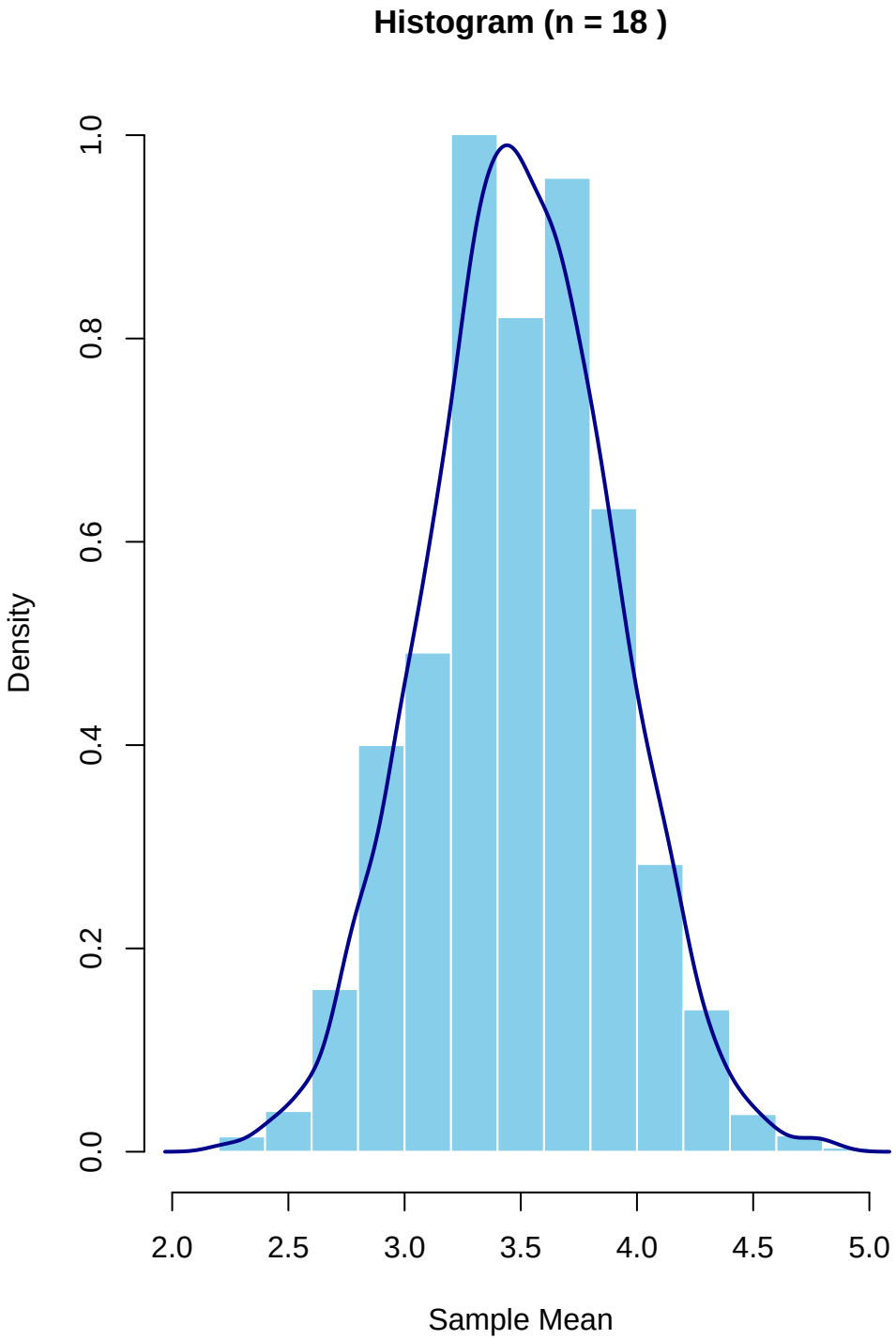
Six−Sided Die — n = 17

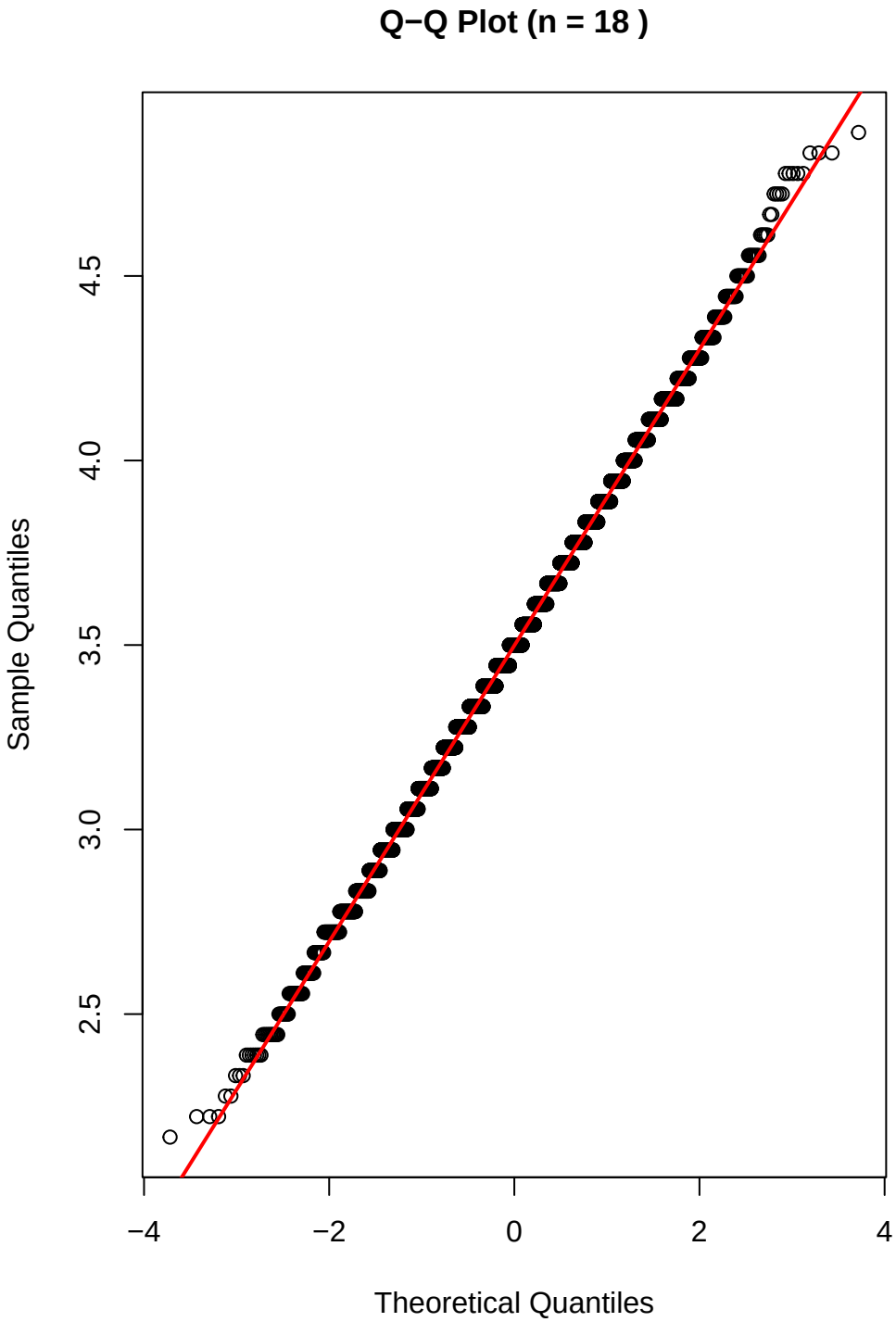




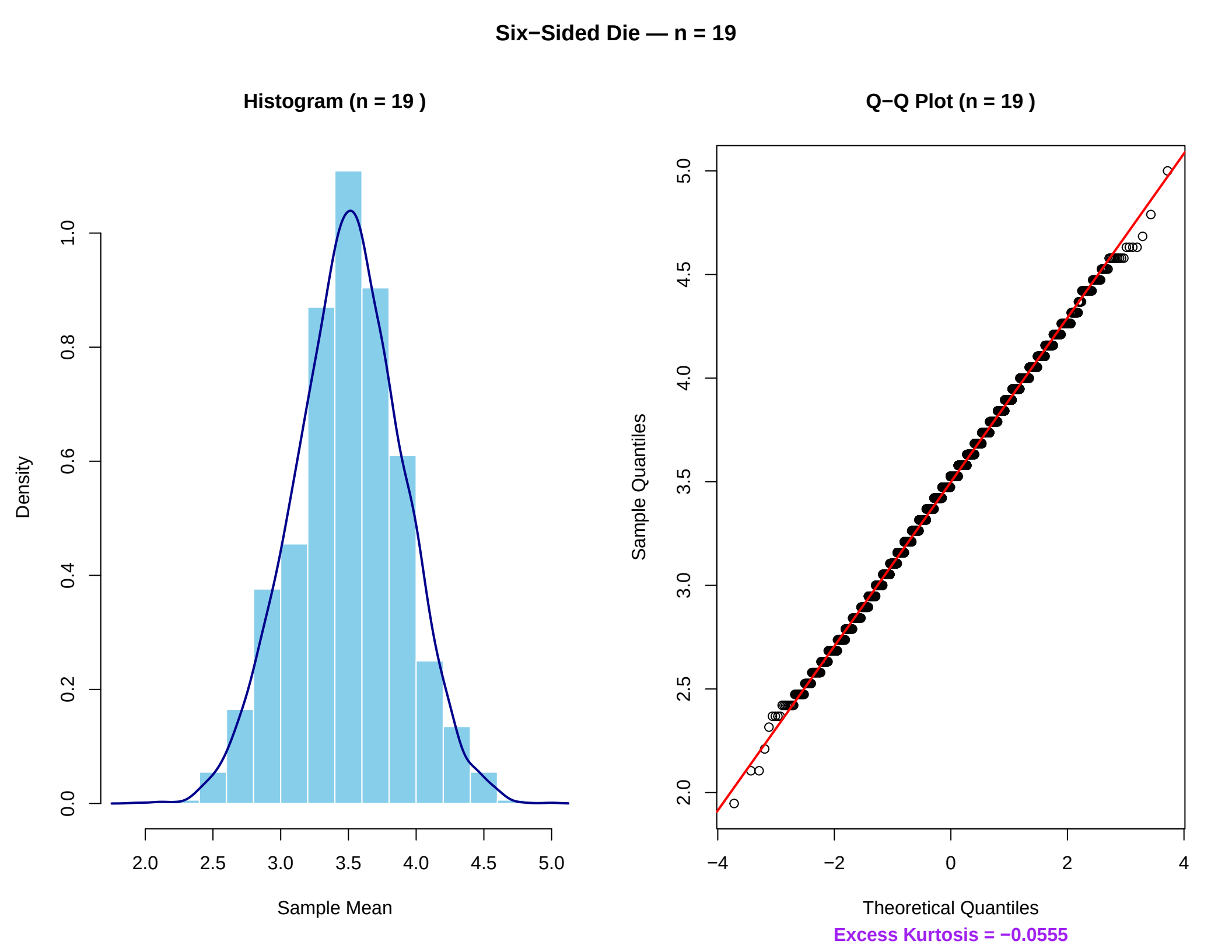
Excess Kurtosis = −0.052

Six−Sided Die — n = 18

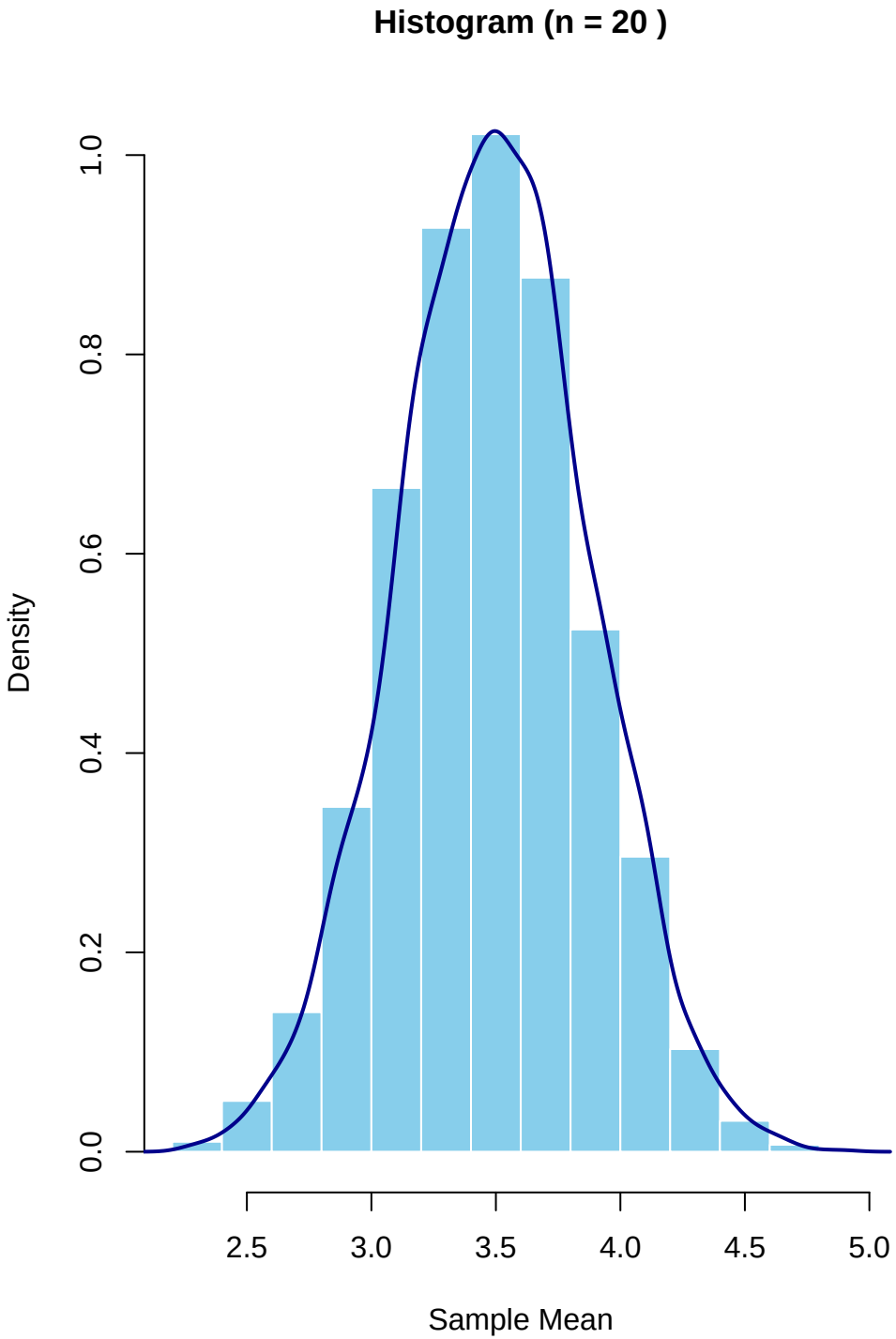


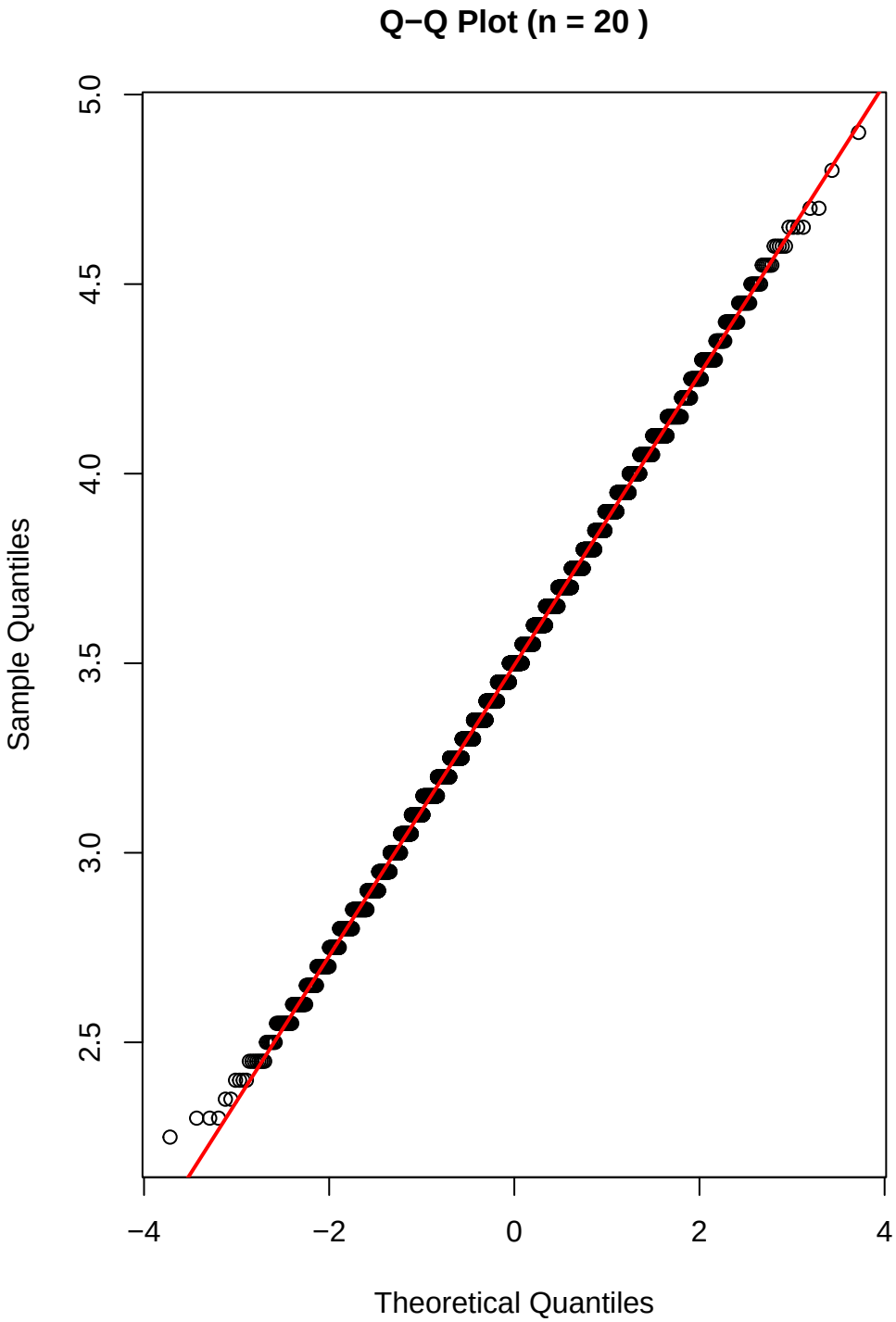


Excess Kurtosis = −0.0145

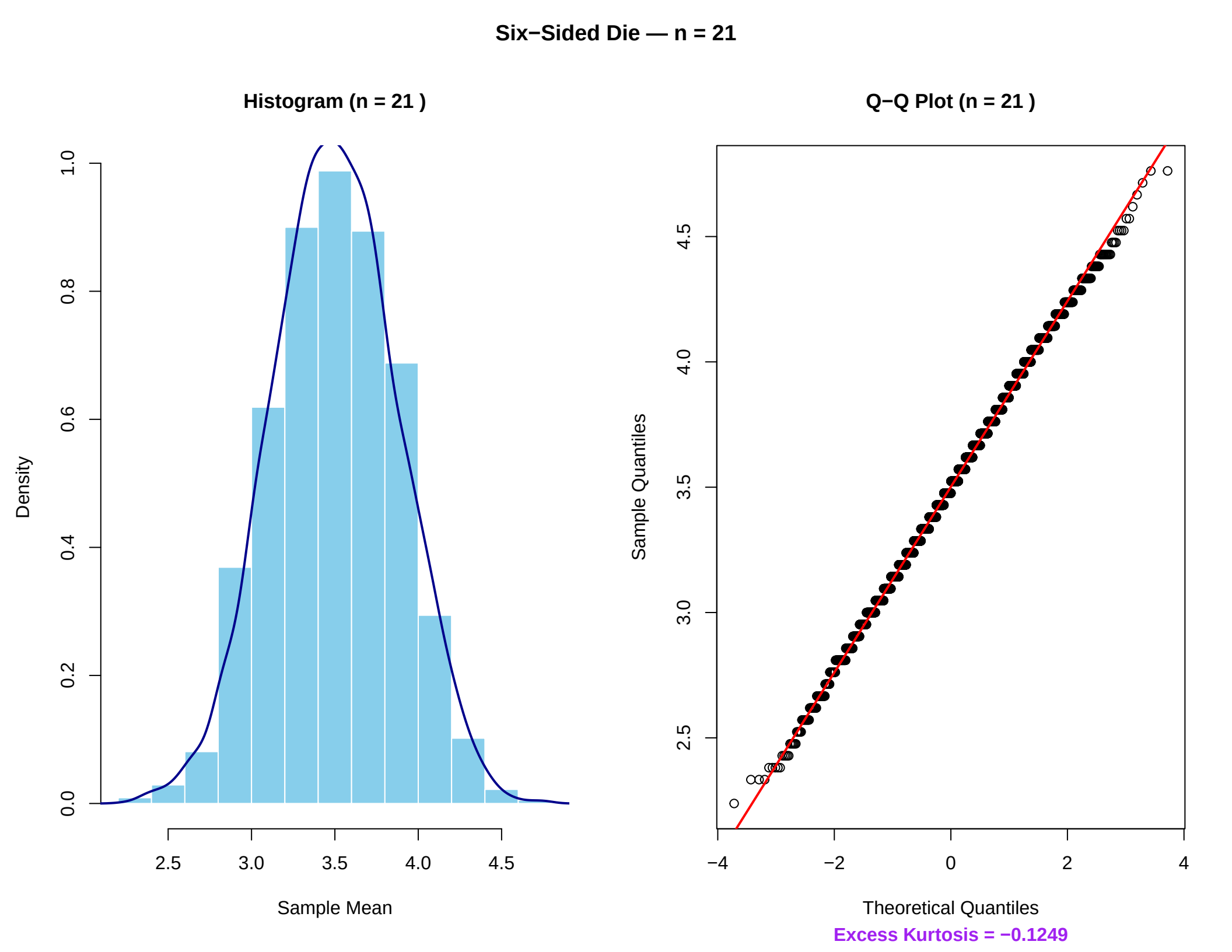

Six−Sided Die — n = 19
Histogram (n = 19)
Density
0.0
0.2
0.4
0.6
0.8
1.0
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample Mean
Q−Q Plot (n = 19)
Sample Quantiles
2.0
2.5
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0555

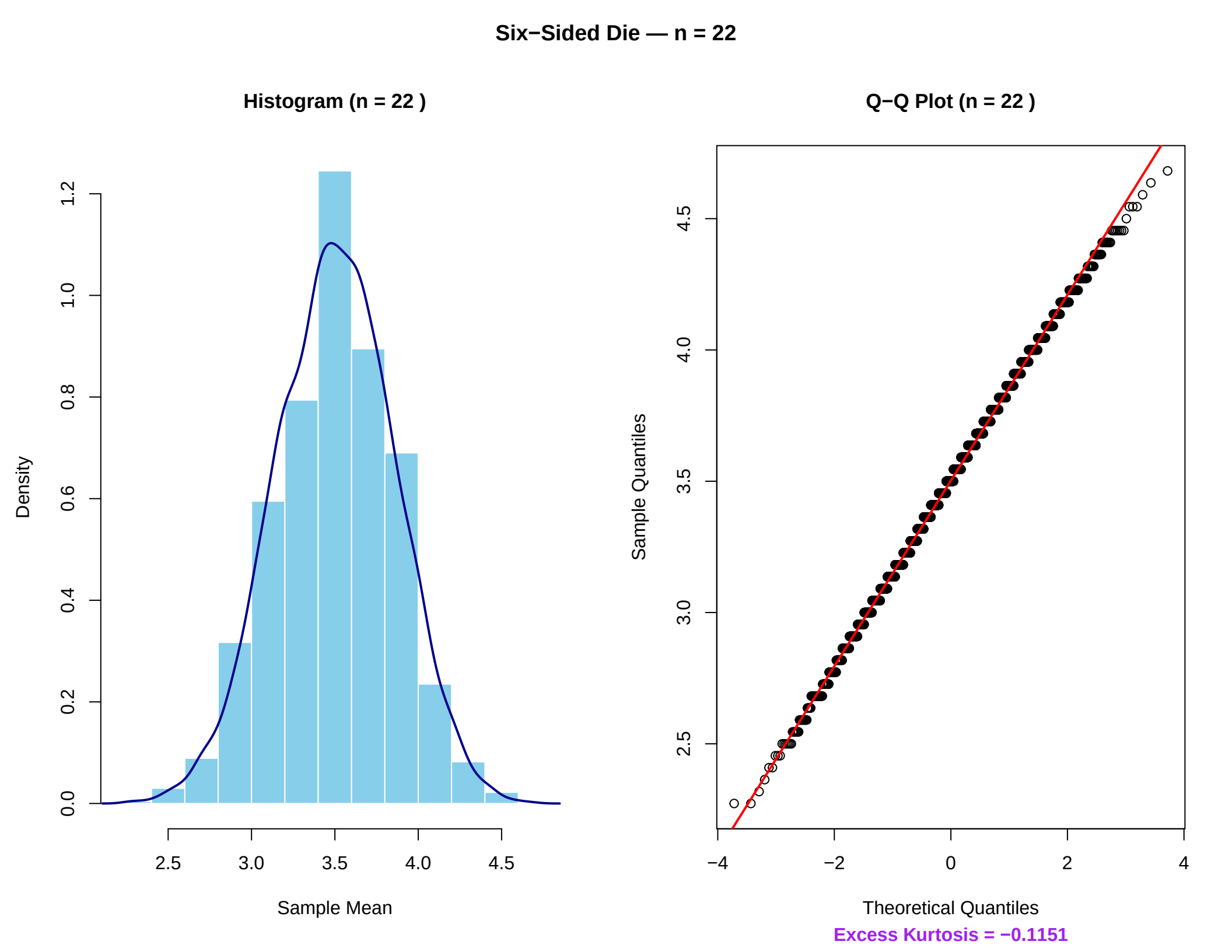
Six−Sided Die — n = 20

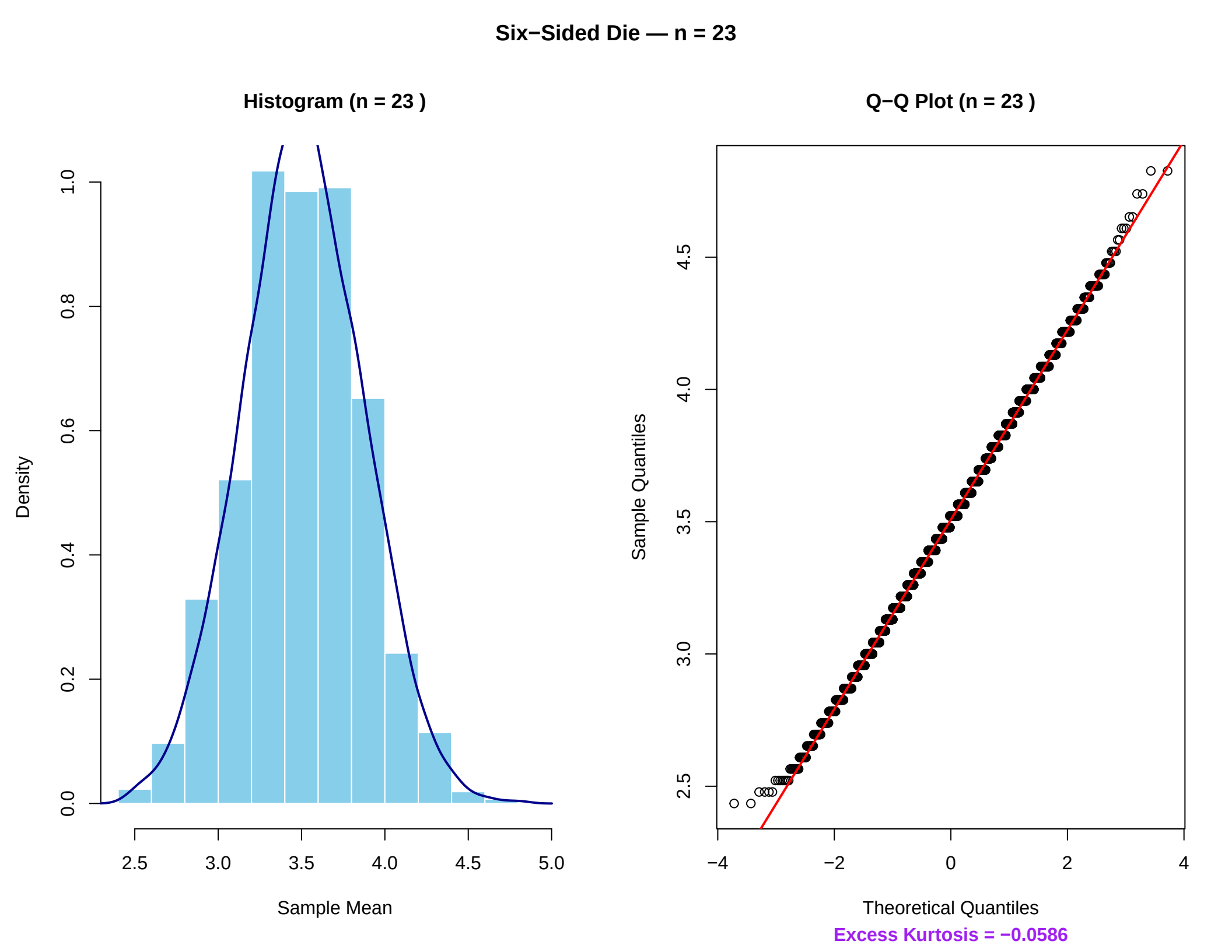


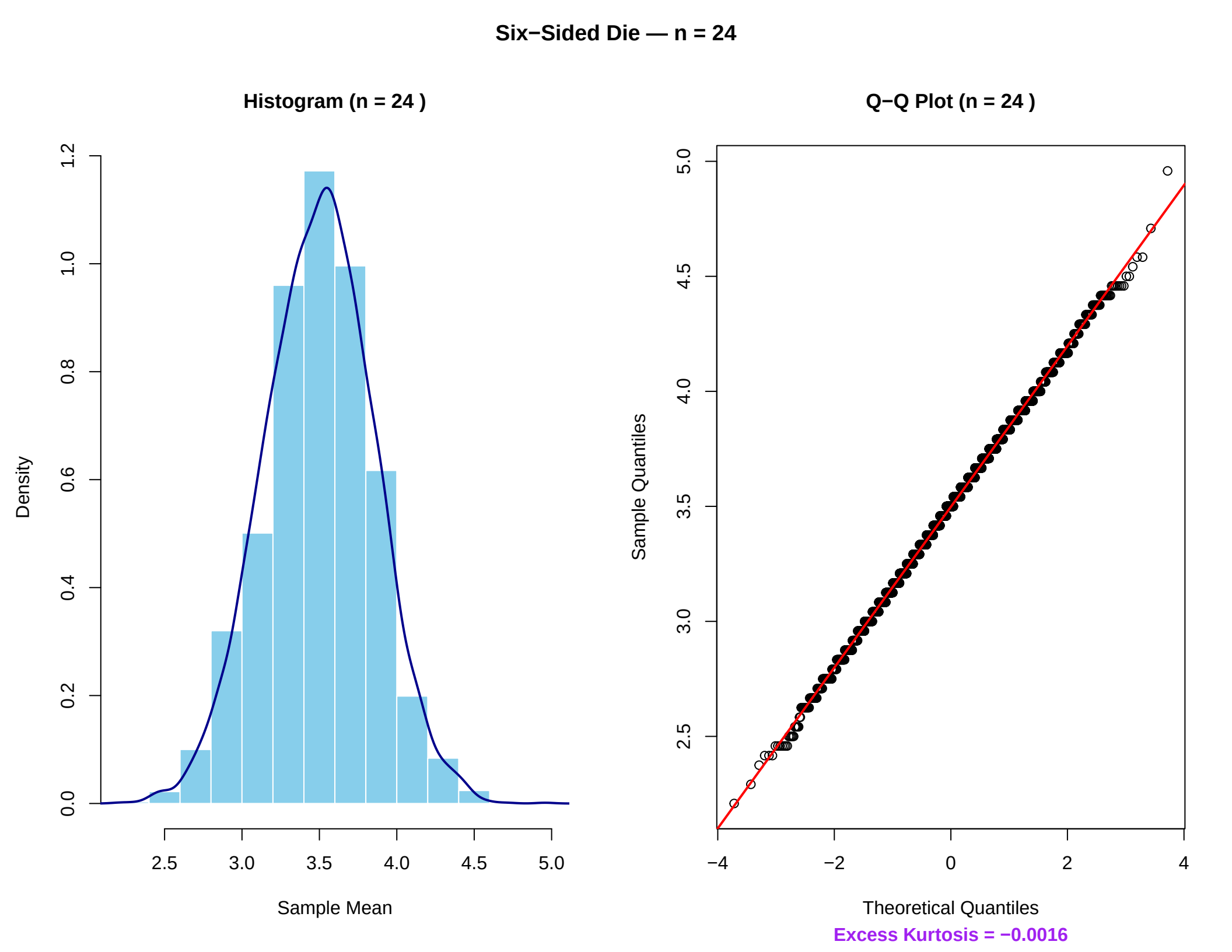


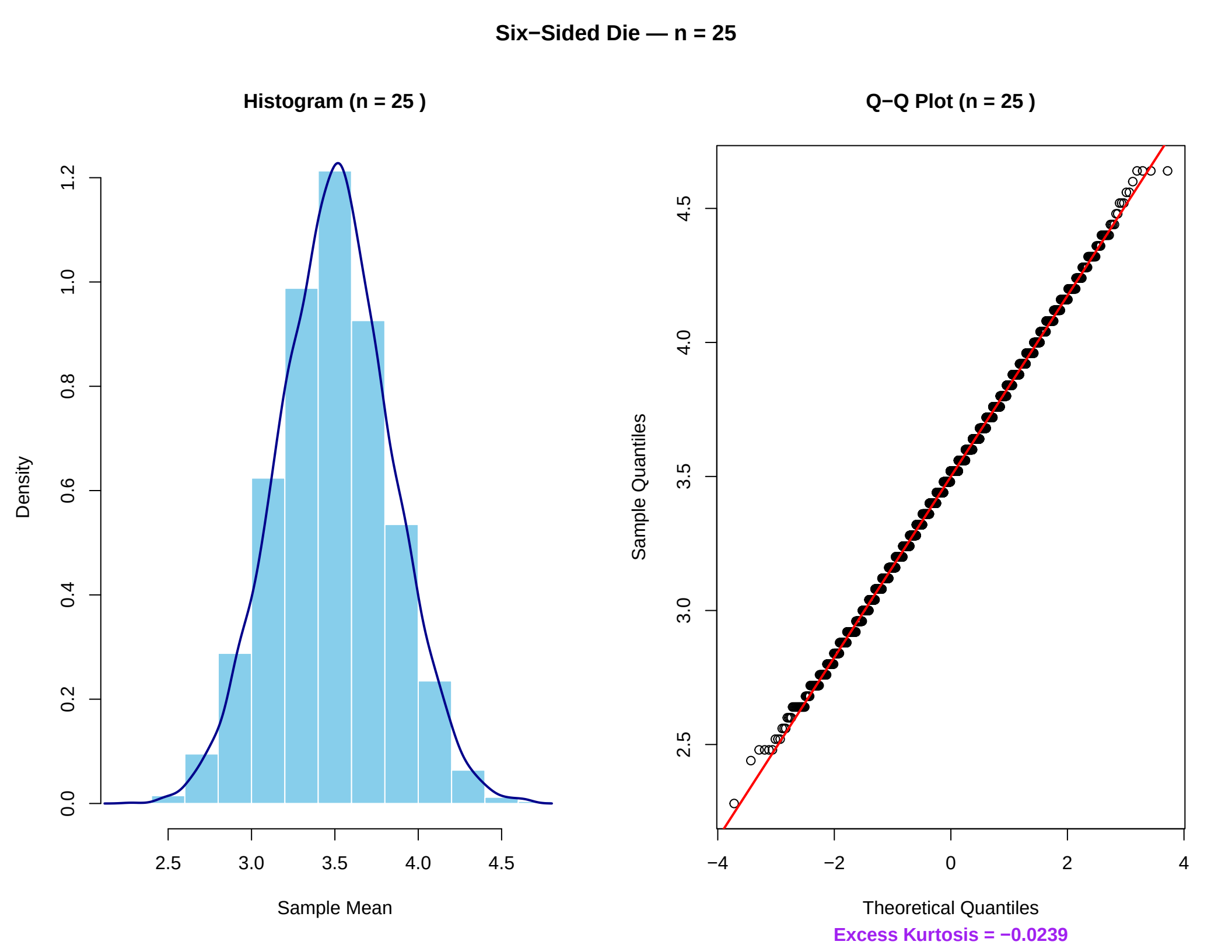
Excess Kurtosis = −0.0539

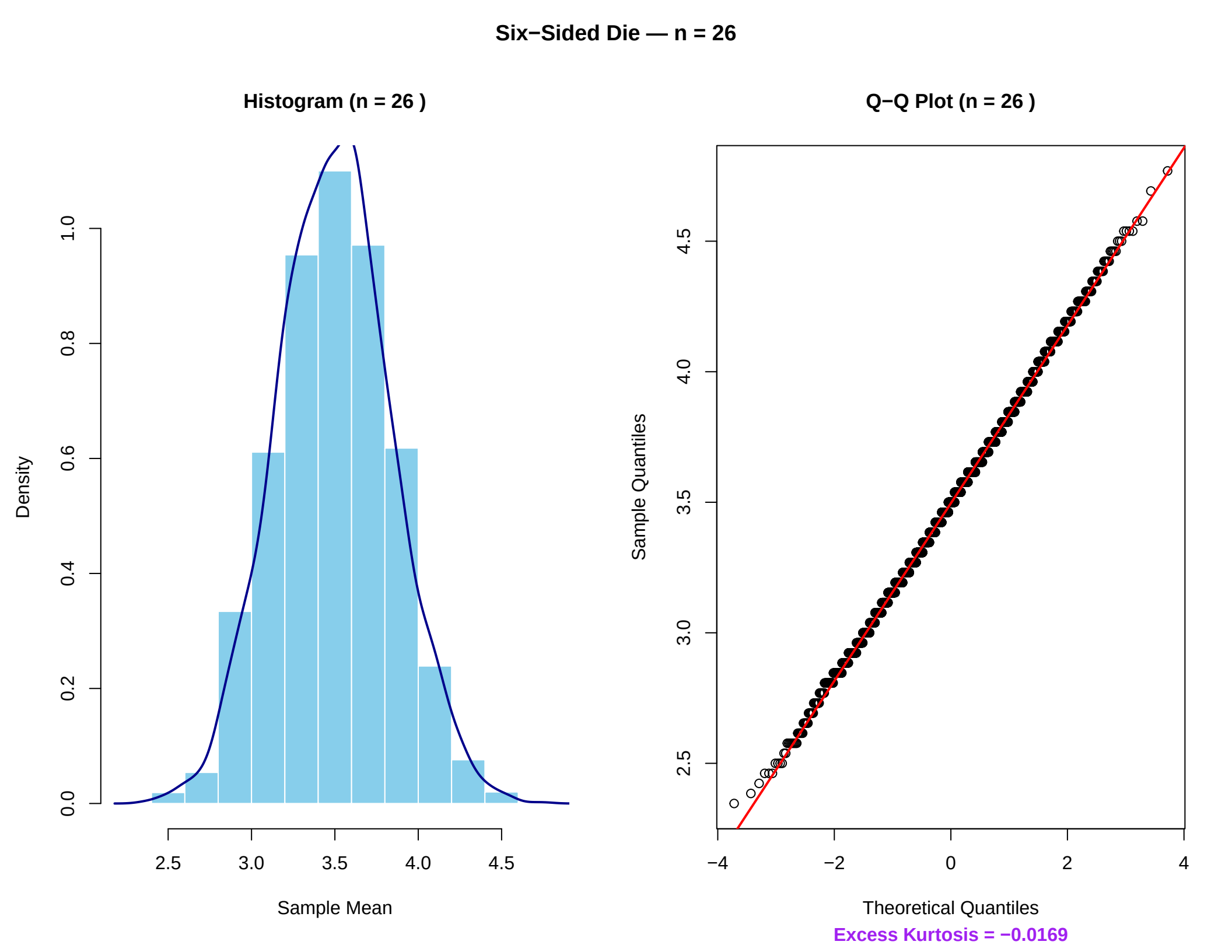

Six−Sided Die — n = 21
Histogram (n = 21)
Density
0.0
0.2
0.4
0.6
0.8
1.0
2.5
3.0
3.5
4.0
4.5
Sample Mean
Q−Q Plot (n = 21)
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.1249

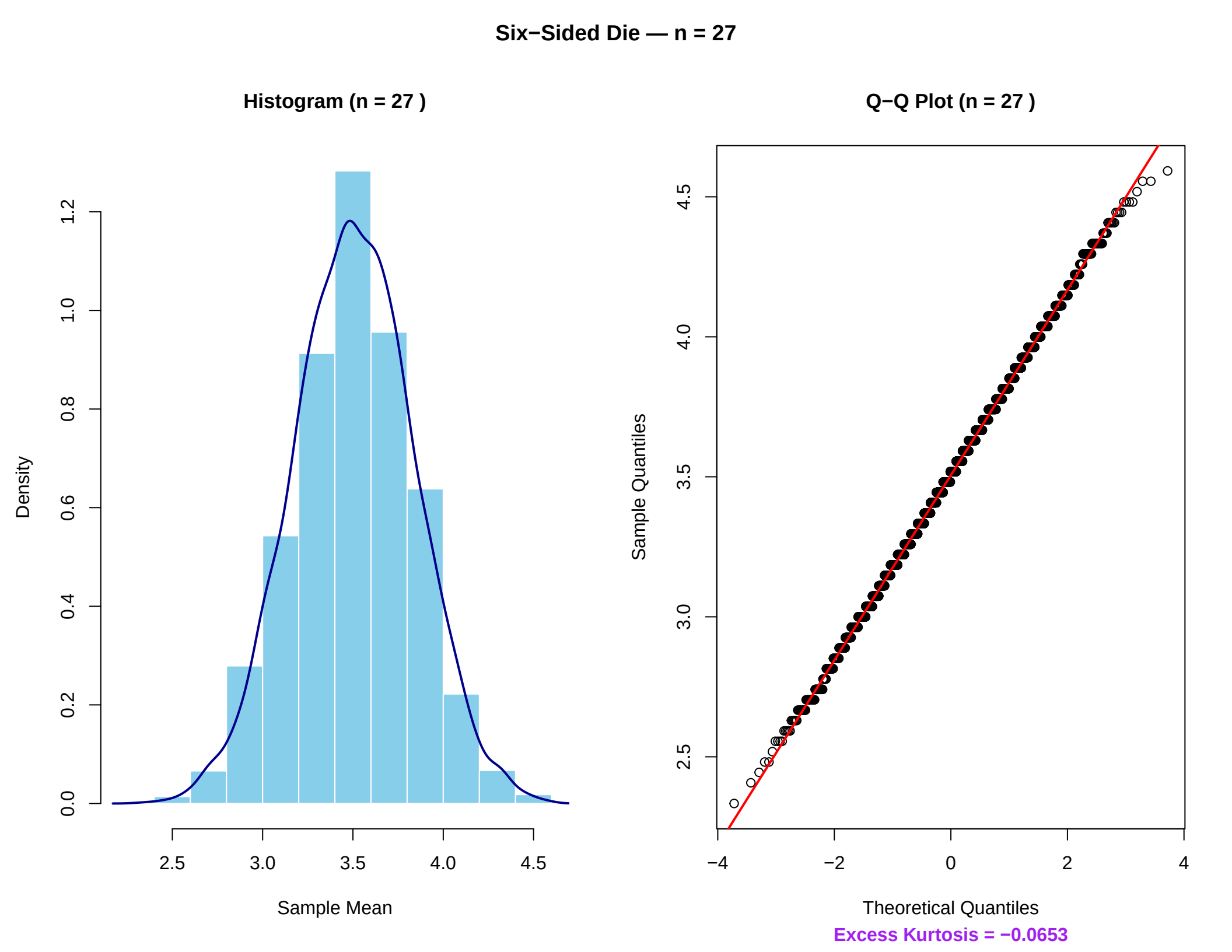

Six−Sided Die — n = 22
Histogram (n = 22)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.5
3.0
3.5
4.0
4.5
Sample Mean
Q−Q Plot (n = 22)
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.1151


Six−Sided Die — n = 23
Histogram (n = 23)
Density
0.0
0.2
0.4
0.6
0.8
1.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample Mean
Q−Q Plot (n = 23)
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0586

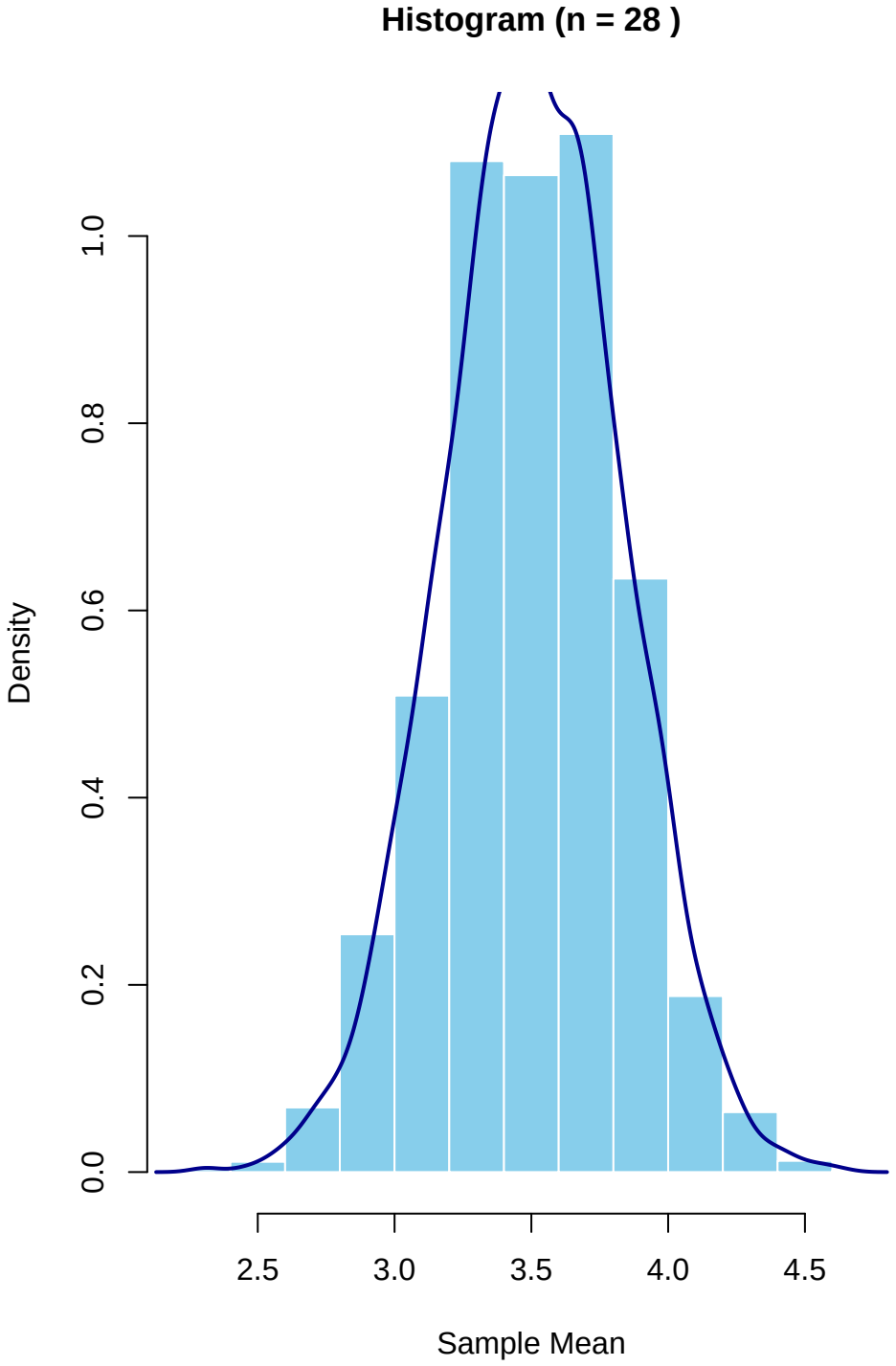

Six−Sided Die — n = 24
Histogram (n = 24)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.5
3.0
3.5
4.0
4.5
5.0
Sample Mean
Q−Q Plot (n = 24)
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0016

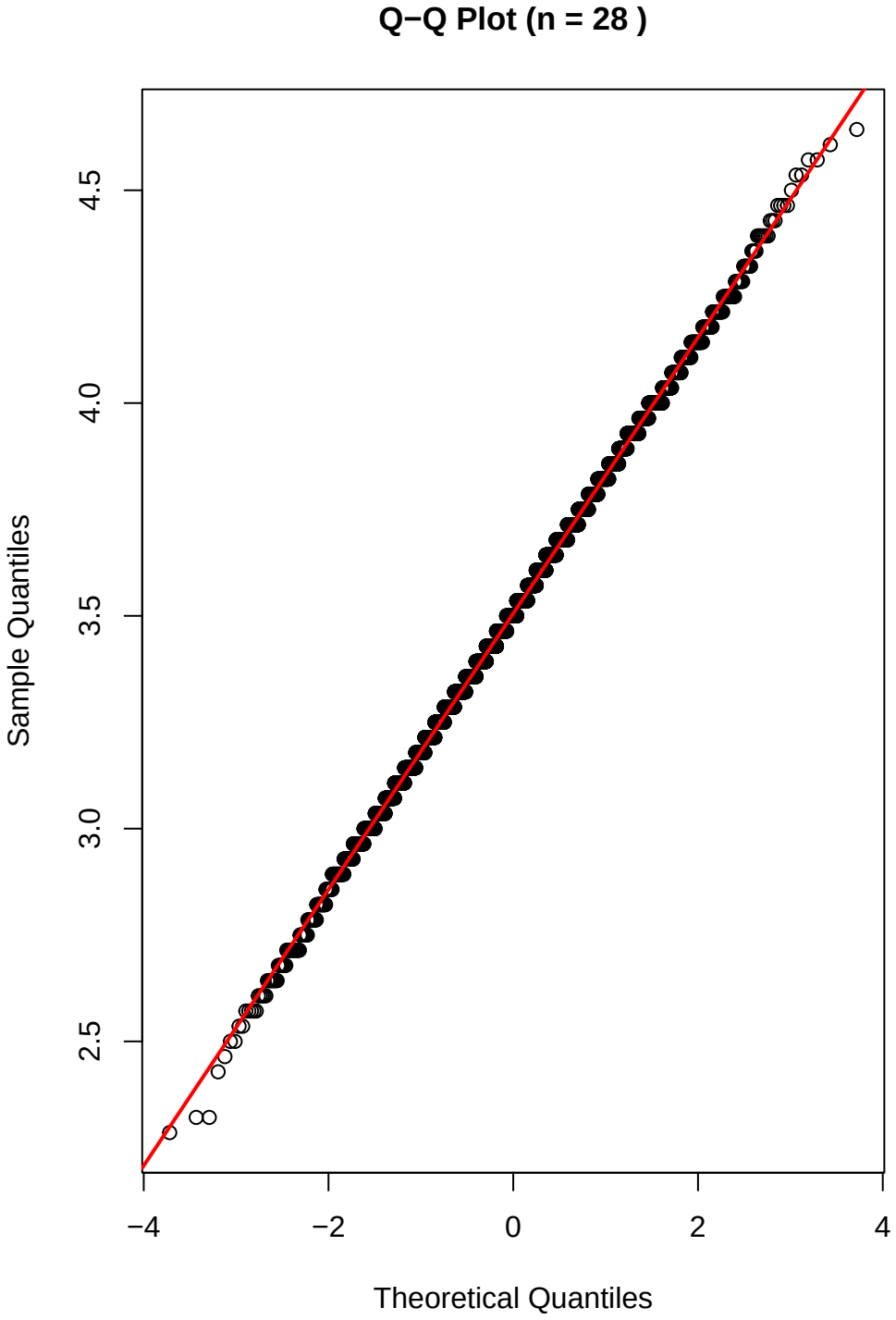

Six−Sided Die — n = 25
Histogram (n = 25)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.5
3.0
3.5
4.0
4.5
Sample Mean
Q−Q Plot (n = 25)
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0239


Six−Sided Die — n = 26
Histogram (n = 26)
Density
0.0
0.2
0.4
0.6
0.8
1.0
2.5
3.0
3.5
4.0
4.5
Sample Mean
Q−Q Plot (n = 26)
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0169


Six−Sided Die — n = 27
Histogram (n = 27)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2.5
3.0
3.5
4.0
4.5
Sample Mean
Q−Q Plot (n = 27)
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = −0.0653

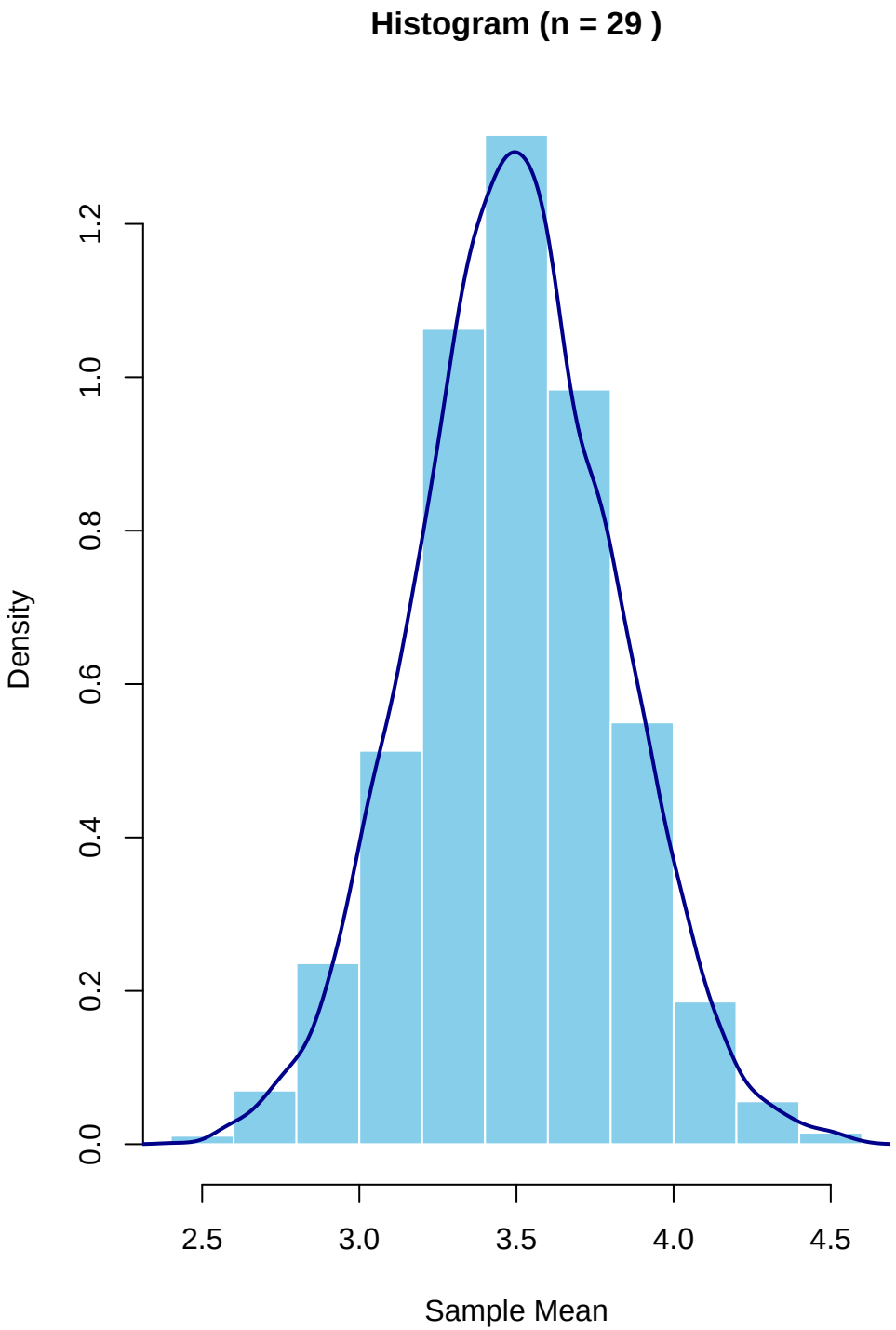
Six−Sided Die — n = 28

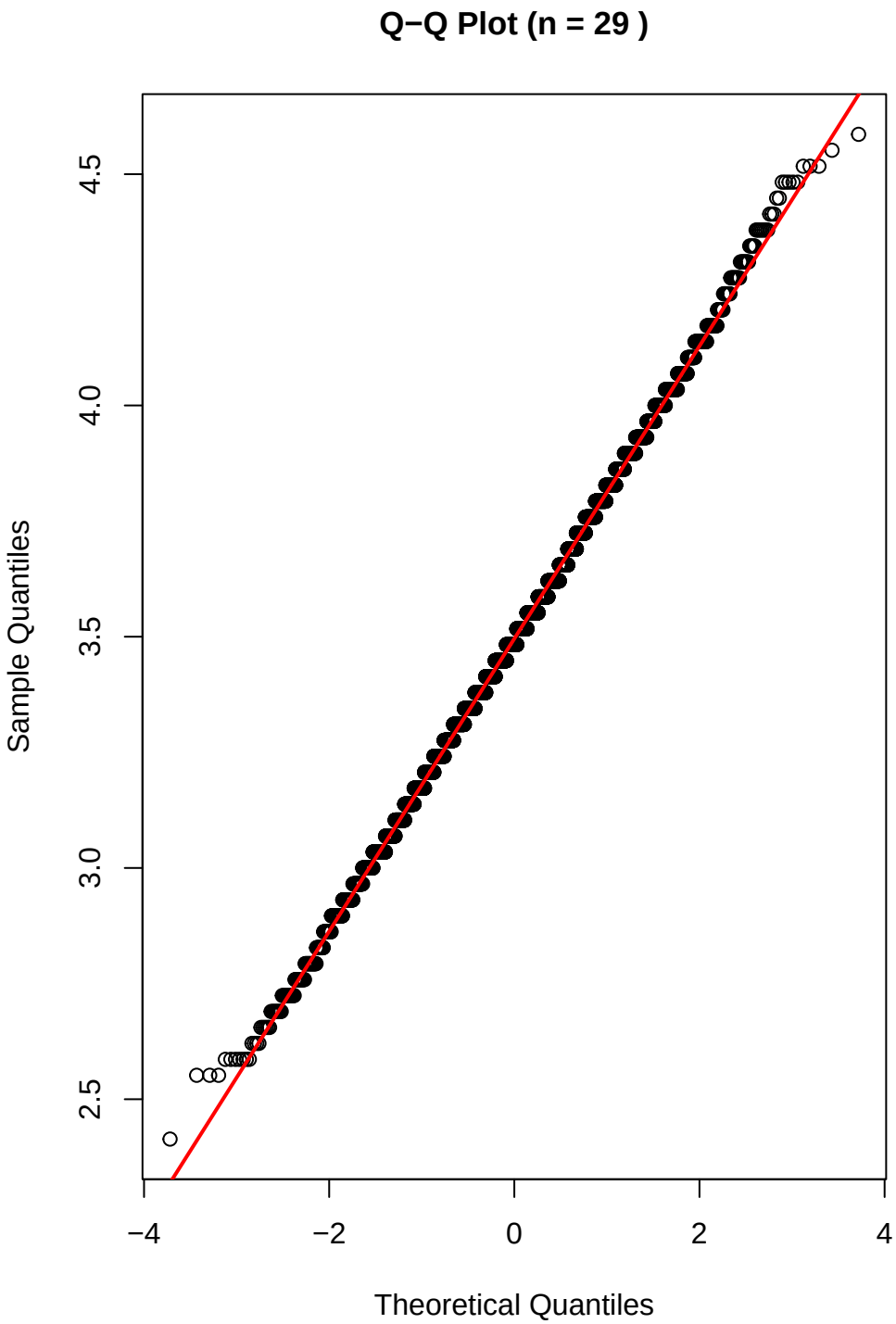




Excess Kurtosis = 0.0344

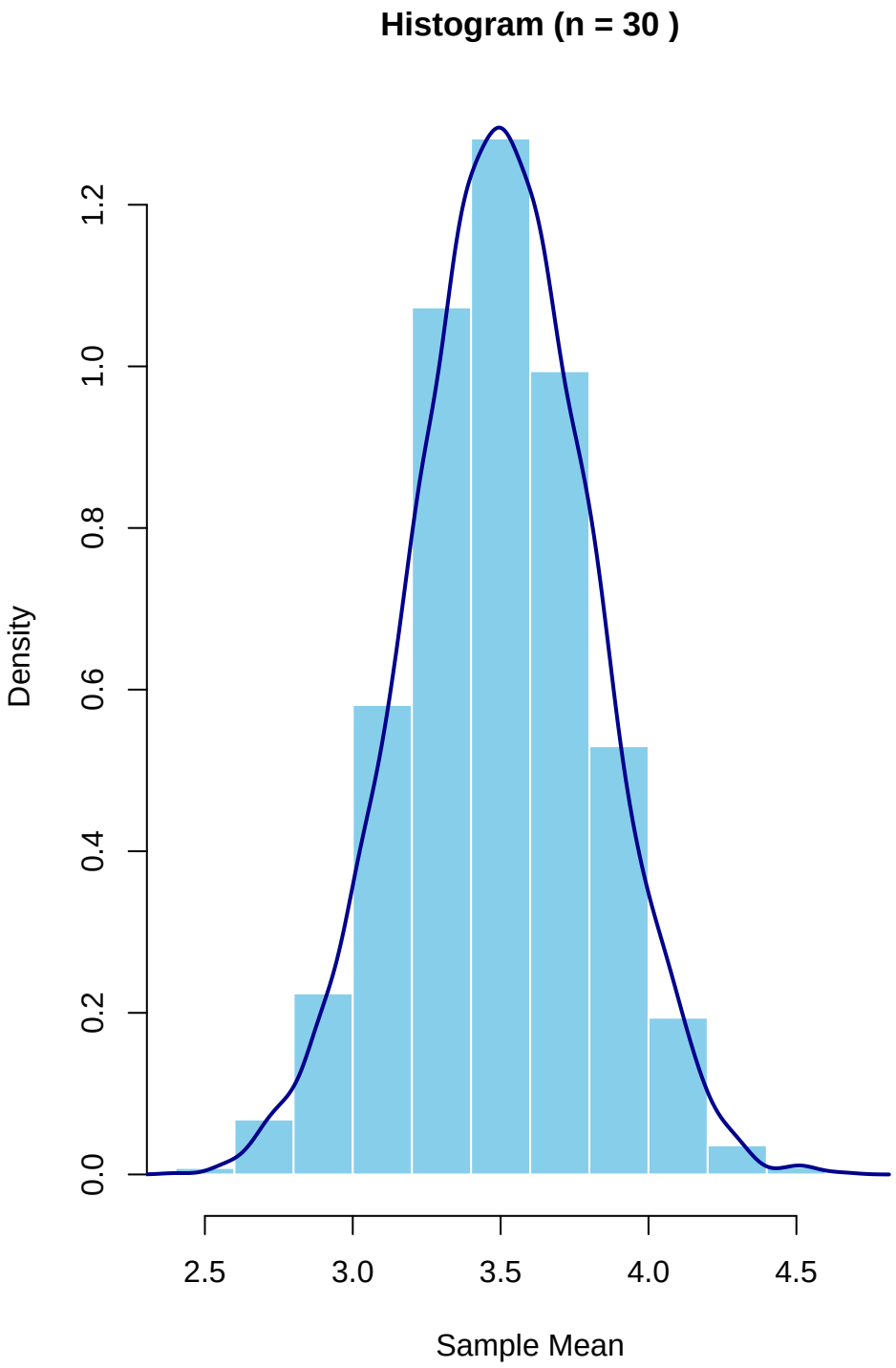
Six−Sided Die — n = 29

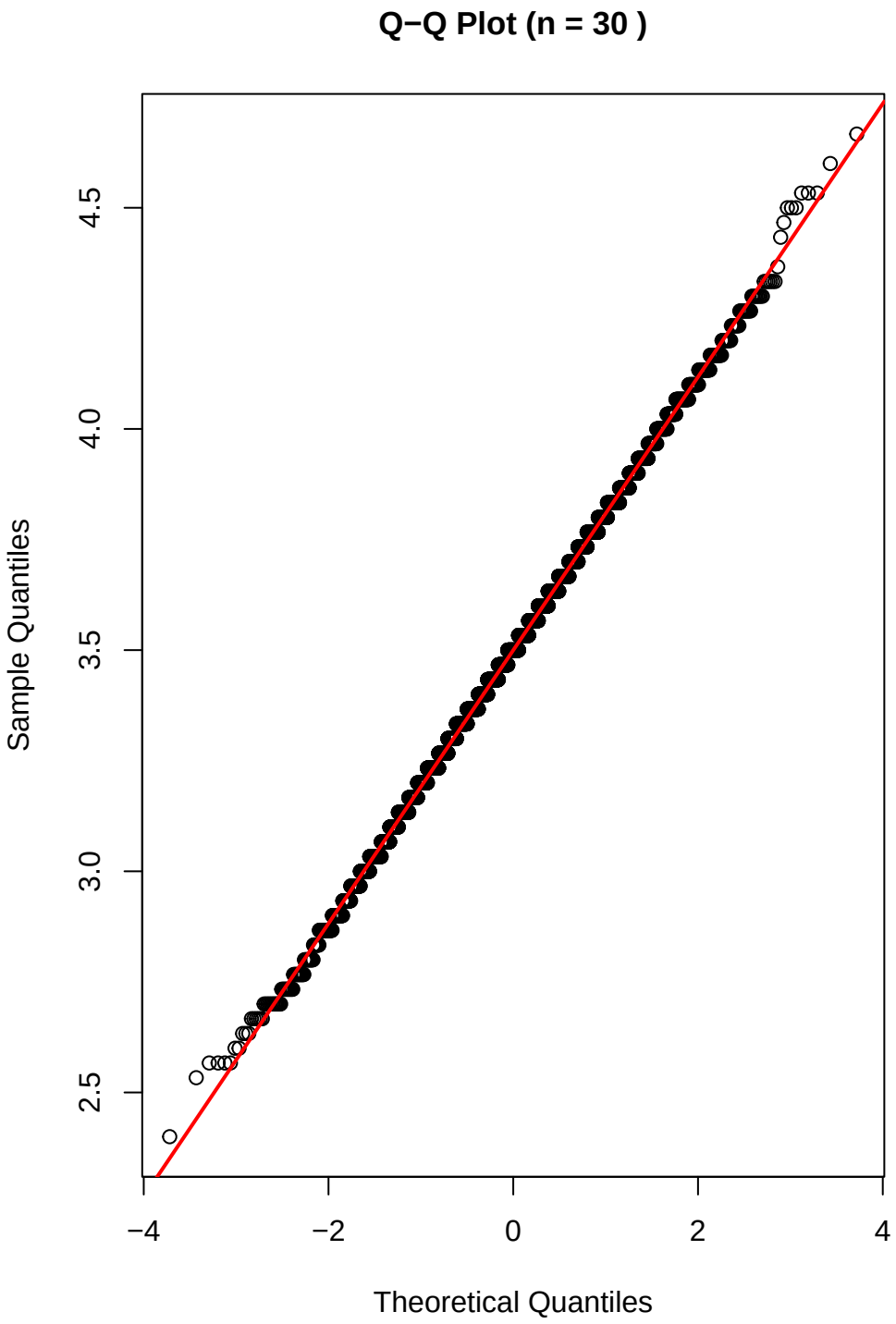




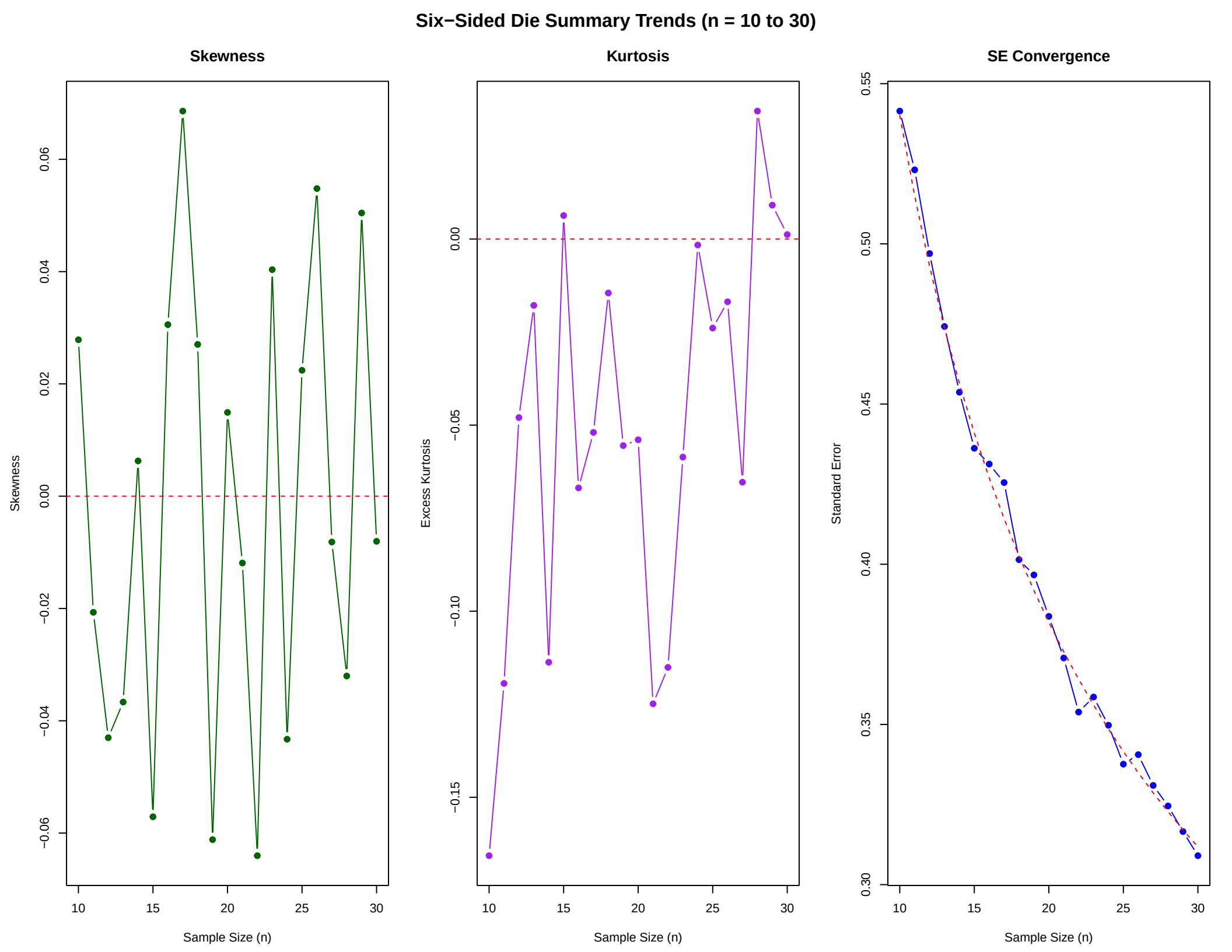
Excess Kurtosis = 0.0091

Six−Sided Die — n = 30

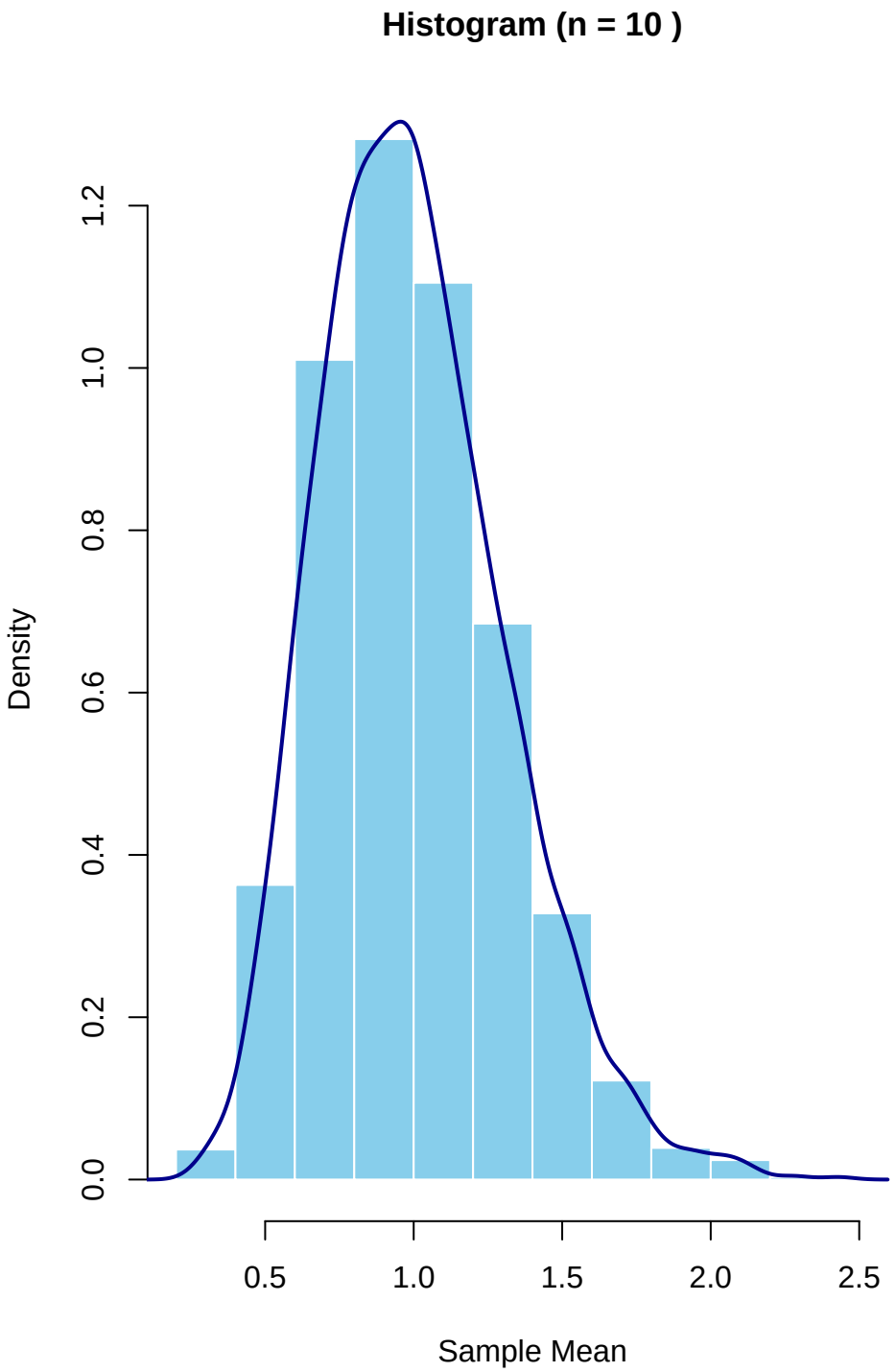


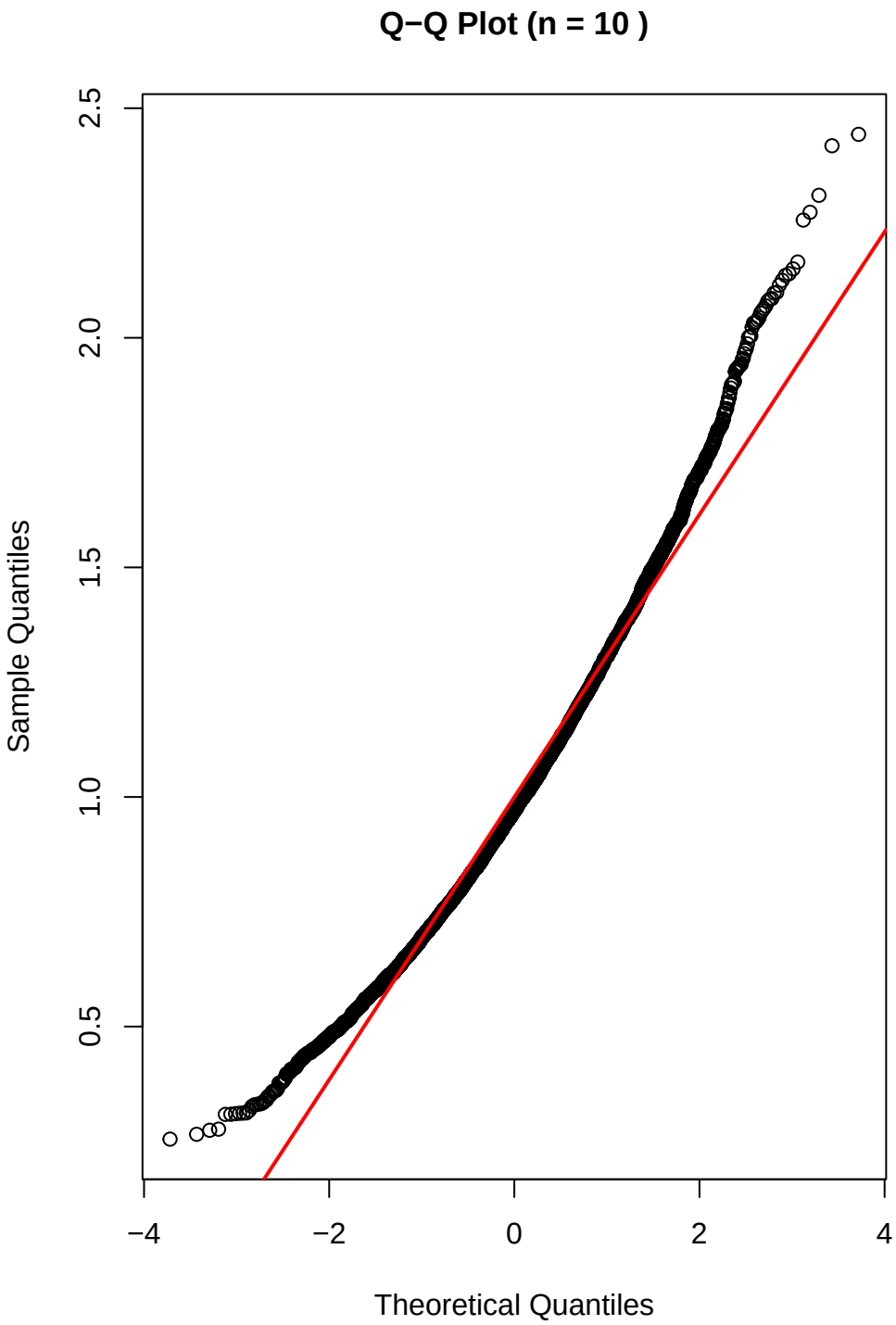


Excess Kurtosis = 0.0012

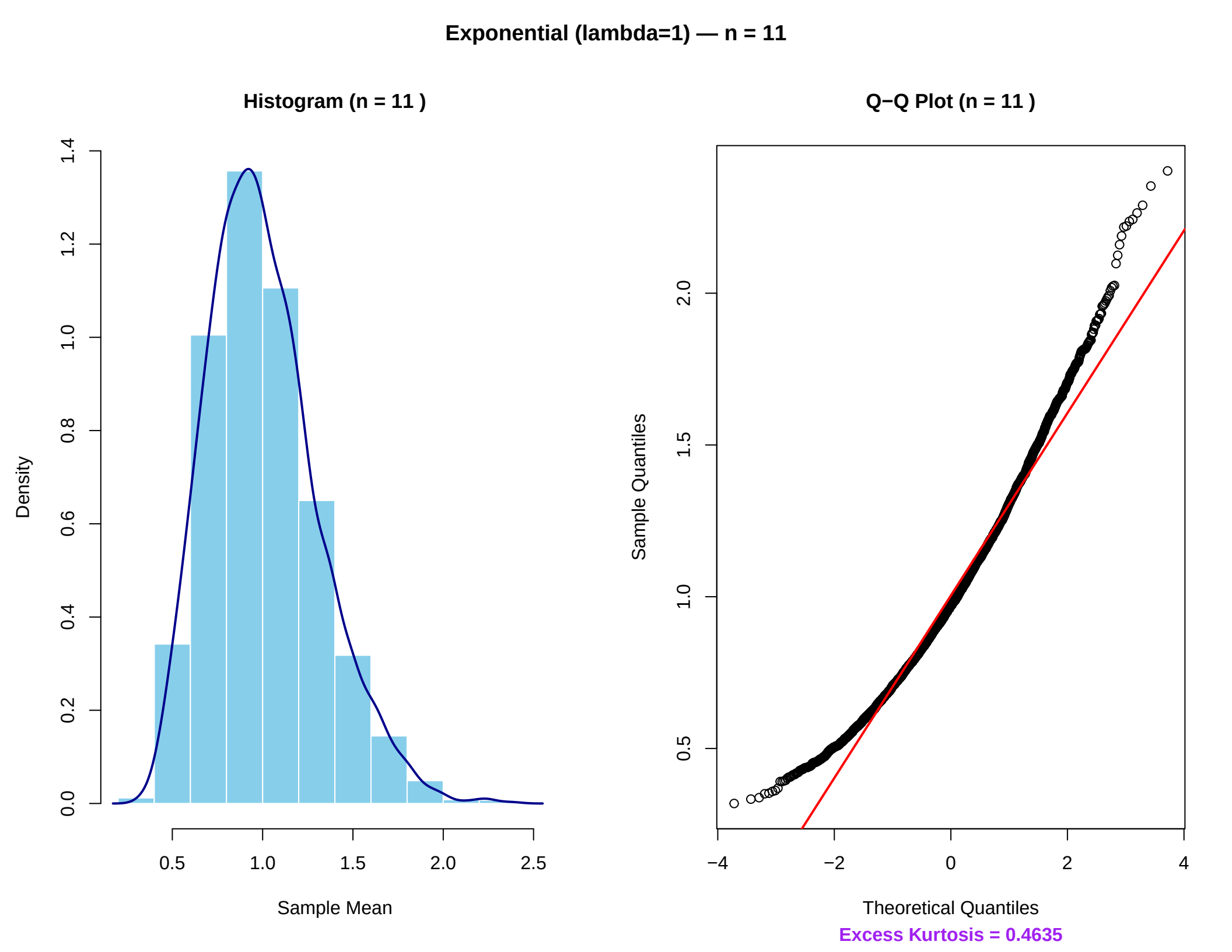

Six−Sided Die Summary Trends (n = 10 to 30)
Skewness
Skewness
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
10
15
20
25
30
Sample Size (n)
Kurtosis
Excess Kurtosis
0.00
−0.05
−0.10
−0.15
10
15
20
25
30
Sample Size (n)
SE Convergence
Standard Error
0.55
0.50
0.45
0.40
0.35
0.30
10
15
20
25
30
Sample Size (n)

Exponential (lambda=1) — n = 10

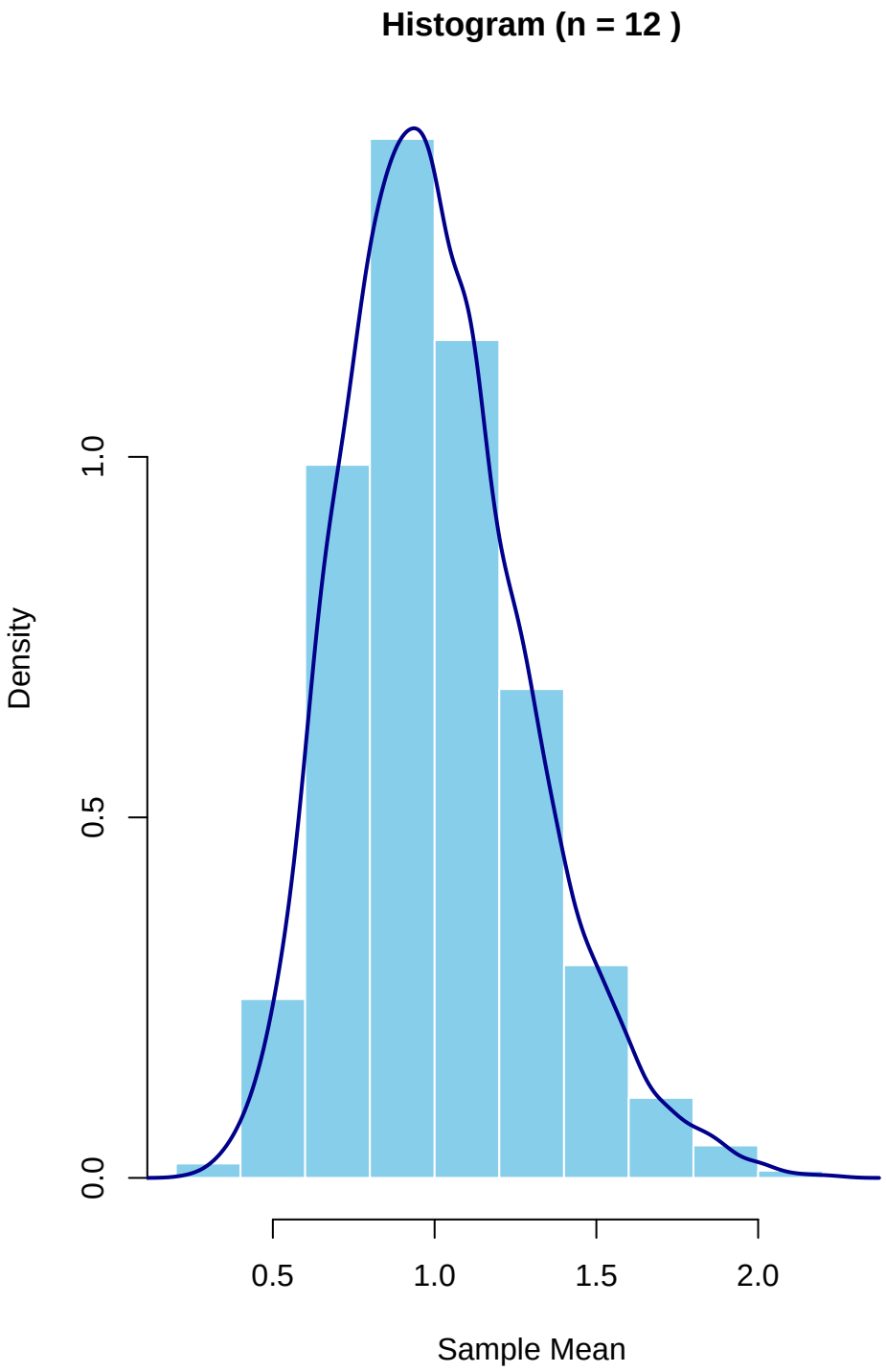


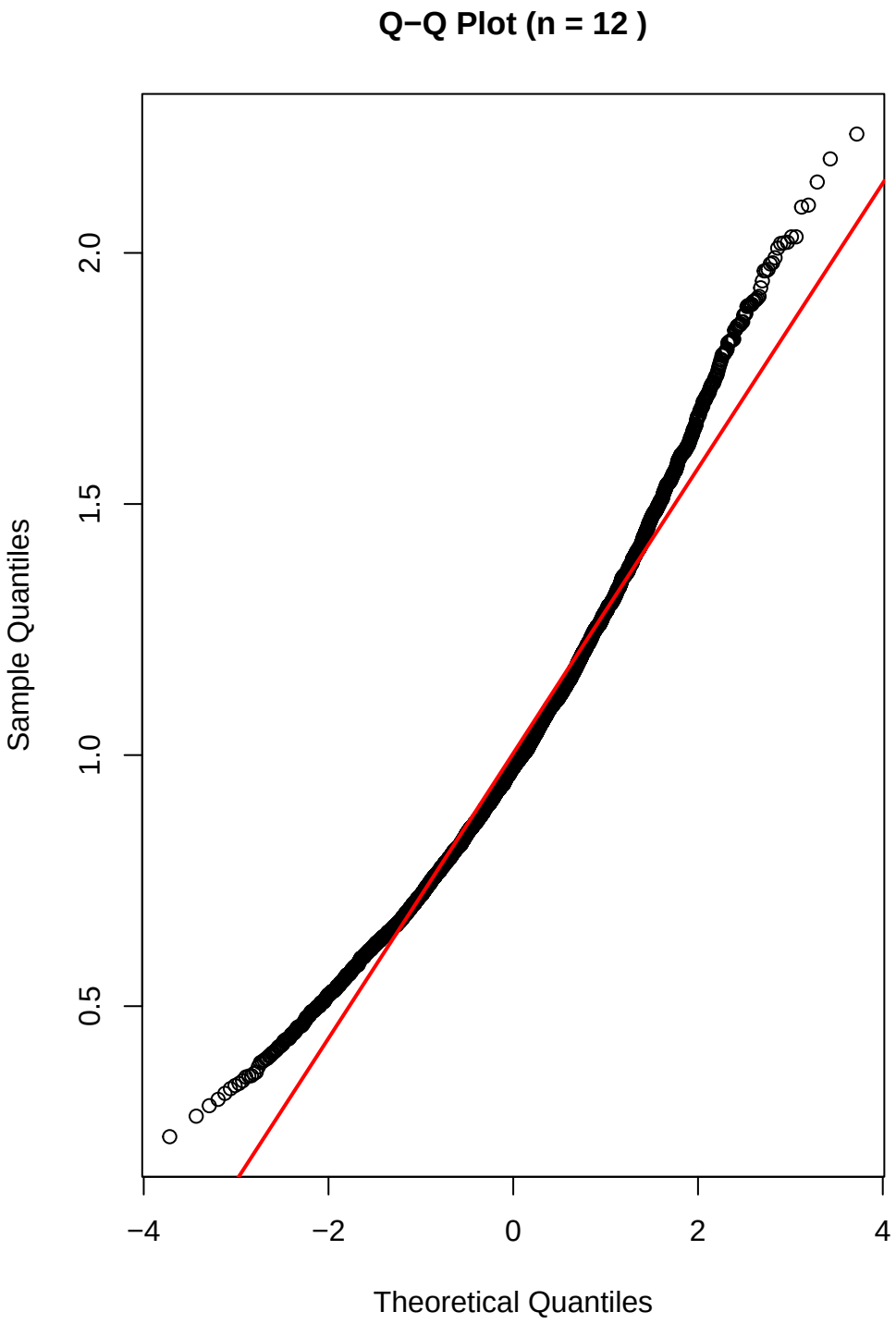


Excess Kurtosis = 0.5352

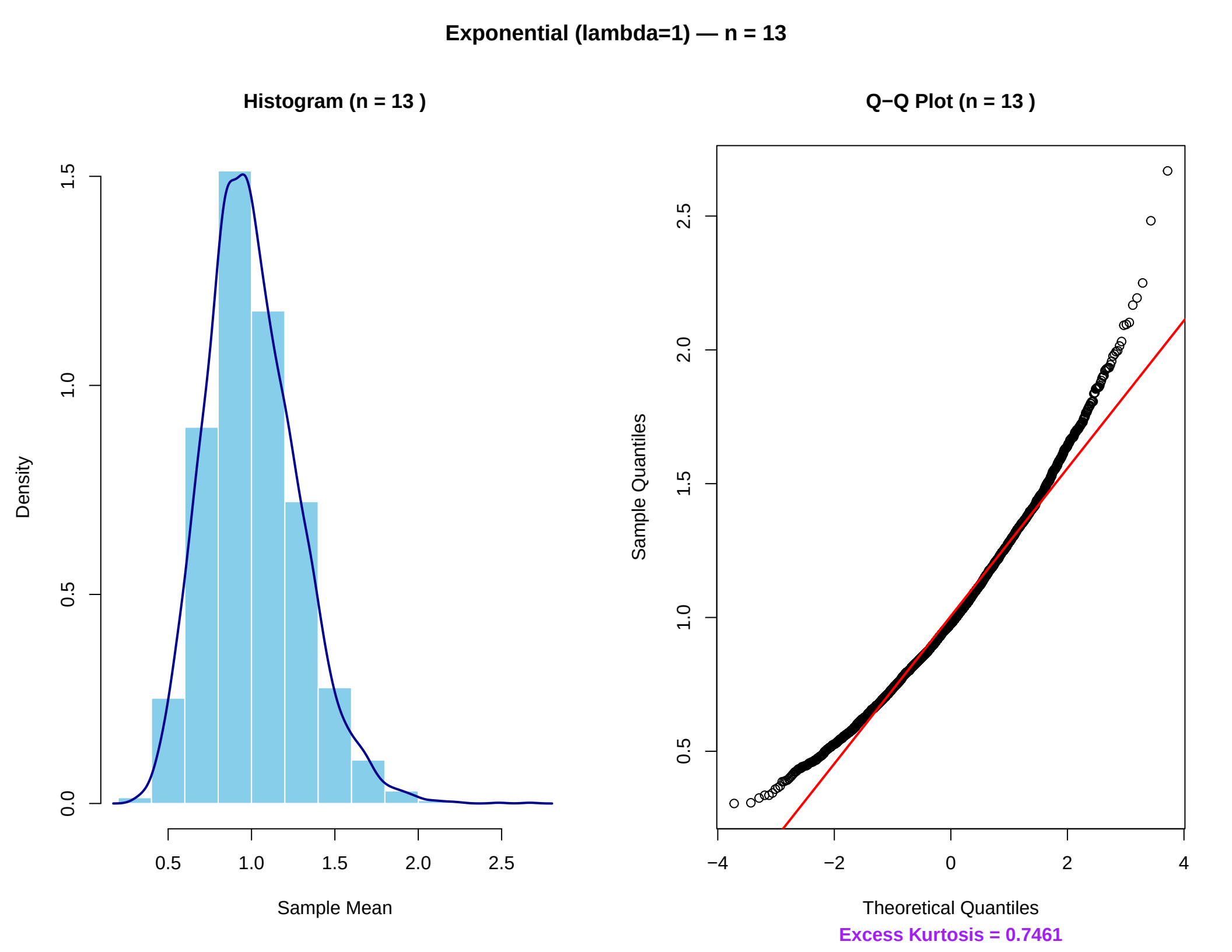

Exponential (lambda=1) — n = 11
Histogram (n = 11)
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.5
1.0
1.5
2.0
2.5
Sample Mean
Q−Q Plot (n = 11)
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.4635

Exponential (lambda=1) — n = 12

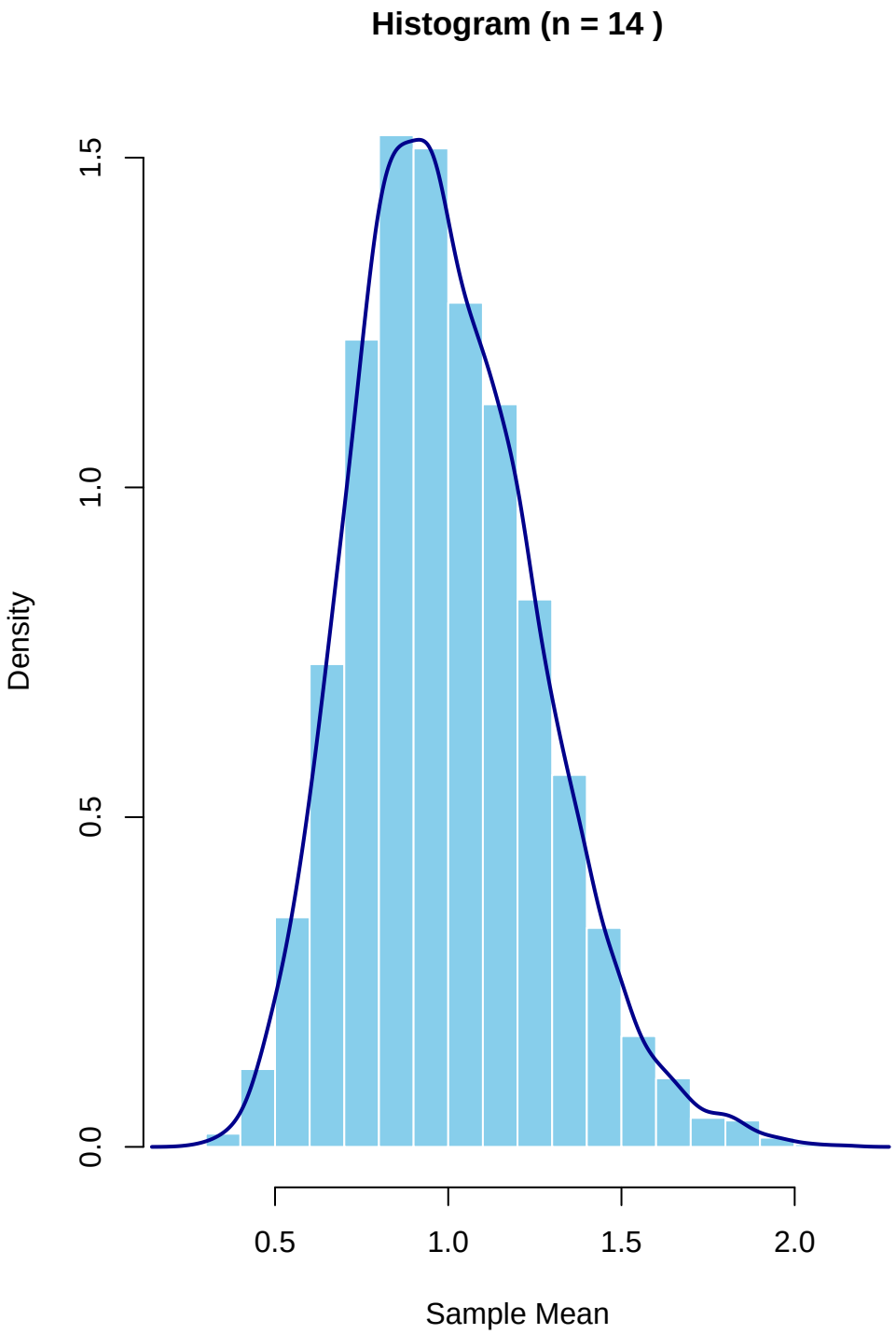


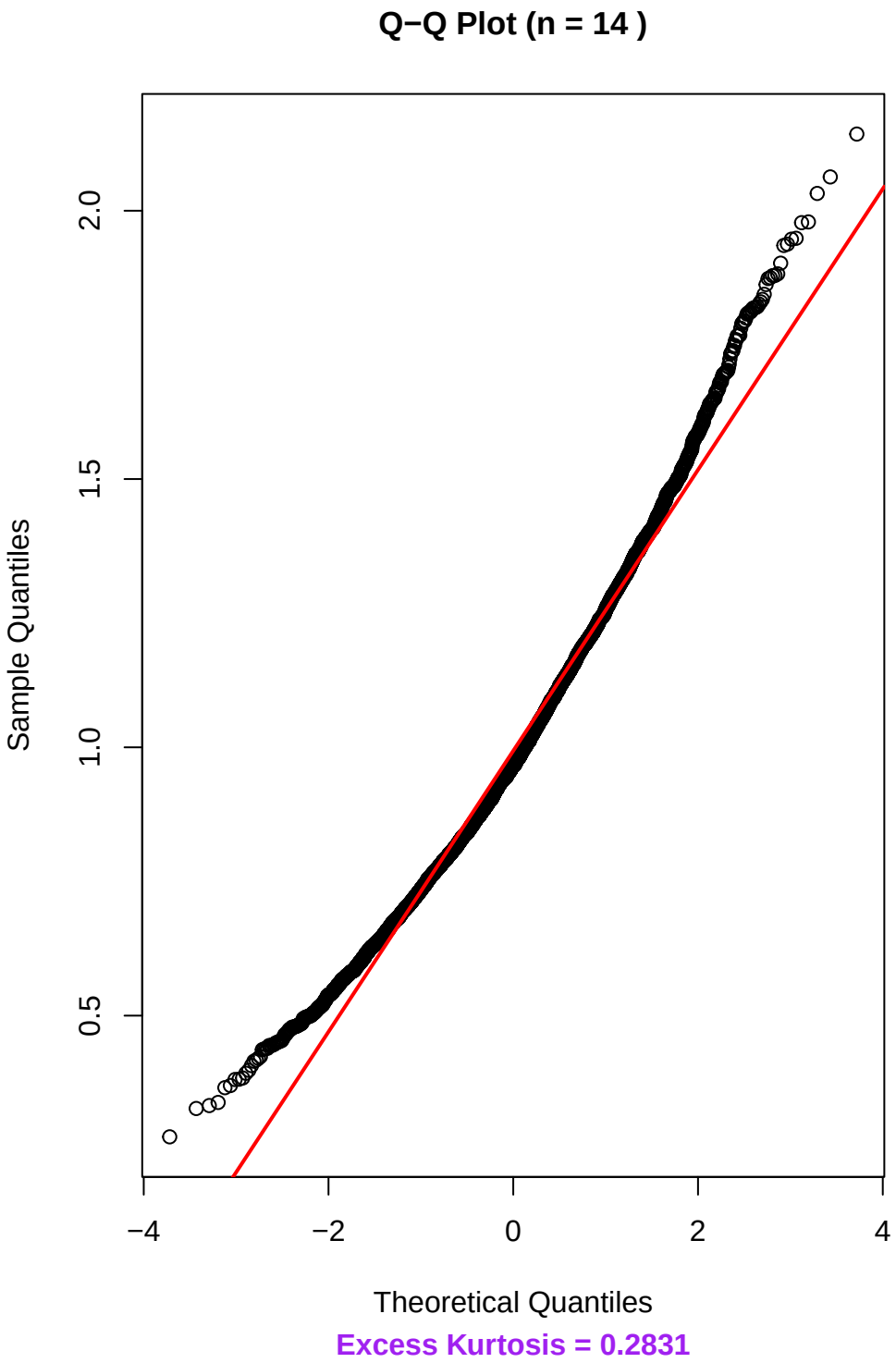


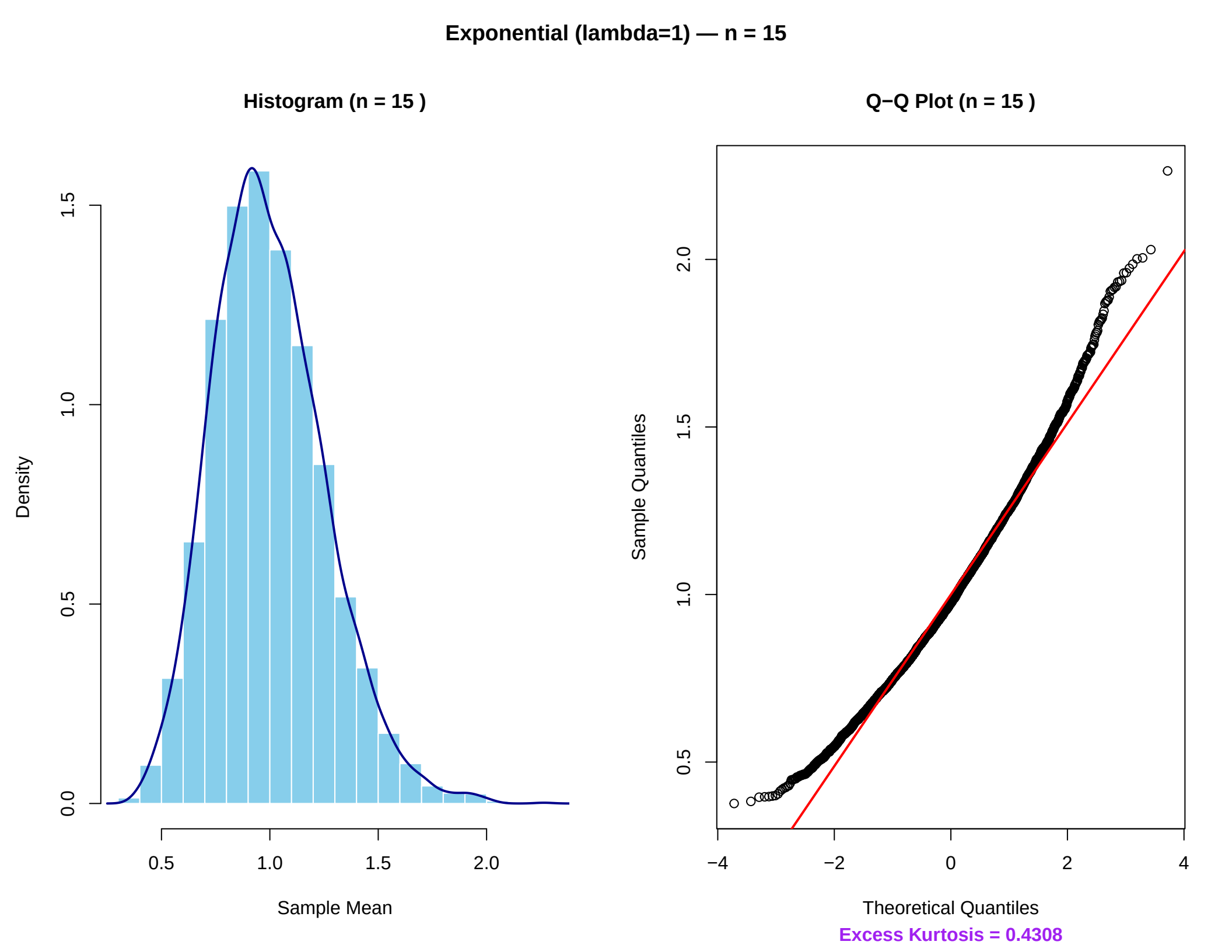
Excess Kurtosis = 0.4433

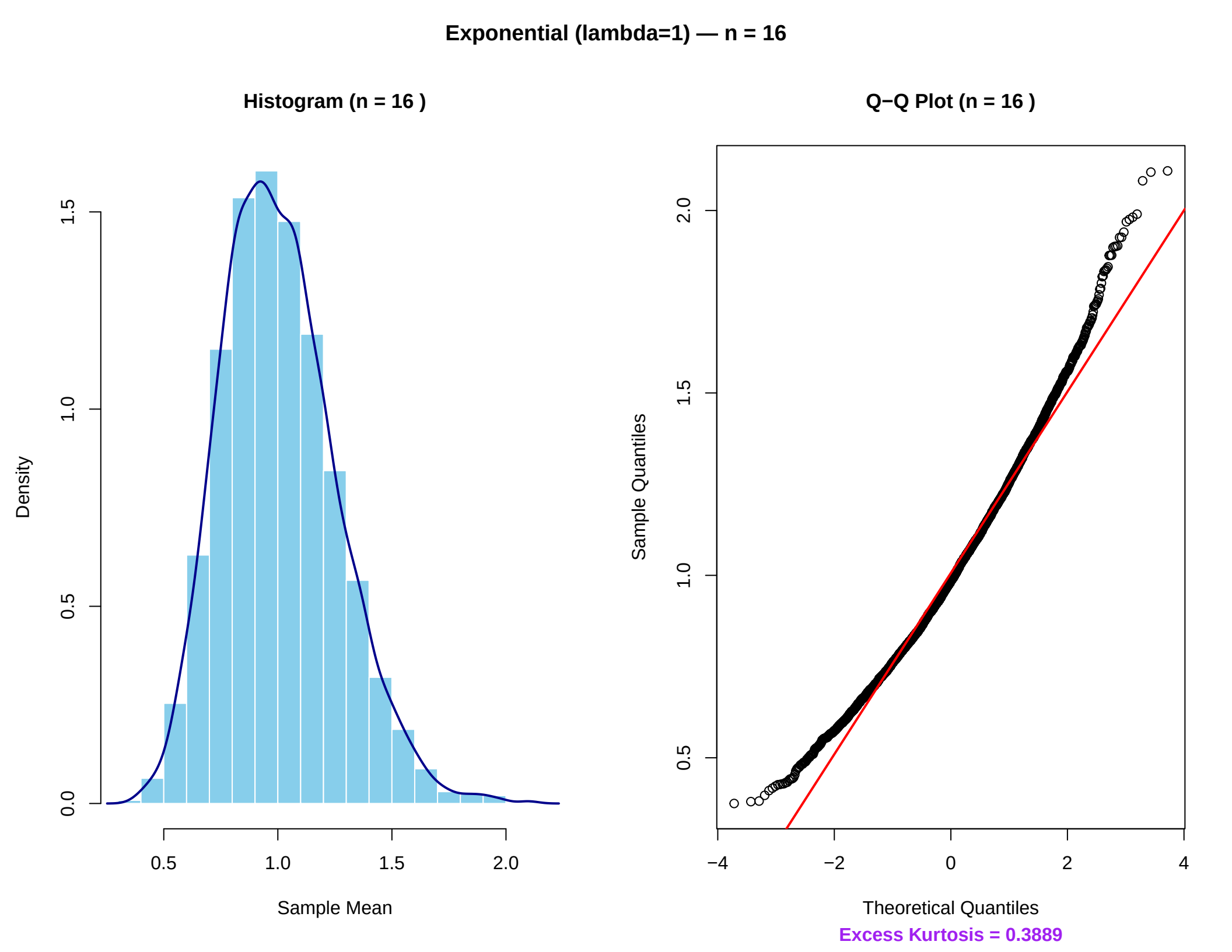

Exponential (lambda=1) — n = 13
Histogram (n = 13)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
2.5
Sample Mean
Q−Q Plot (n = 13)
Sample Quantiles
0.5
1.0
1.5
2.0
2.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.7461

Exponential (lambda=1) — n = 14

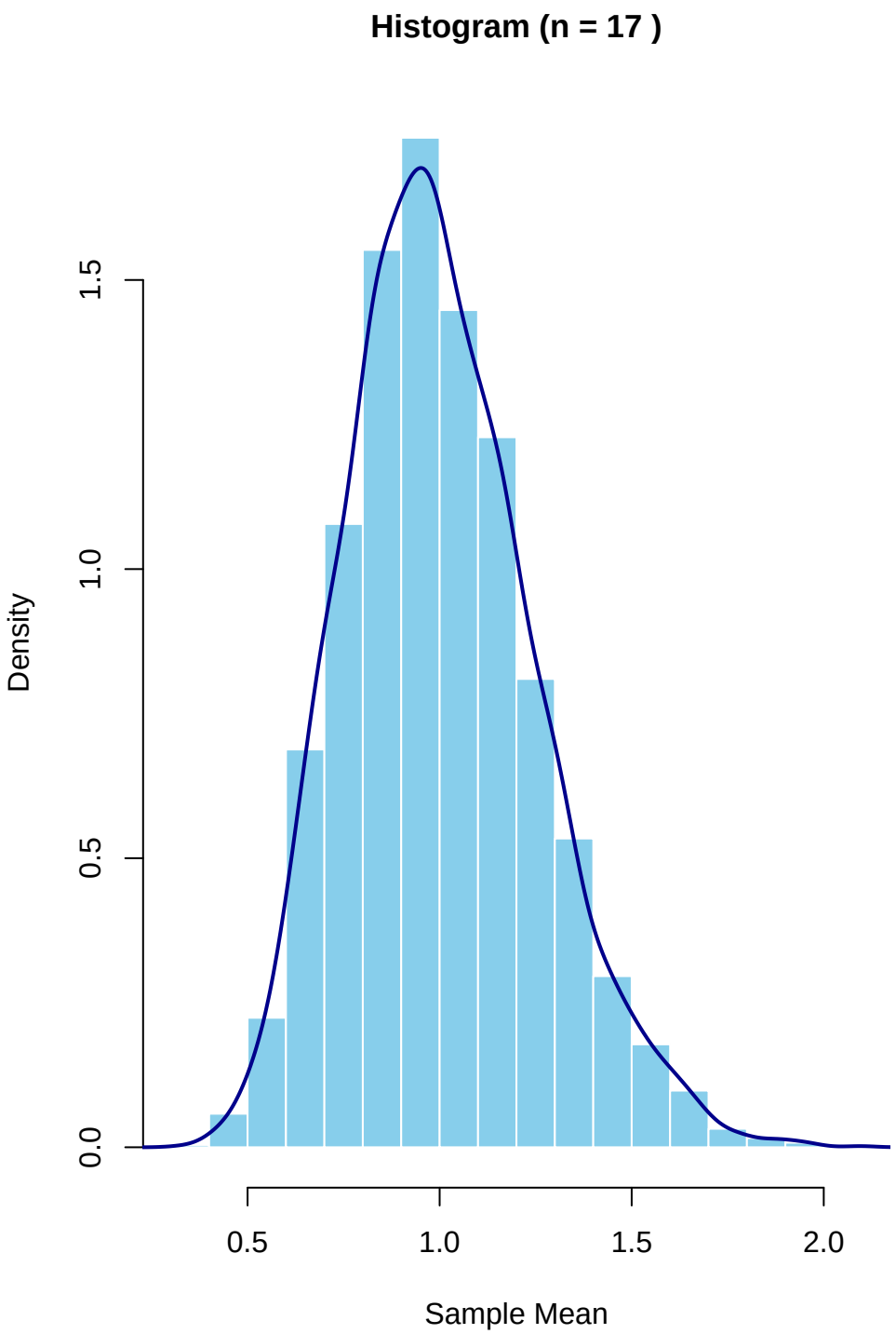


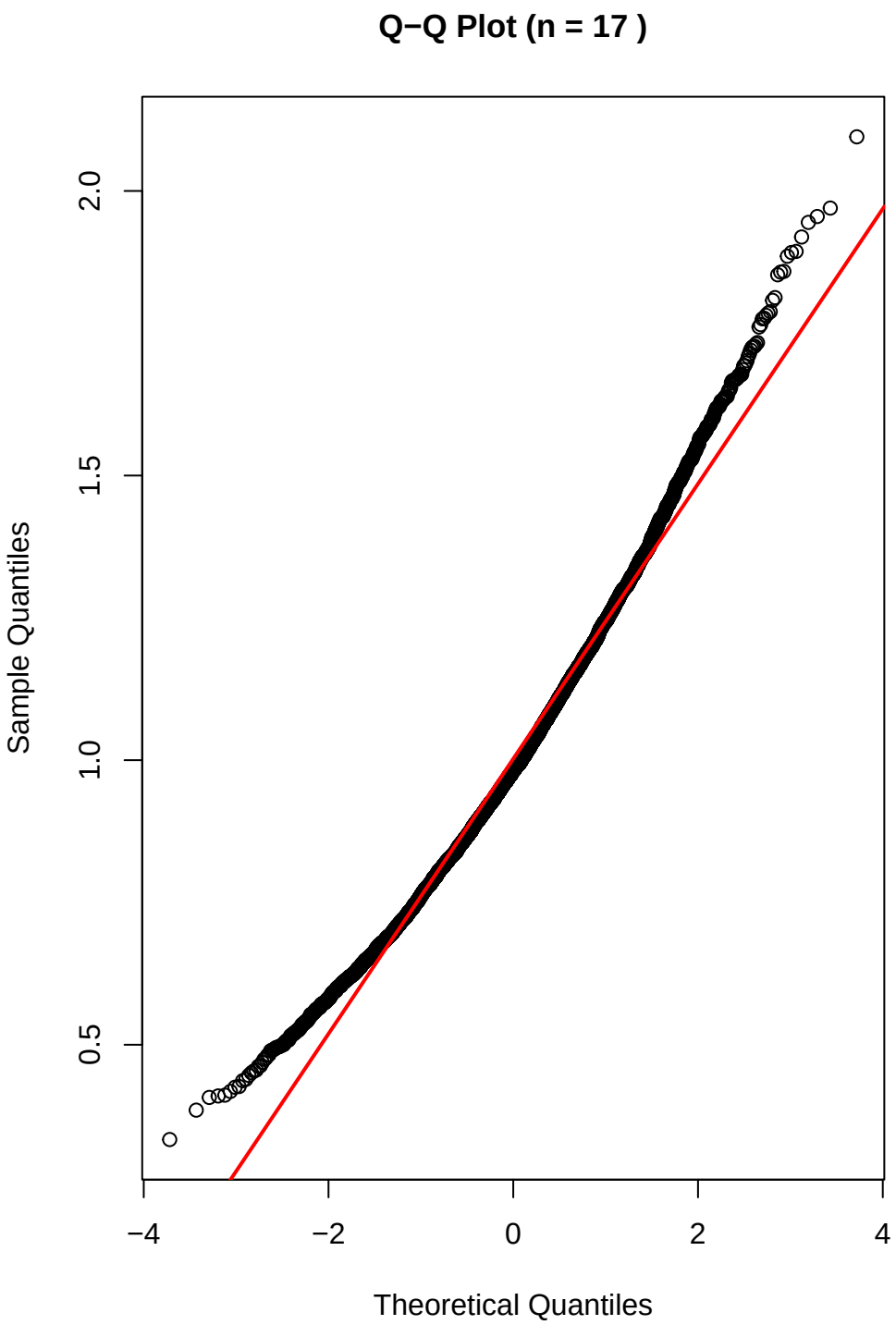



Exponential (lambda=1) — n = 15
Histogram (n = 15)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample Mean
Q−Q Plot (n = 15)
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.4308

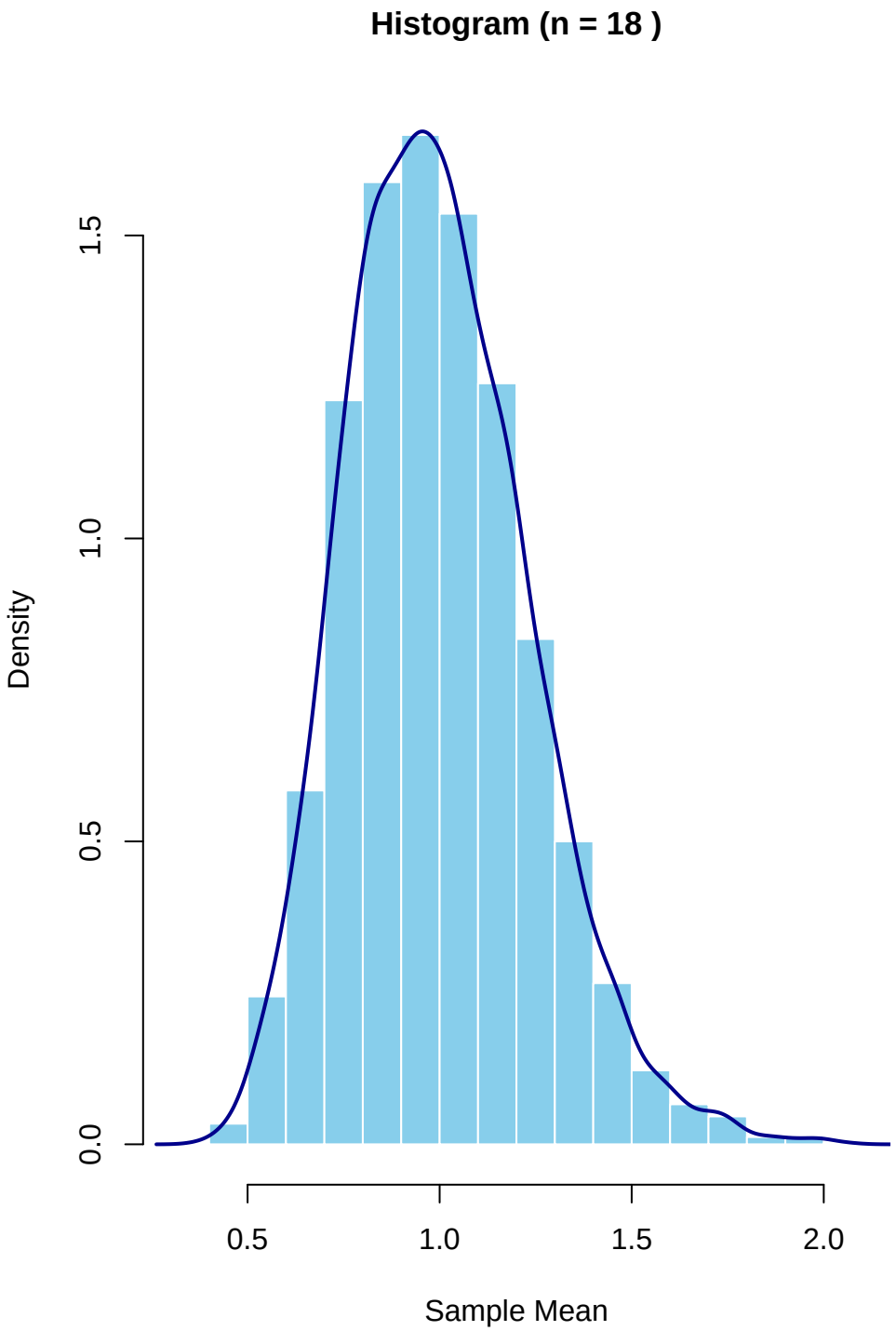

Exponential (lambda=1) — n = 16
Histogram (n = 16)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample Mean
Q−Q Plot (n = 16)
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.3889

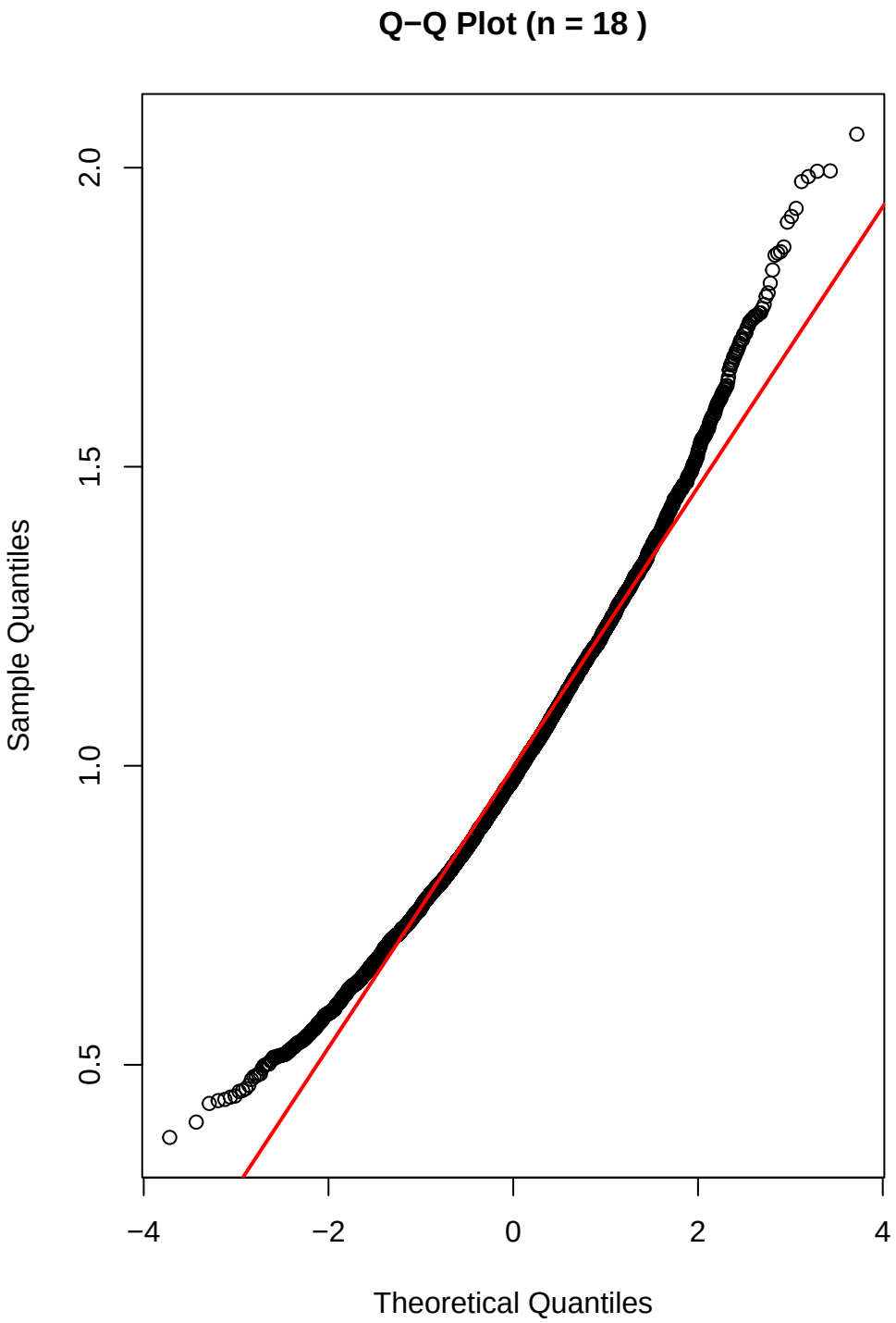
Exponential (lambda=1) — n = 17



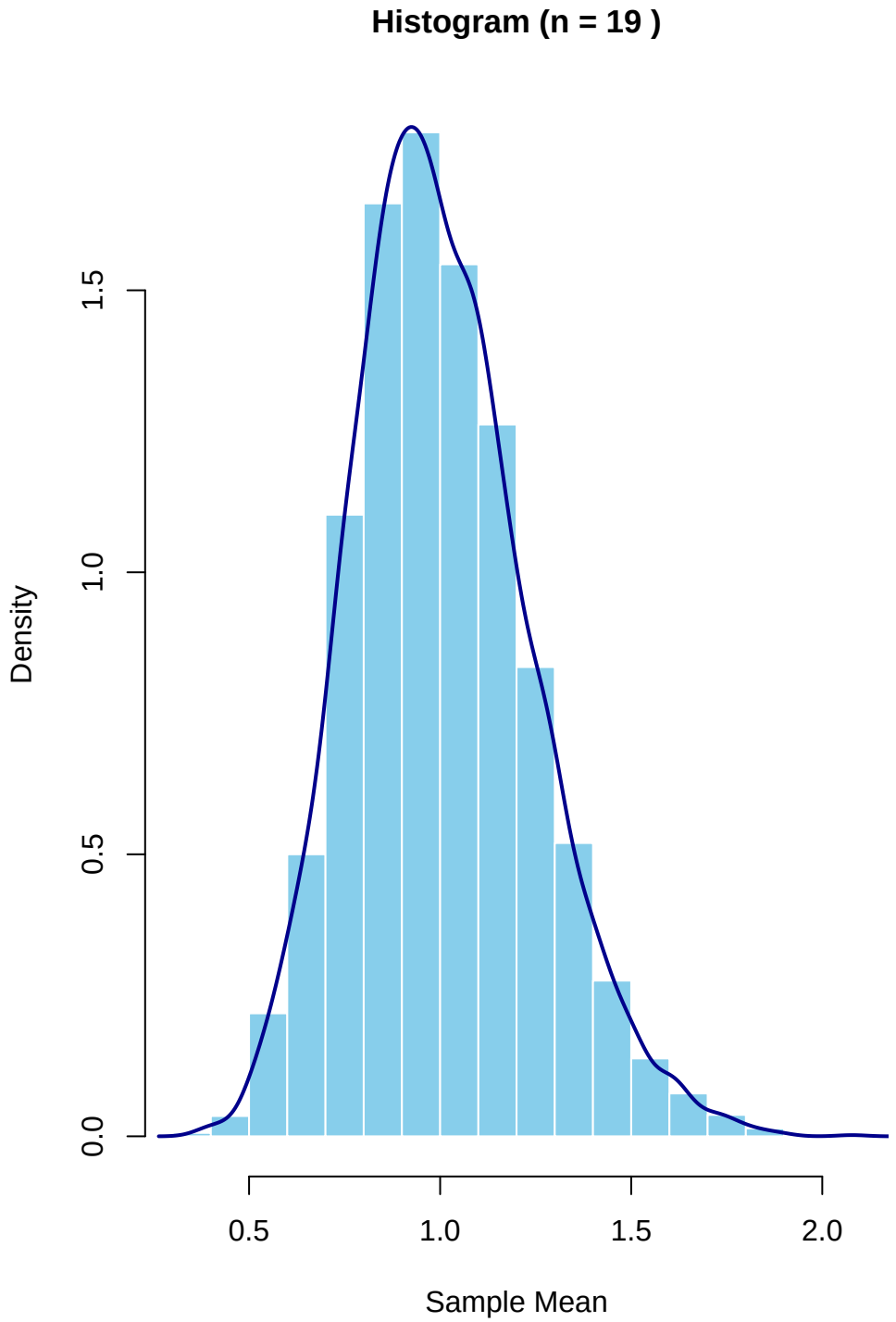


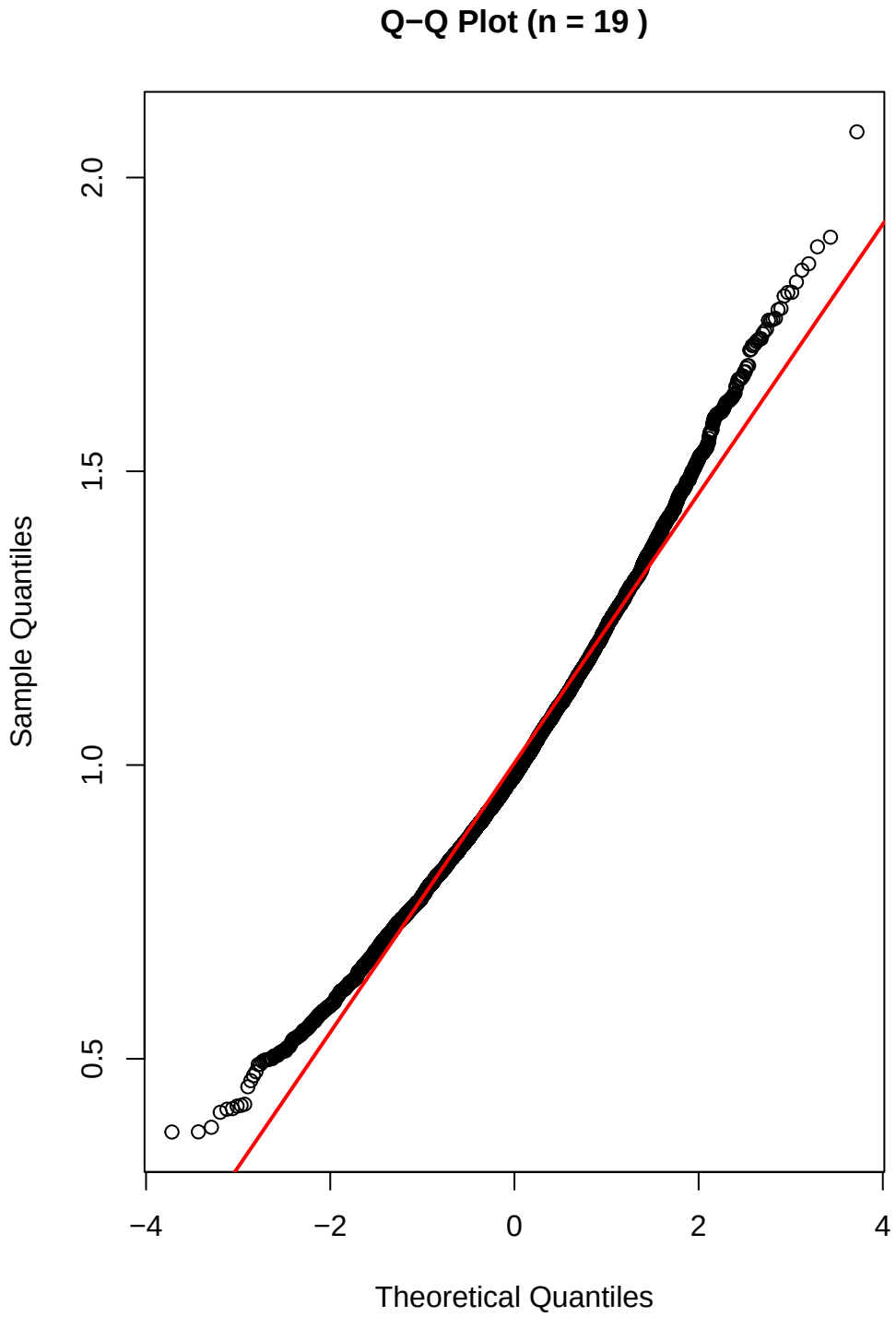
Exponential (lambda=1) — n = 18



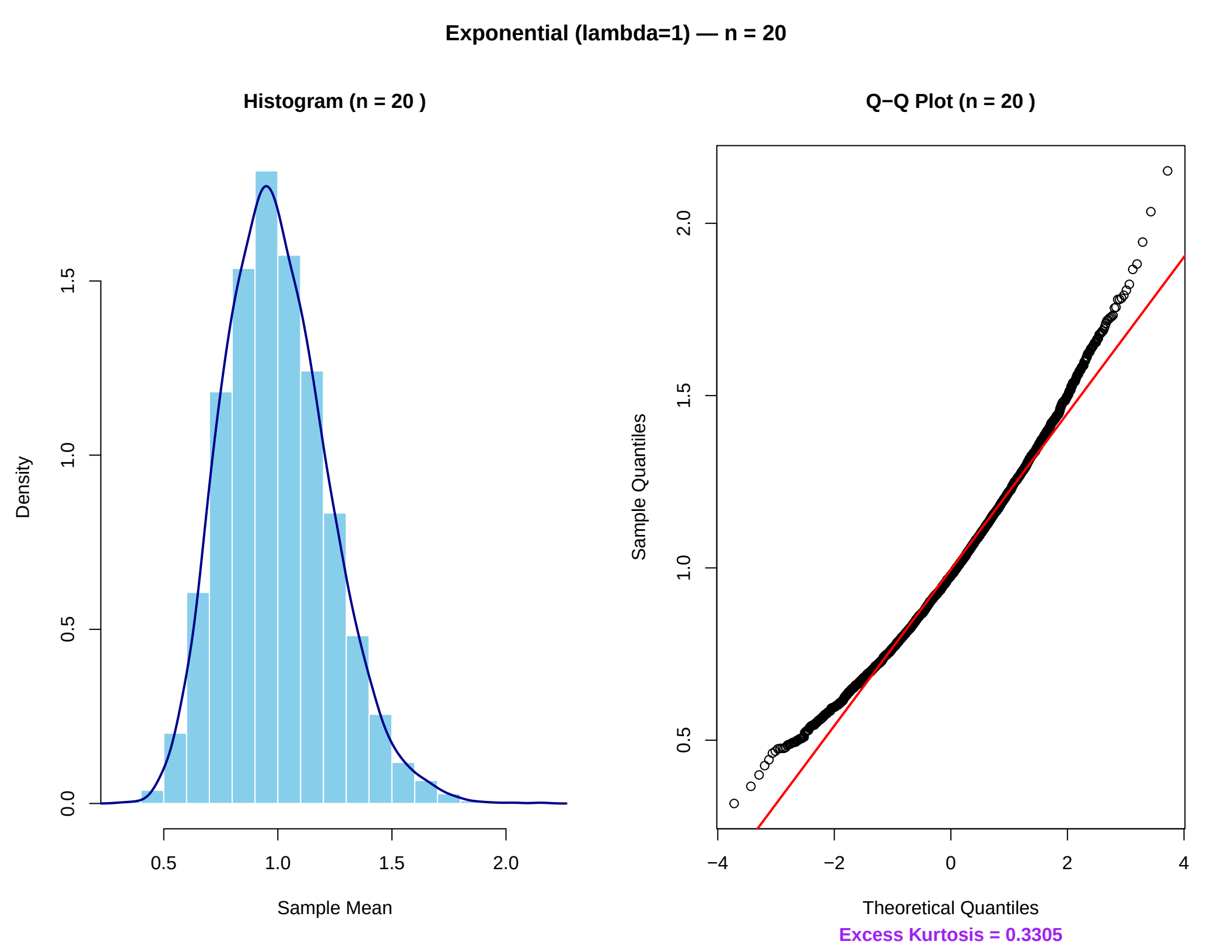


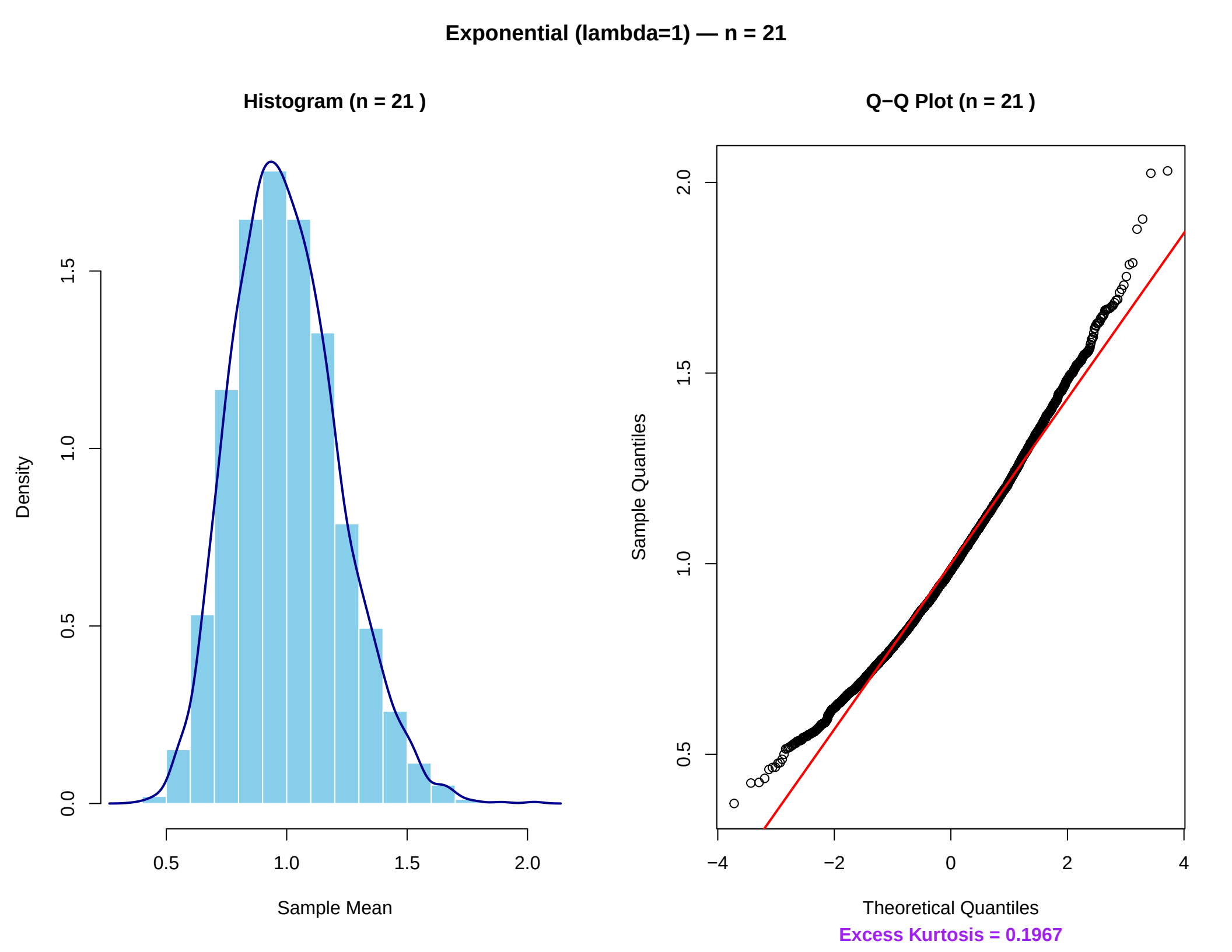
Exponential (lambda=1) — n = 19

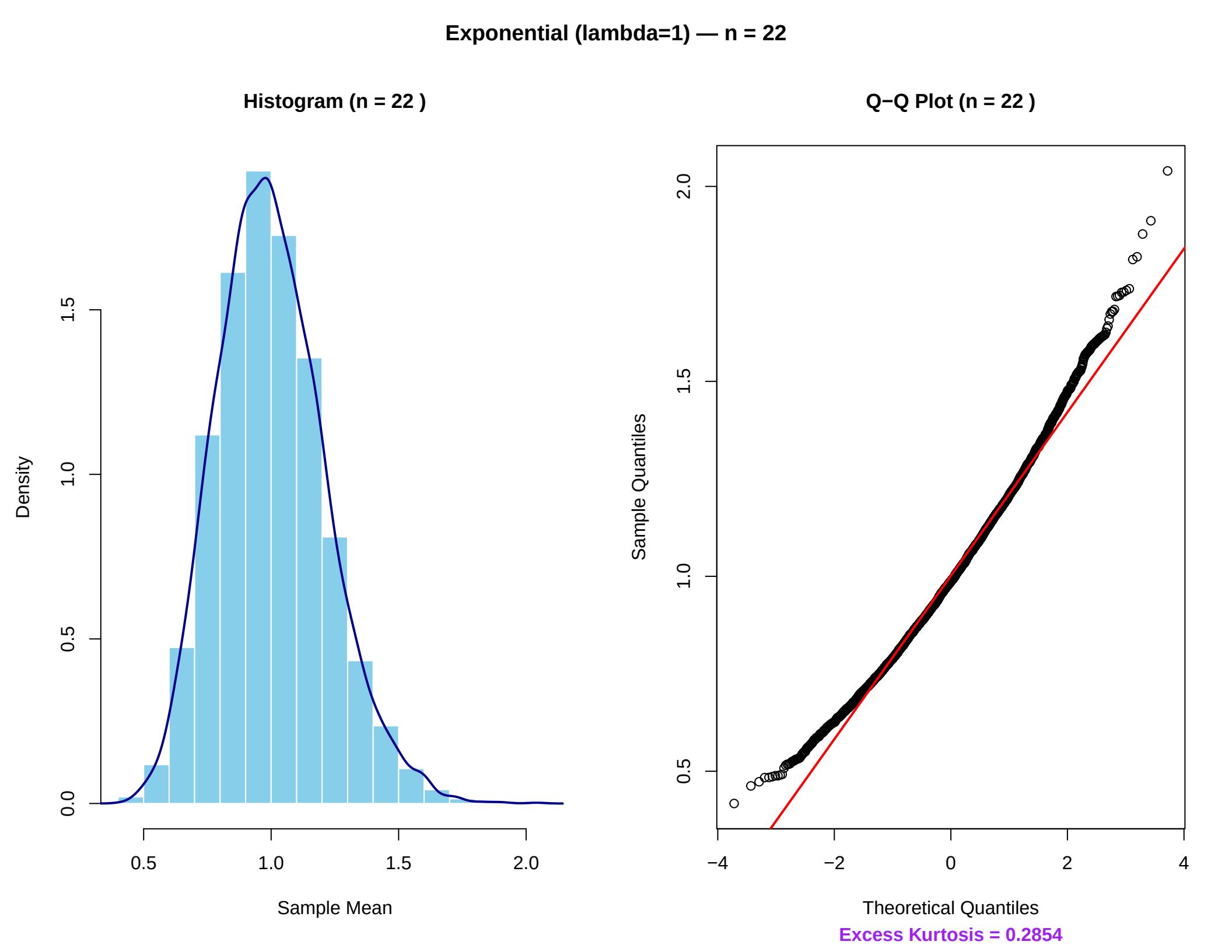


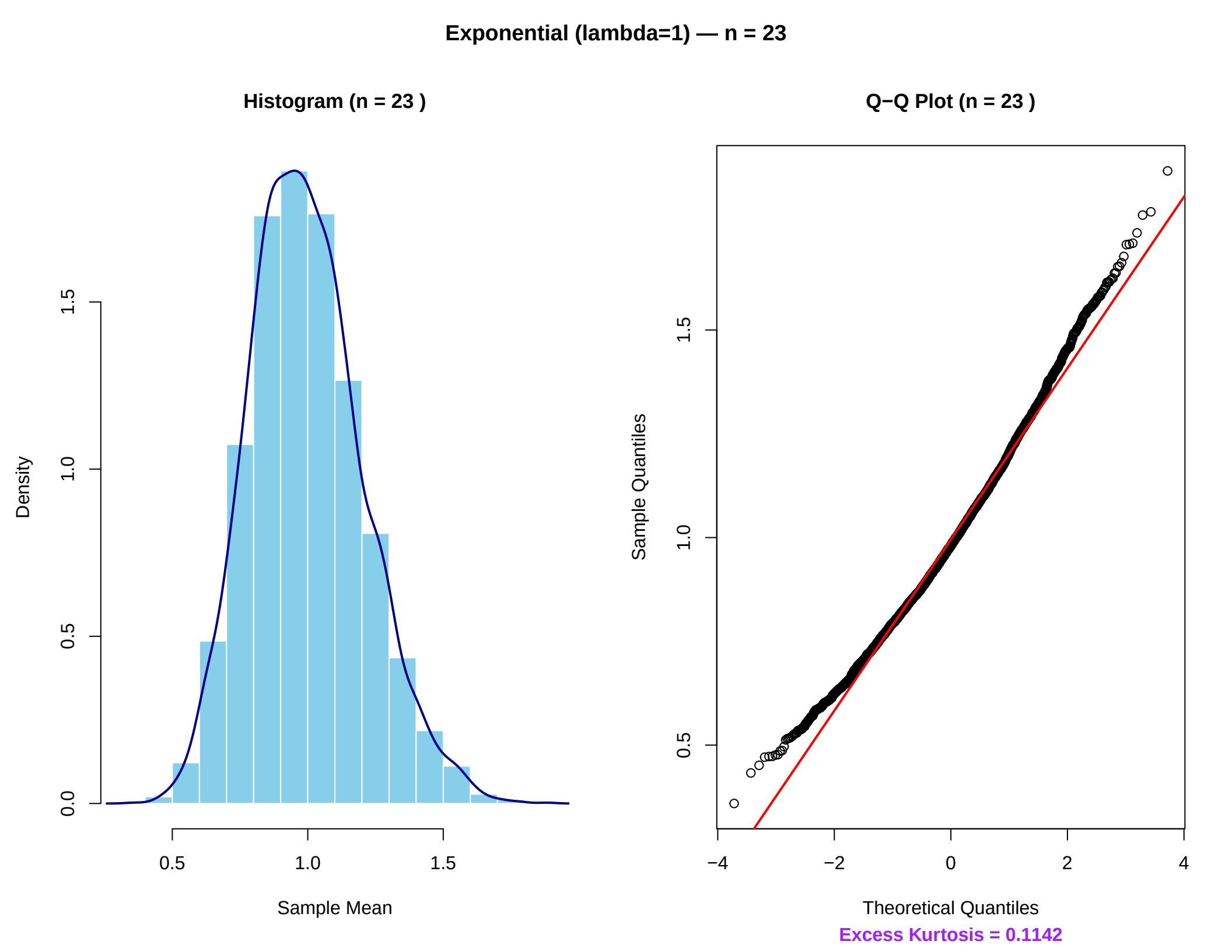


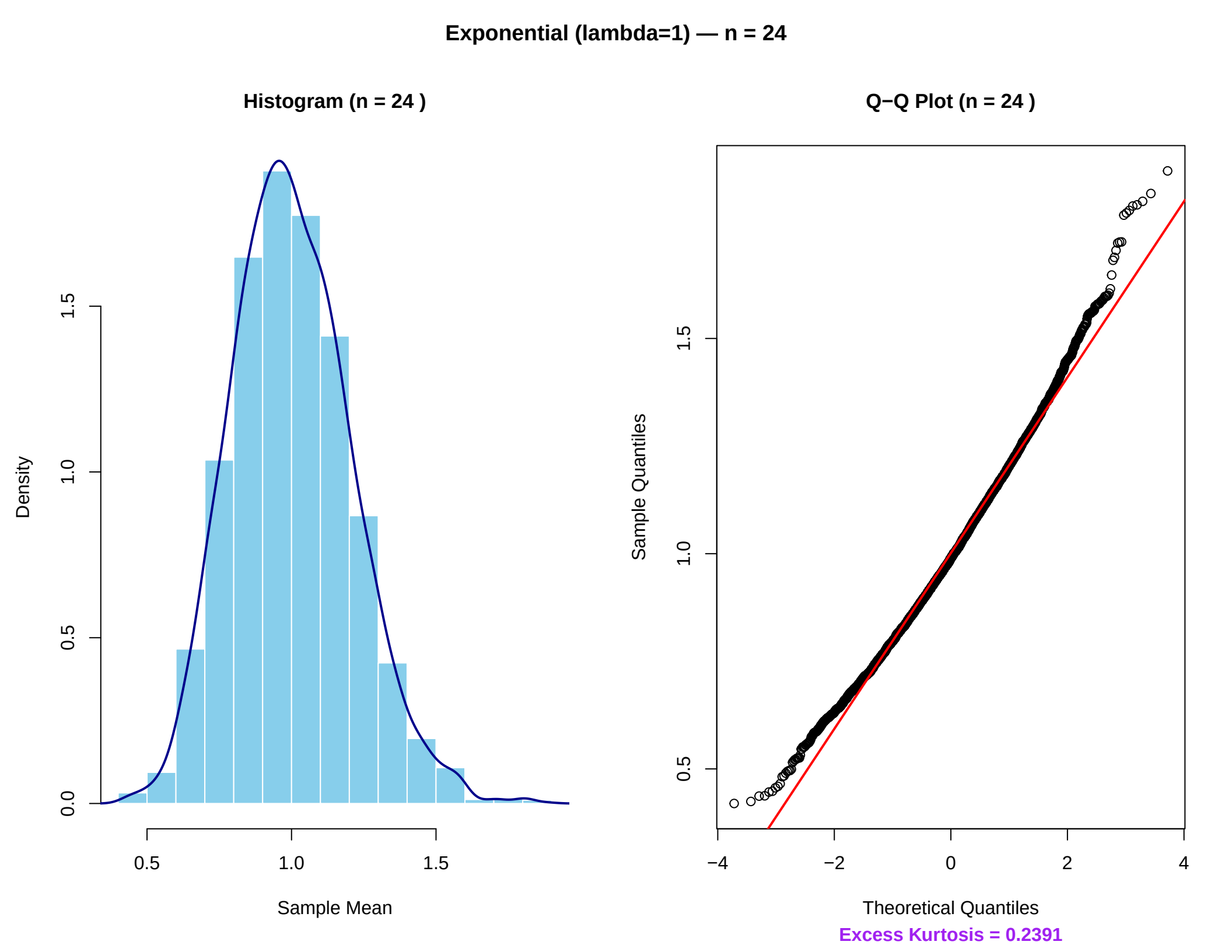
Excess Kurtosis = 0.2795

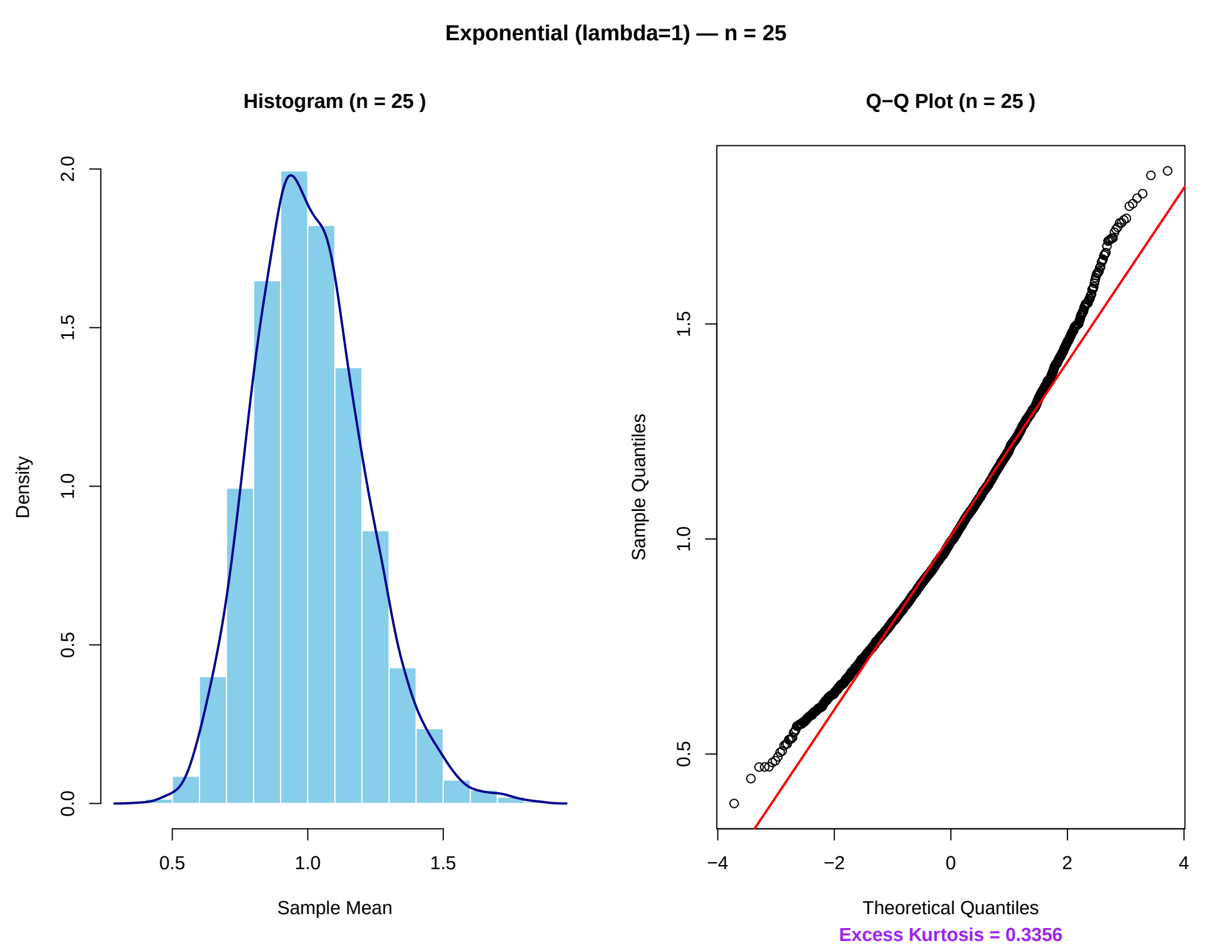

Exponential (lambda=1) — n = 20
Histogram (n = 20)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample Mean
Q−Q Plot (n = 20)
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.3305

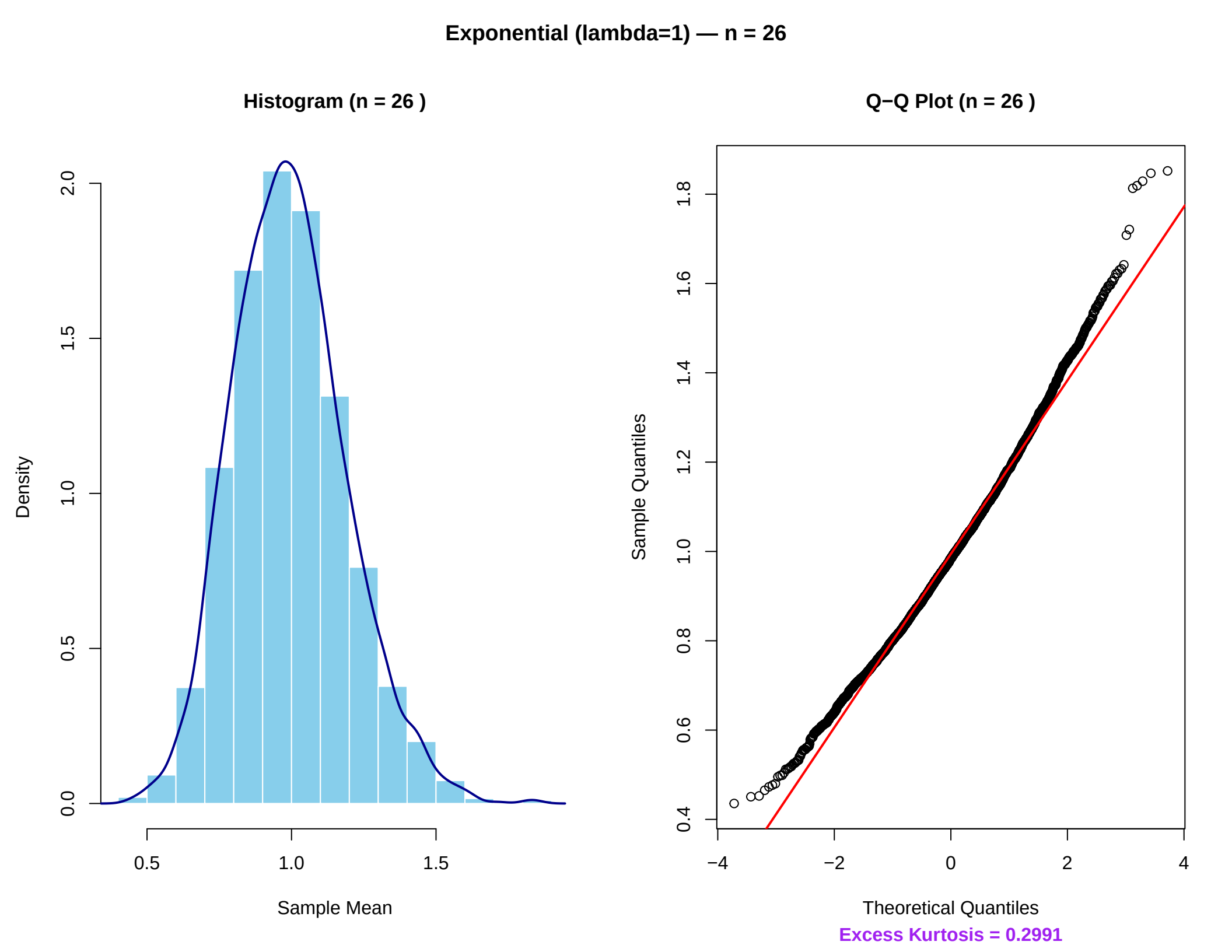

Exponential (lambda=1) — n = 21
Histogram (n = 21)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample Mean
Q−Q Plot (n = 21)
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.1967

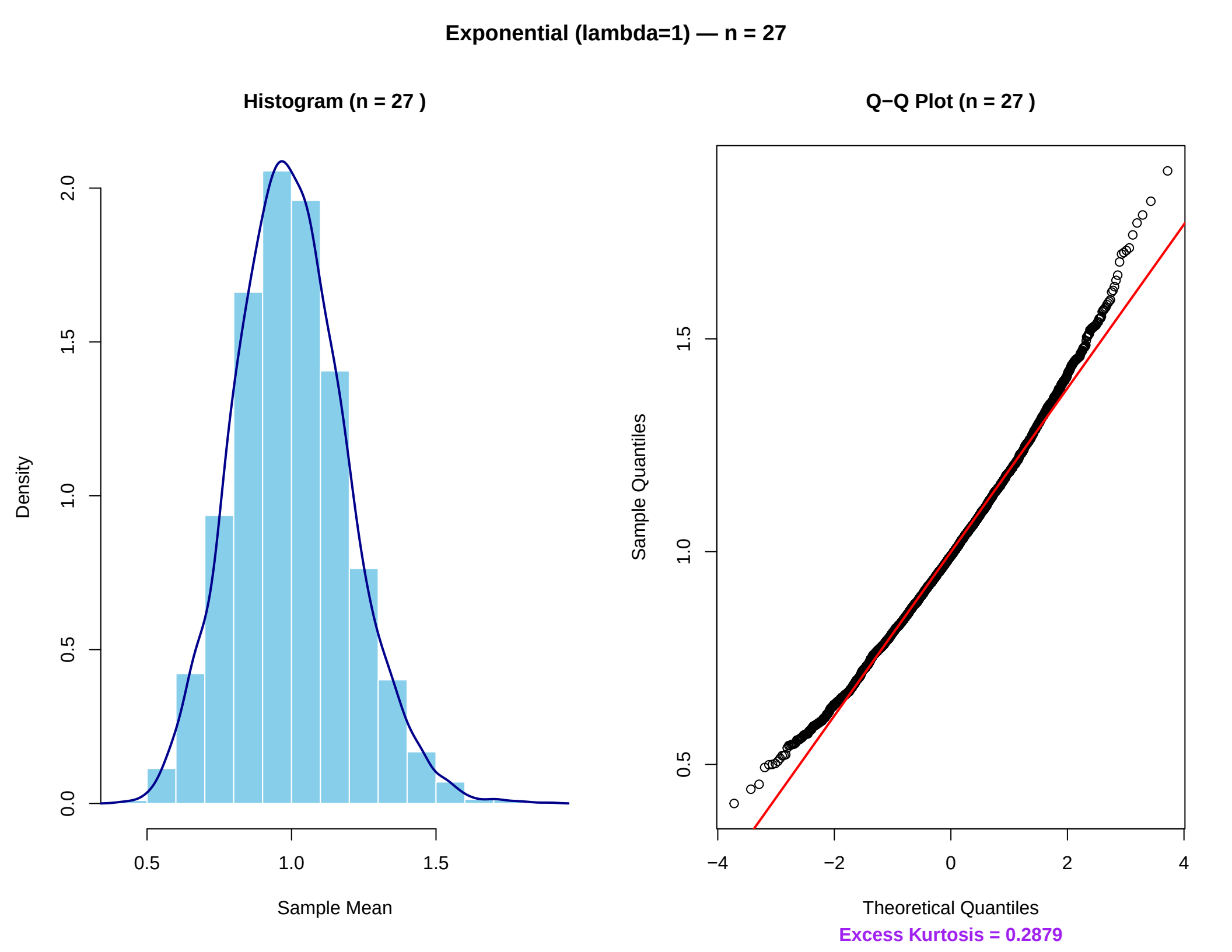

Exponential (lambda=1) — n = 22
Histogram (n = 22)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample Mean
Q−Q Plot (n = 22)
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.2854

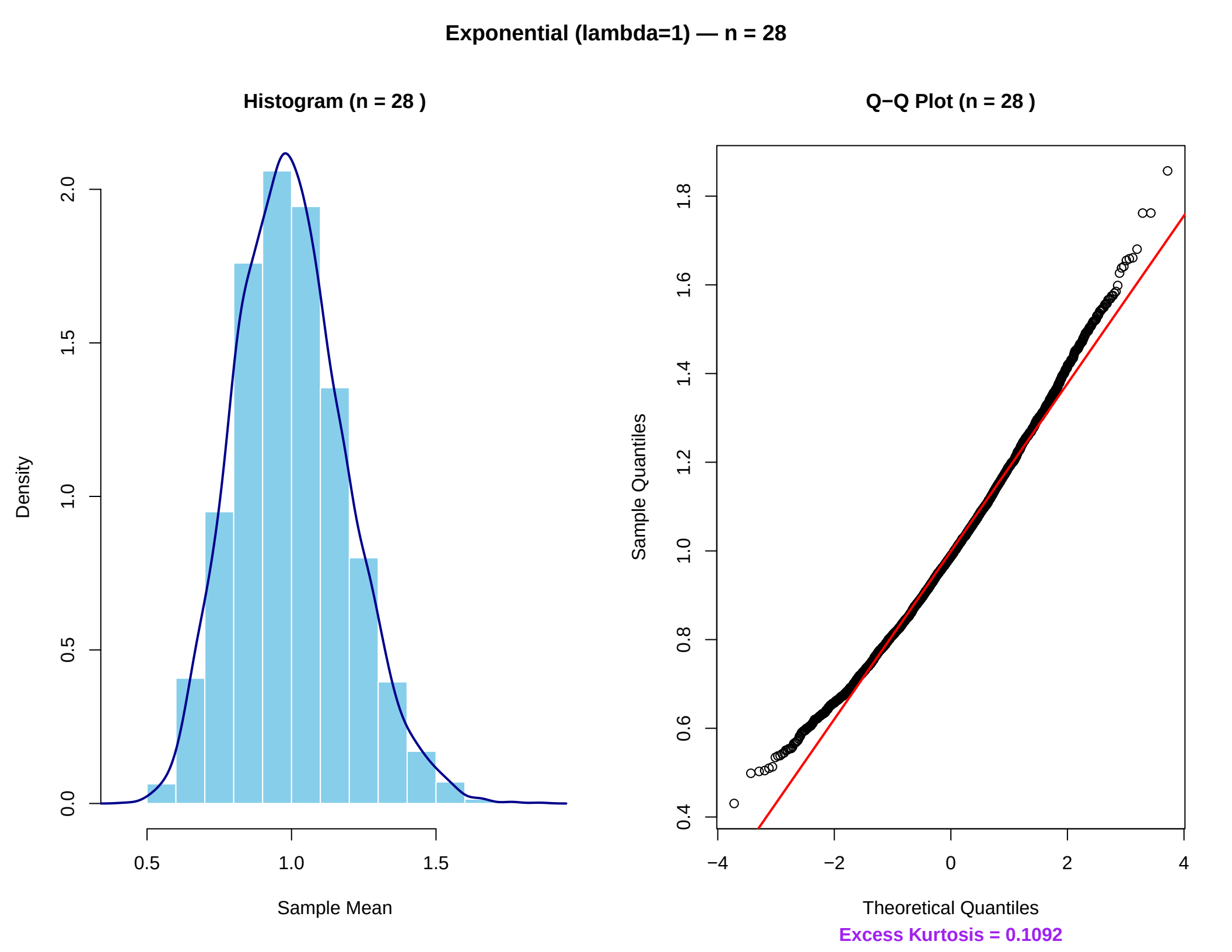

Exponential (lambda=1) — n = 23
Histogram (n = 23)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 23)
Sample Quantiles
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.1142

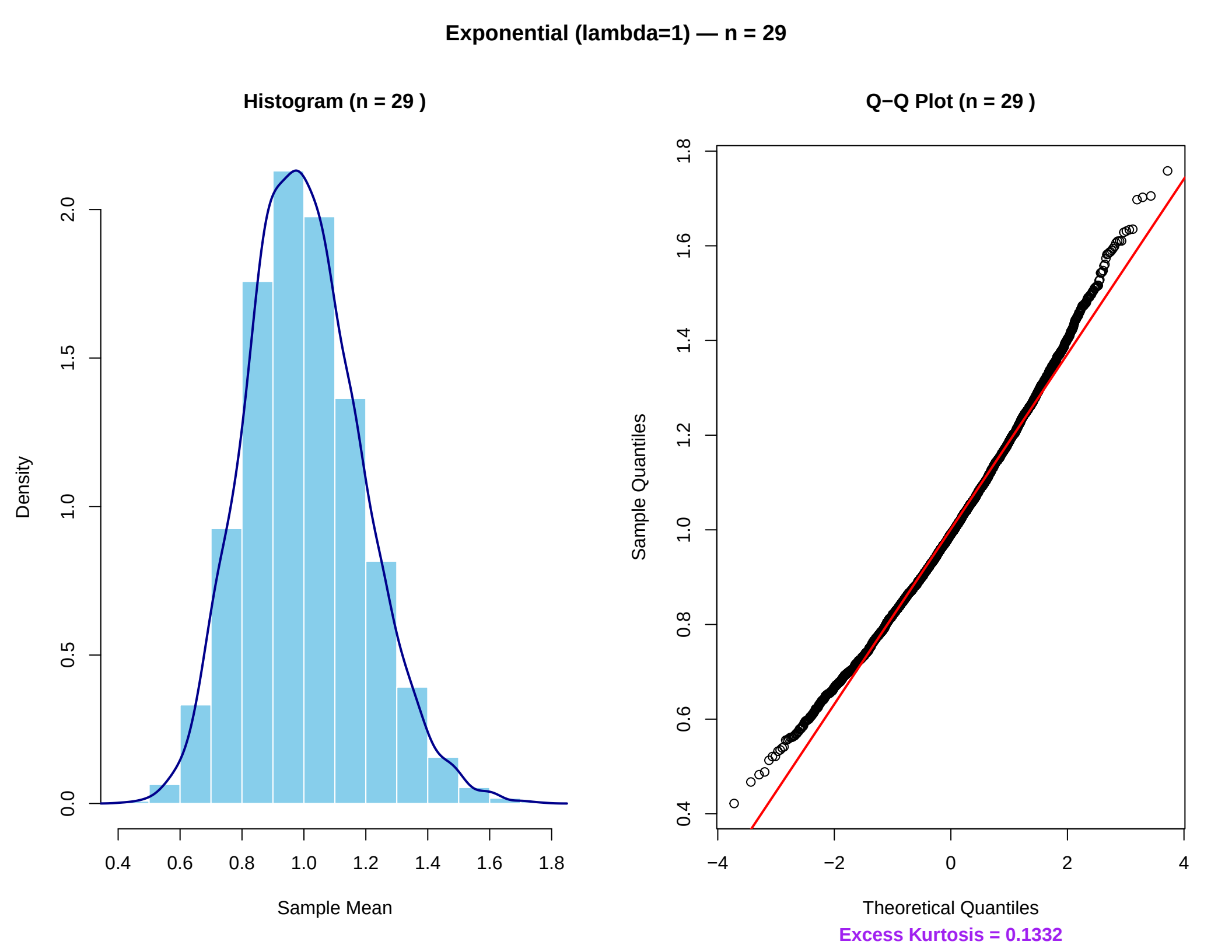

Exponential (lambda=1) — n = 24
Histogram (n = 24)
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 24)
Sample Quantiles
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.2391

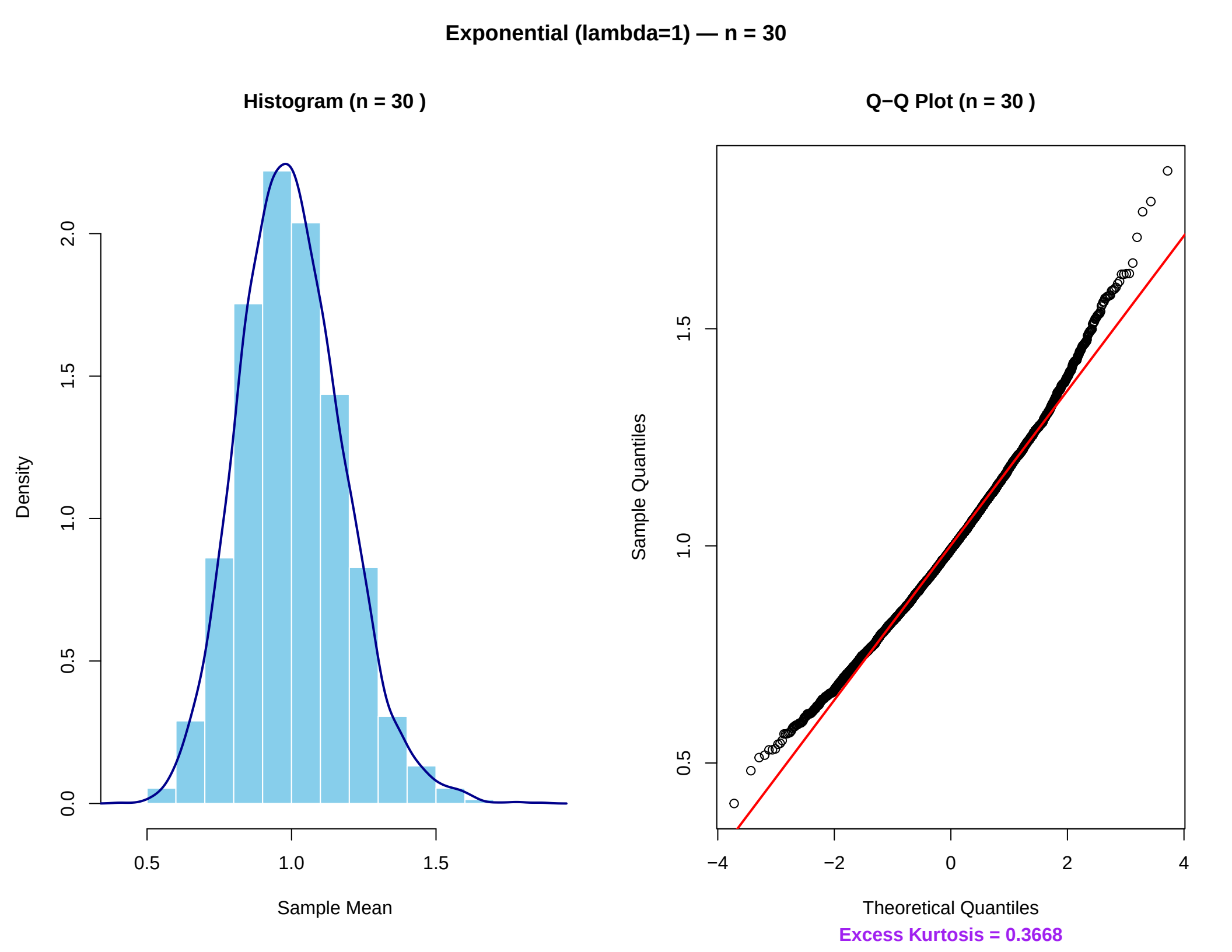

Exponential (lambda=1) — n = 25
Histogram (n = 25)
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 25)
Sample Quantiles
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.3356

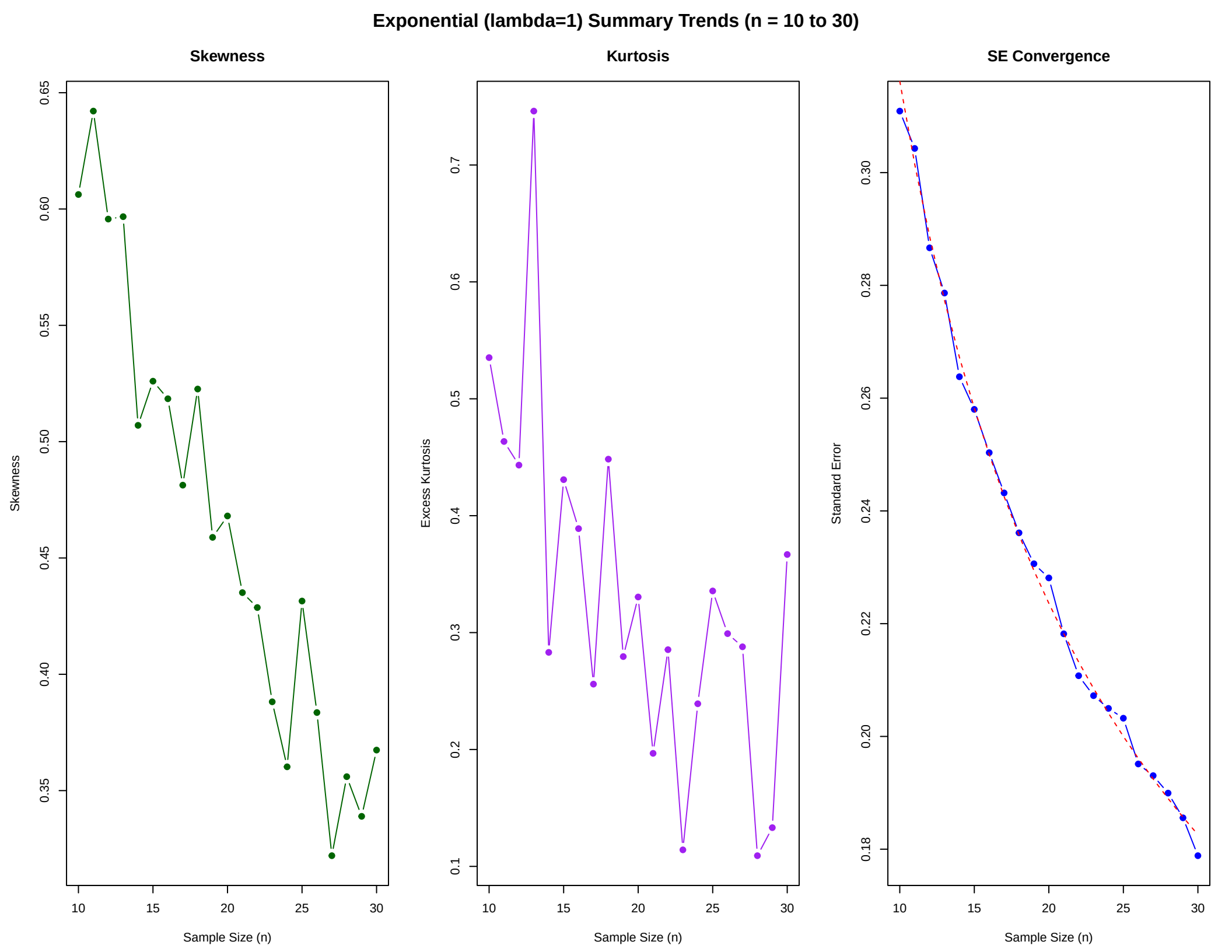

Exponential (lambda=1) — n = 26
Histogram (n = 26)
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 26)
Sample Quantiles
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.2991


Exponential (lambda=1) — n = 27
Histogram (n = 27)
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 27)
Sample Quantiles
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.2879

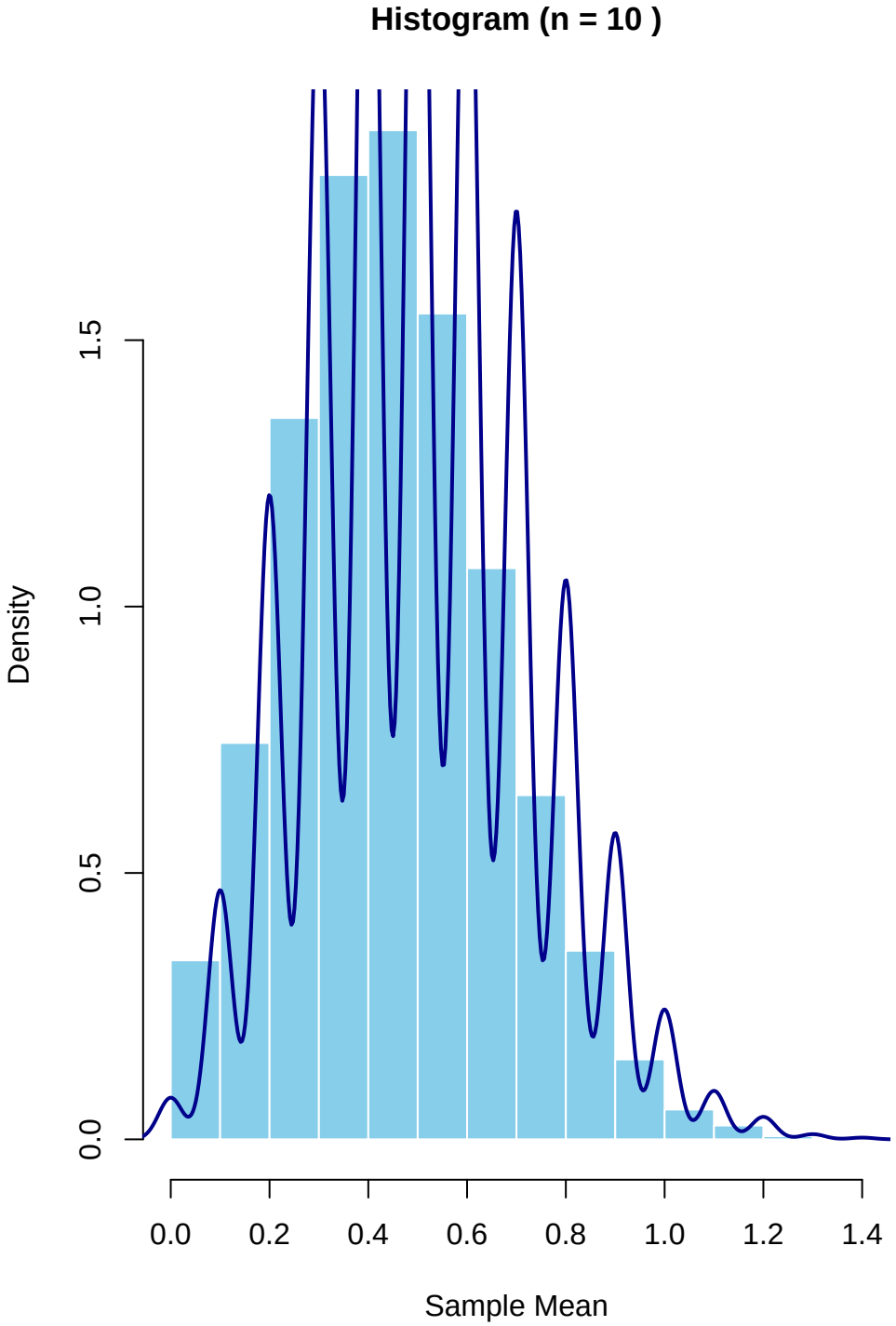

Exponential (lambda=1) — n = 28
Histogram (n = 28)
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 28)
Sample Quantiles
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.1092

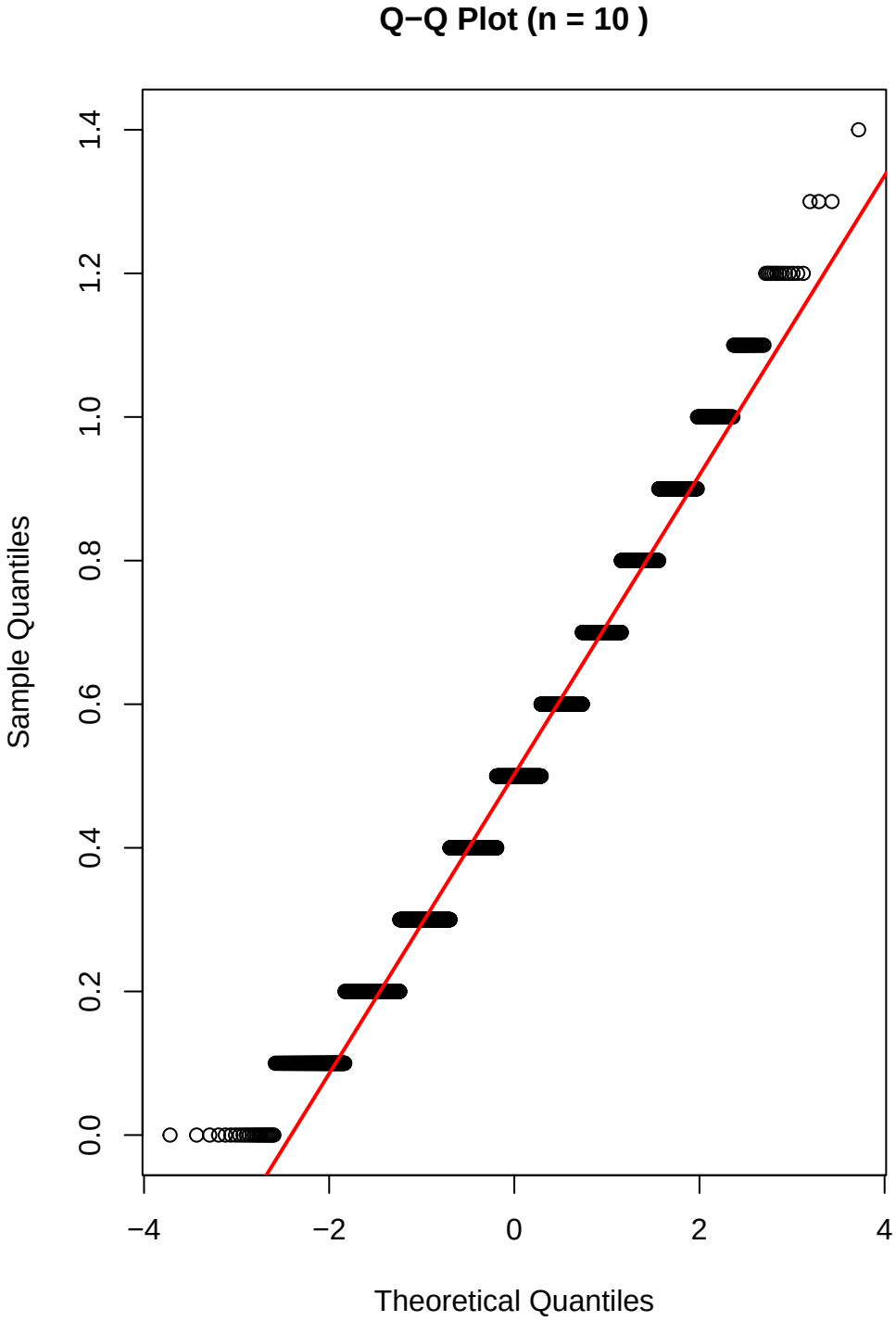

Exponential (lambda=1) — n = 29
Histogram (n = 29)
Density
0.0
0.5
1.0
1.5
2.0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Sample Mean
Q−Q Plot (n = 29)
Sample Quantiles
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.1332

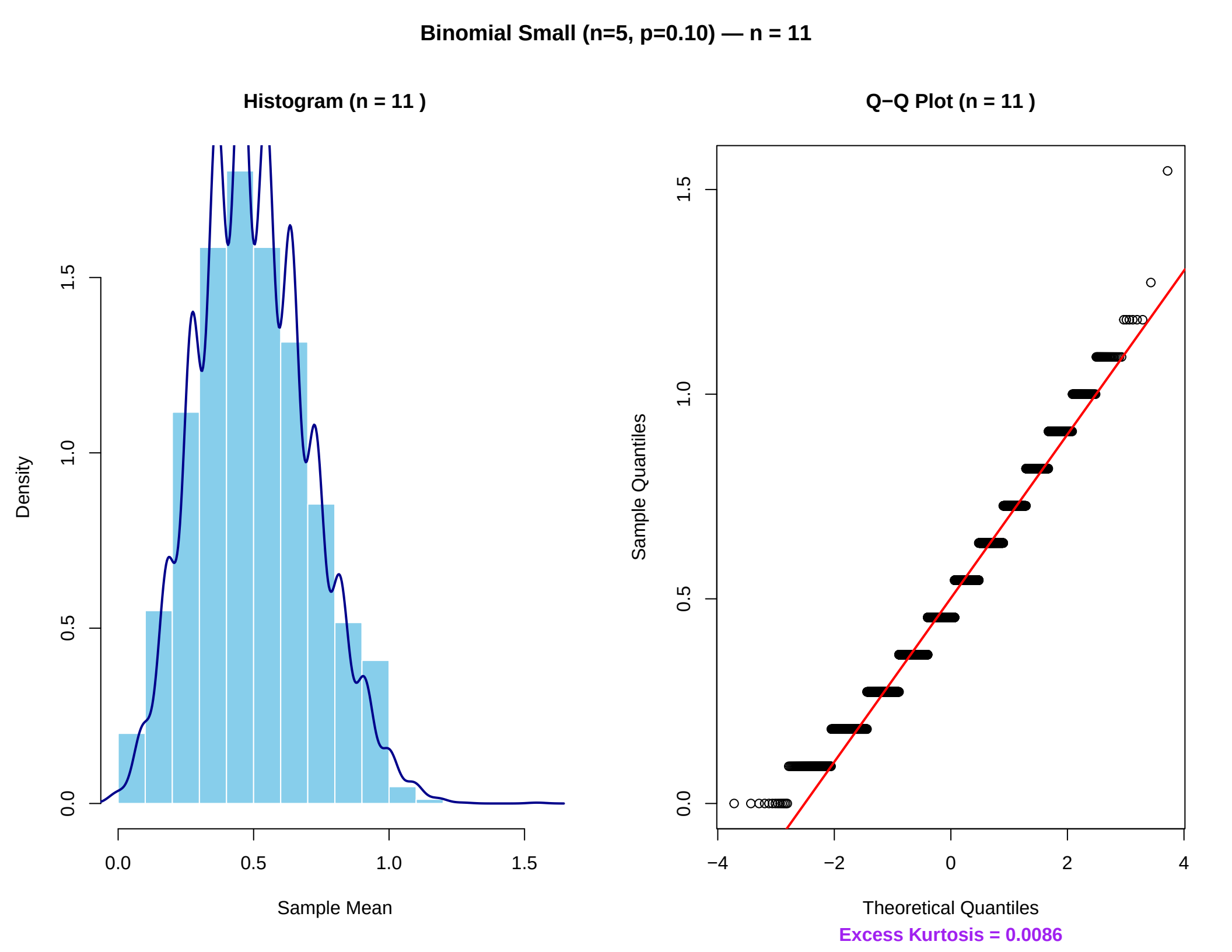

Exponential (lambda=1) — n = 30
Histogram (n = 30)
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 30)
Sample Quantiles
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.3668


Exponential (lambda=1) Summary Trends (n = 10 to 30)
Skewness
Skewness
0.65
0.60
0.55
0.50
0.45
0.40
0.35
10
15
20
25
30
Sample Size (n)
Kurtosis
Excess Kurtosis
0.7
0.6
0.5
0.4
0.3
0.2
0.1
10
15
20
25
30
Sample Size (n)
SE Convergence
Standard Error
0.30
0.28
0.26
0.24
0.22
0.20
0.18
10
15
20
25
30
Sample Size (n)

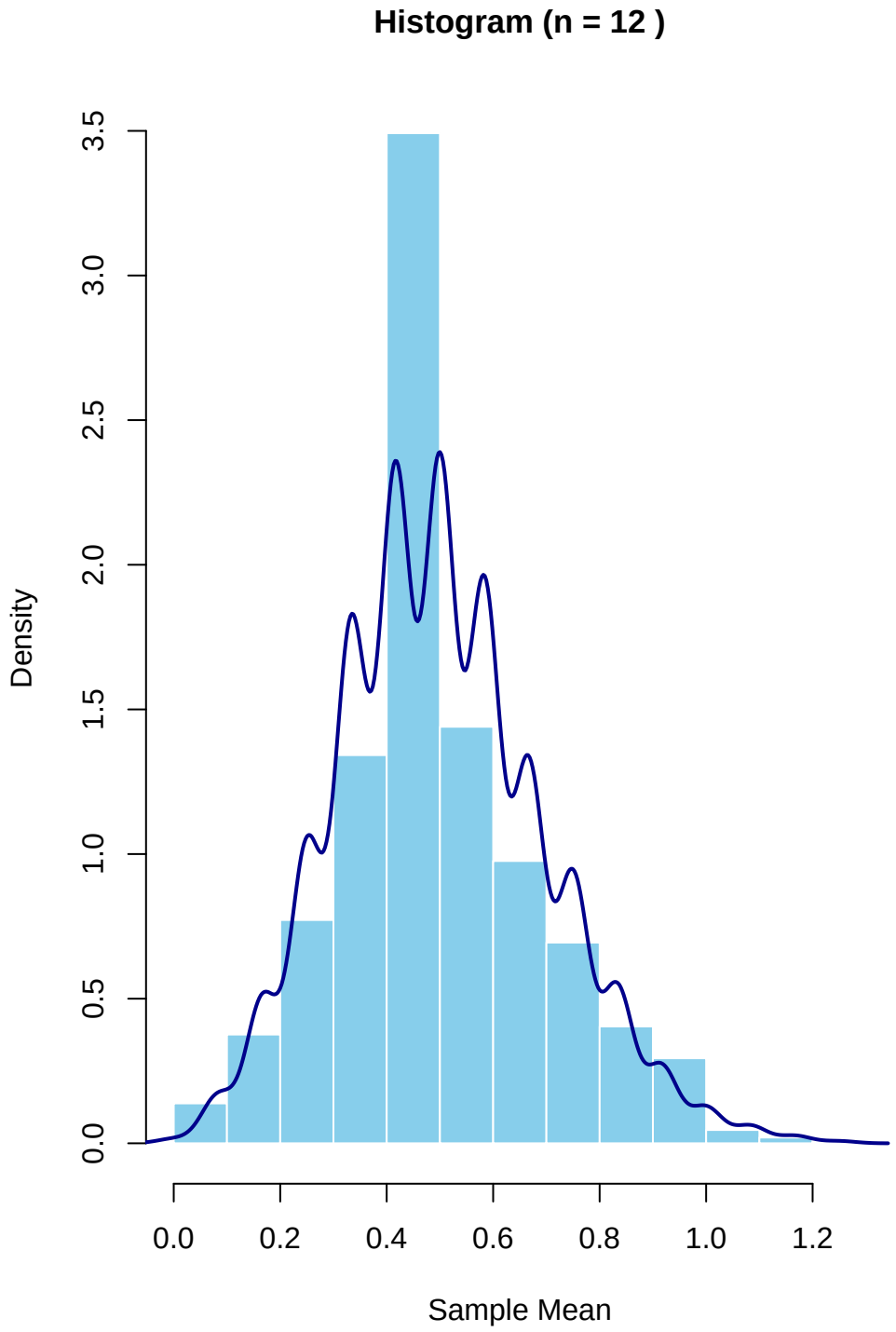
Binomial Small (n=5, p=0.10) — n = 10

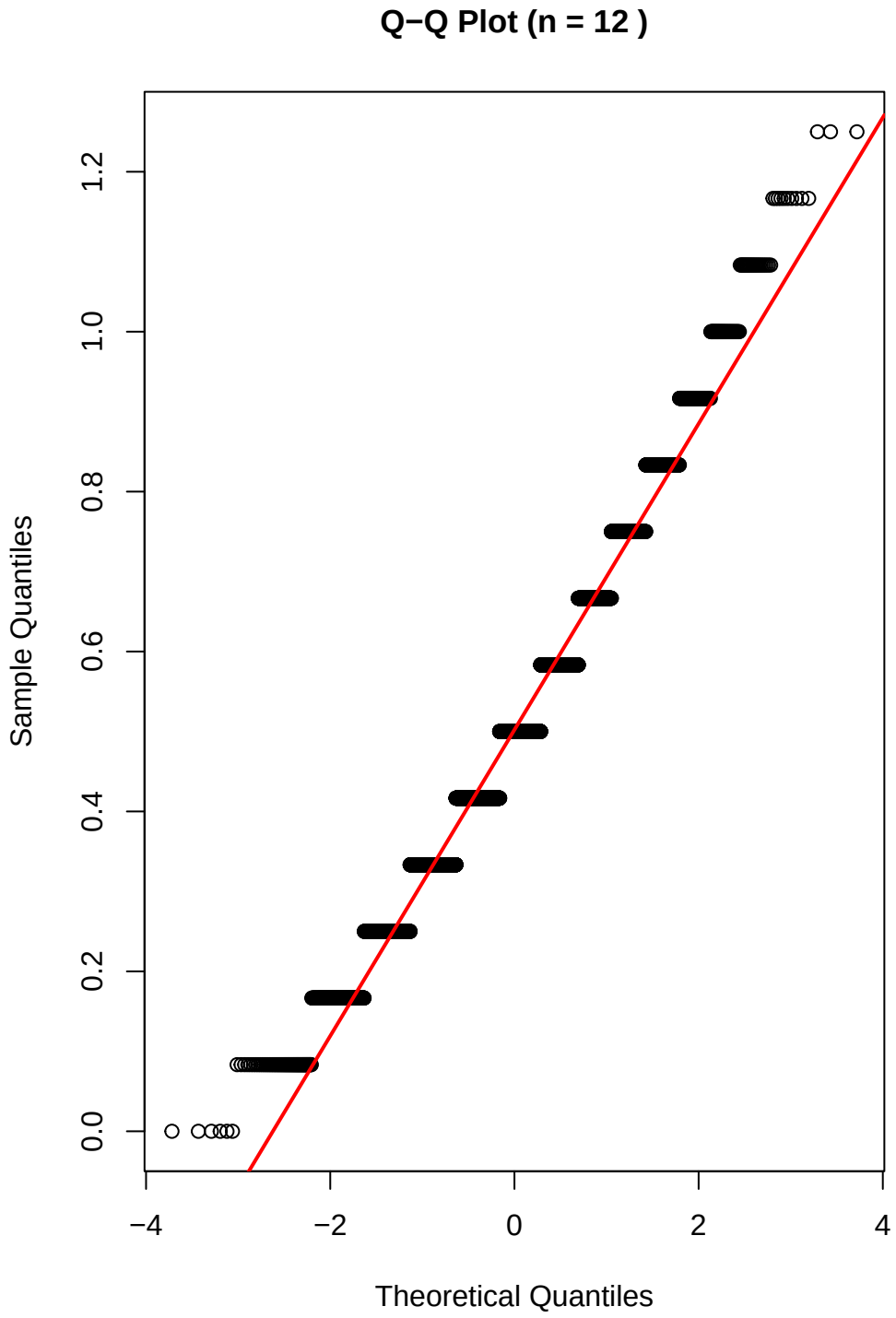





Binomial Small (n=5, p=0.10) — n = 11
Histogram (n = 11)
Density
0.0
0.5
1.0
1.5
0.0
0.5
1.0
1.5
Sample Mean
Q−Q Plot (n = 11)
Sample Quantiles
0.0
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0086

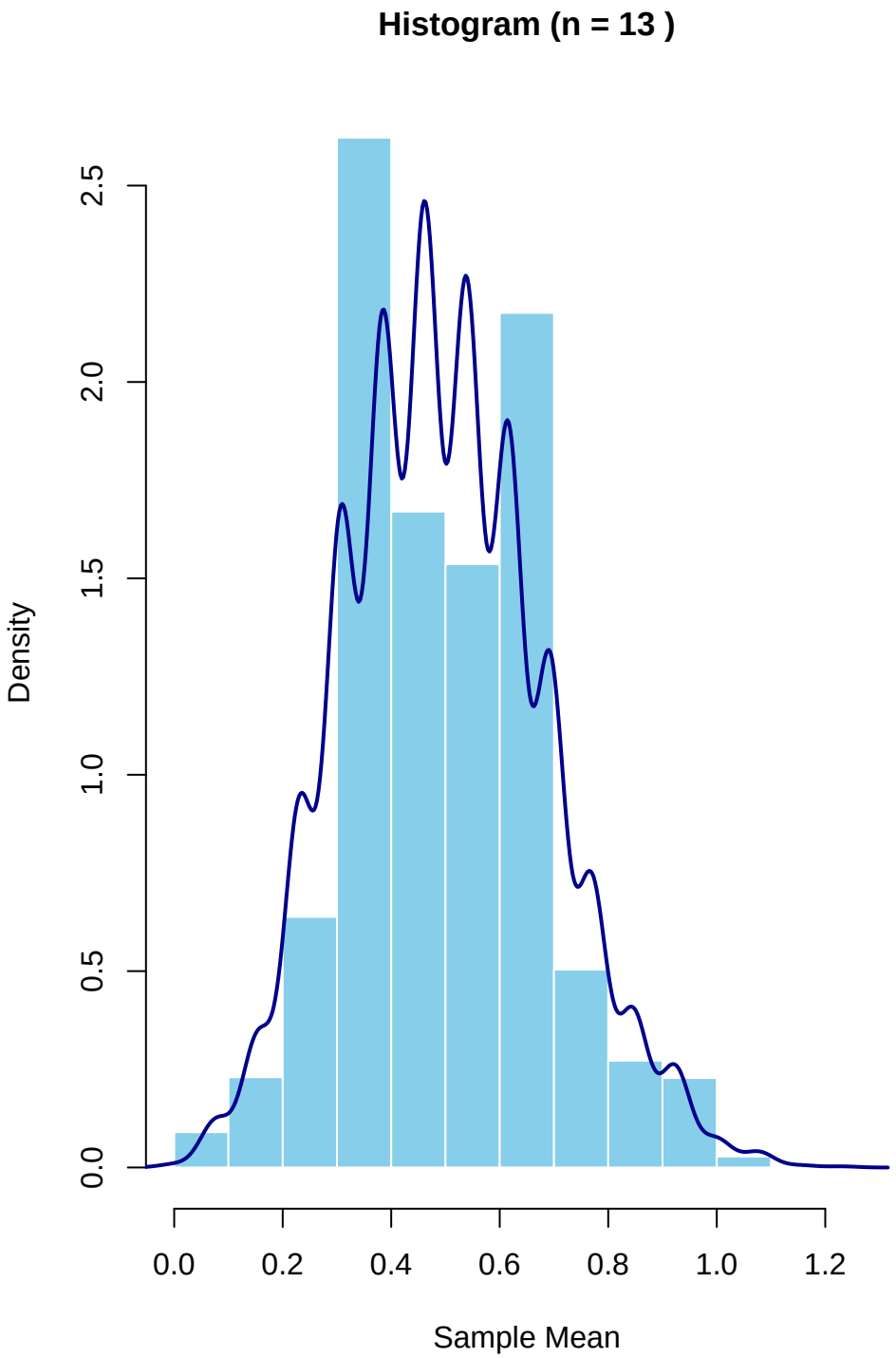
Binomial Small (n=5, p=0.10) — n = 12

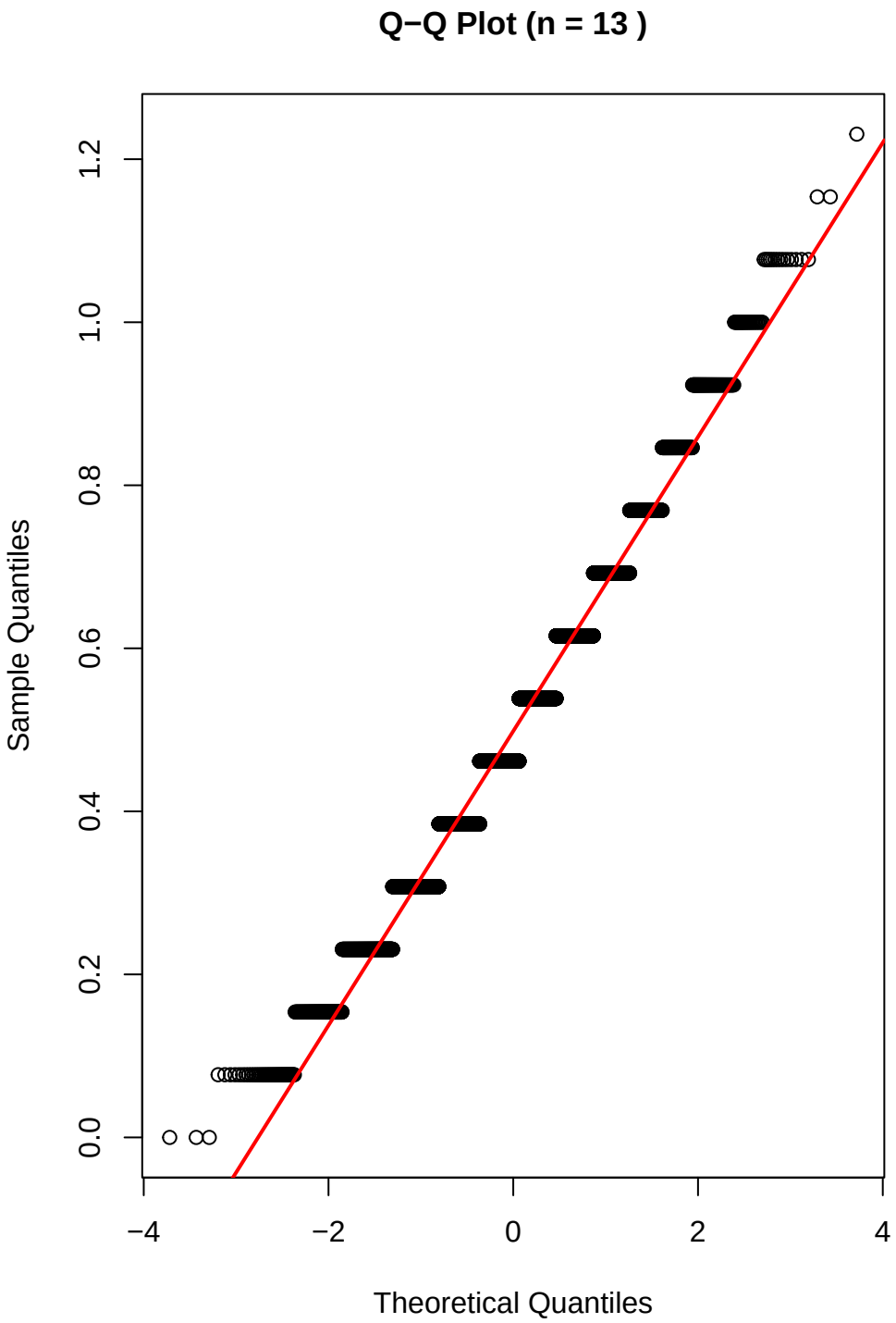




Excess Kurtosis = 0.1639

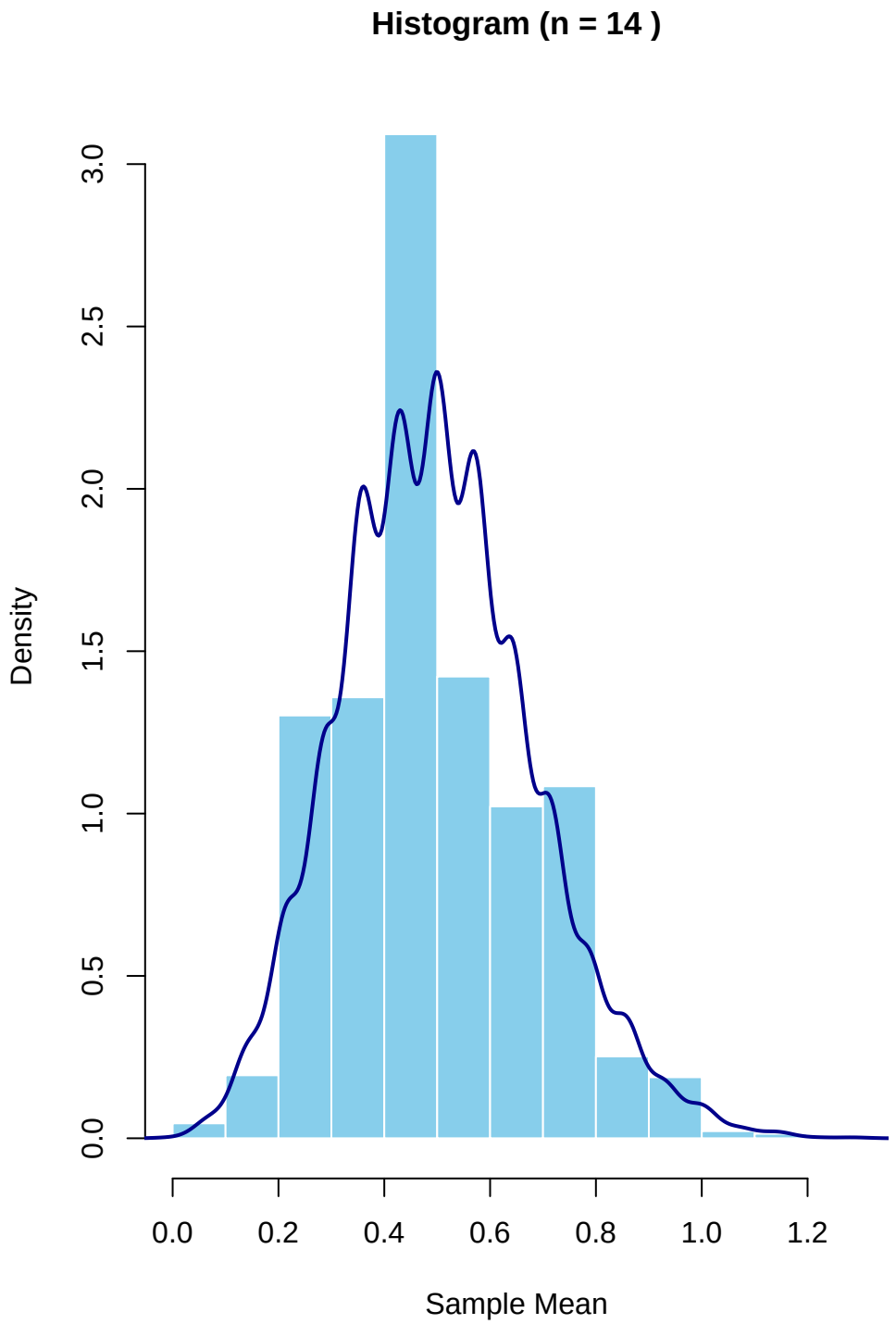
Binomial Small (n=5, p=0.10) — n = 13

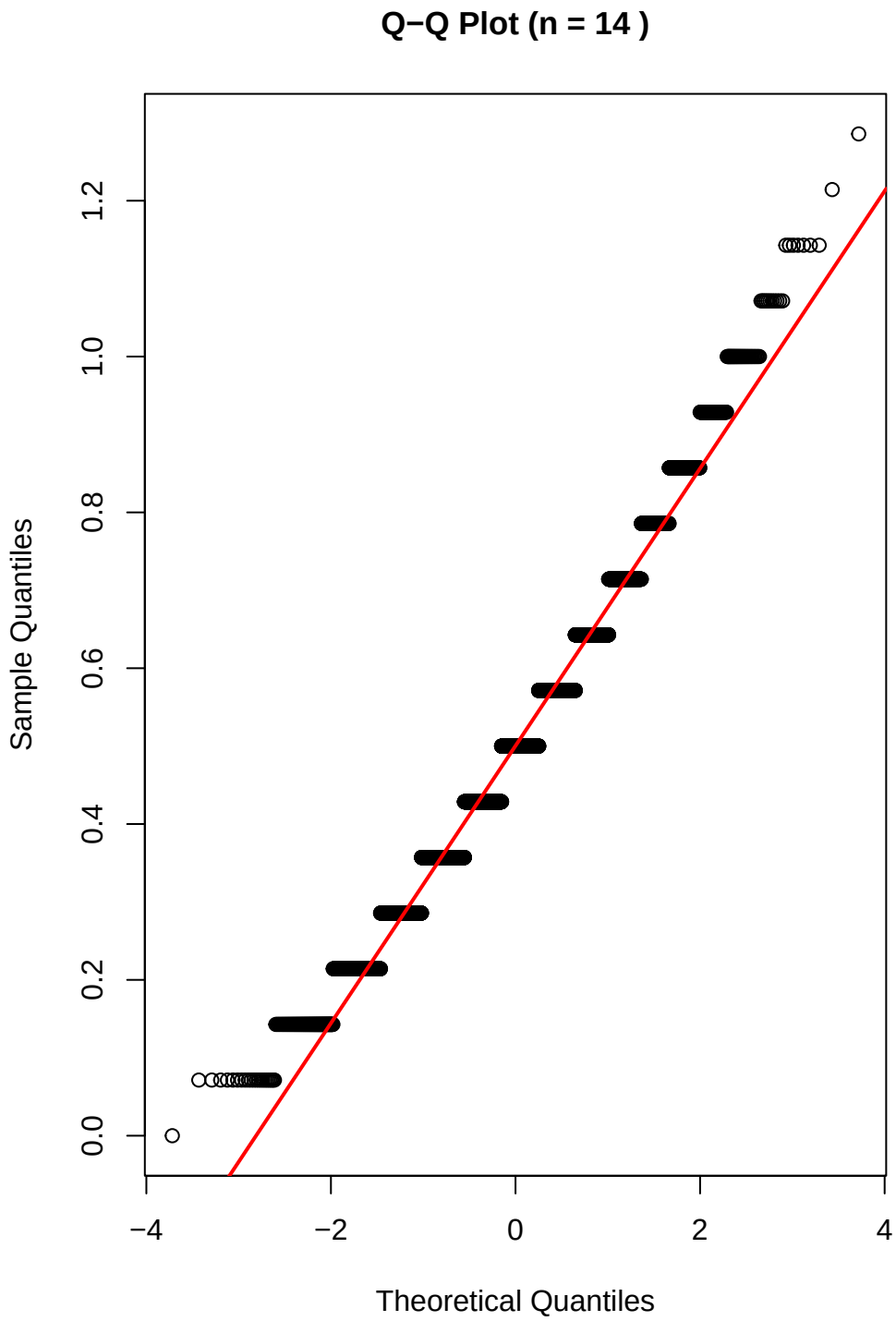




Excess Kurtosis = −0.0279

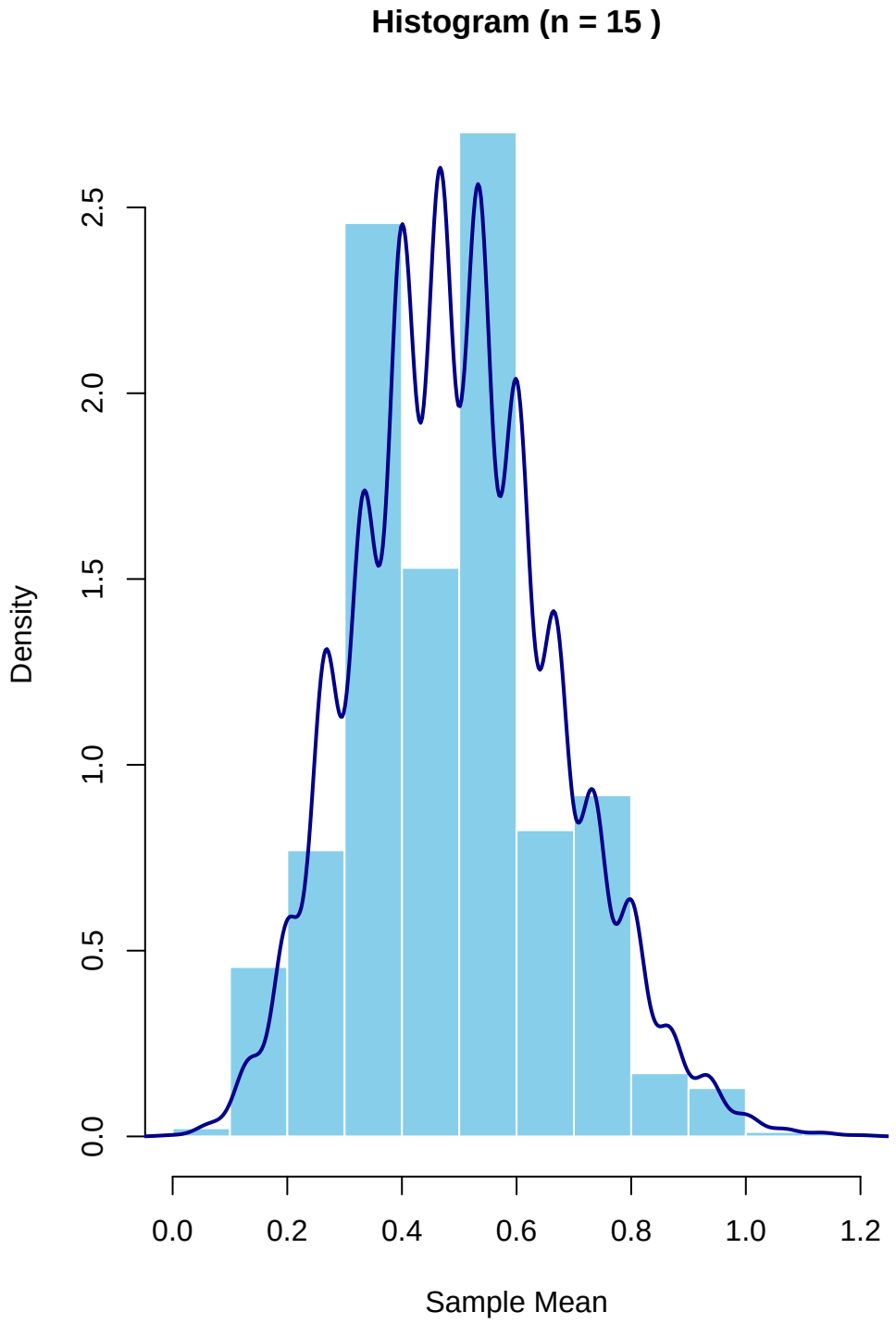
Binomial Small (n=5, p=0.10) — n = 14

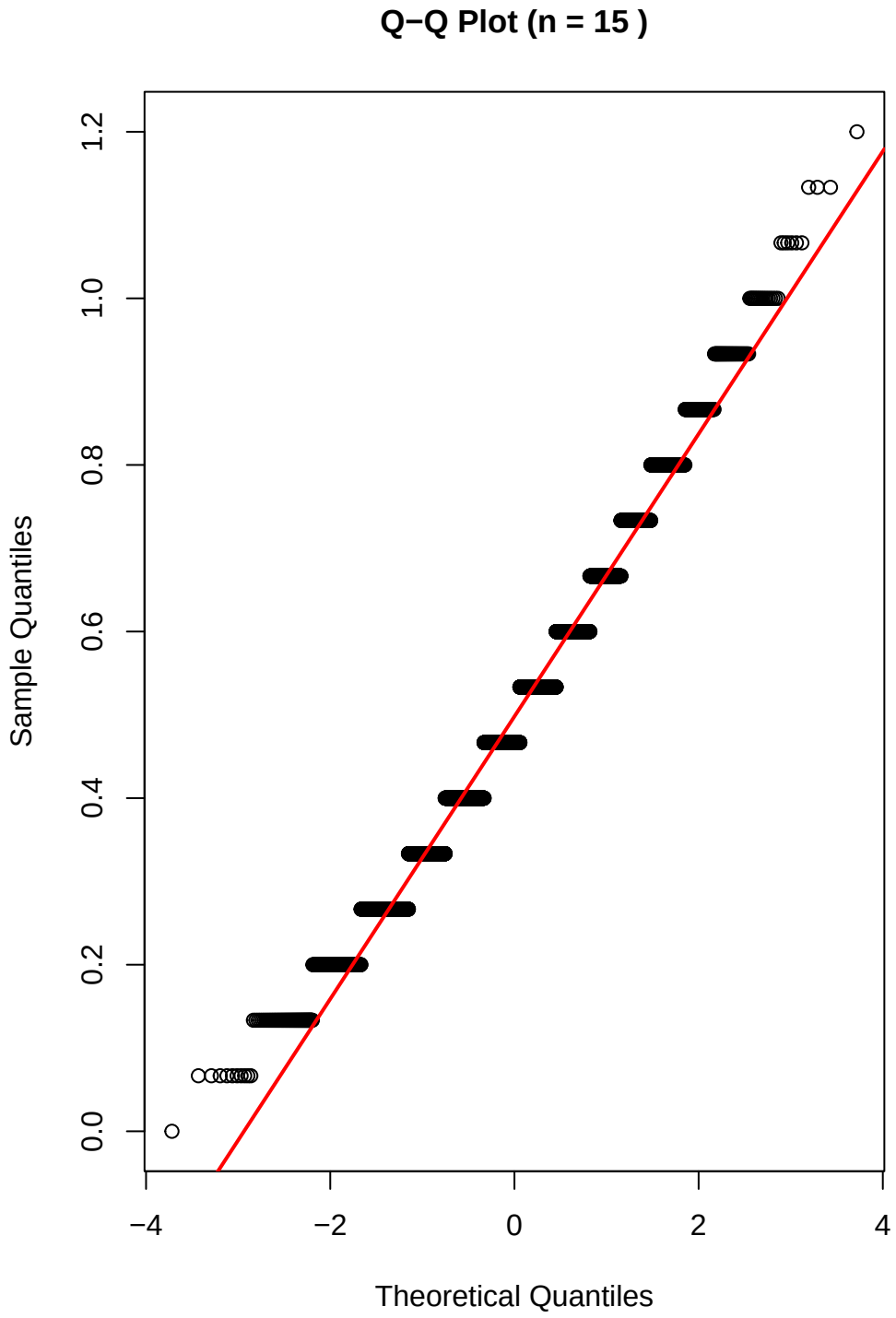




Excess Kurtosis = 0.1703

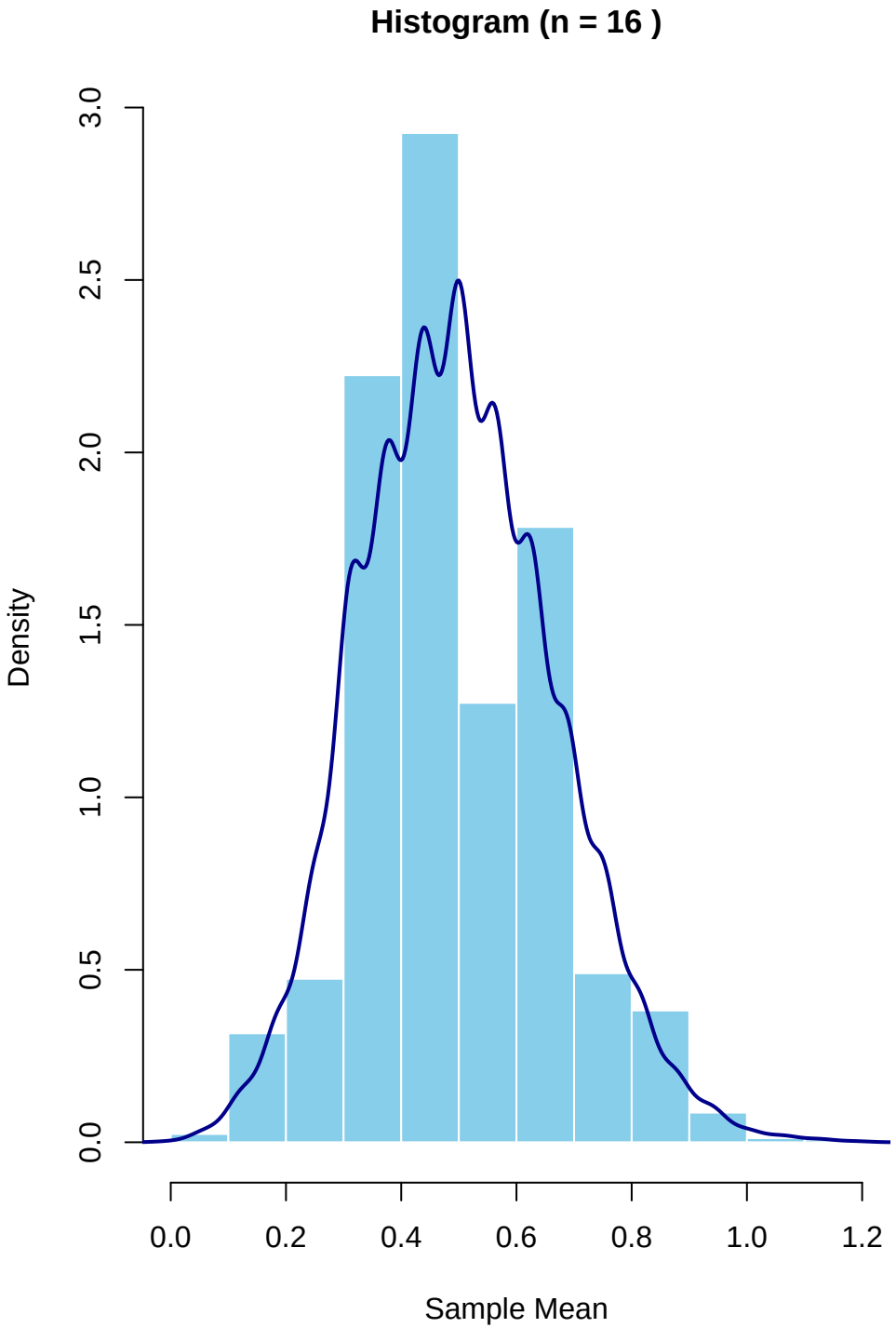
Binomial Small (n=5, p=0.10) — n = 15

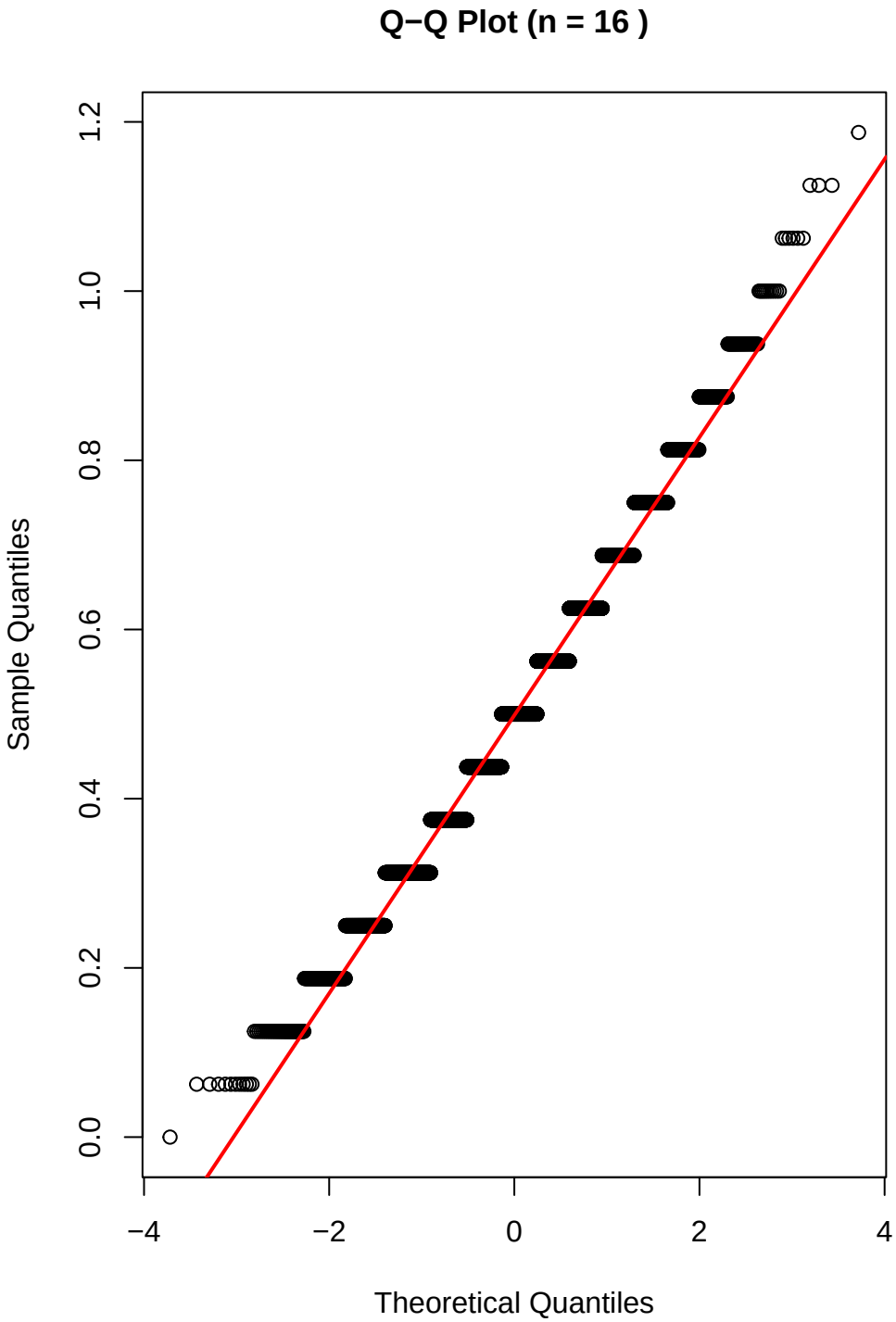




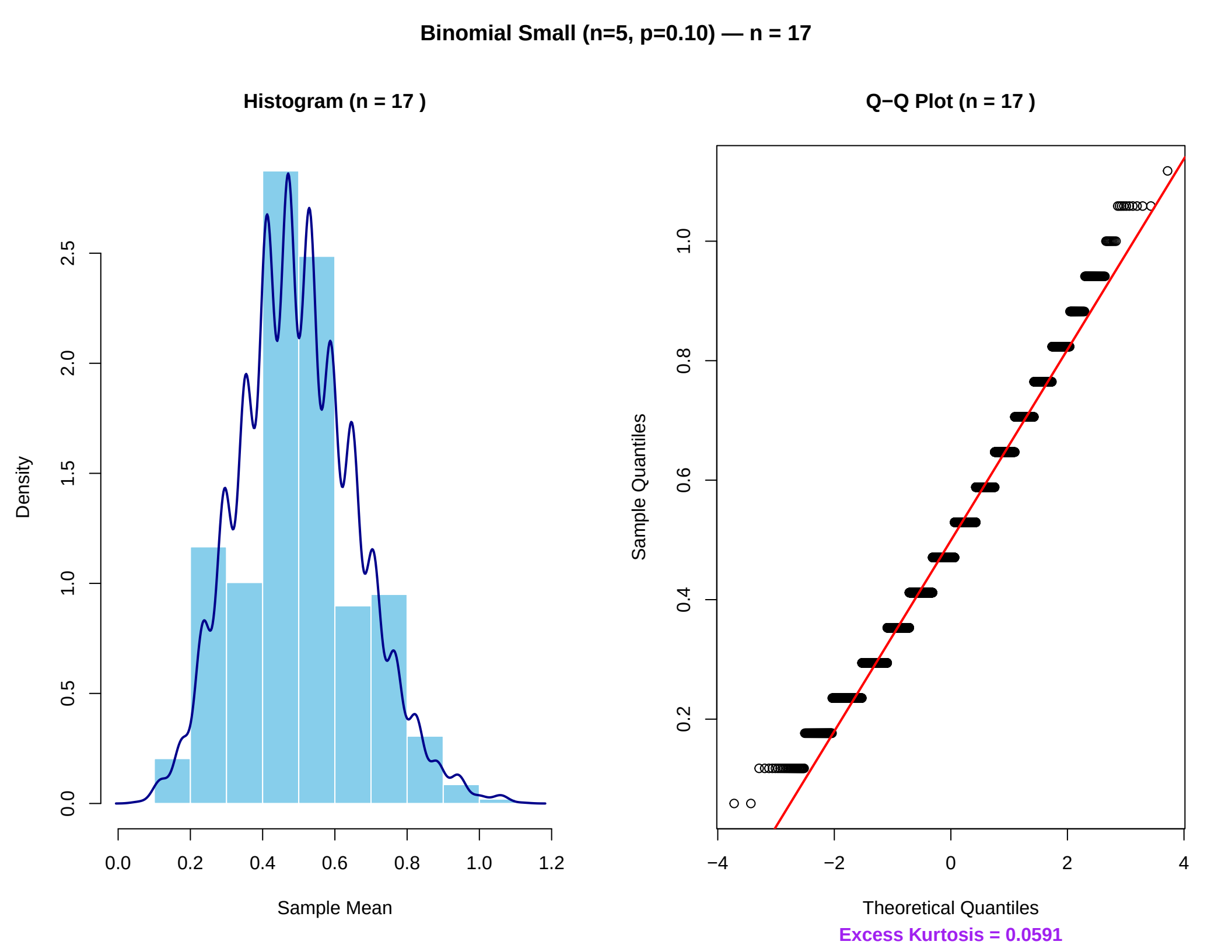
Excess Kurtosis = −0.0102

Binomial Small (n=5, p=0.10) — n = 16

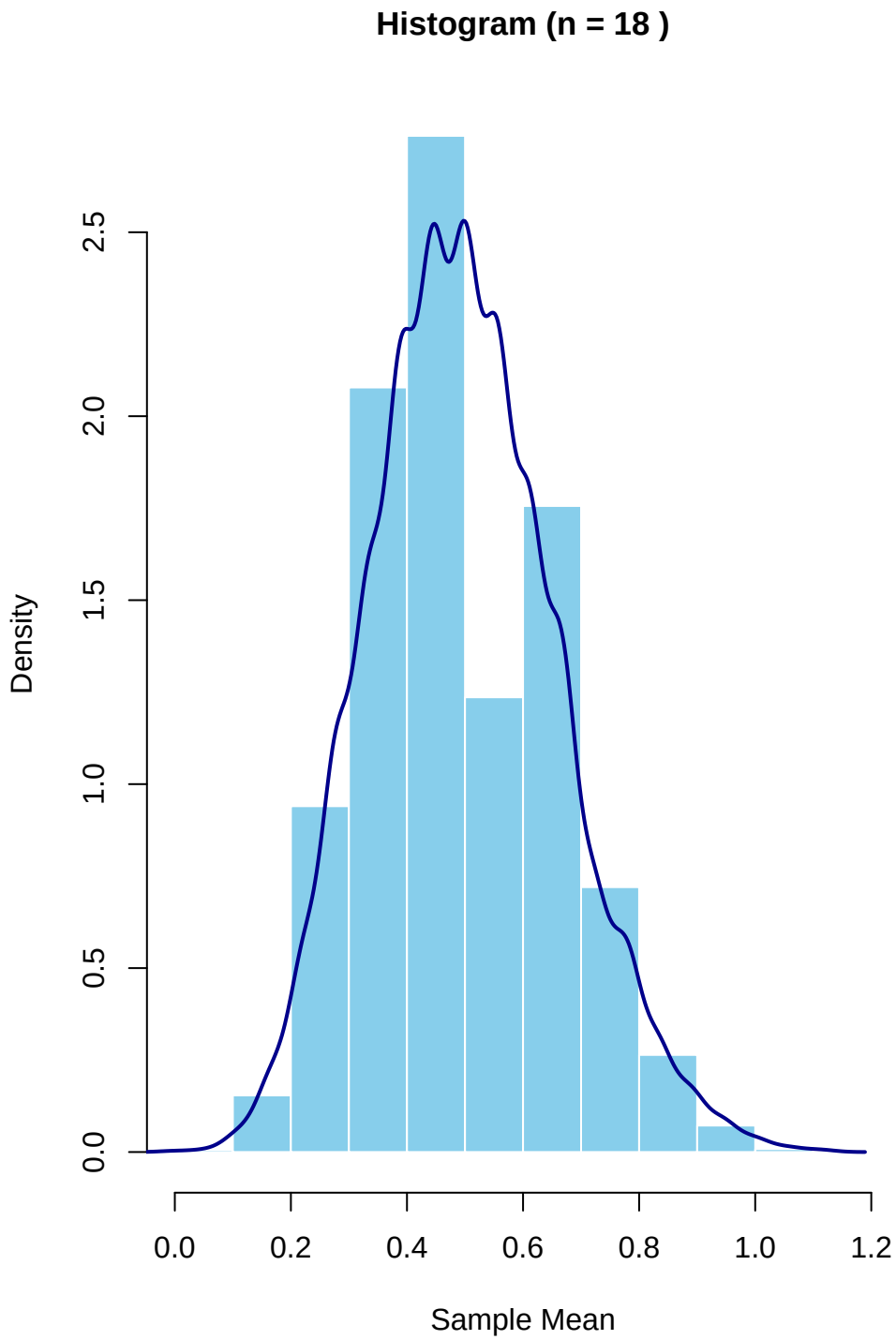


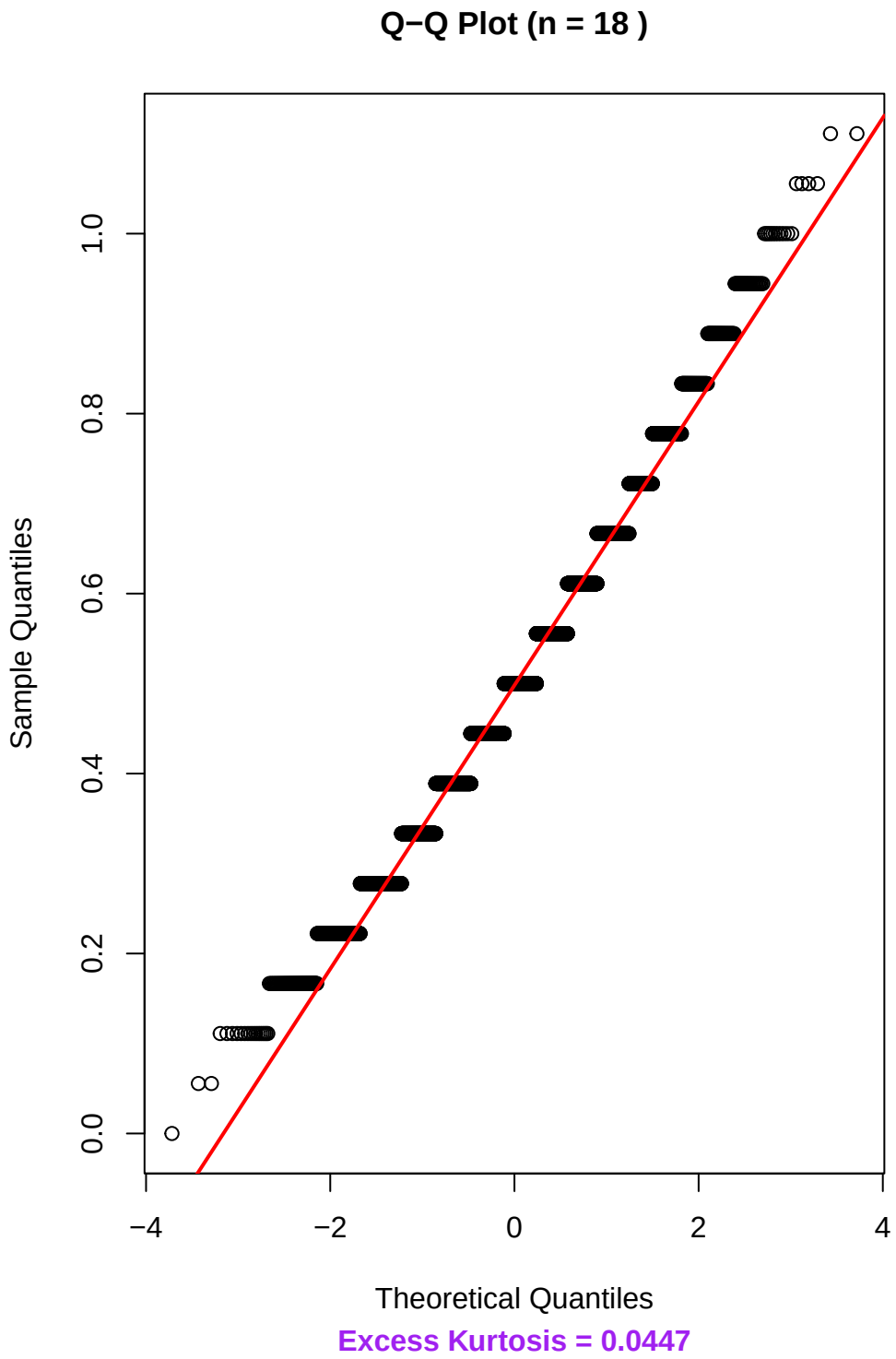


Excess Kurtosis = 0.0074

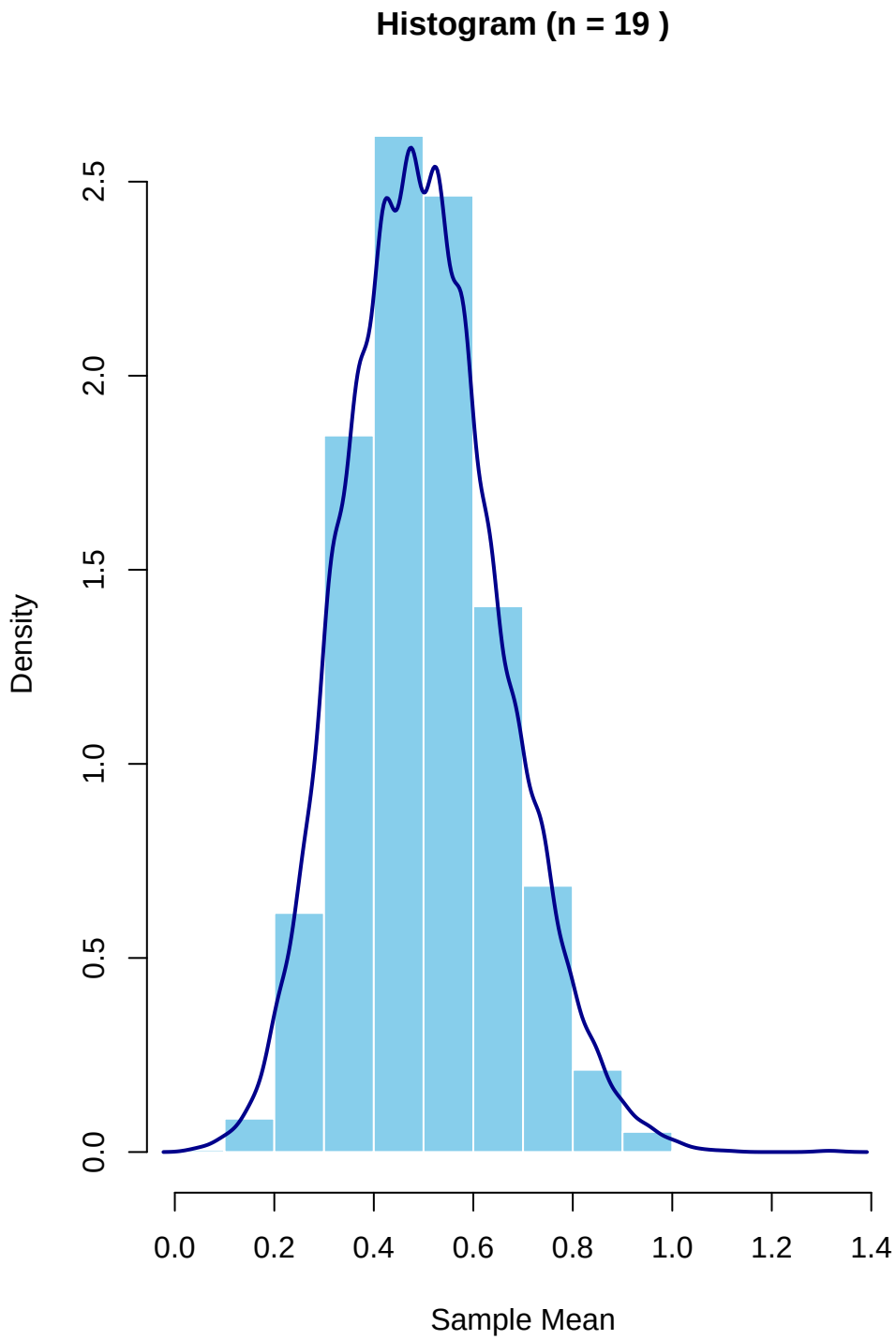

Binomial Small (n=5, p=0.10) — n = 17
Histogram (n = 17)
Density
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample Mean
Q−Q Plot (n = 17)
Sample Quantiles
0.2
0.4
0.6
0.8
1.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.0591

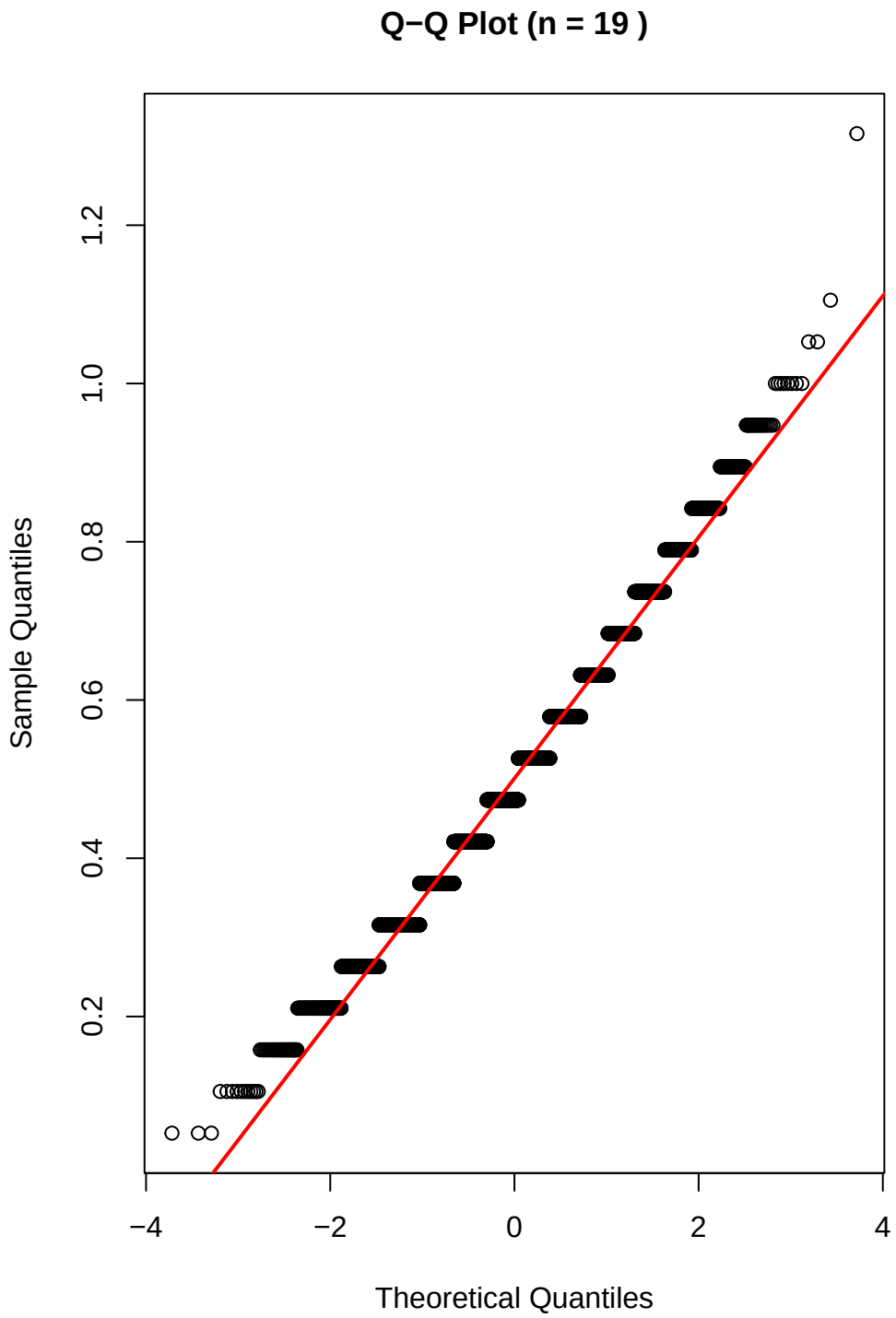
Binomial Small (n=5, p=0.10) — n = 18





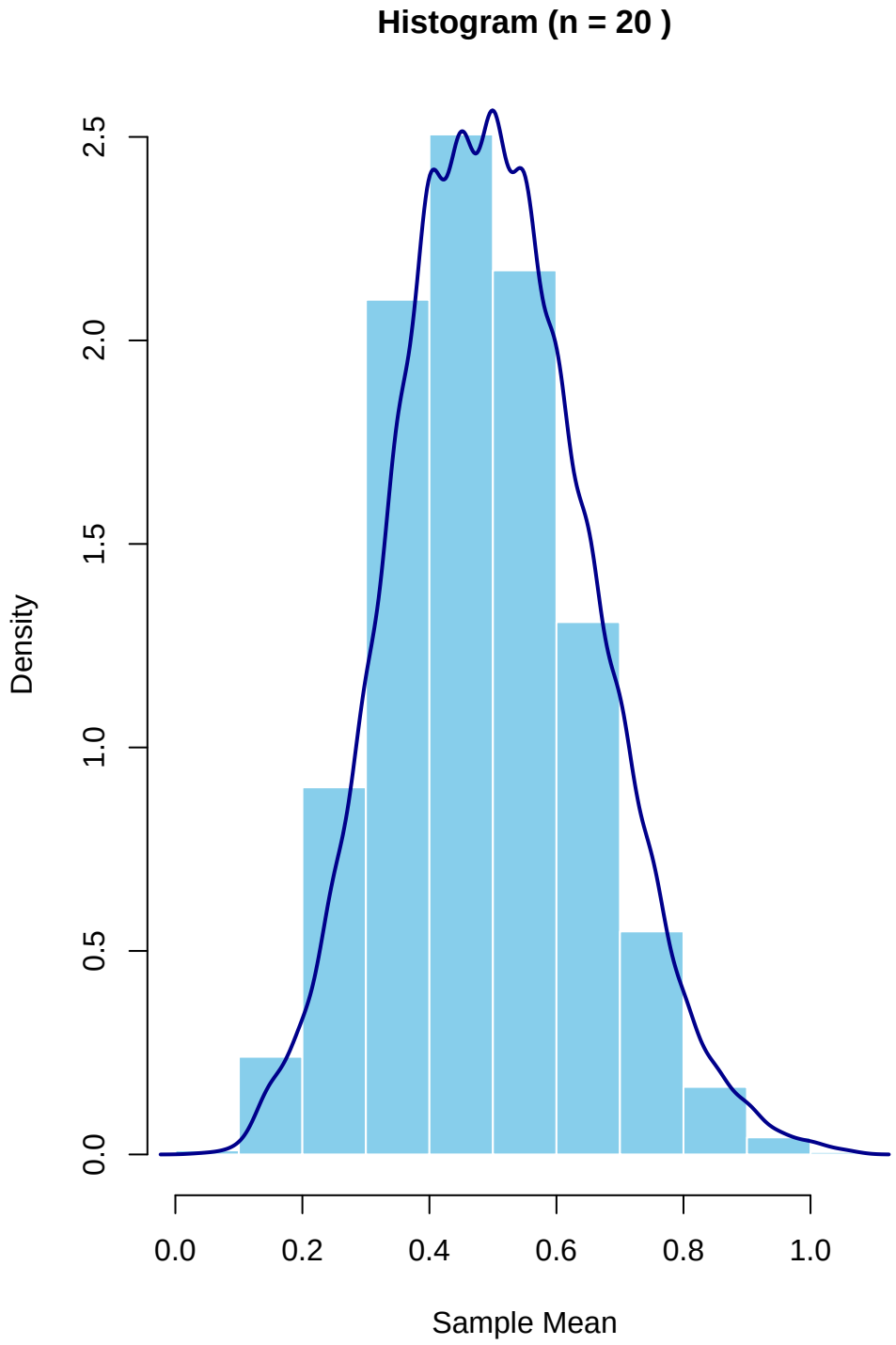
Binomial Small (n=5, p=0.10) — n = 19

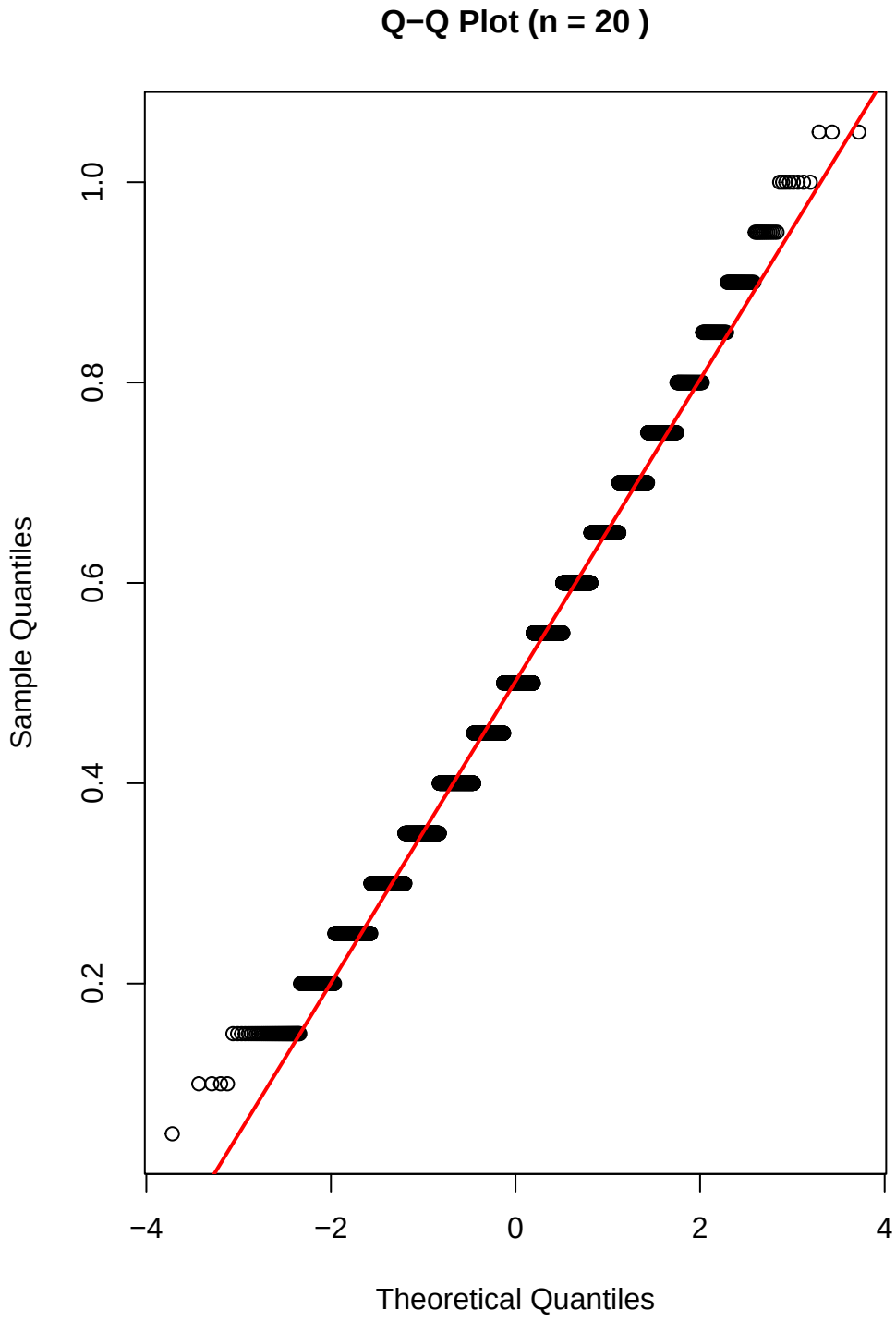


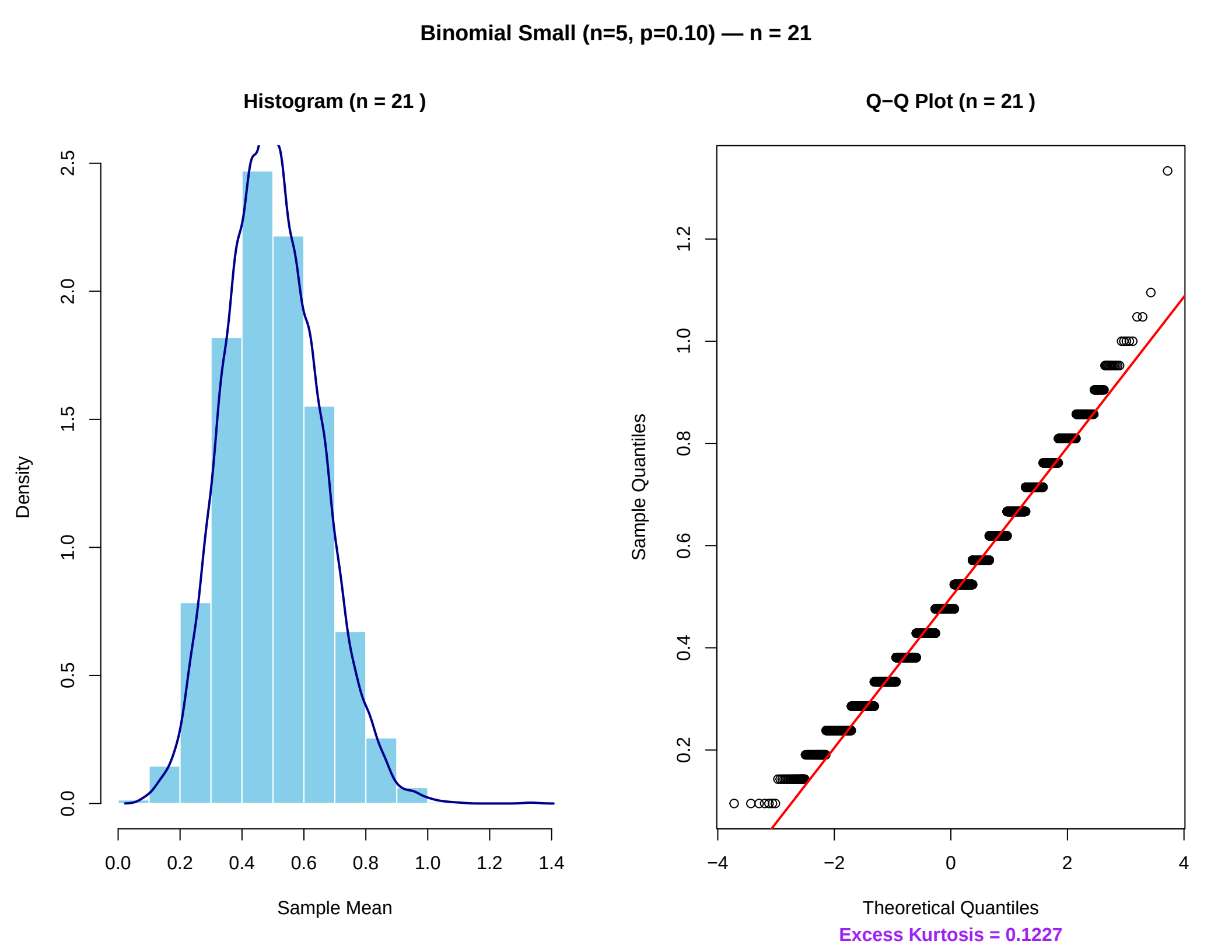


Excess Kurtosis = 0.1045

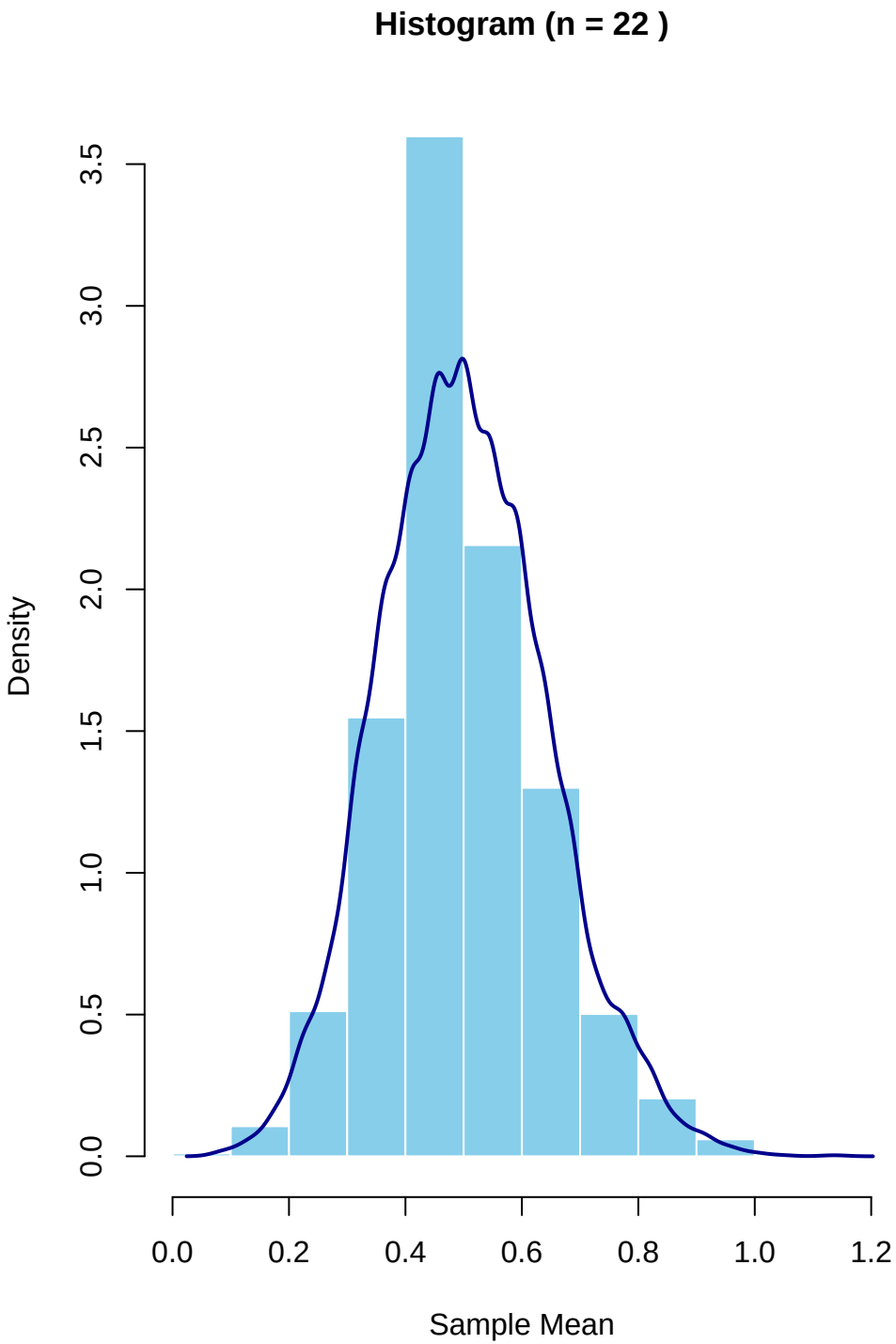
Binomial Small (n=5, p=0.10) — n = 20

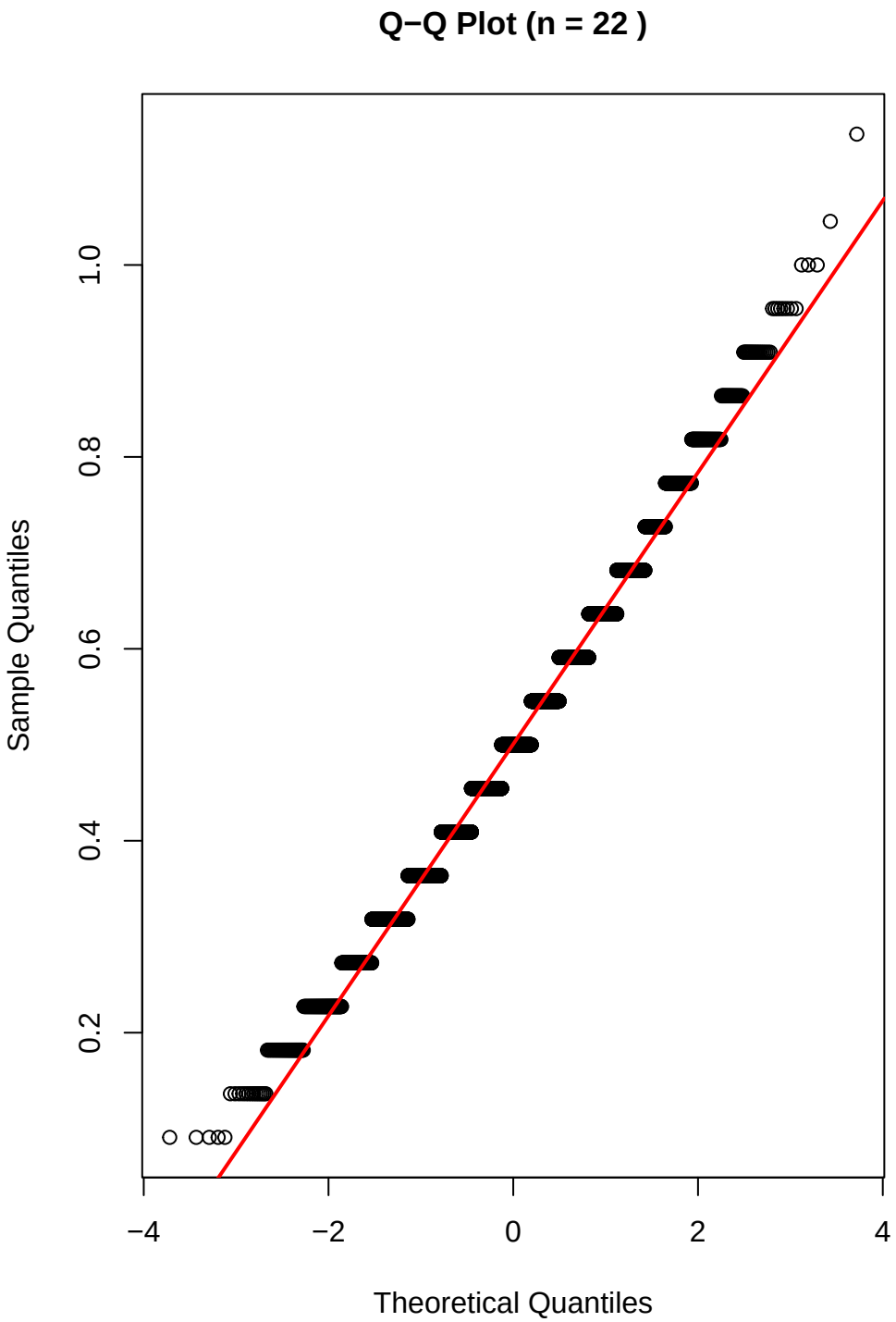





Binomial Small (n=5, p=0.10) — n = 21
Histogram (n = 21)
Density
Sample Mean
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Q−Q Plot (n = 21)
Sample Quantiles
Theoretical Quantiles
0.2
0.4
0.6
0.8
1.0
1.2
−4
−2
0
2
4
Excess Kurtosis = 0.1227

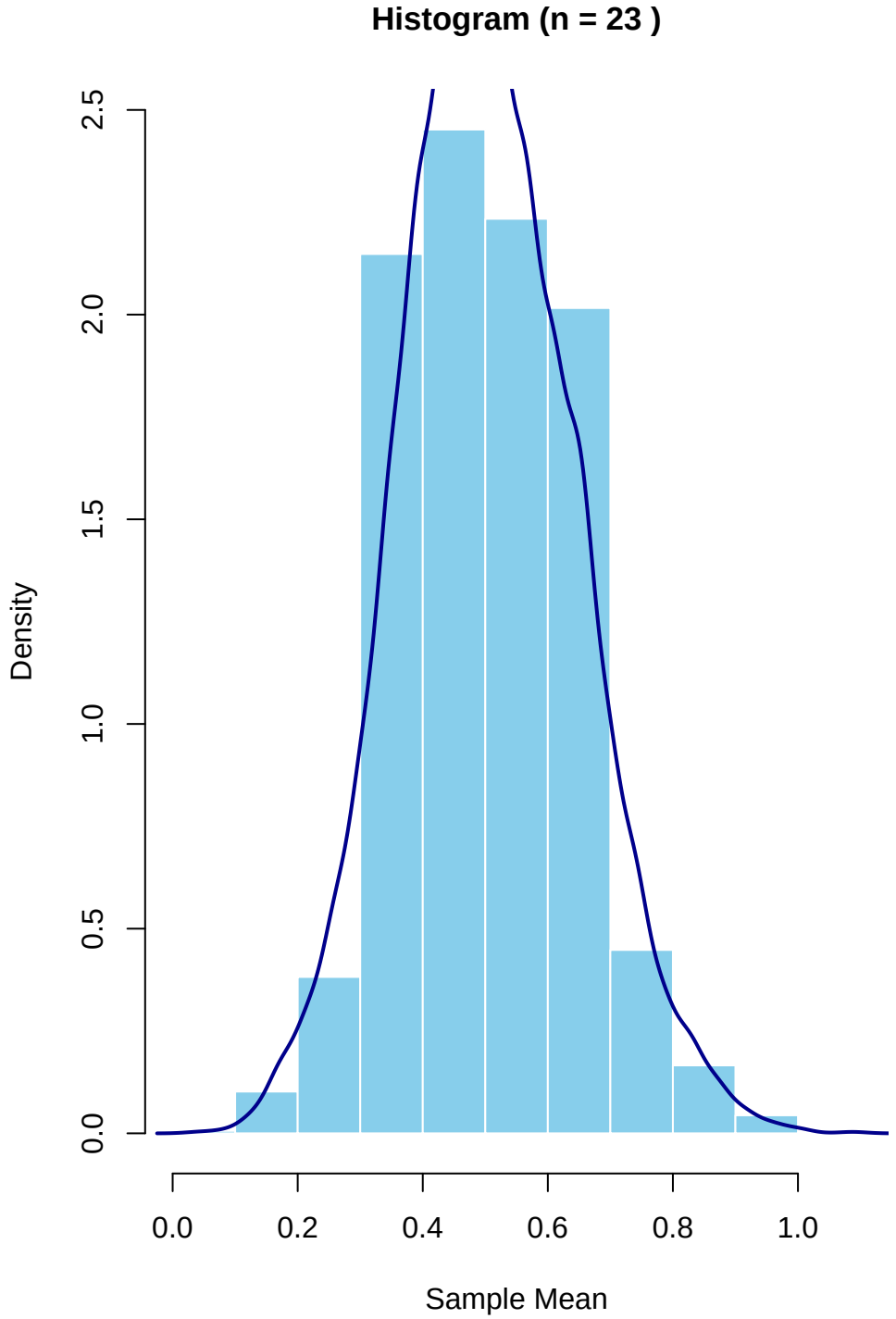
Binomial Small (n=5, p=0.10) — n = 22

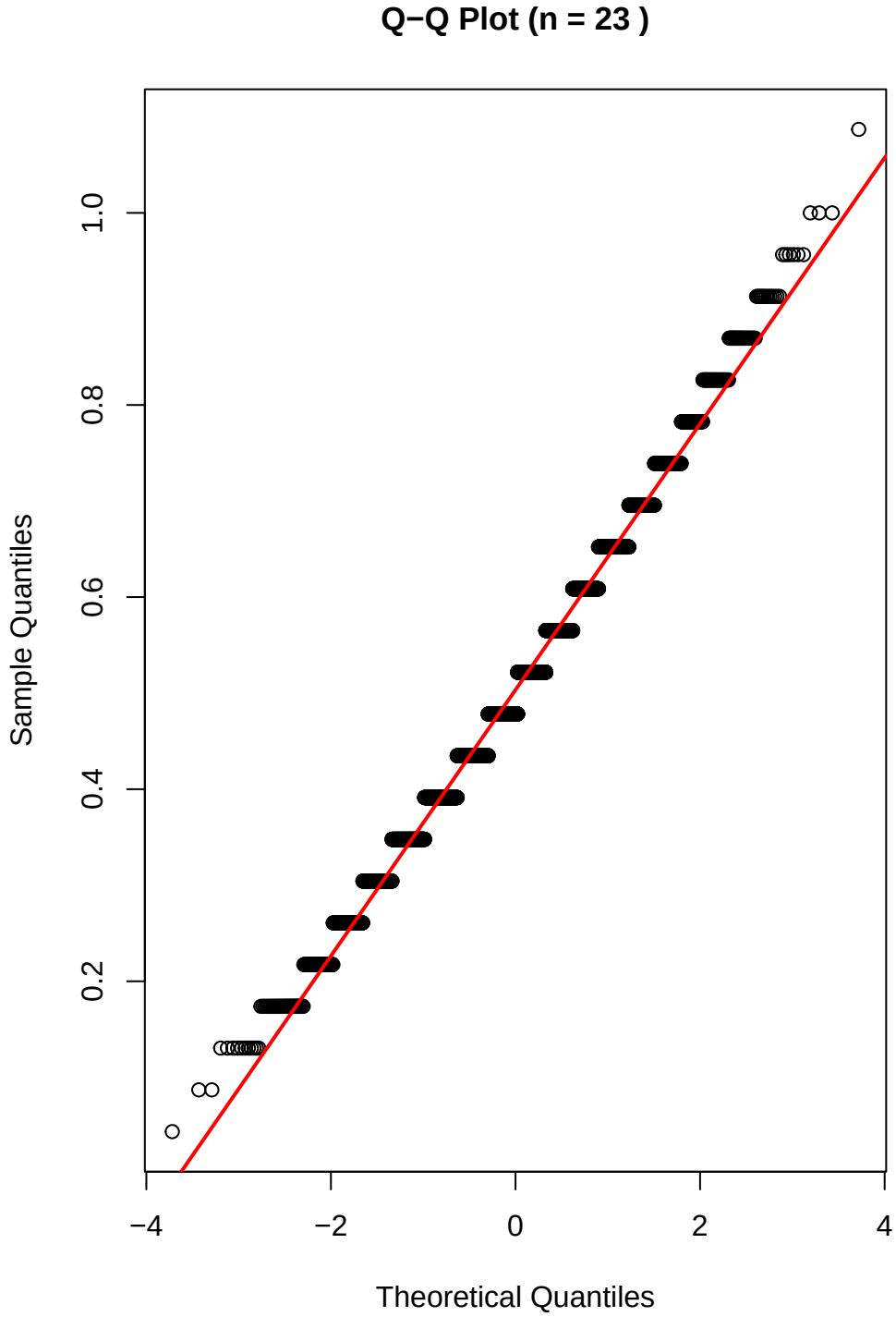




Excess Kurtosis = 0.072

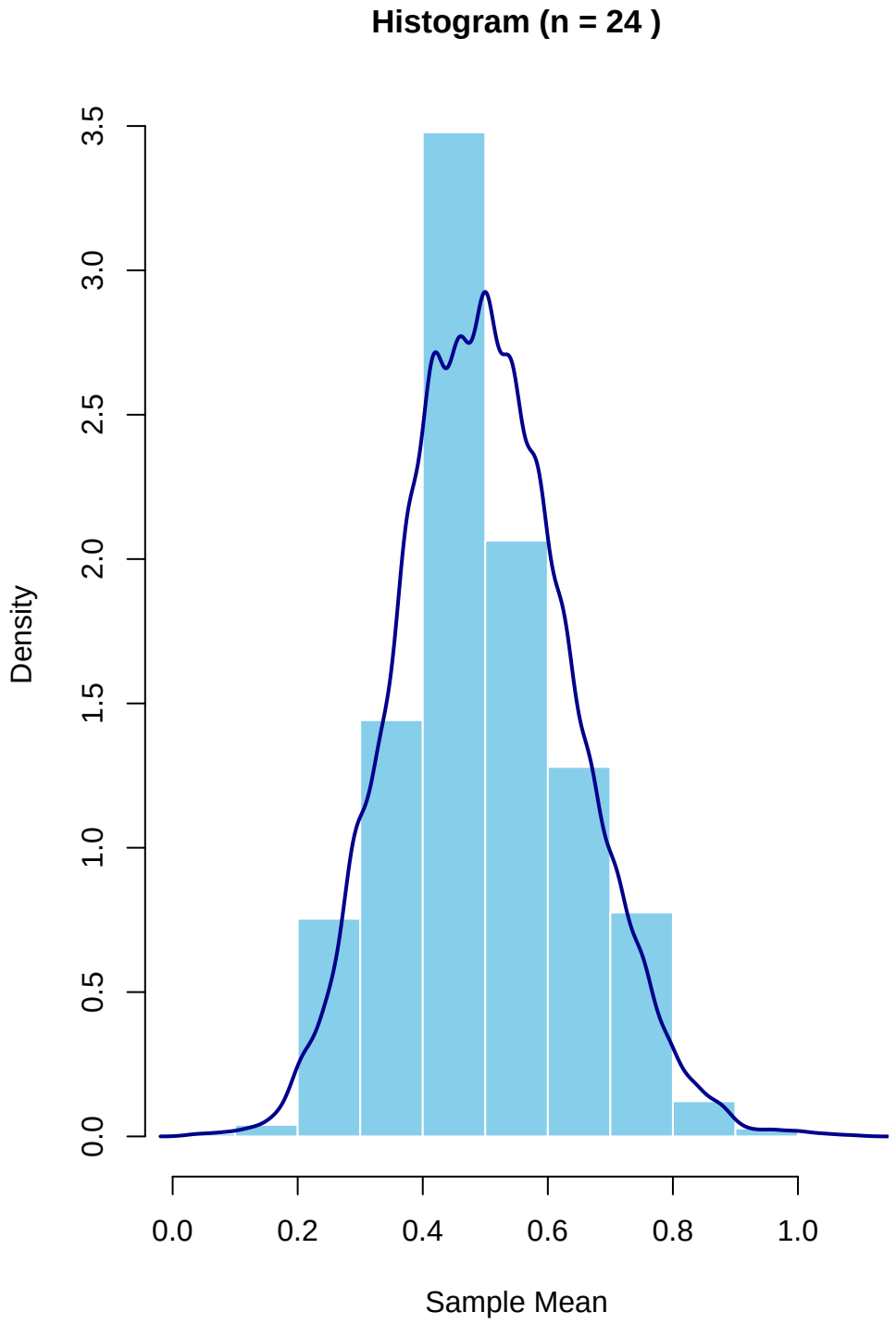
Binomial Small (n=5, p=0.10) — n = 23

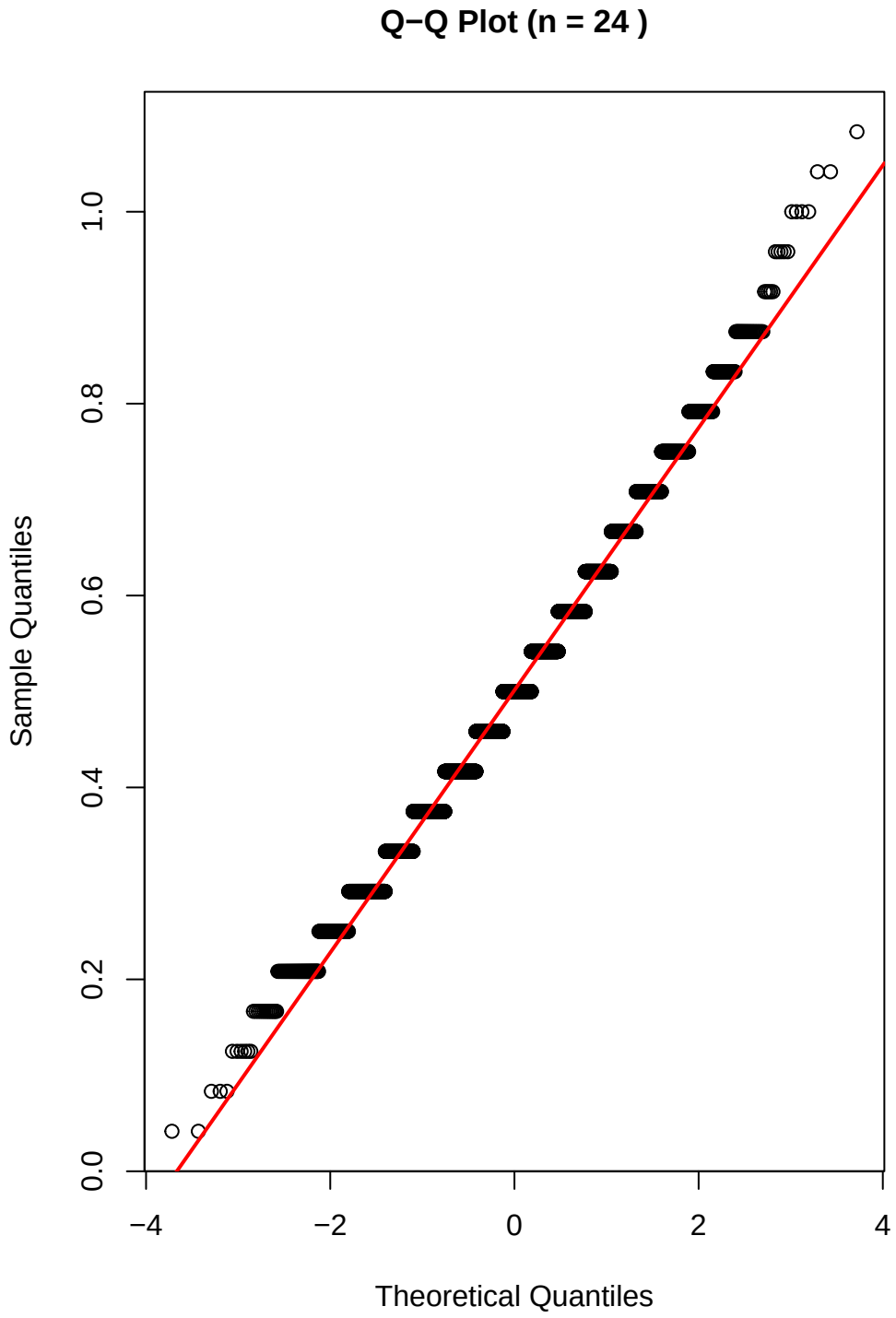




Excess Kurtosis = 0.071

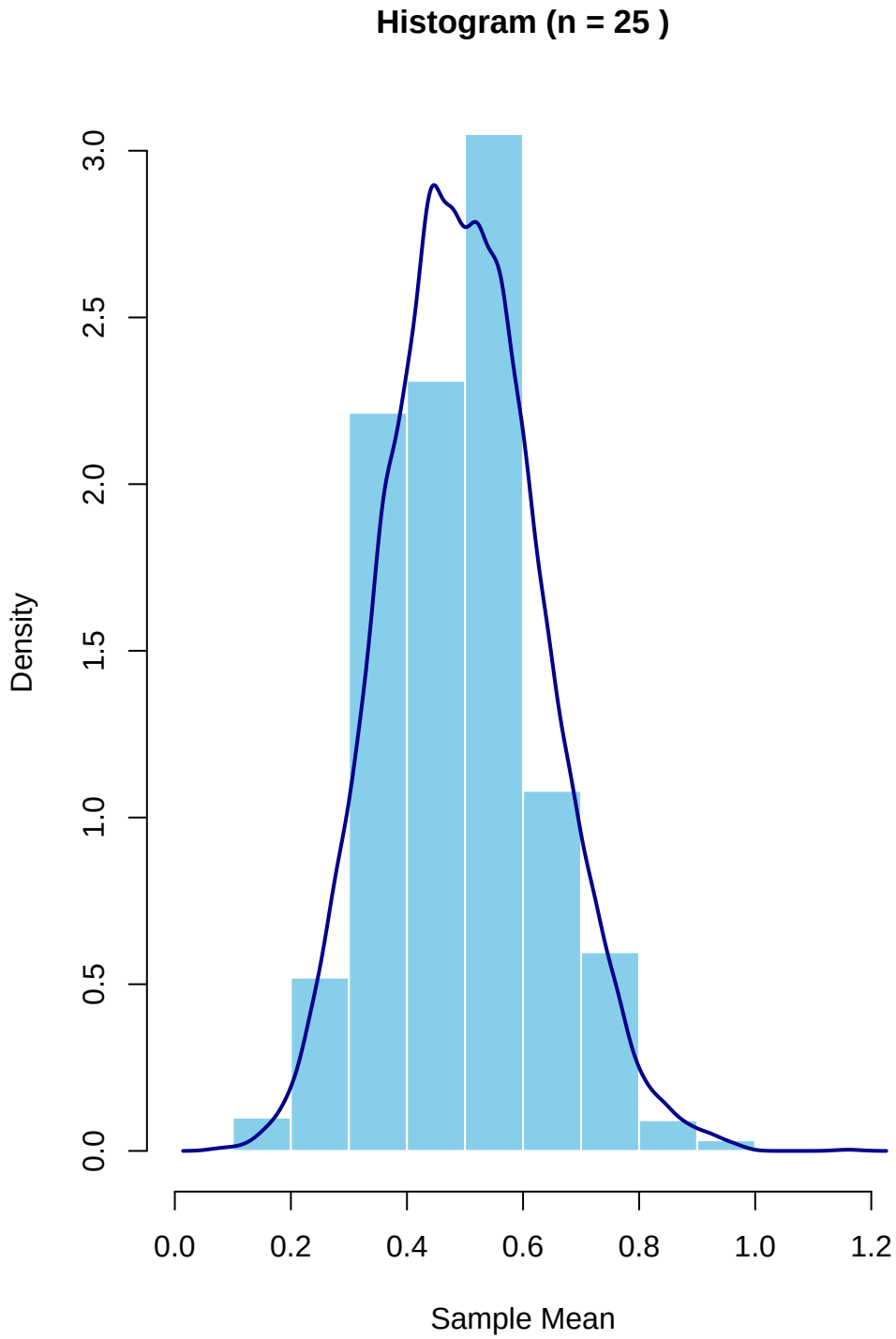
Binomial Small (n=5, p=0.10) — n = 24

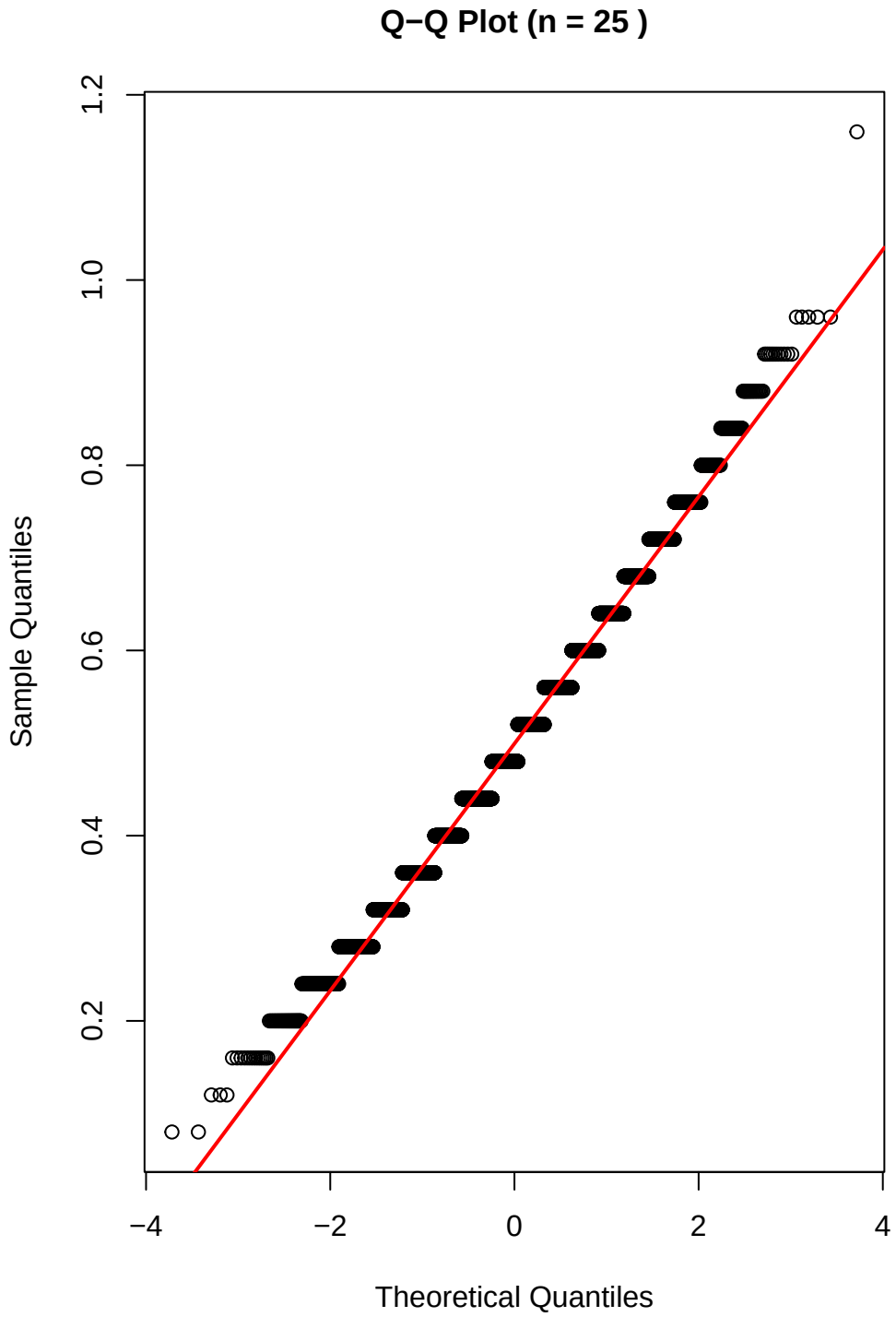




Excess Kurtosis = 0.1001

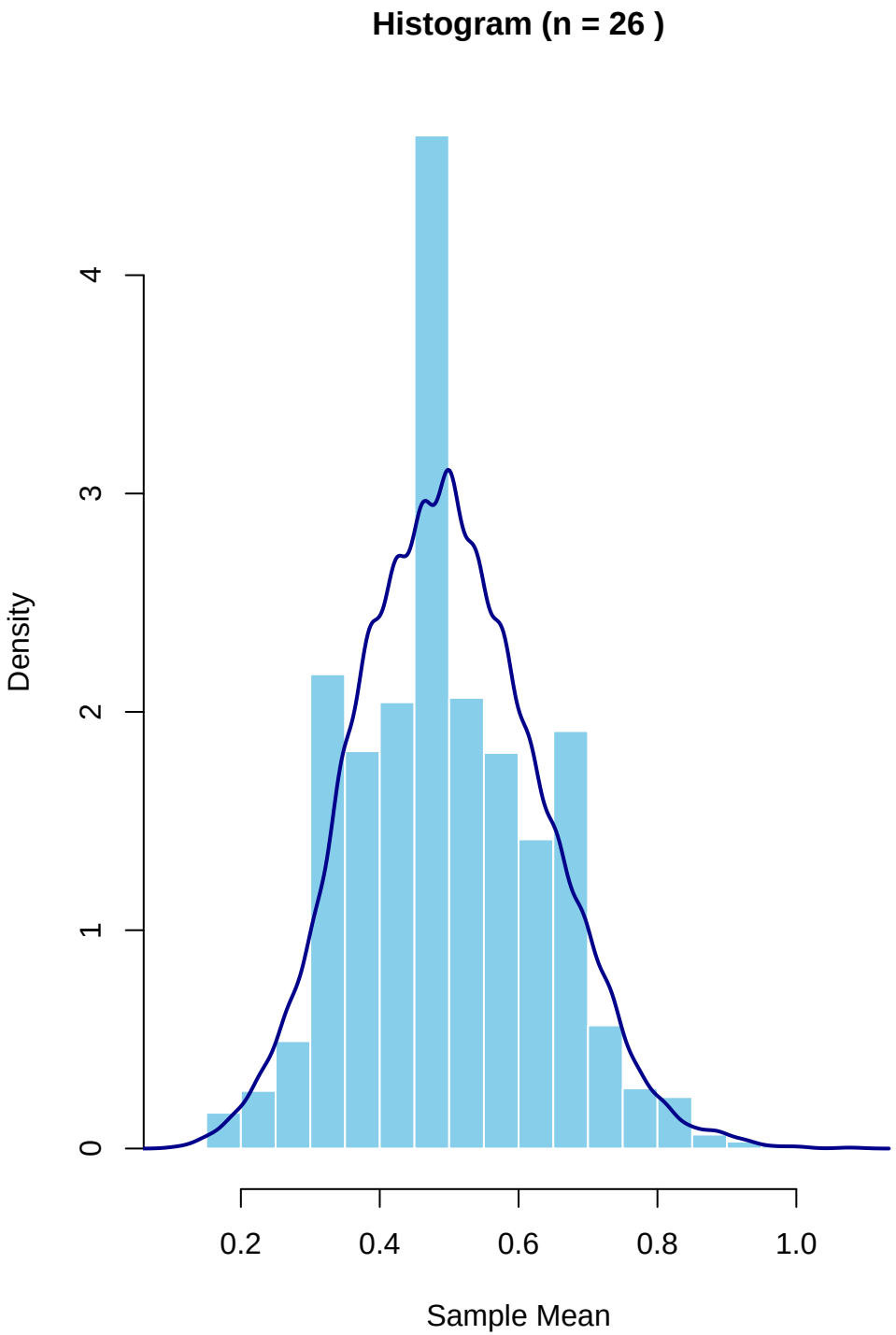
Binomial Small (n=5, p=0.10) — n = 25

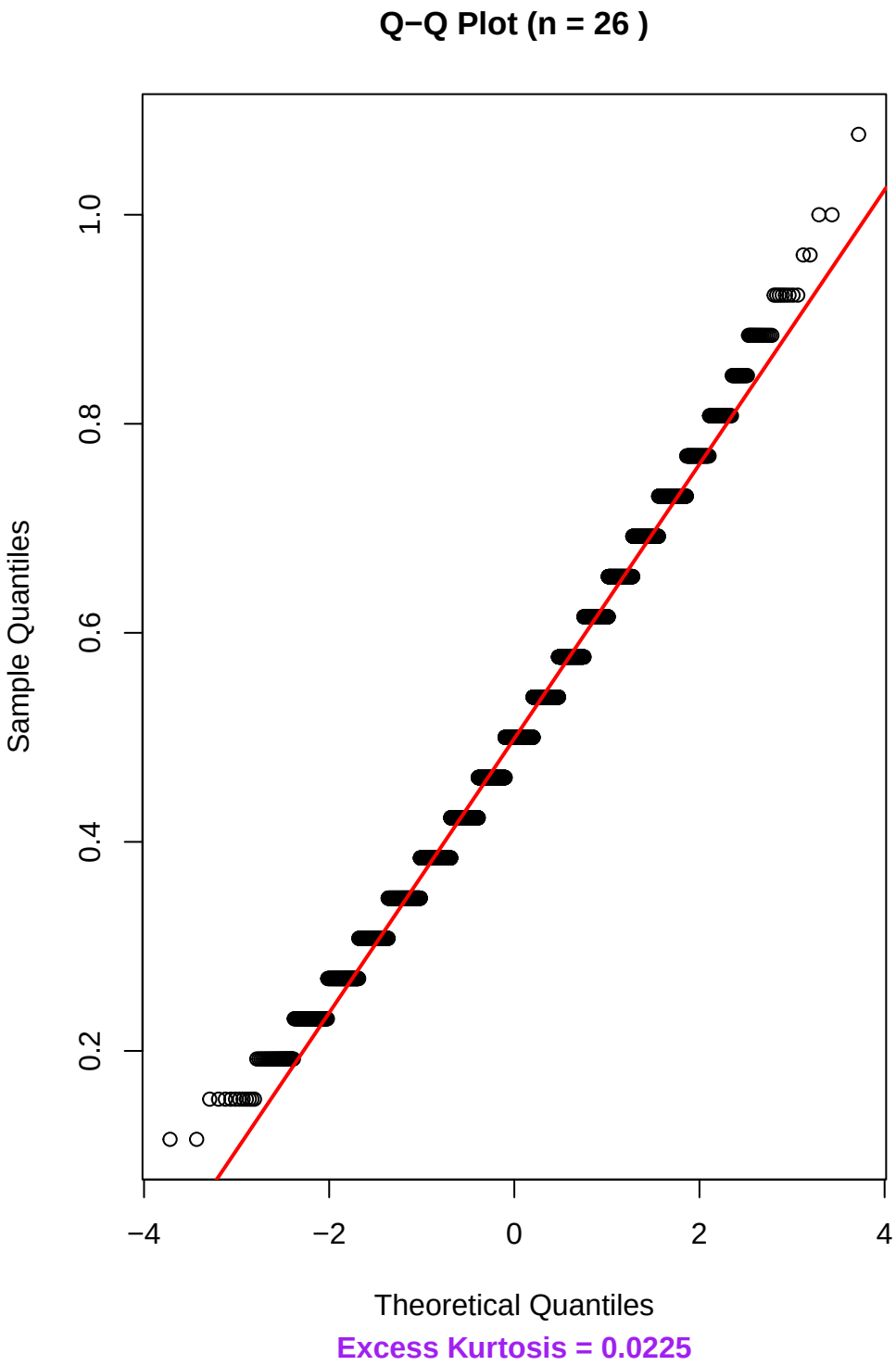




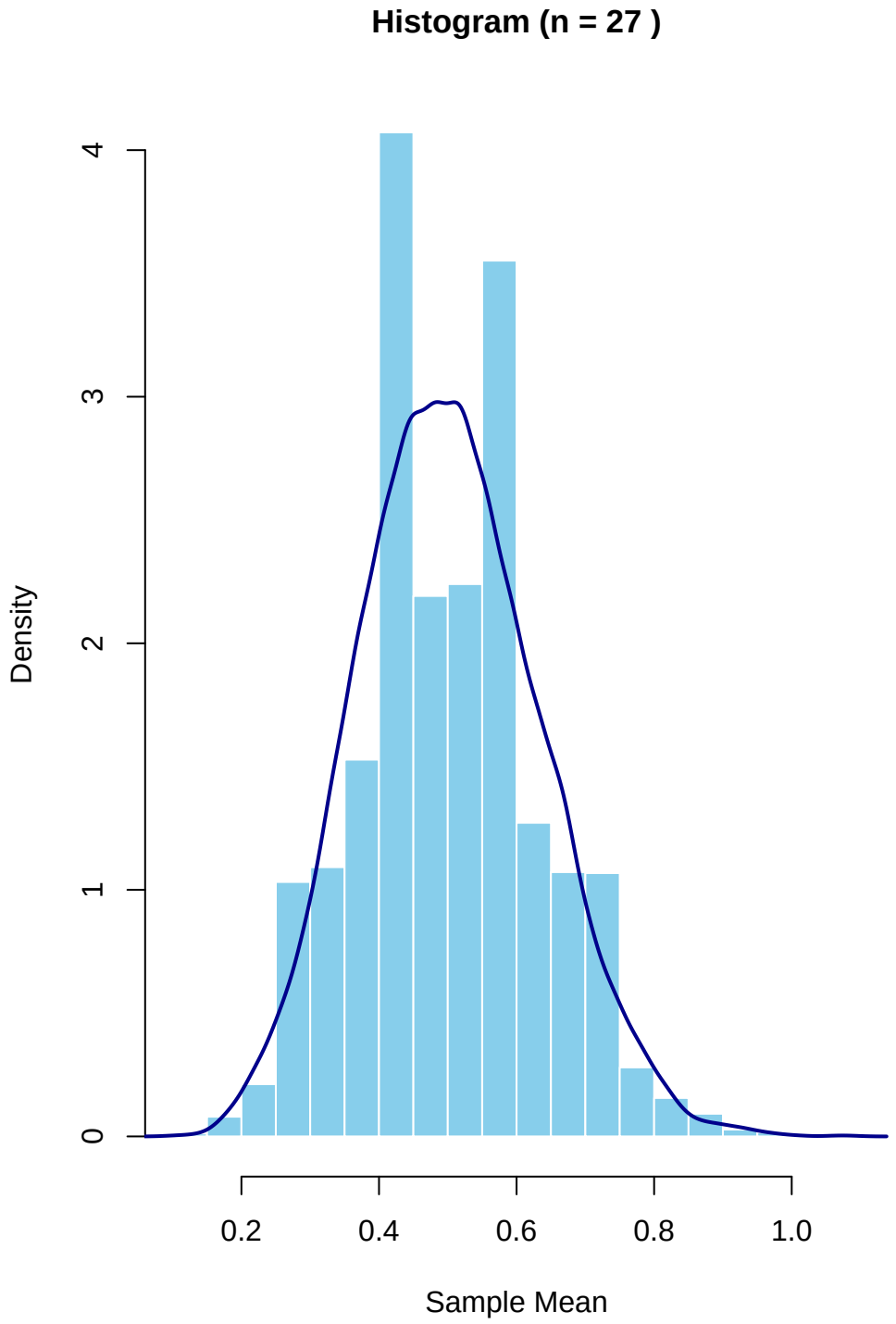
Excess Kurtosis = 0.0522

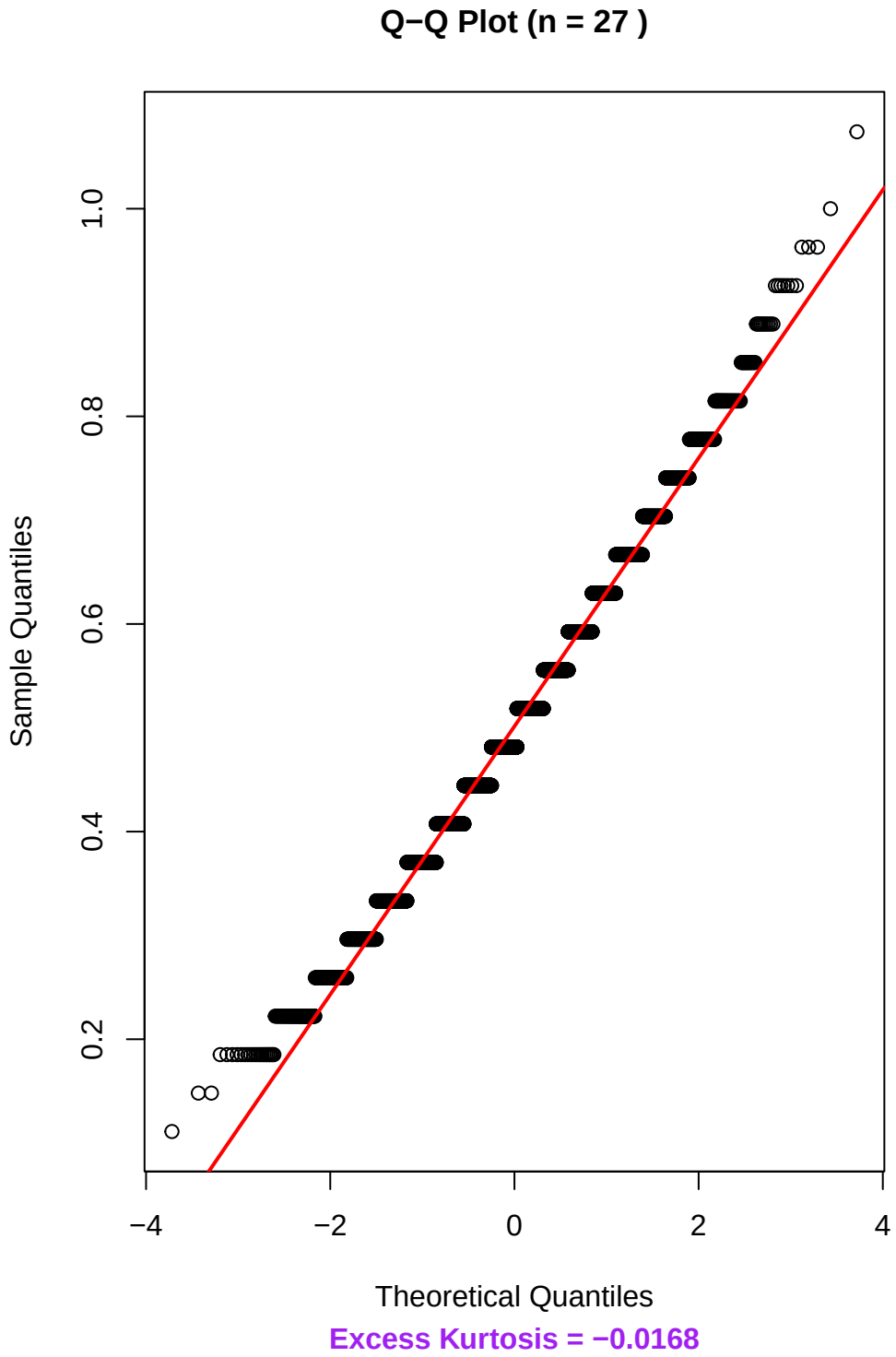
Binomial Small (n=5, p=0.10) — n = 26



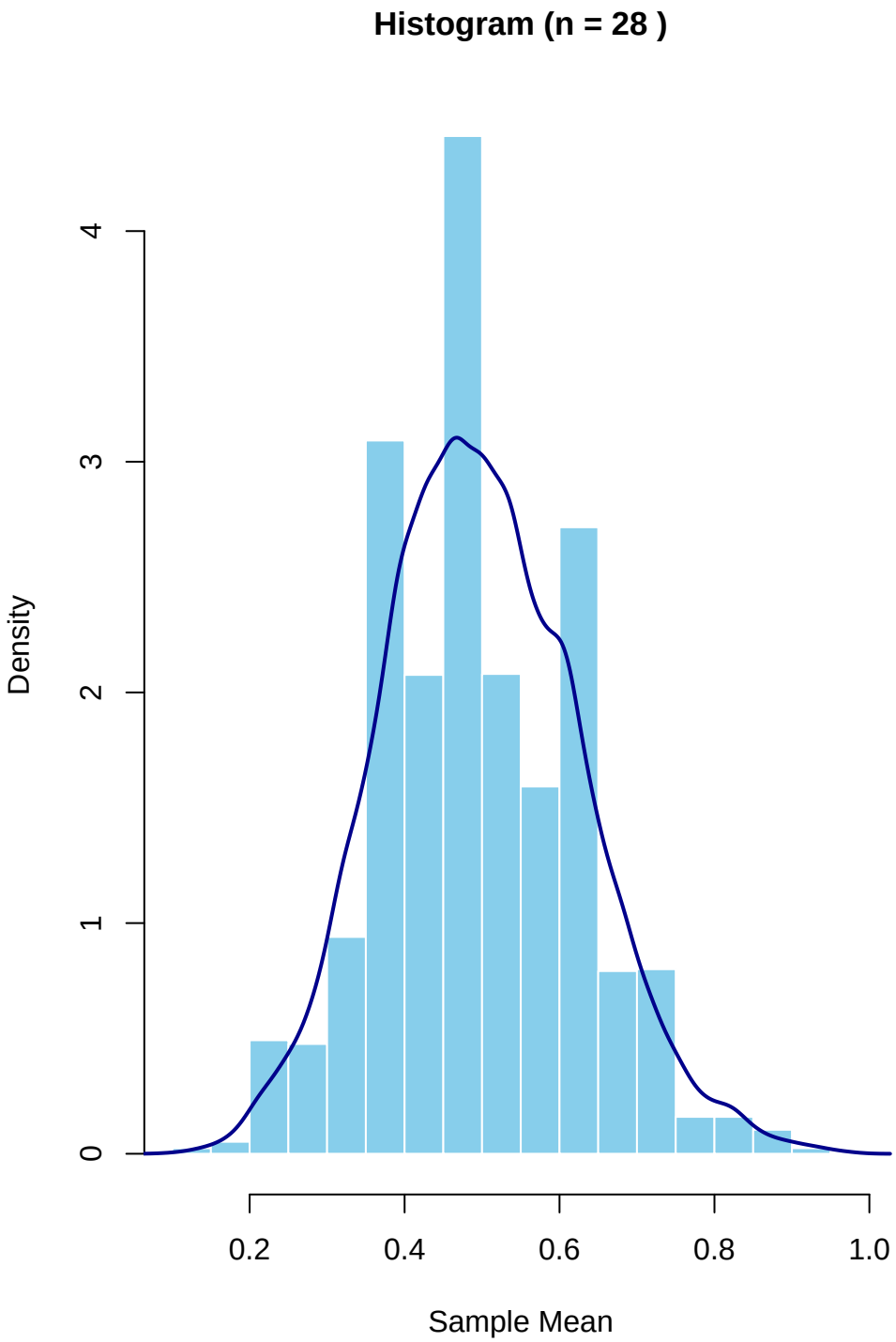


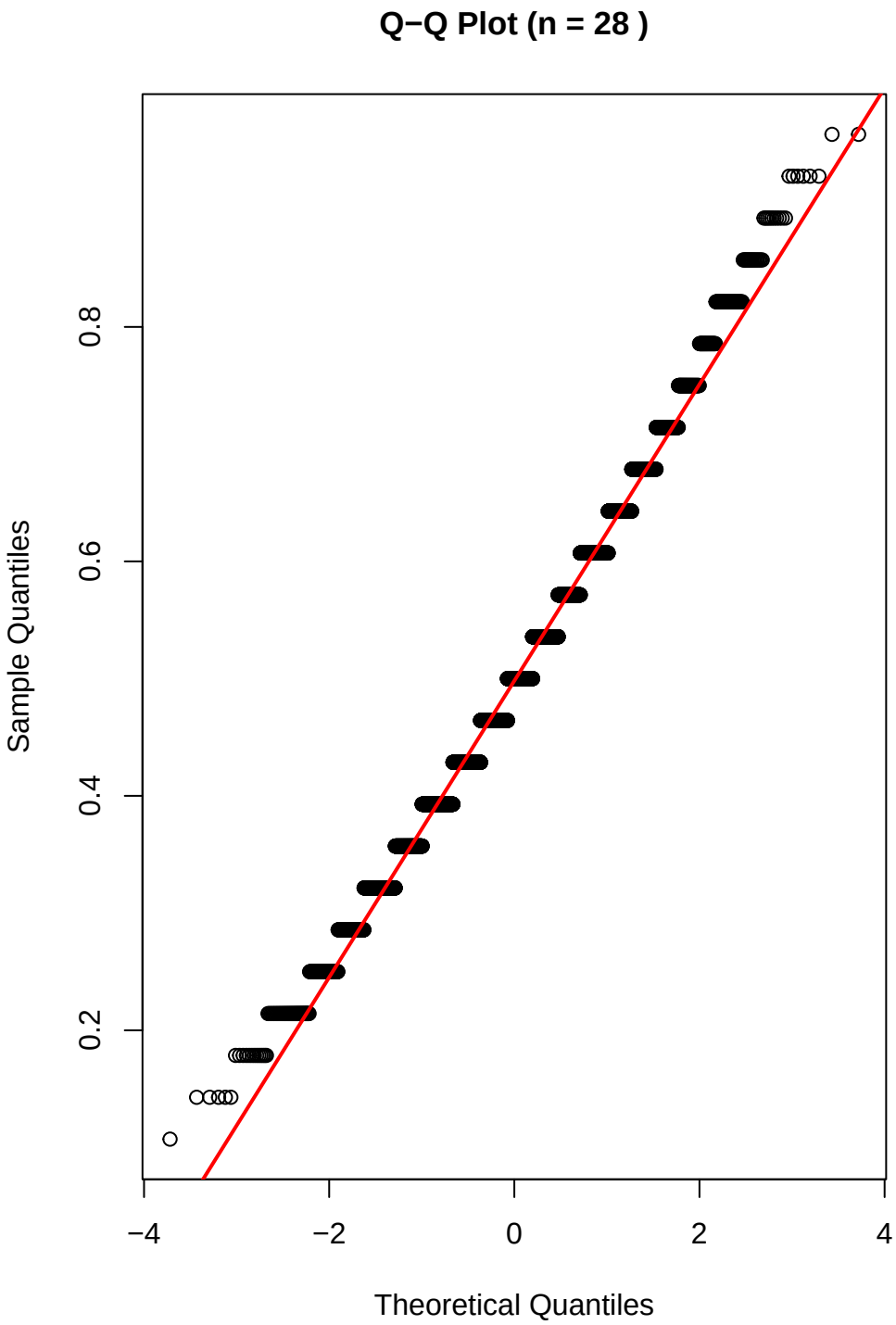
Binomial Small (n=5, p=0.10) — n = 27

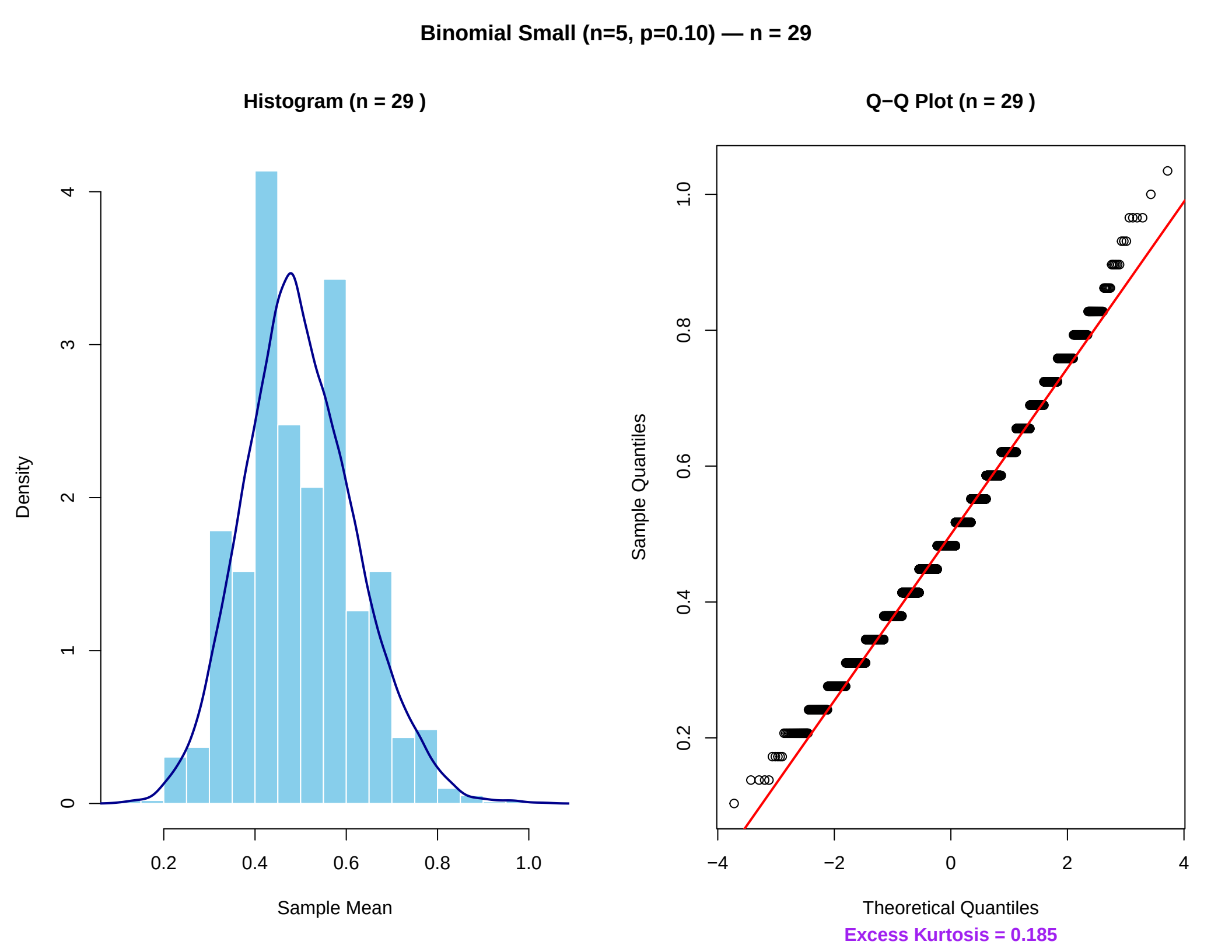




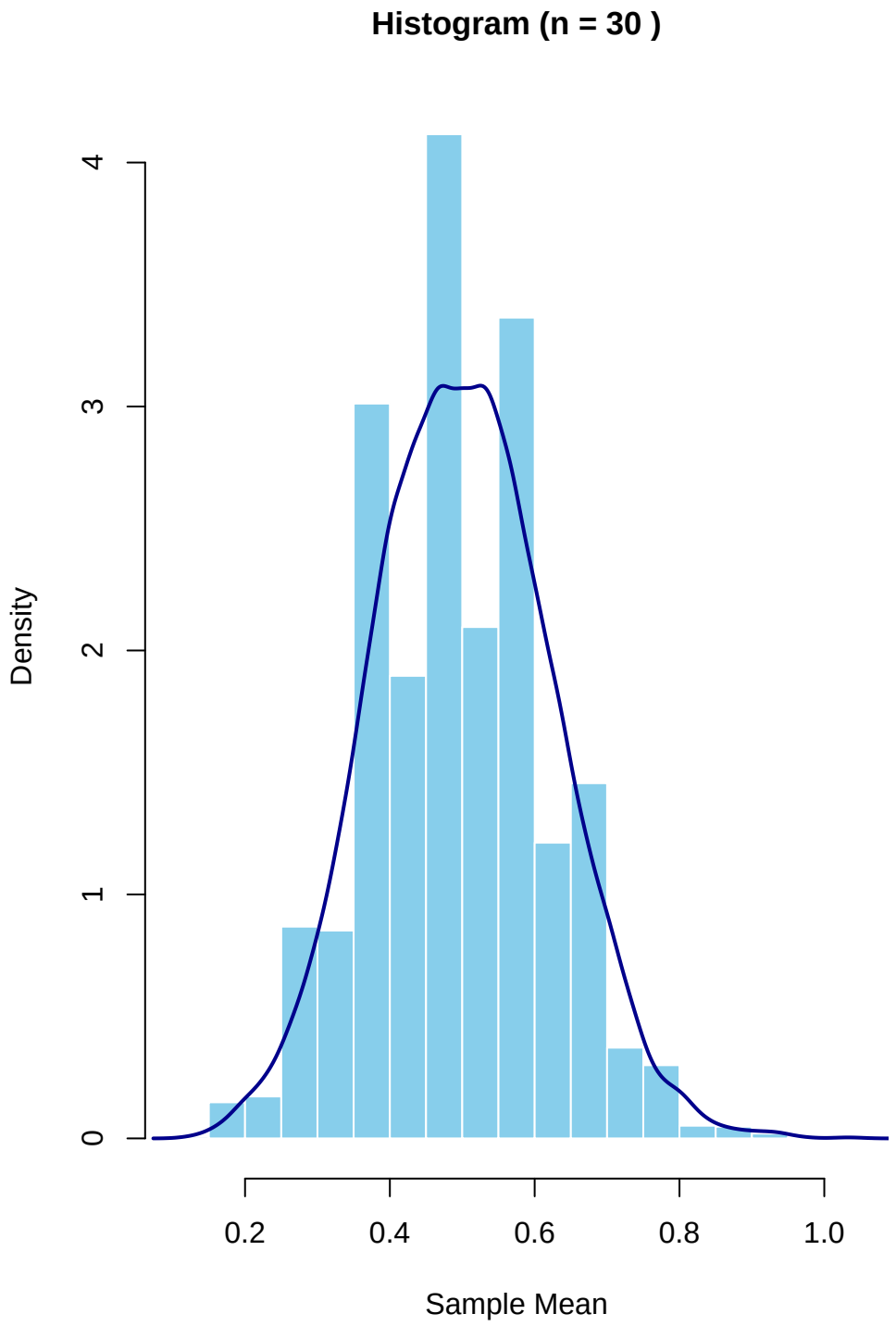
Binomial Small (n=5, p=0.10) — n = 28

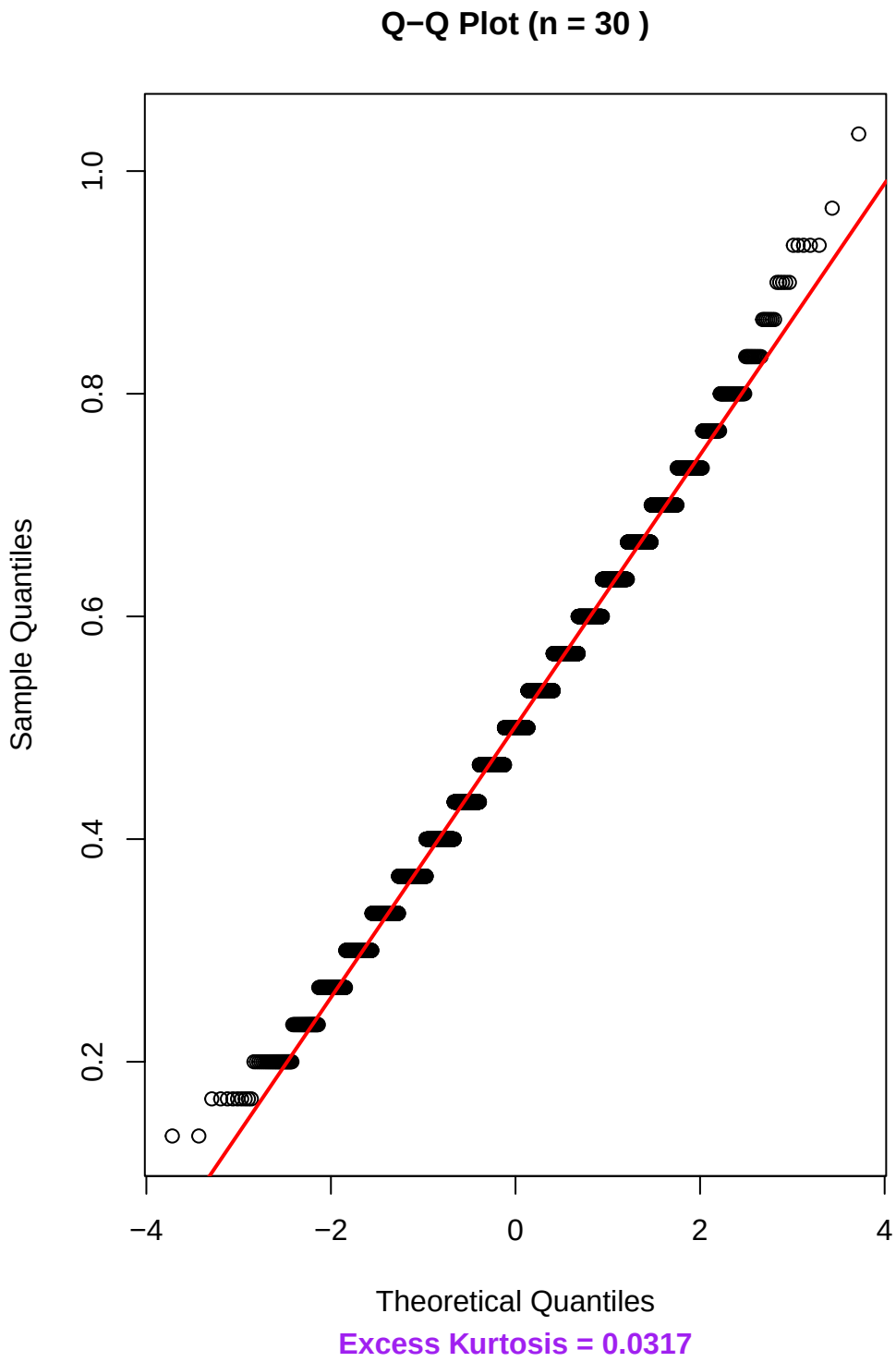


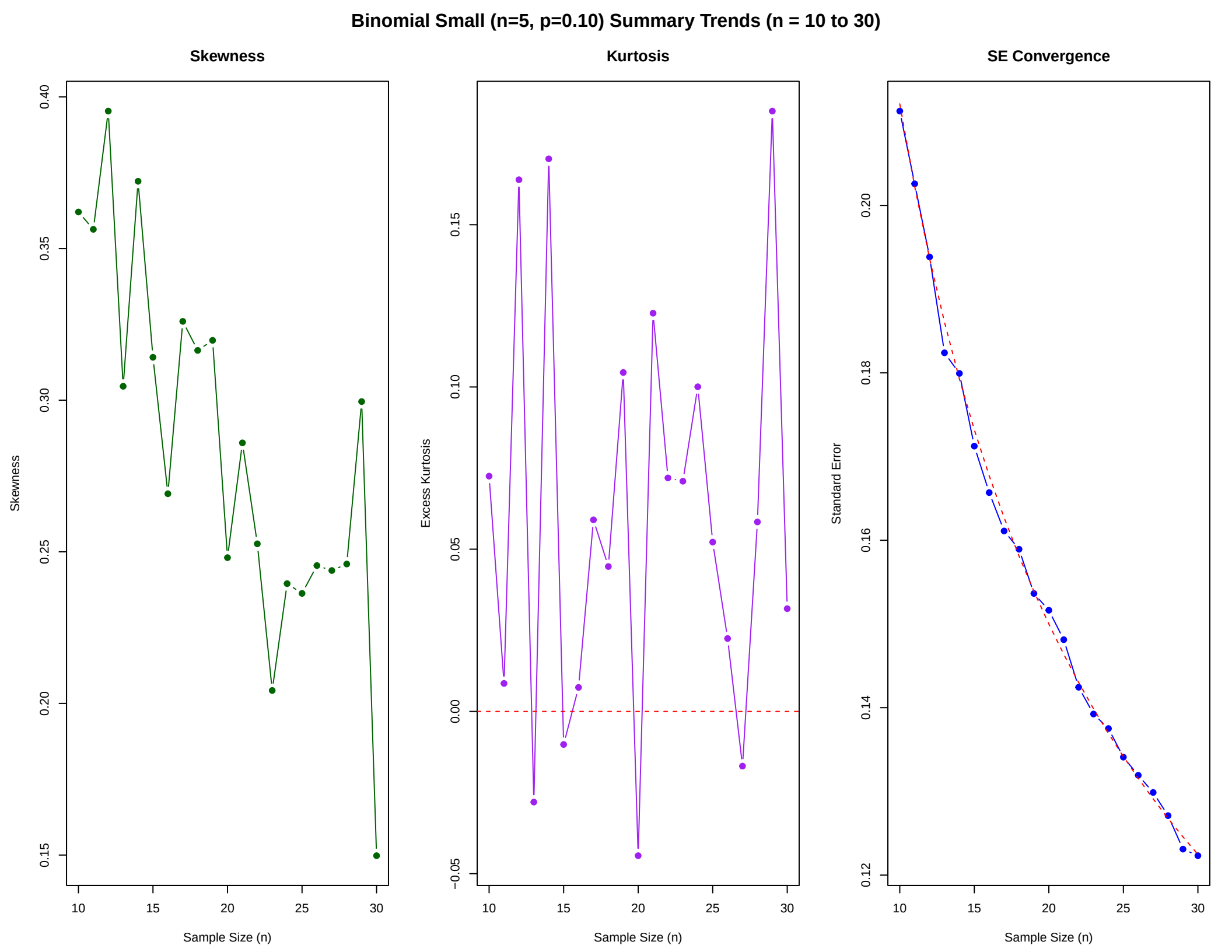



Binomial Small (n=5, p=0.10) — n = 29
Histogram (n = 29)
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample Mean
Q−Q Plot (n = 29)
Sample Quantiles
0.2
0.4
0.6
0.8
1.0
−4
−2
0
2
4
Theoretical Quantiles
Excess Kurtosis = 0.185

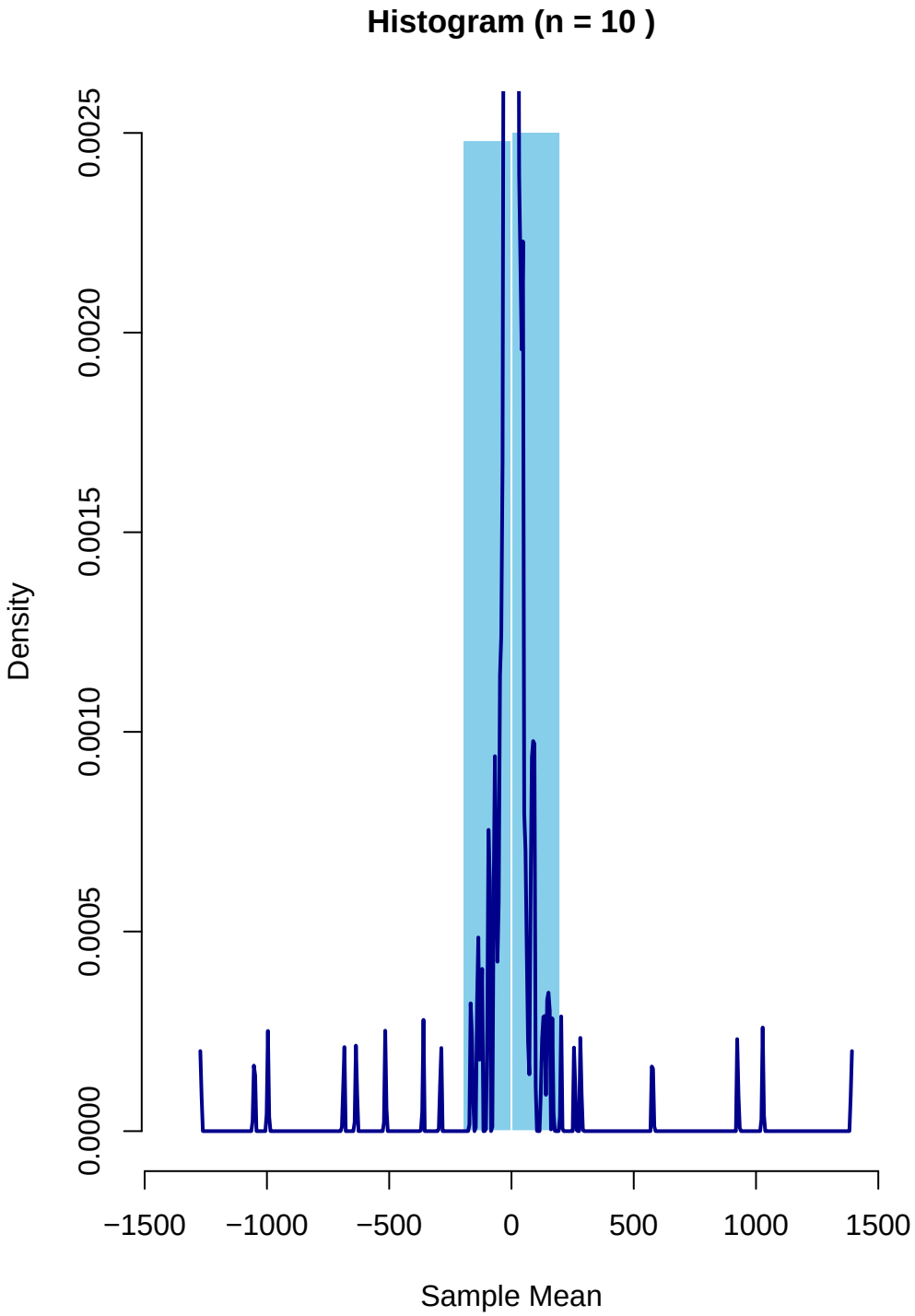
Binomial Small (n=5, p=0.10) — n = 30

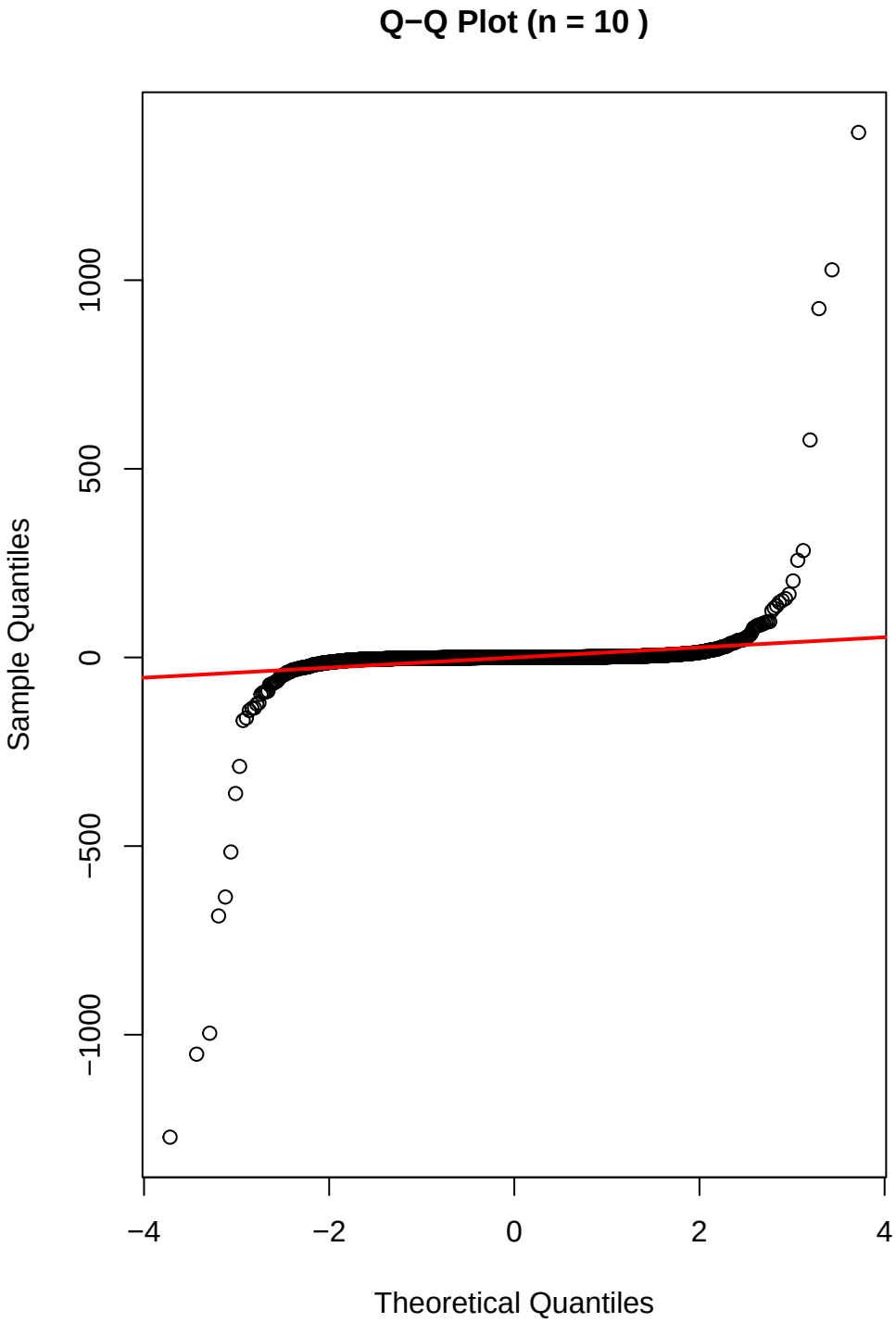





Binomial Small (n=5, p=0.10) Summary Trends (n = 10 to 30)
Skewness
Skewness
0.40
0.35
0.30
0.25
0.20
0.15
10
15
20
25
30
Sample Size (n)
Kurtosis
Excess Kurtosis
0.15
0.10
0.05
0.00
−0.05
10
15
20
25
30
Sample Size (n)
SE Convergence
Standard Error
0.20
0.18
0.16
0.14
0.12
10
15
20
25
30
Sample Size (n)

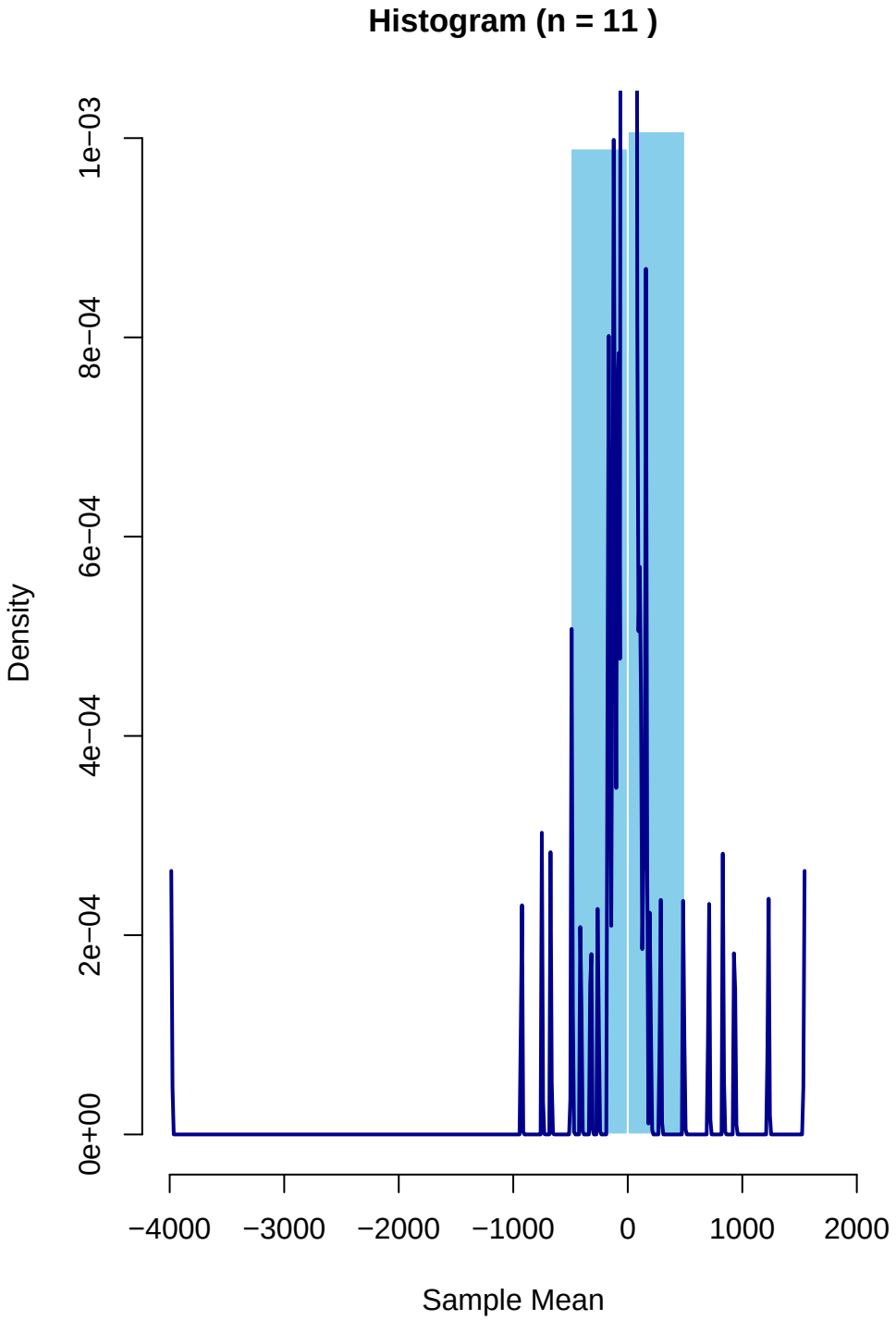
Cauchy (Location=0, Scale=1) — n = 10

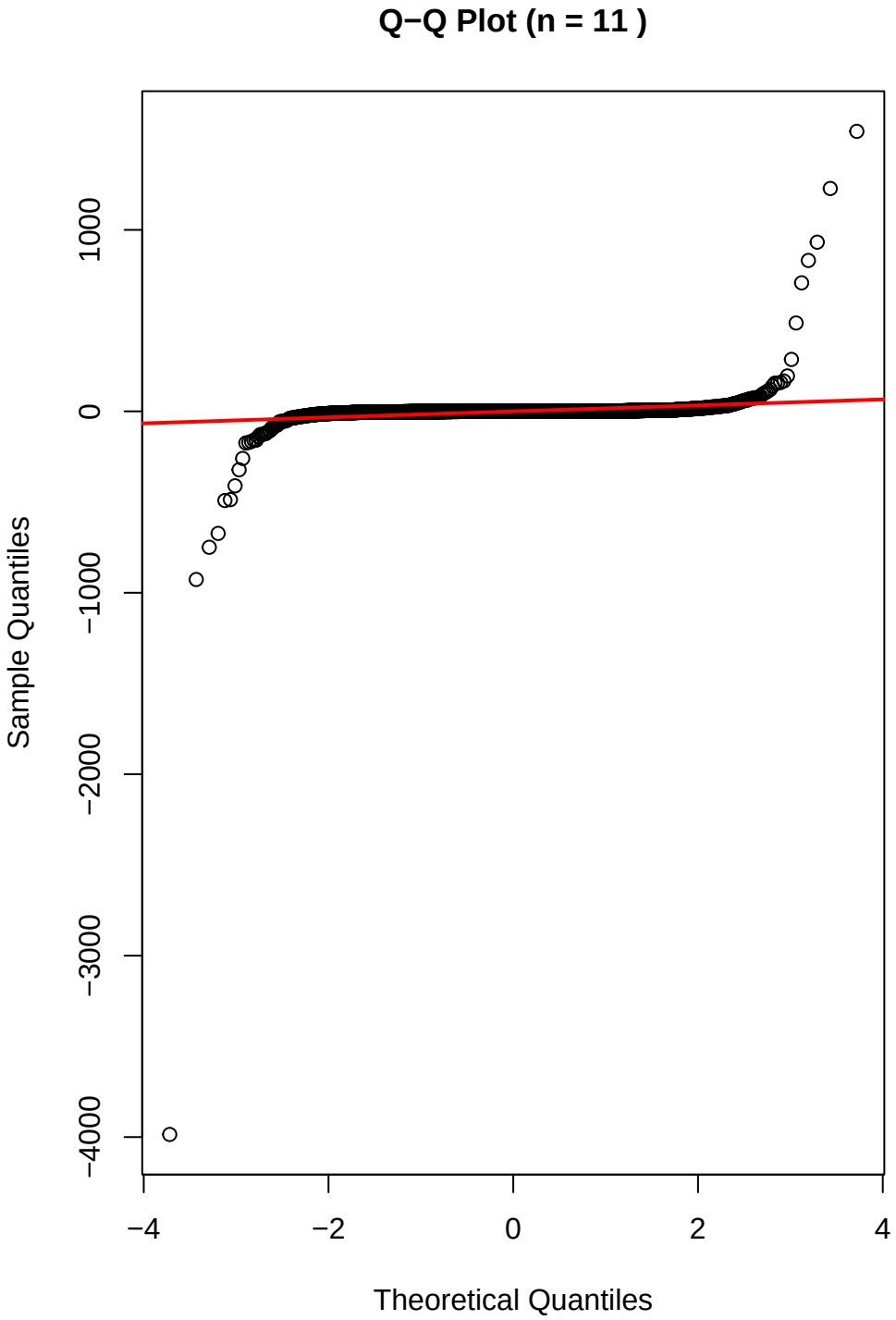




Excess Kurtosis = 541.0525

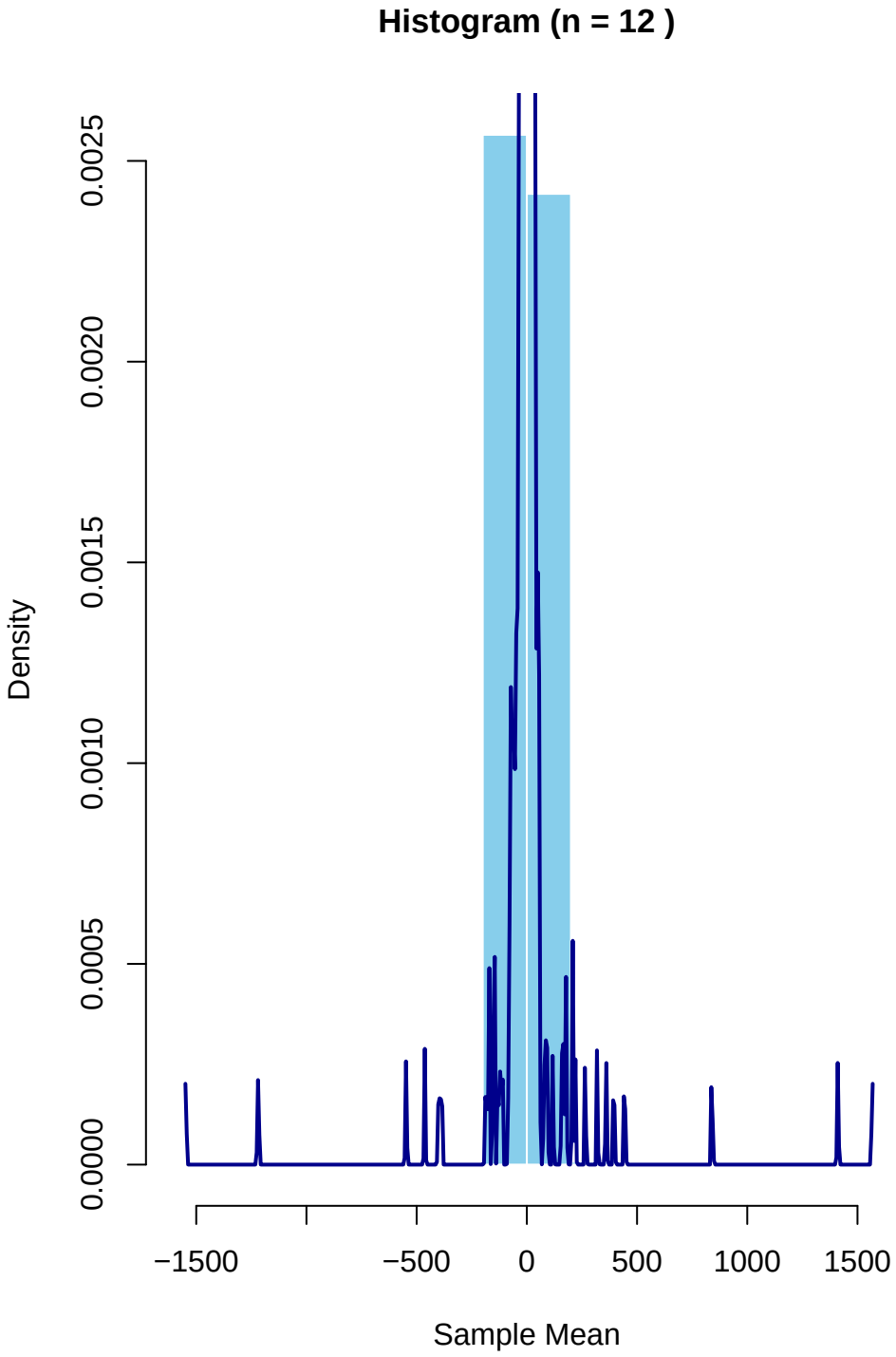
Cauchy (Location=0, Scale=1) — n = 11

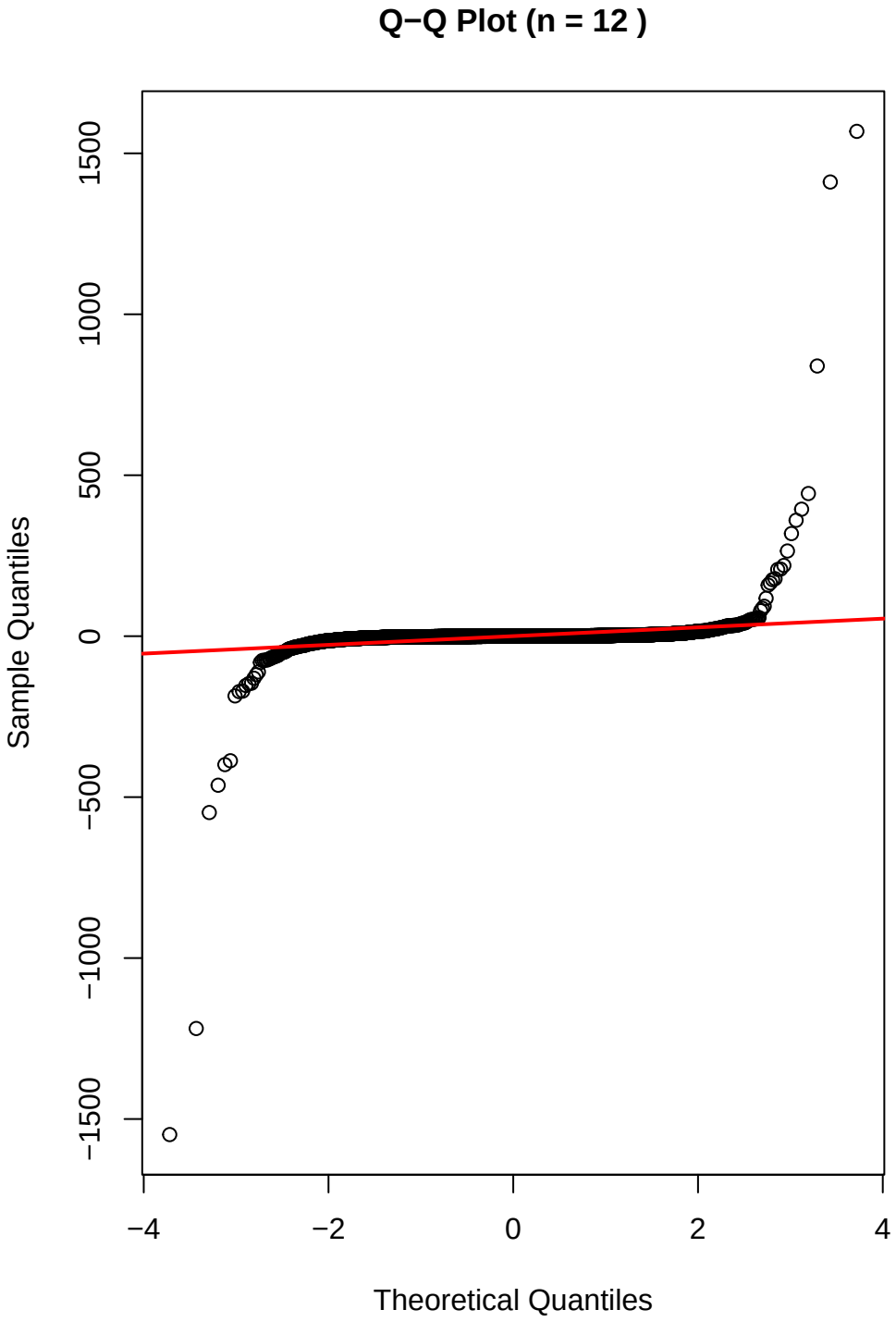




Excess Kurtosis = 2003.4627

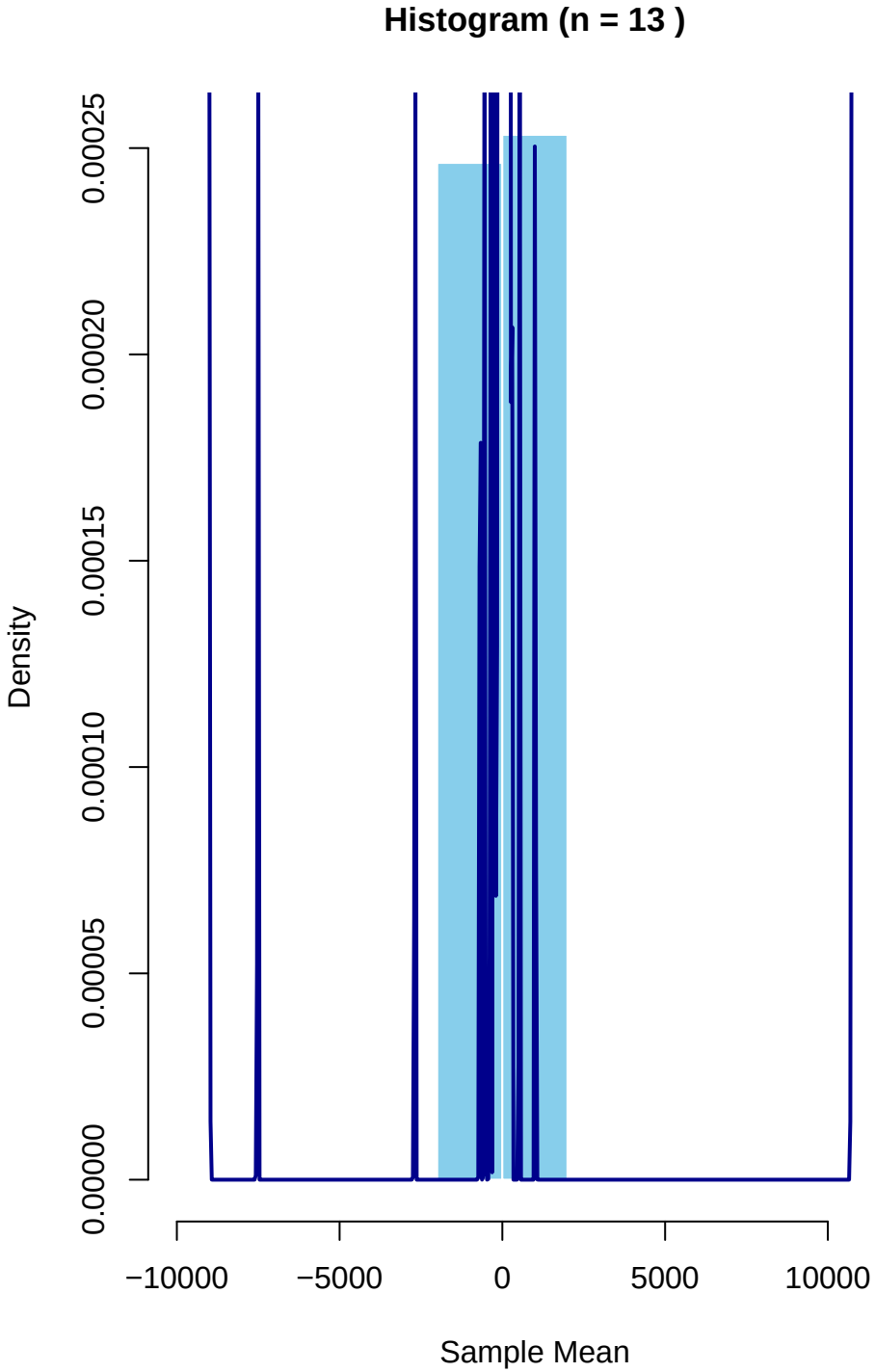
Cauchy (Location=0, Scale=1) — n = 12

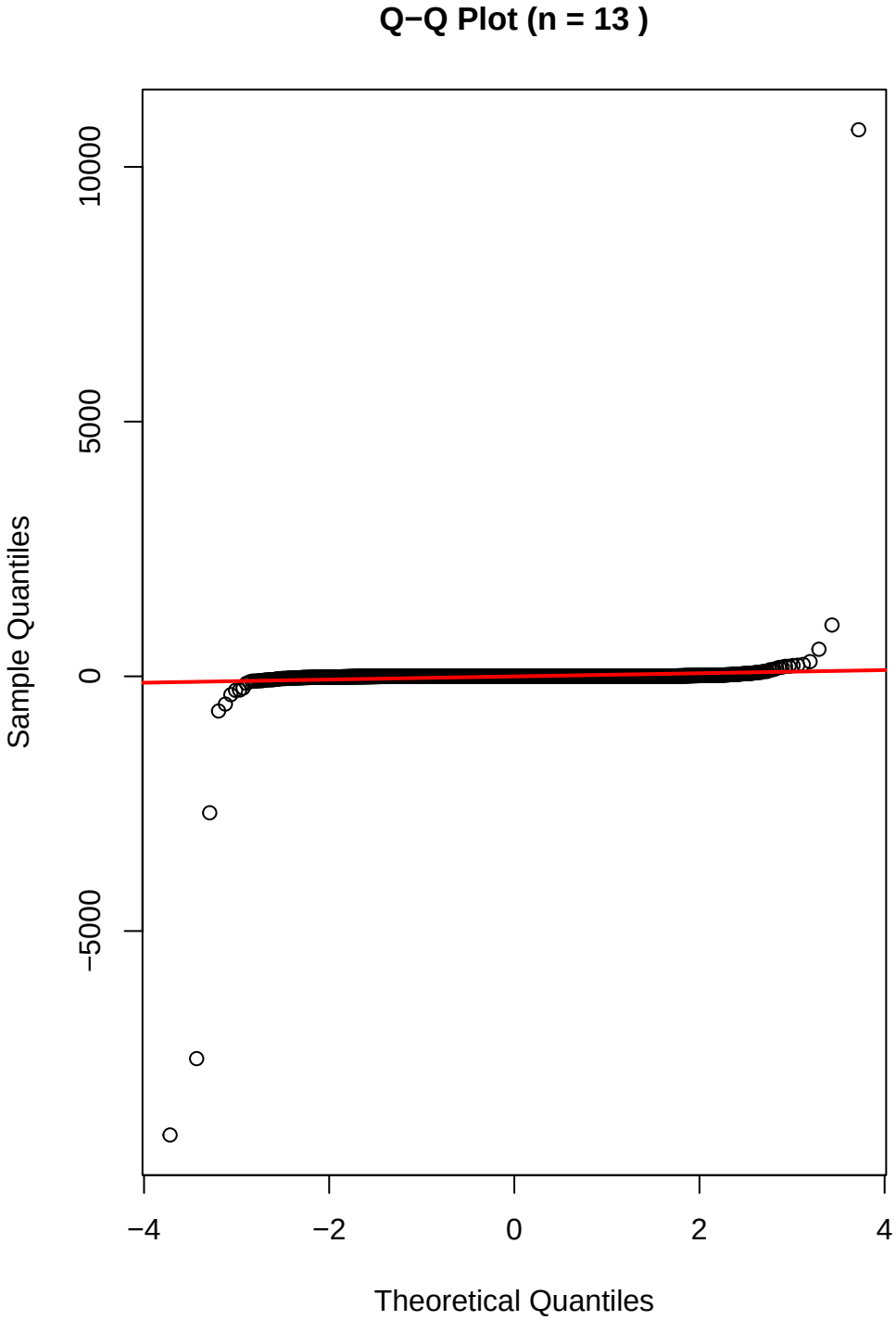




Excess Kurtosis = 736.5359

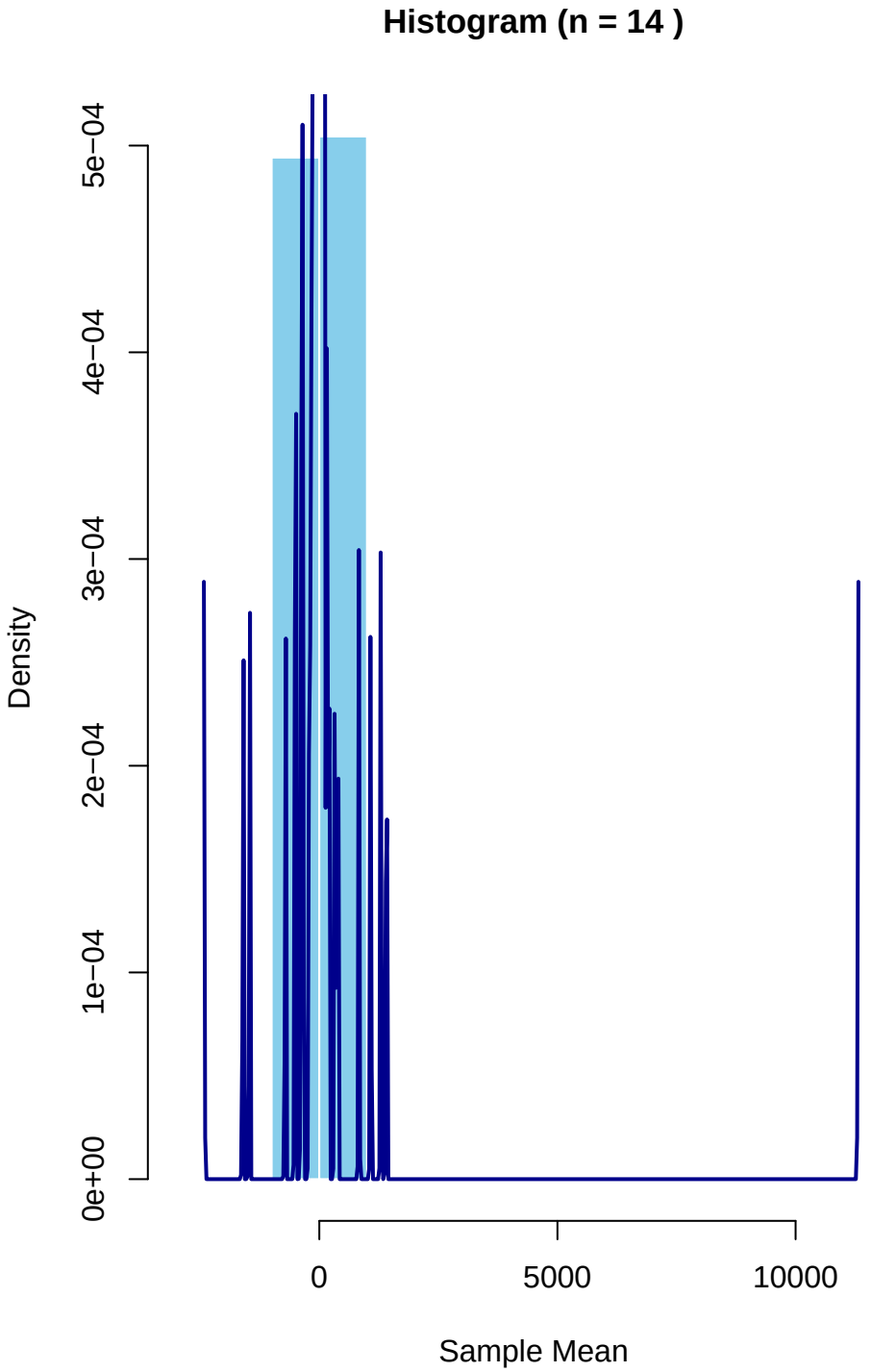
Cauchy (Location=0, Scale=1) — n = 13

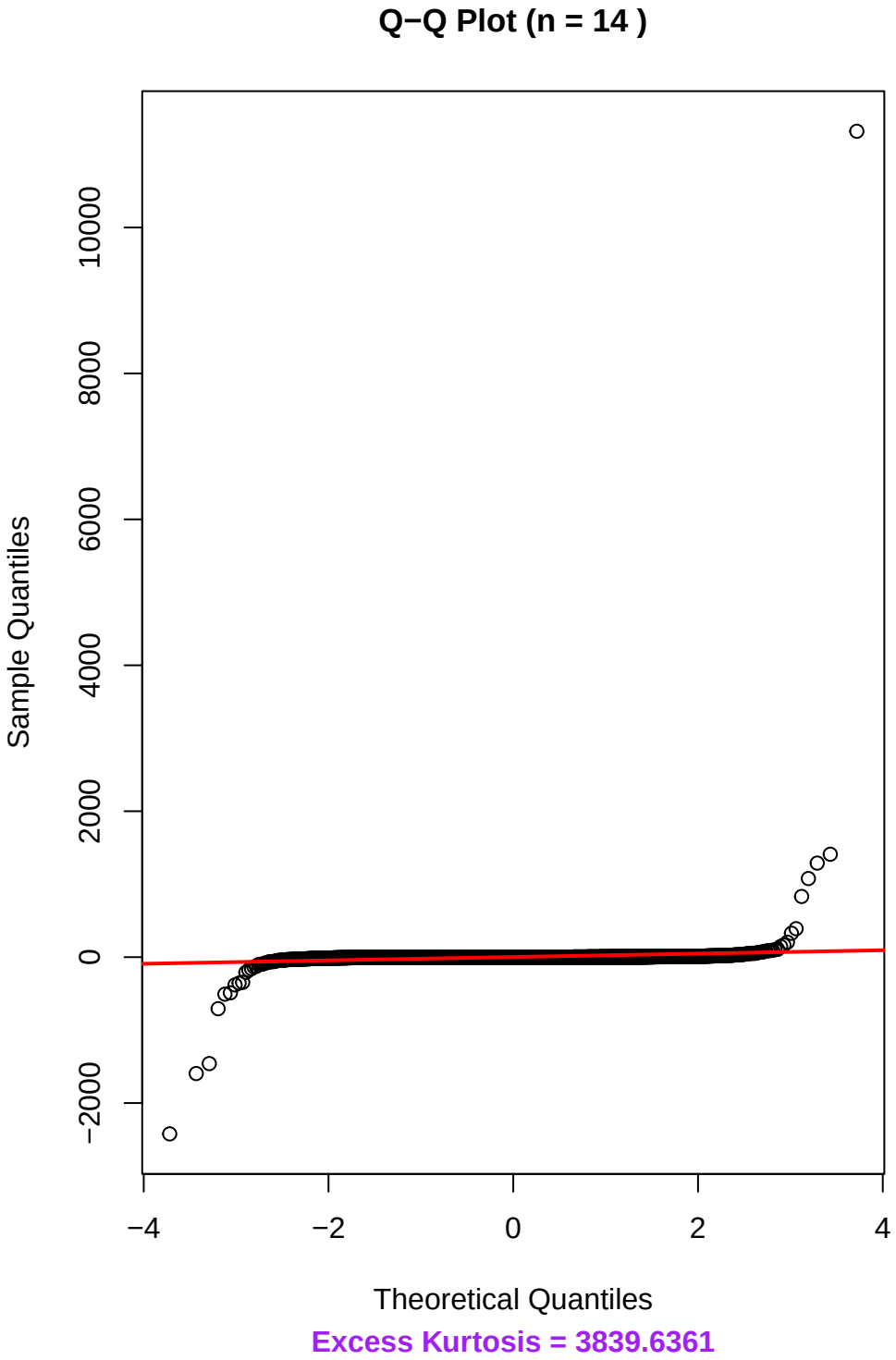




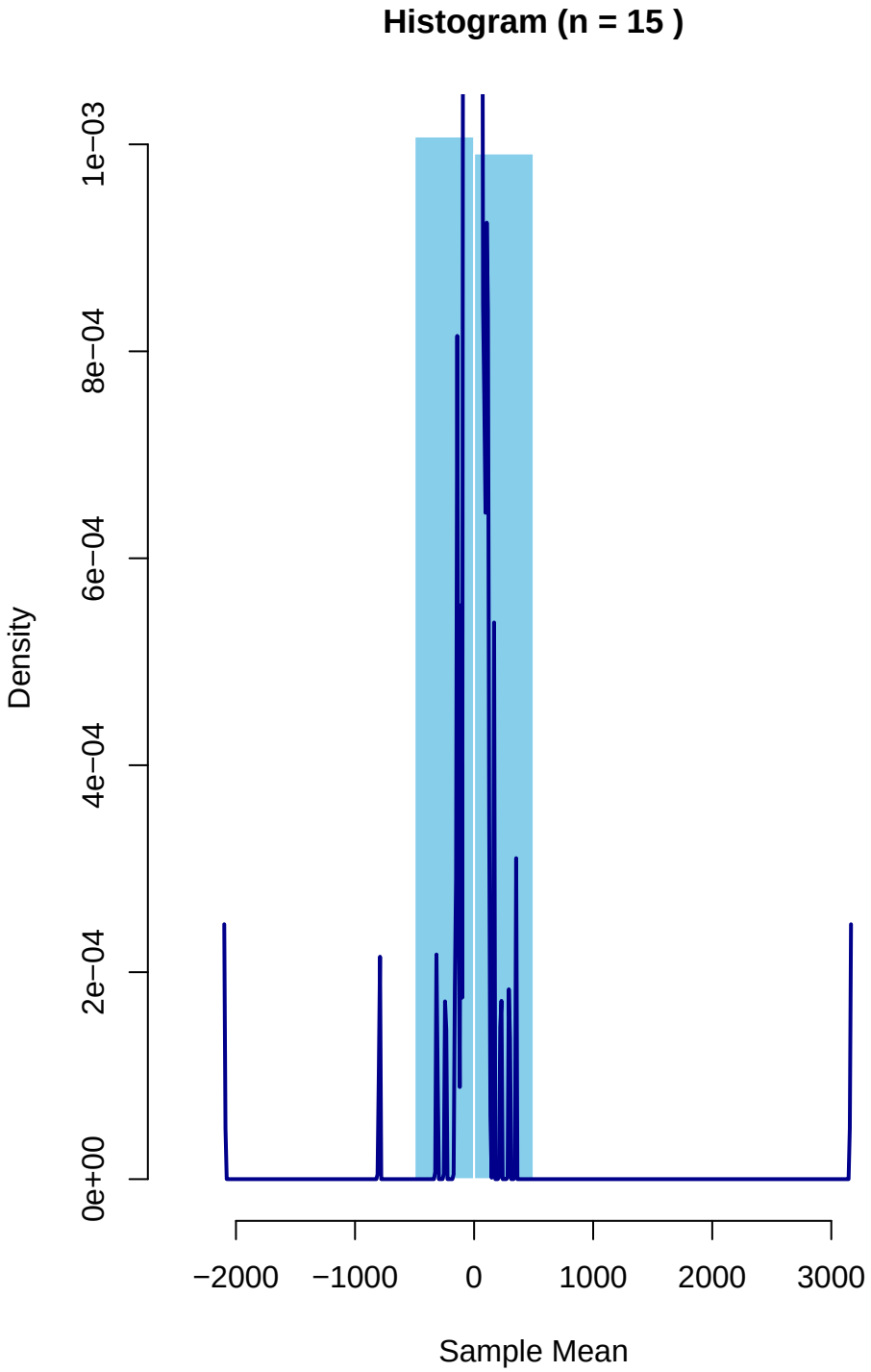
Excess Kurtosis = 1664.0605

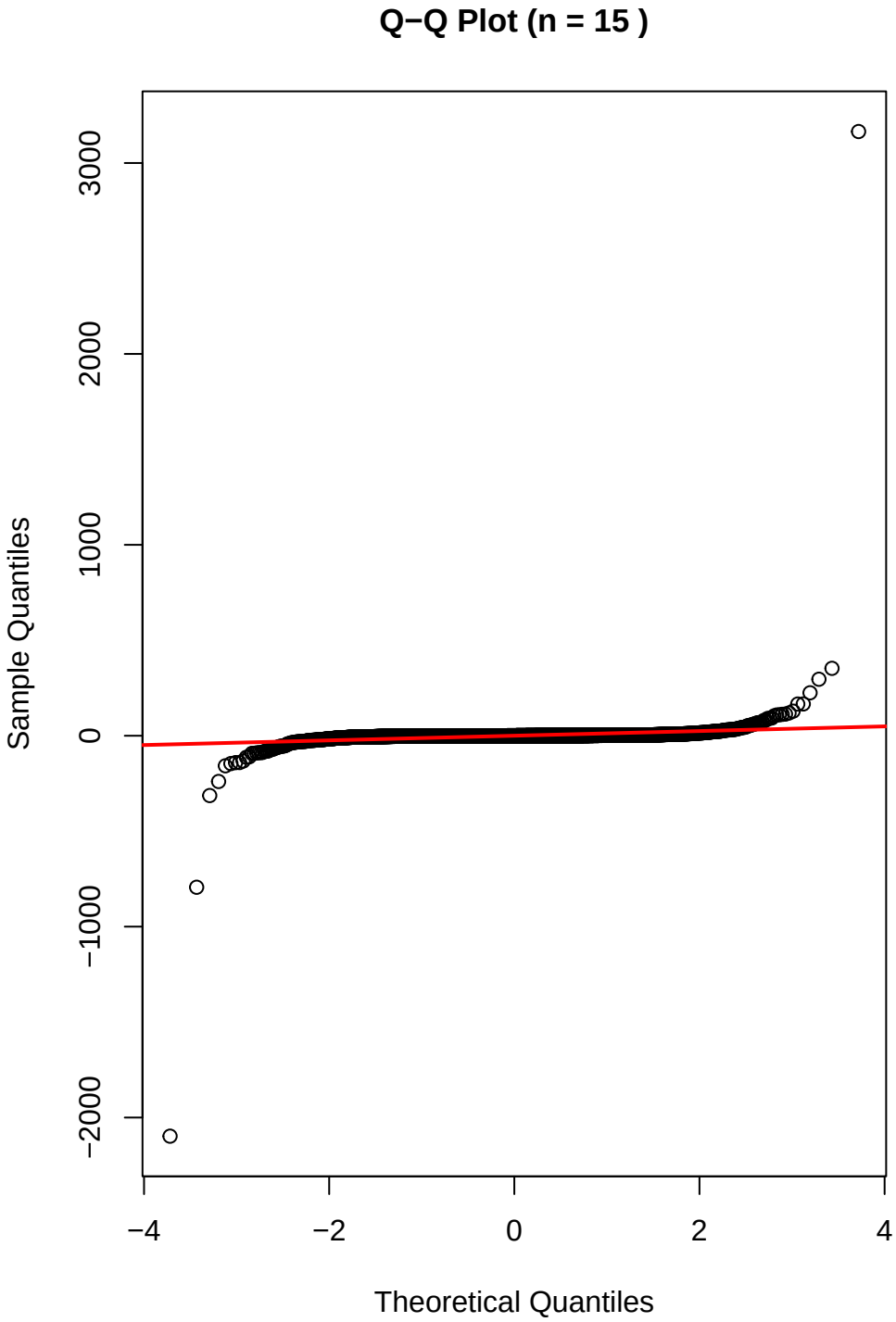
Cauchy (Location=0, Scale=1) — n = 14





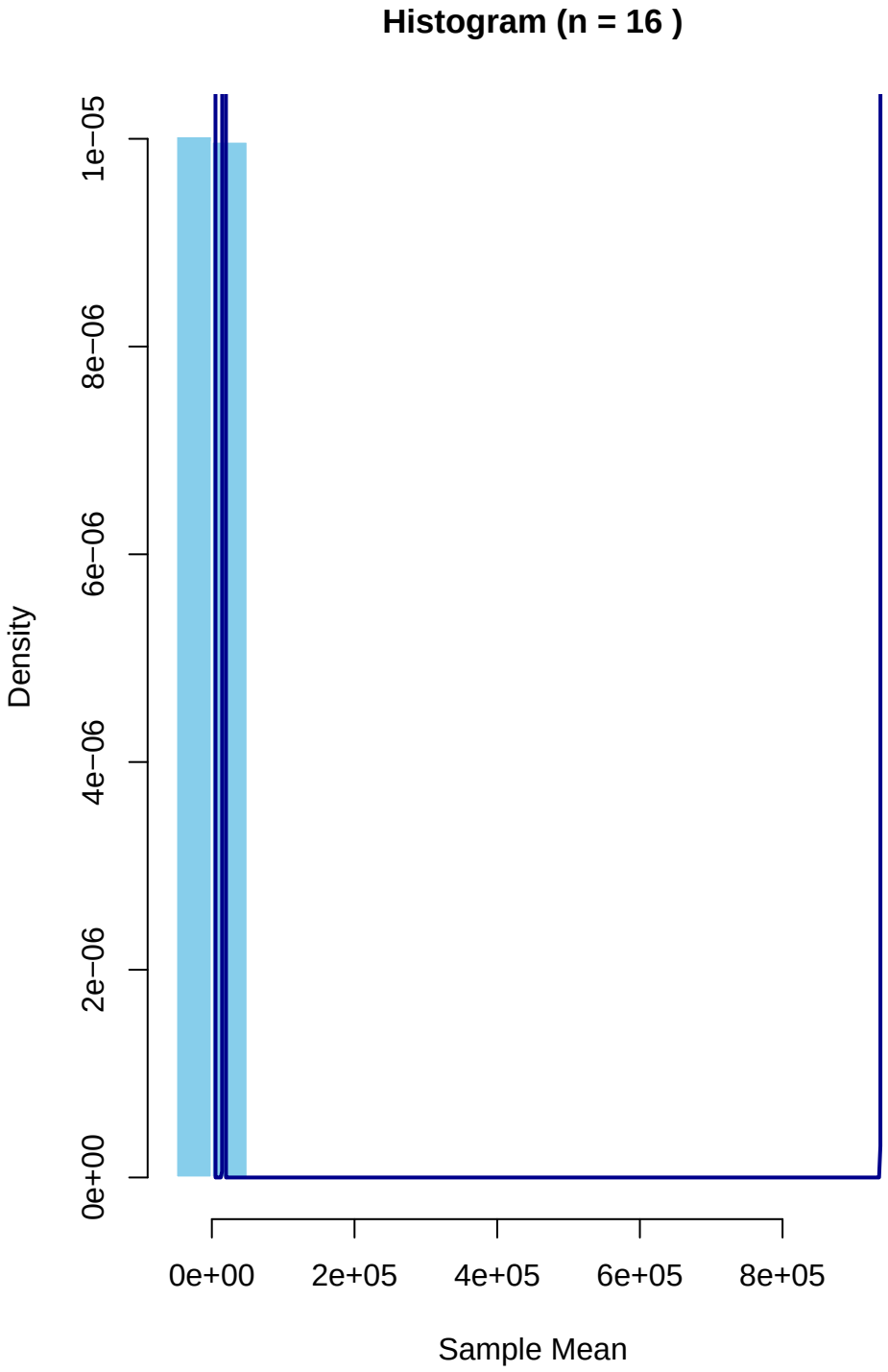
Cauchy (Location=0, Scale=1) — n = 15

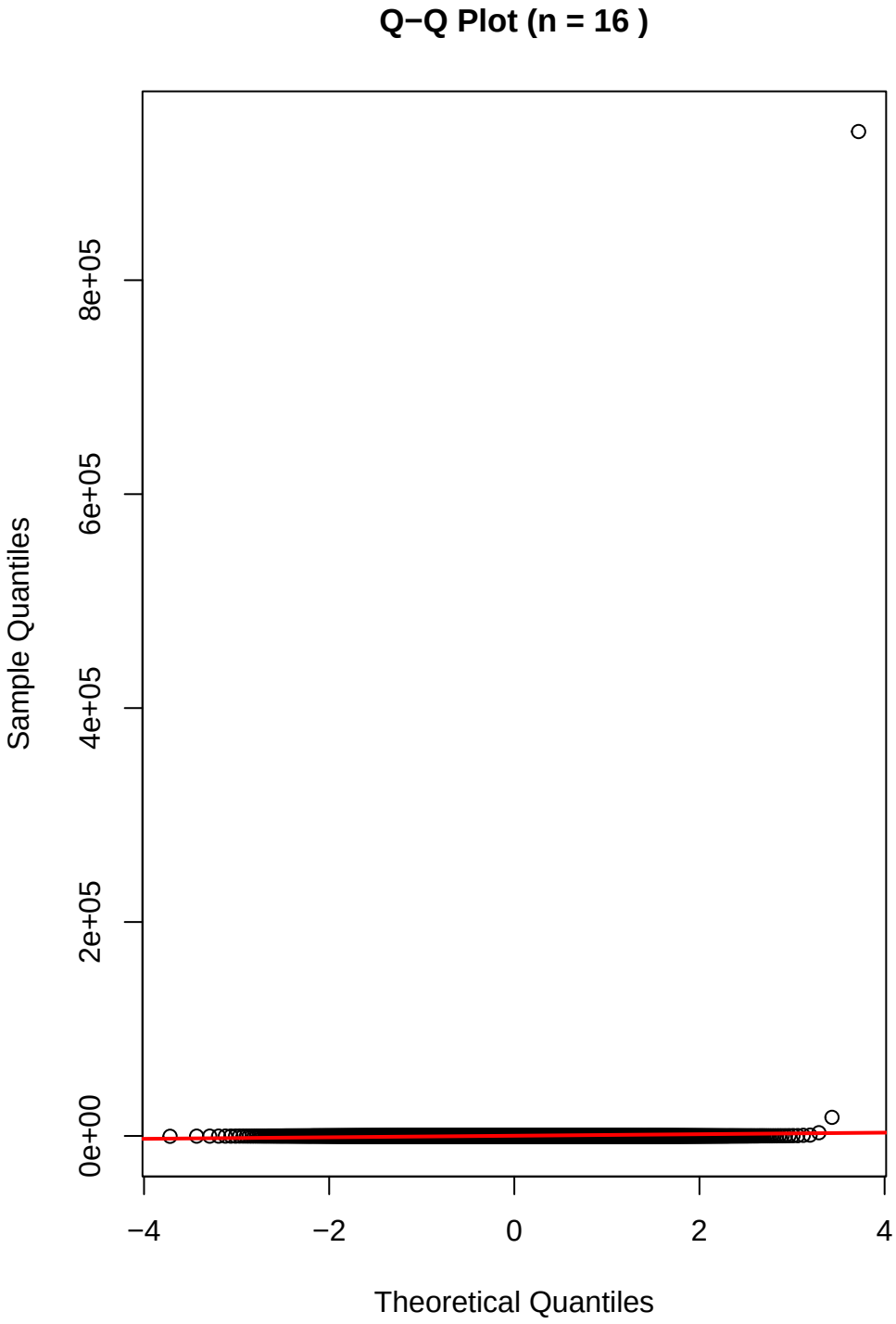




Excess Kurtosis = 2312.8342

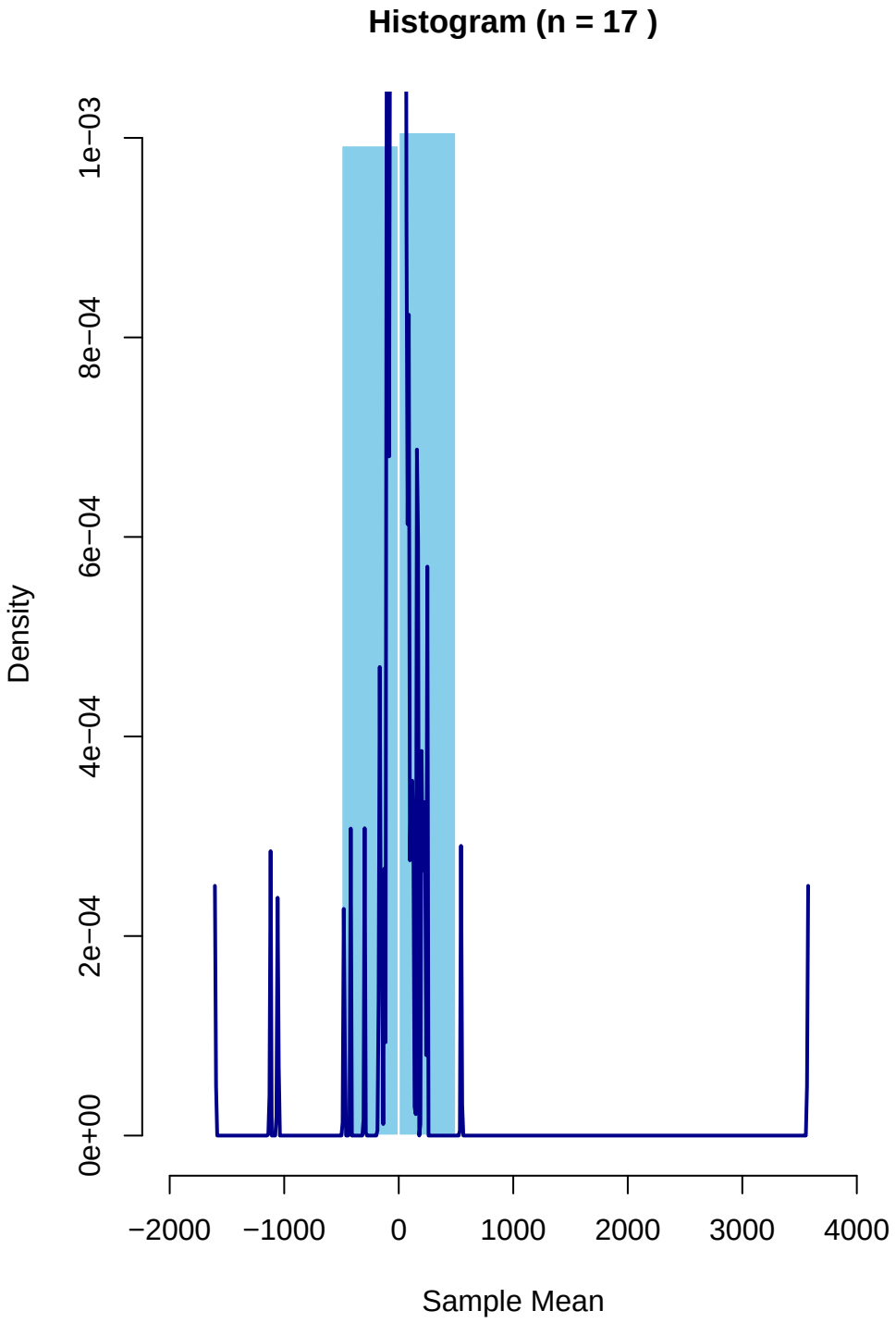
Cauchy (Location=0, Scale=1) — n = 16

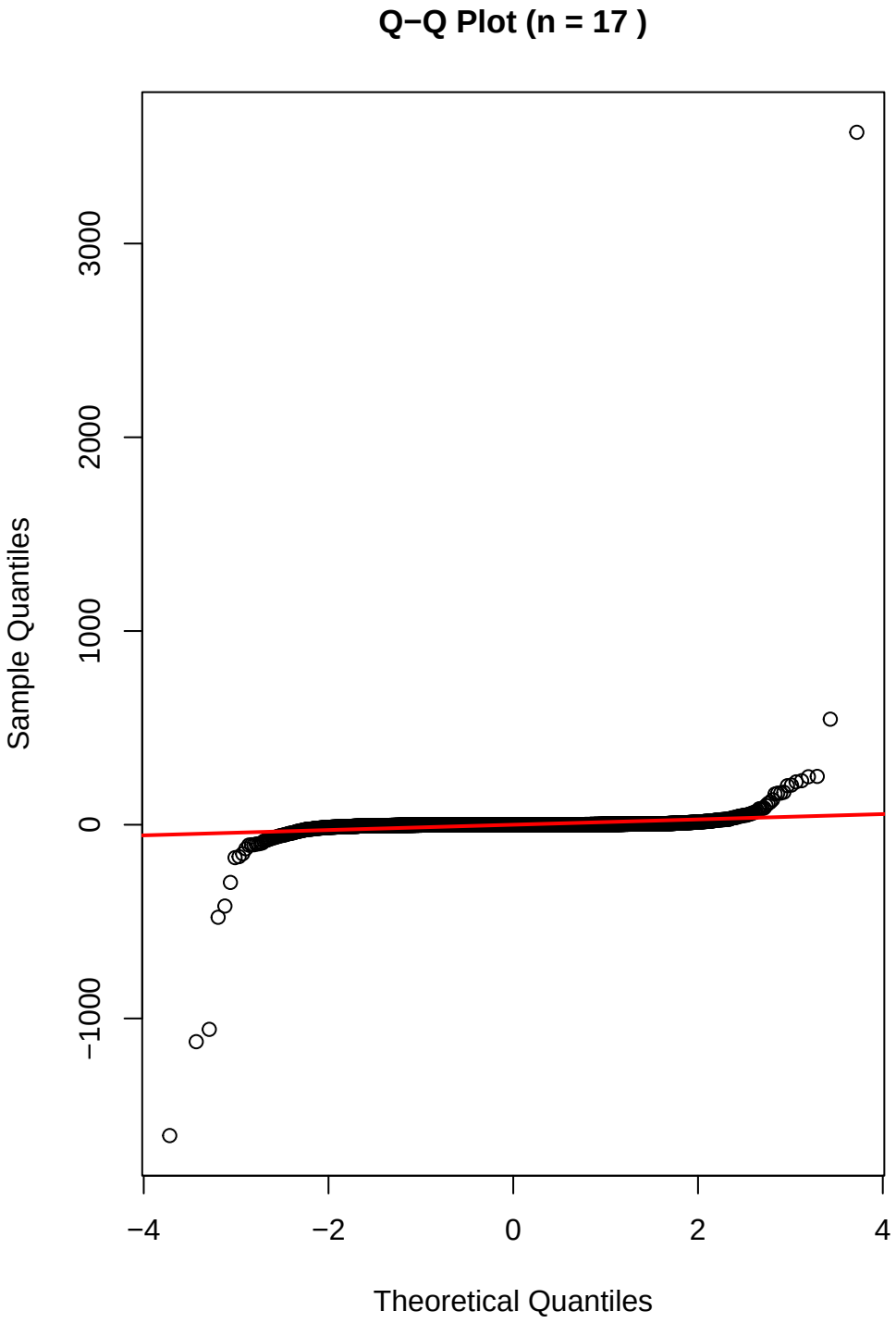




Excess Kurtosis = 4989.4513

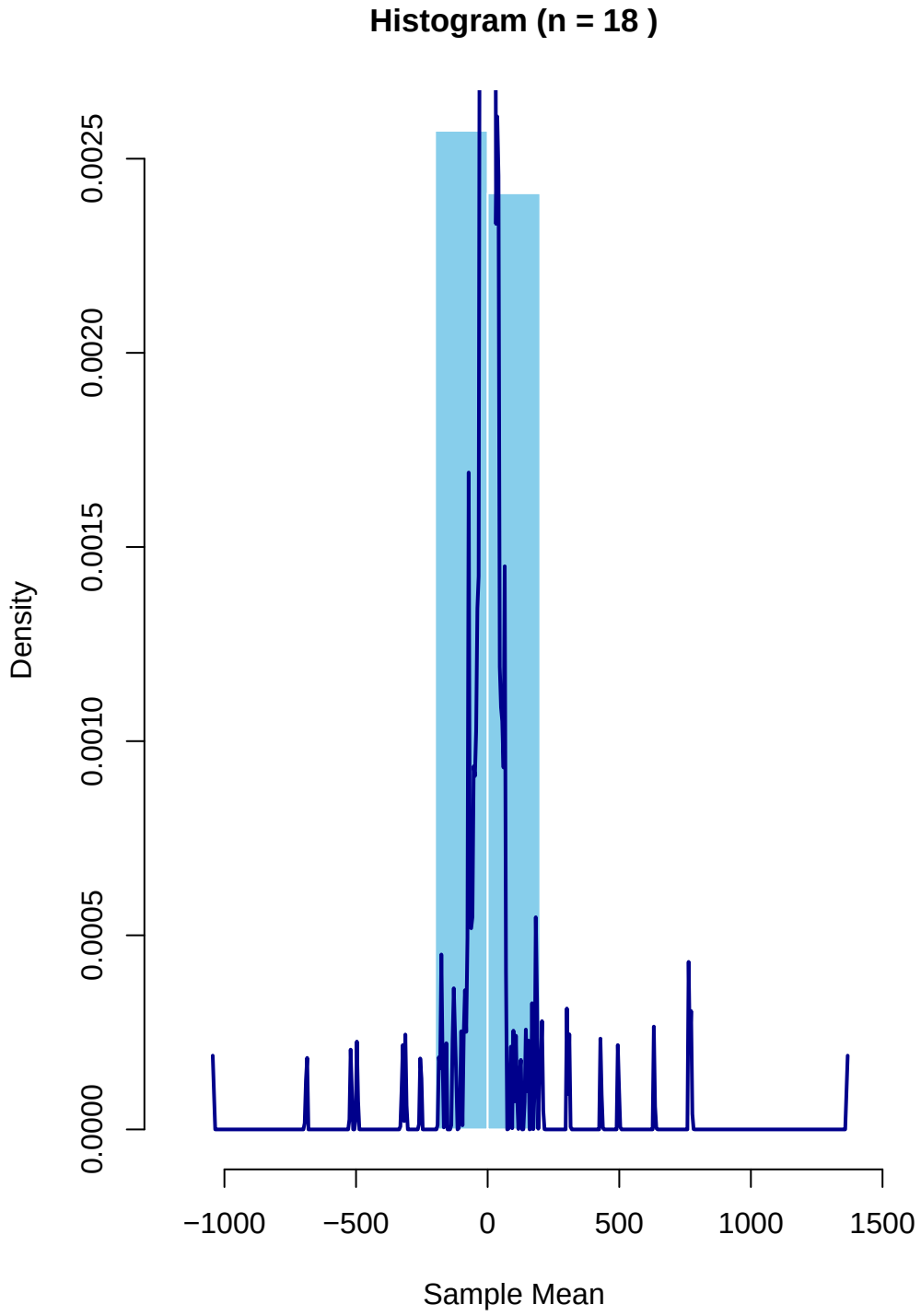
Cauchy (Location=0, Scale=1) — n = 17

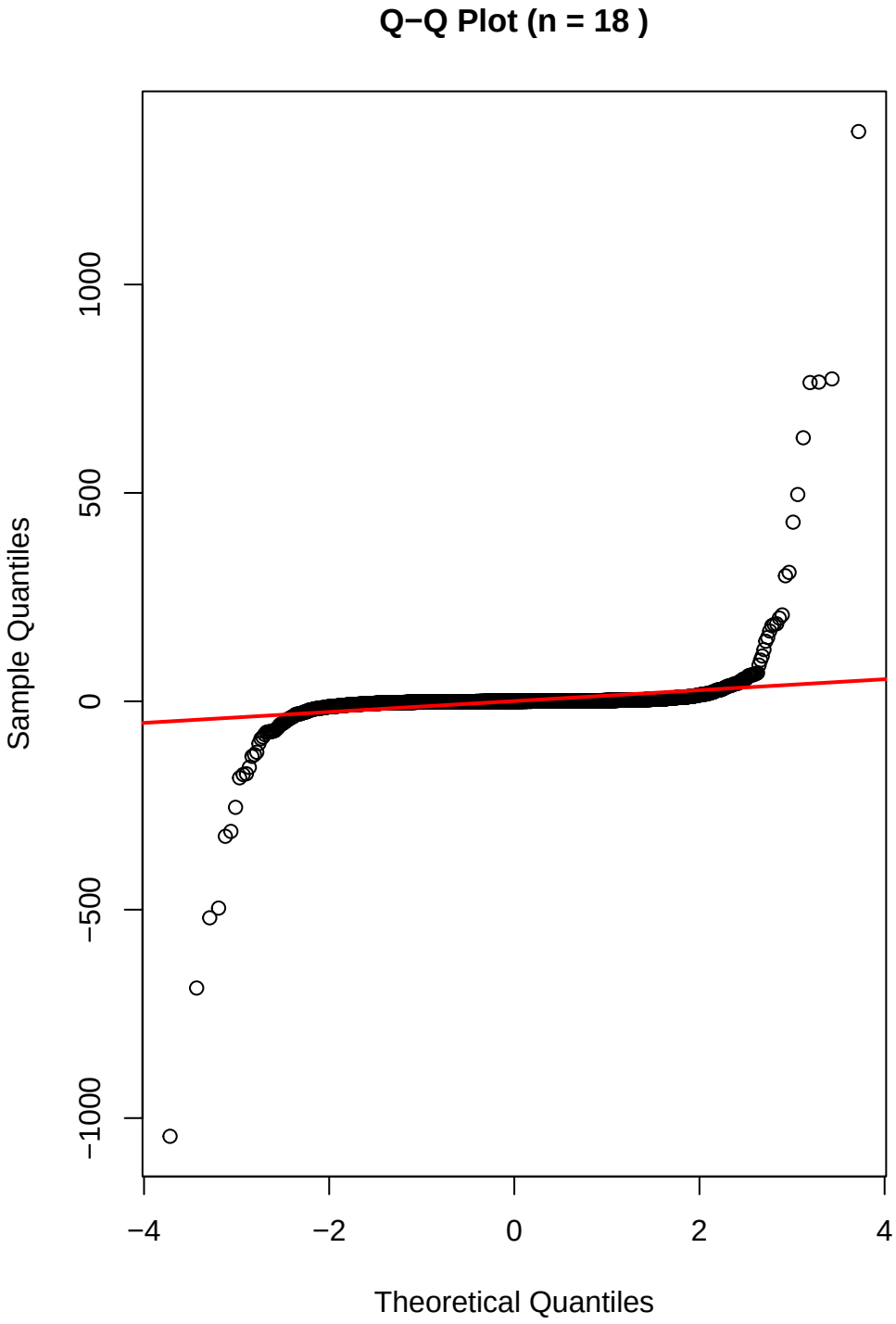




Excess Kurtosis = 2285.3973

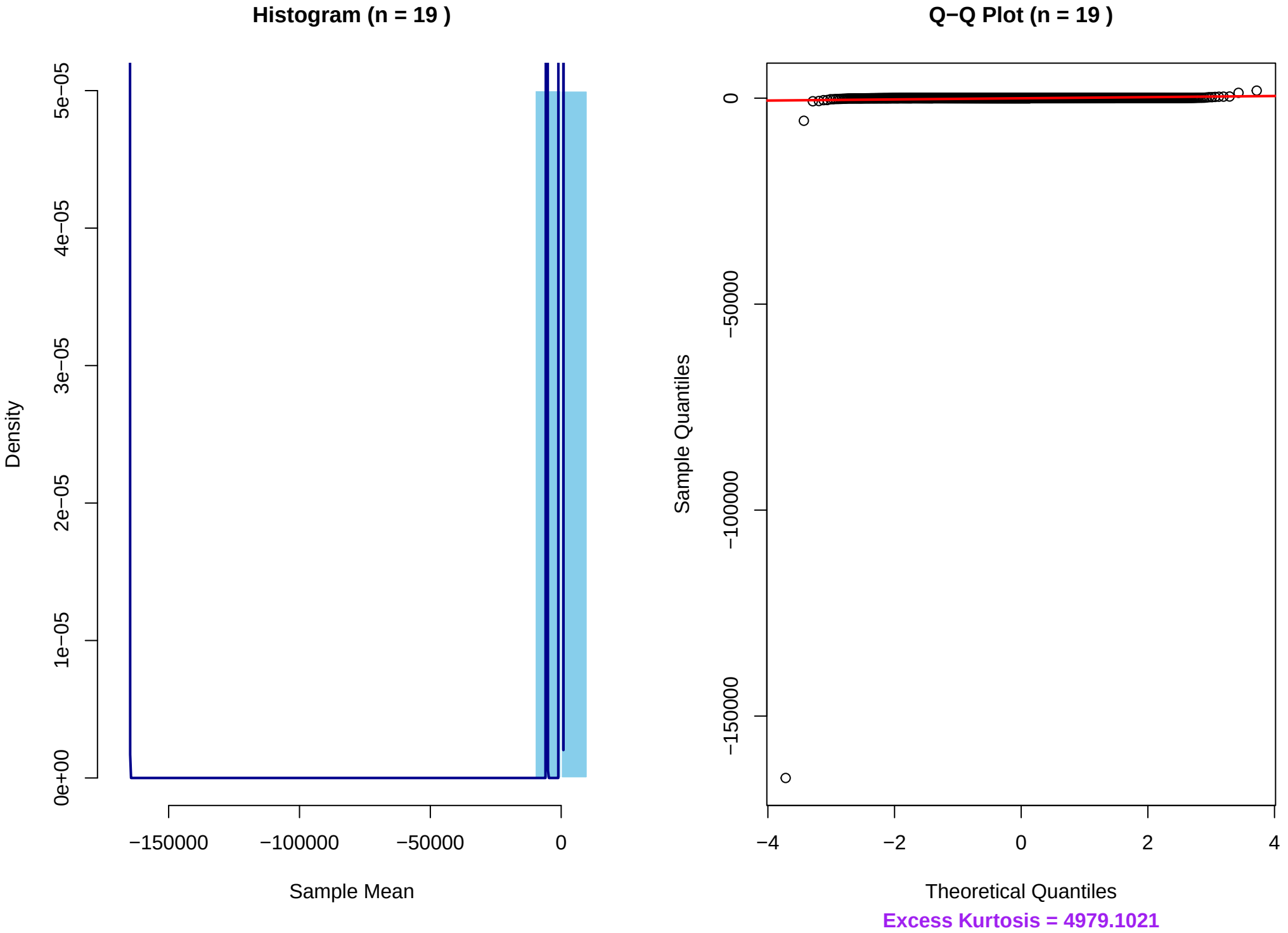
Cauchy (Location=0, Scale=1) — n = 18



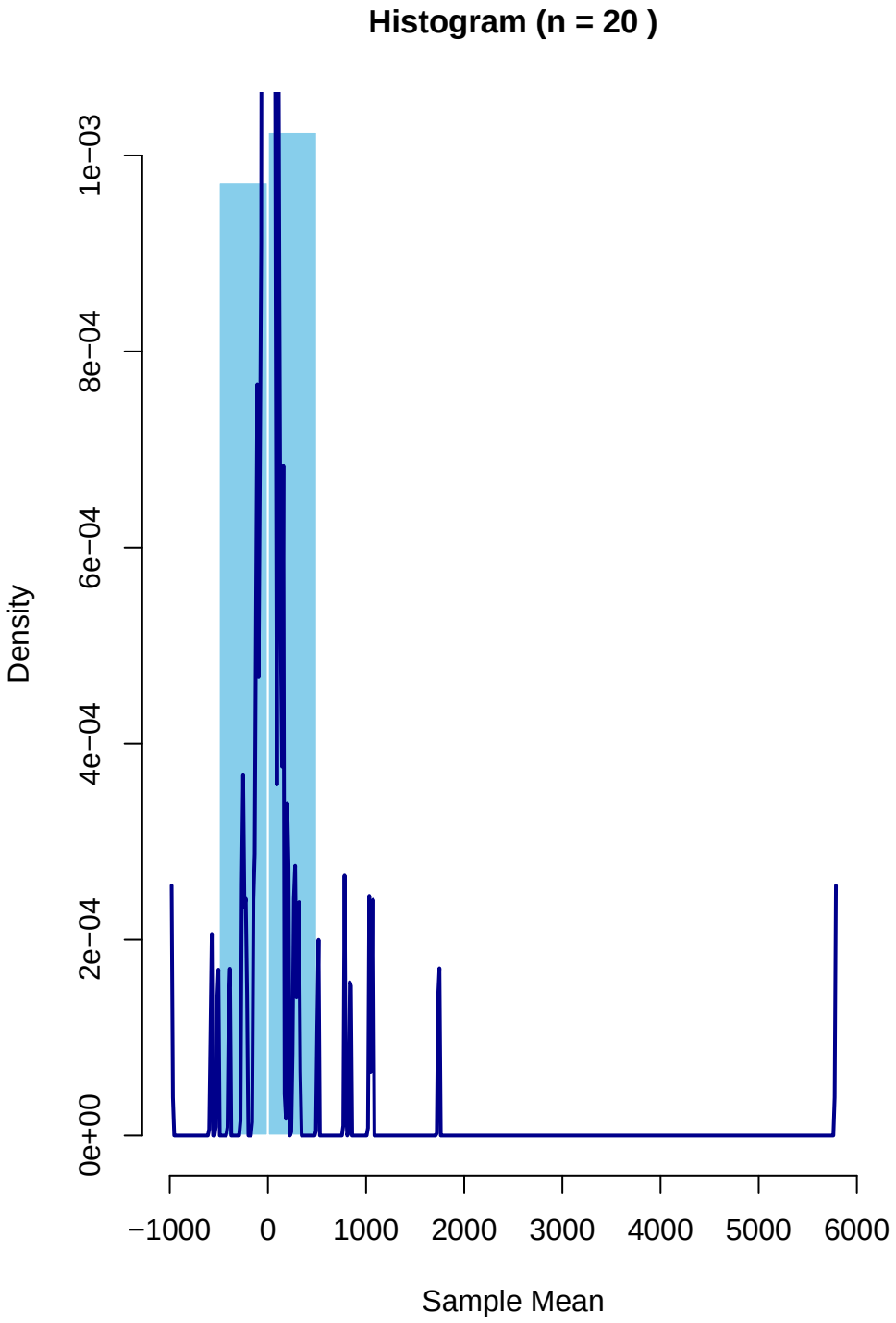


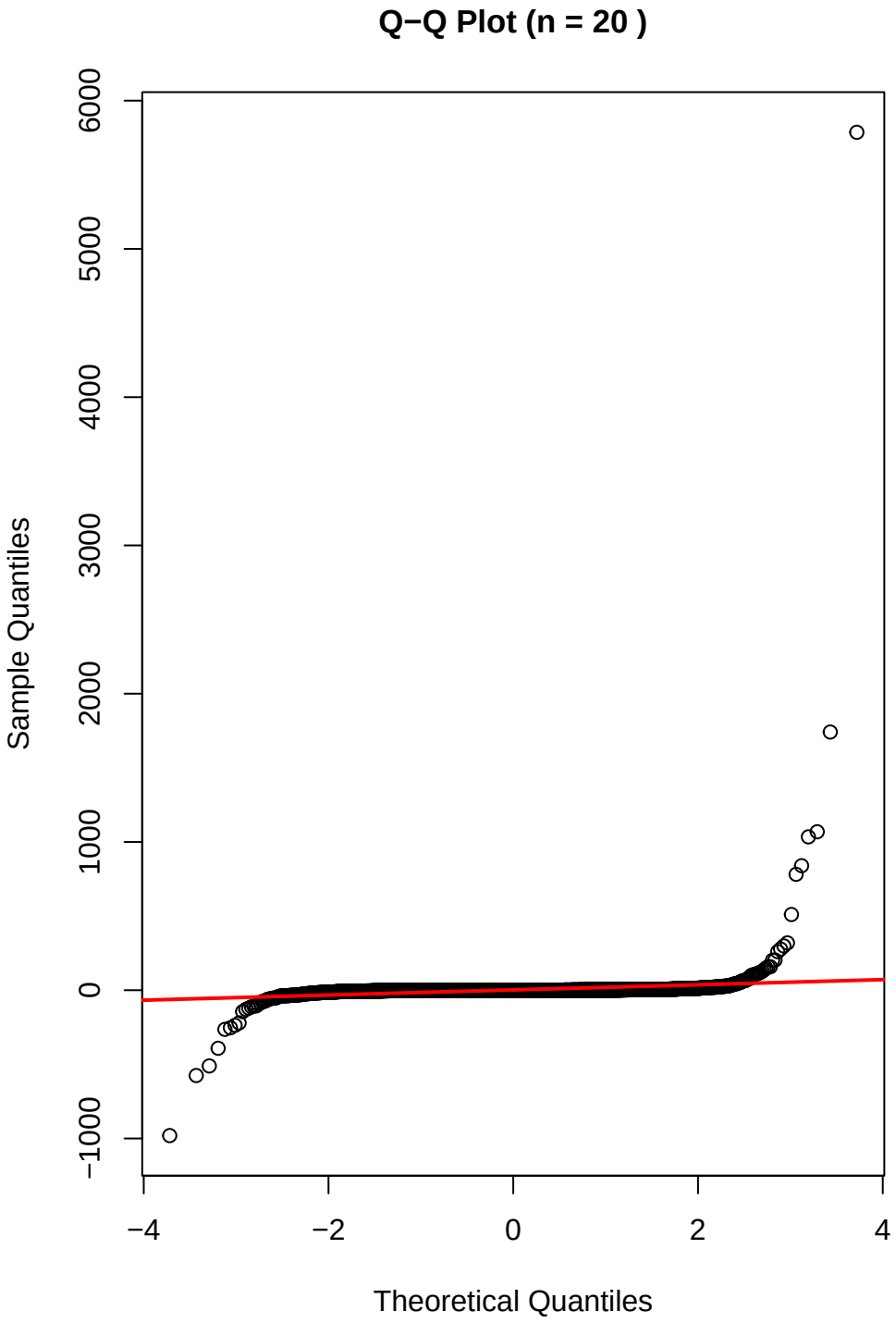
Excess Kurtosis = 525.5887

Cauchy (Location=0, Scale=1) — n = 19



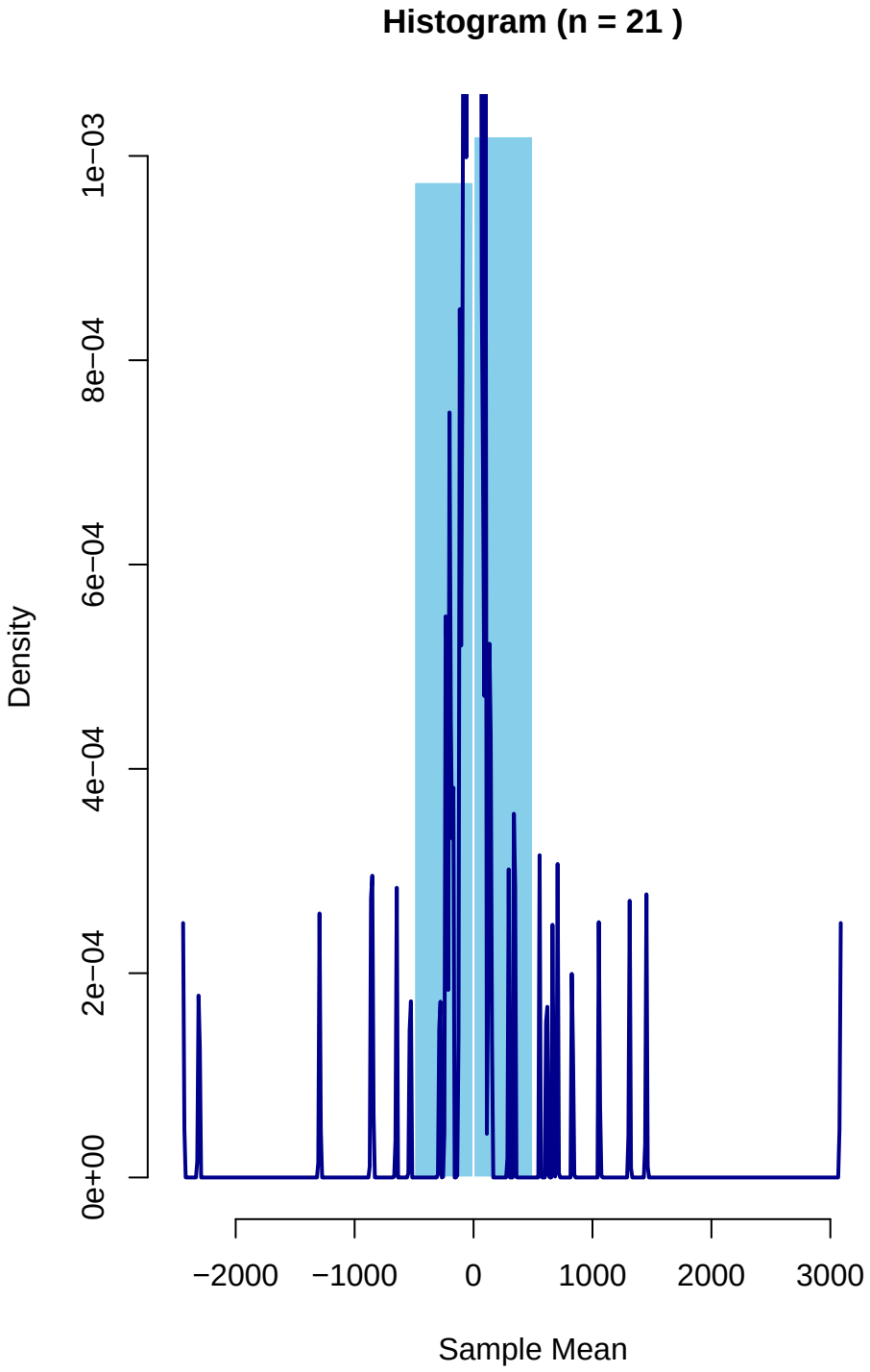
Cauchy (Location=0, Scale=1) — n = 20

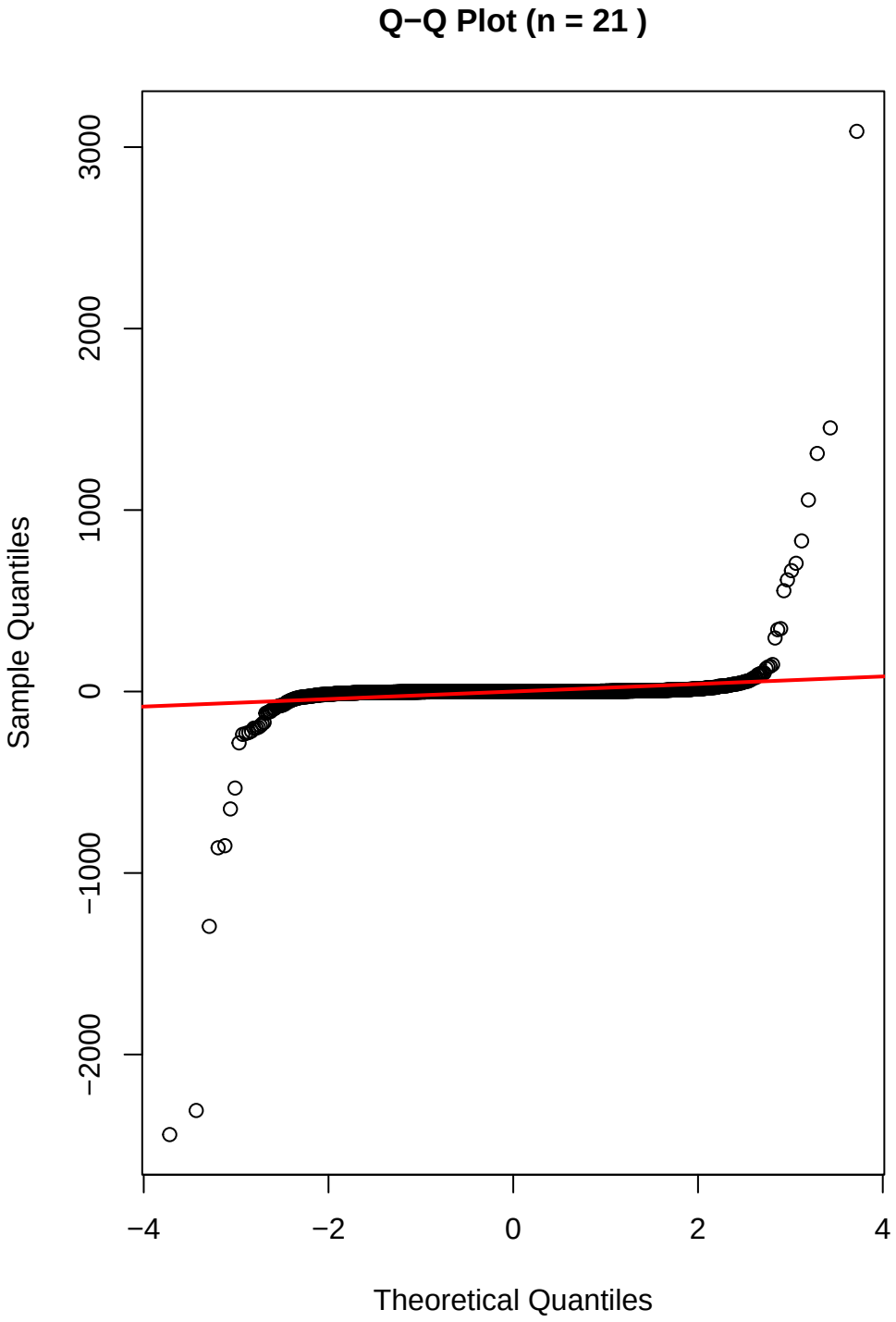




Excess Kurtosis = 3035.295

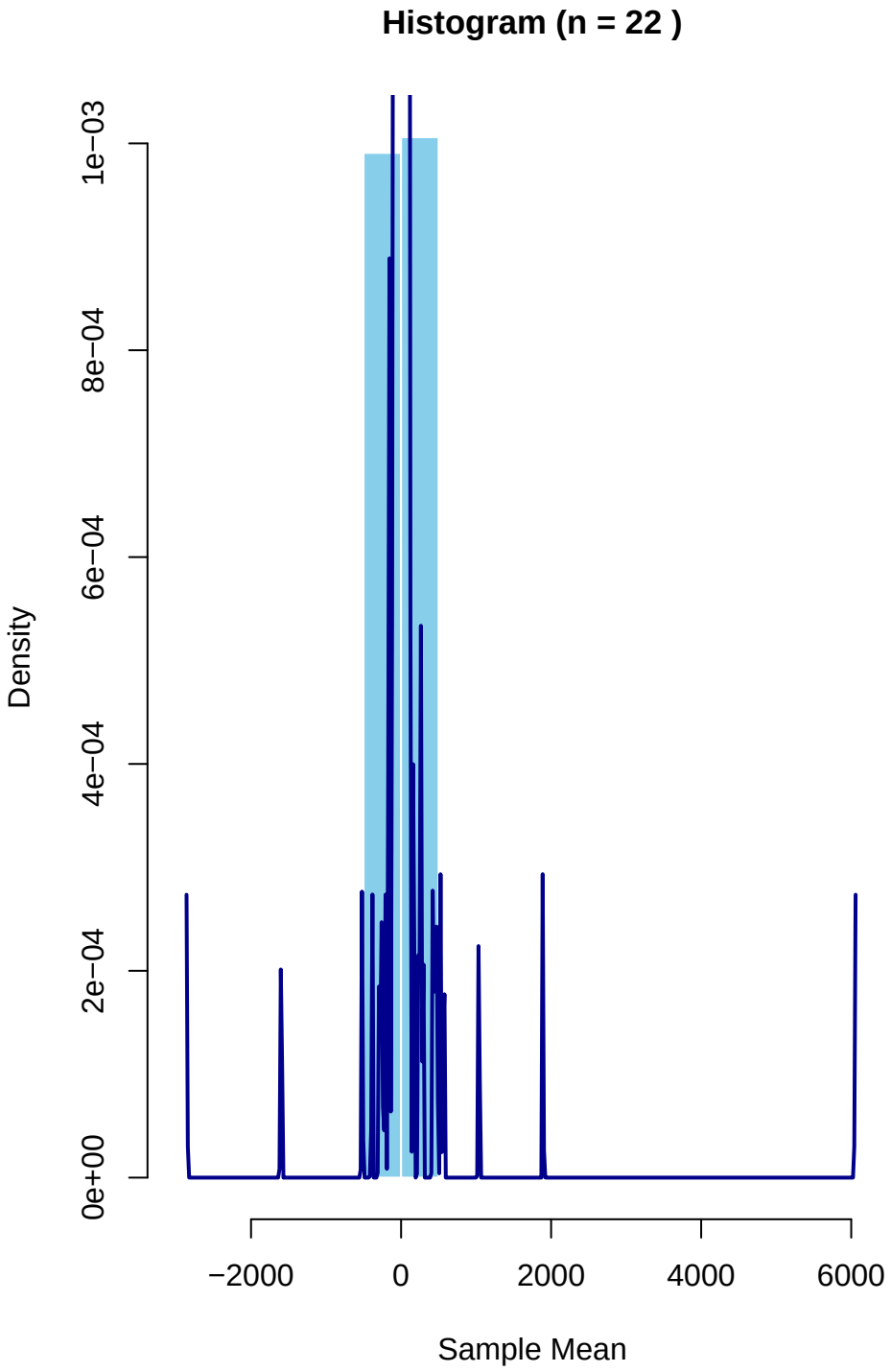
Cauchy (Location=0, Scale=1) — n = 21

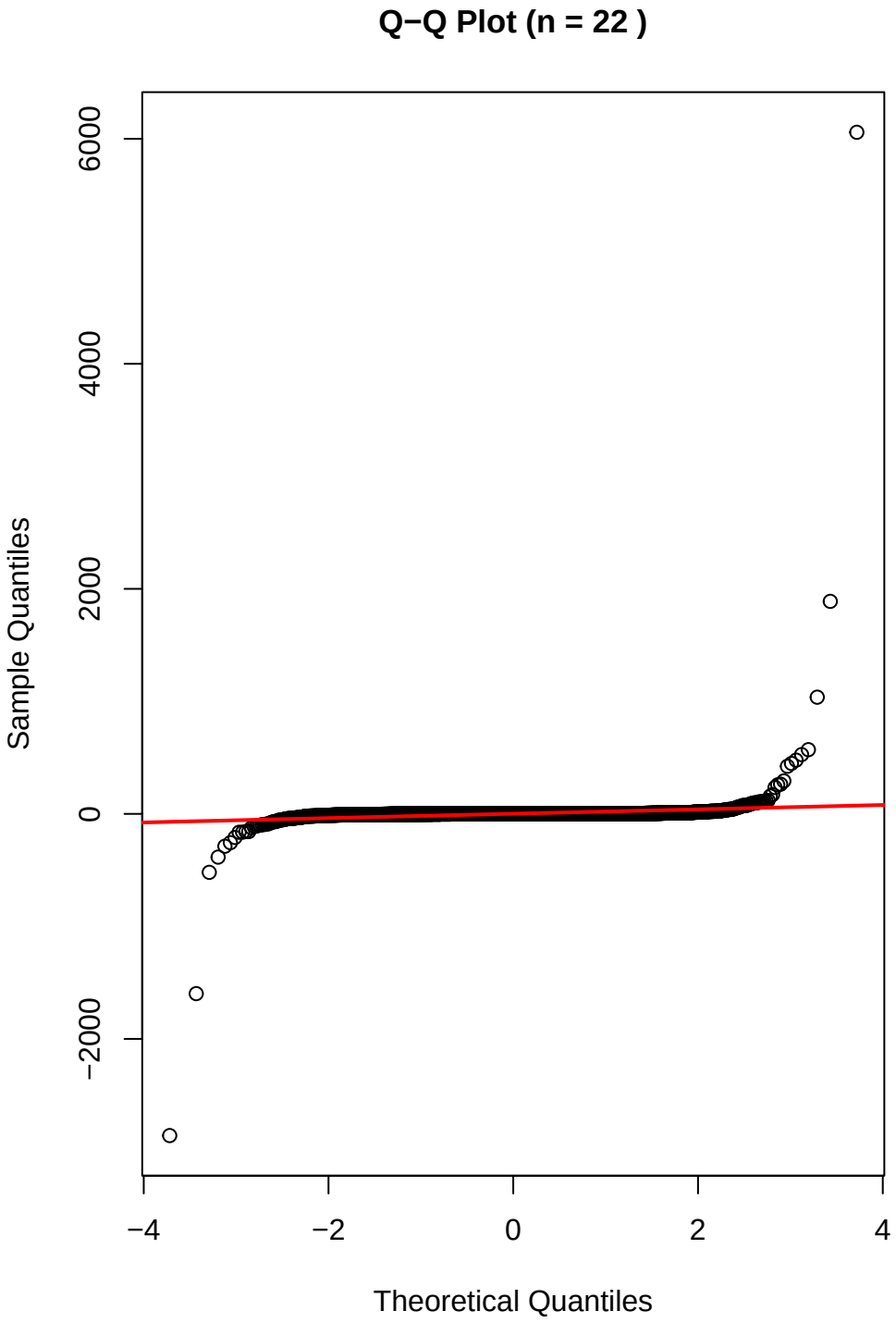




Excess Kurtosis = 760.5657

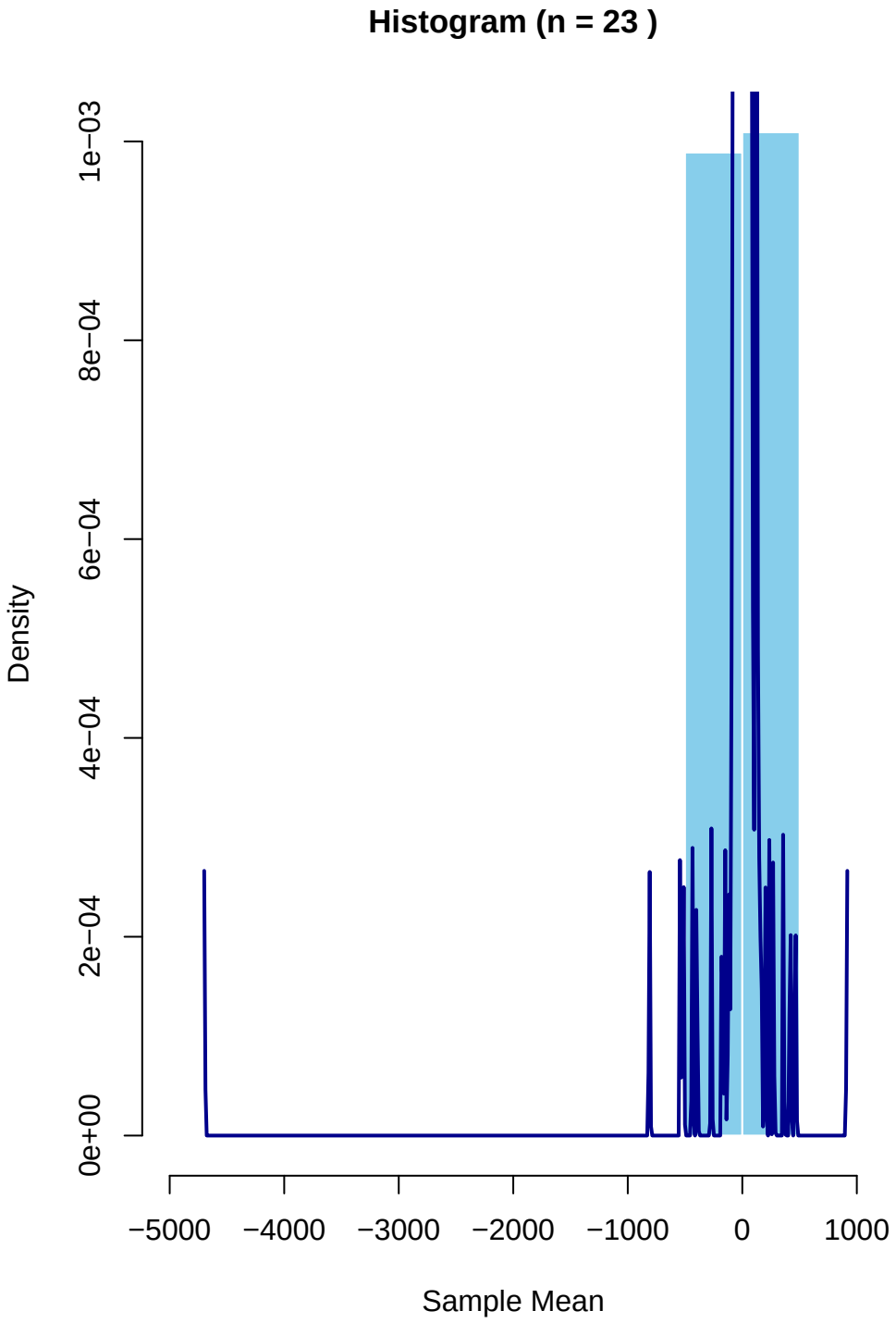
Cauchy (Location=0, Scale=1) — n = 22

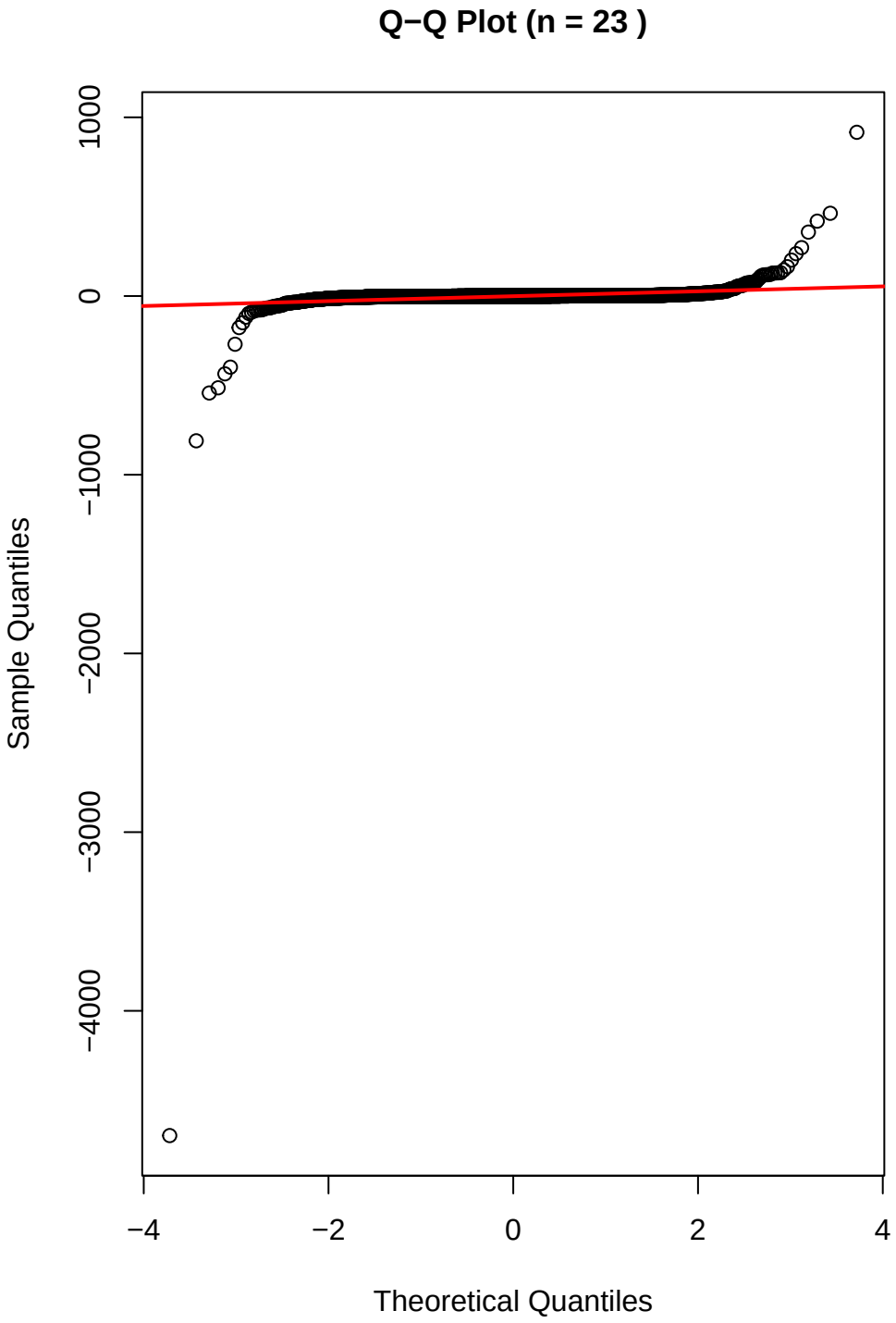




Excess Kurtosis = 2381.8304

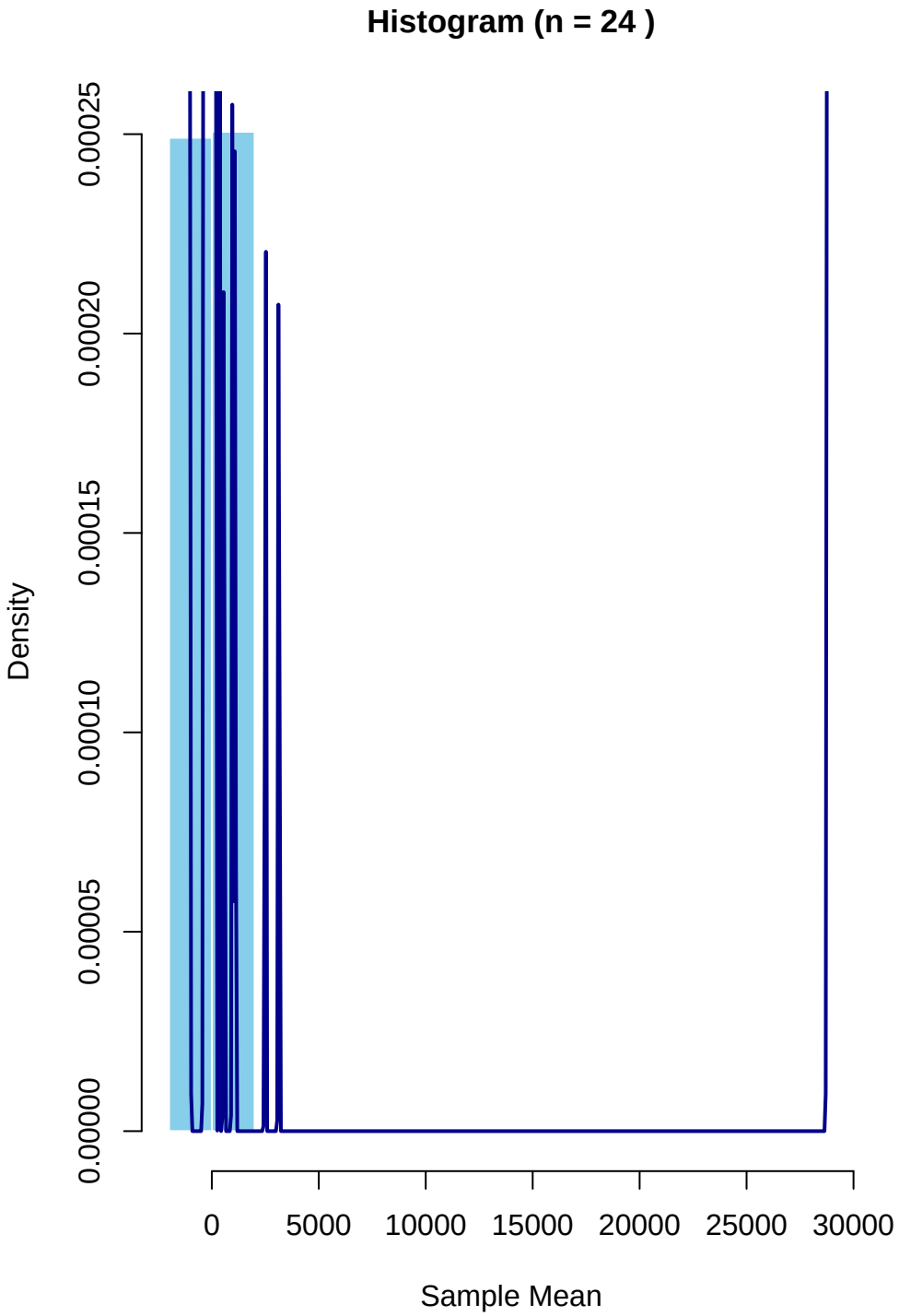
Cauchy (Location=0, Scale=1) — n = 23

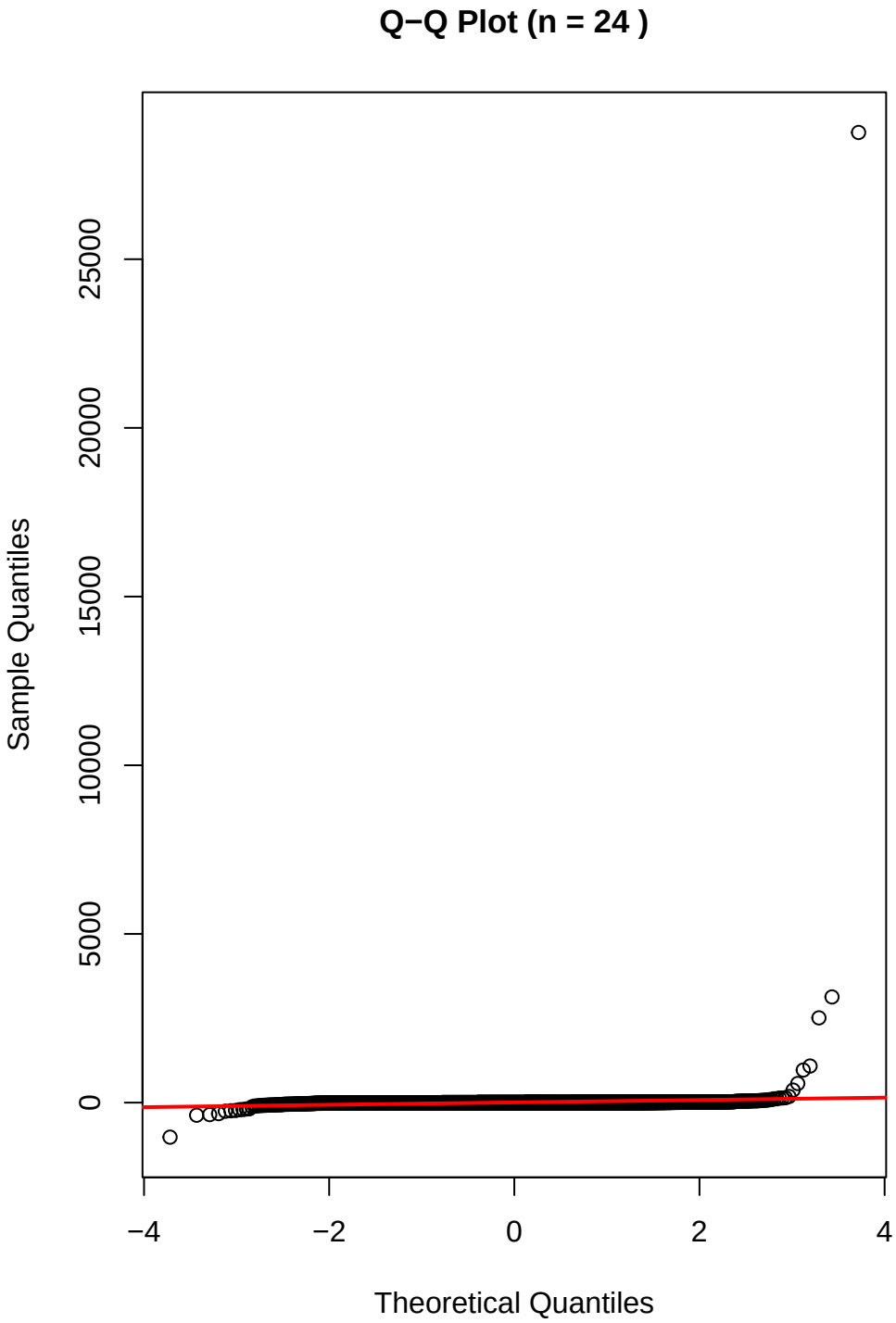




Excess Kurtosis = 3653.3233

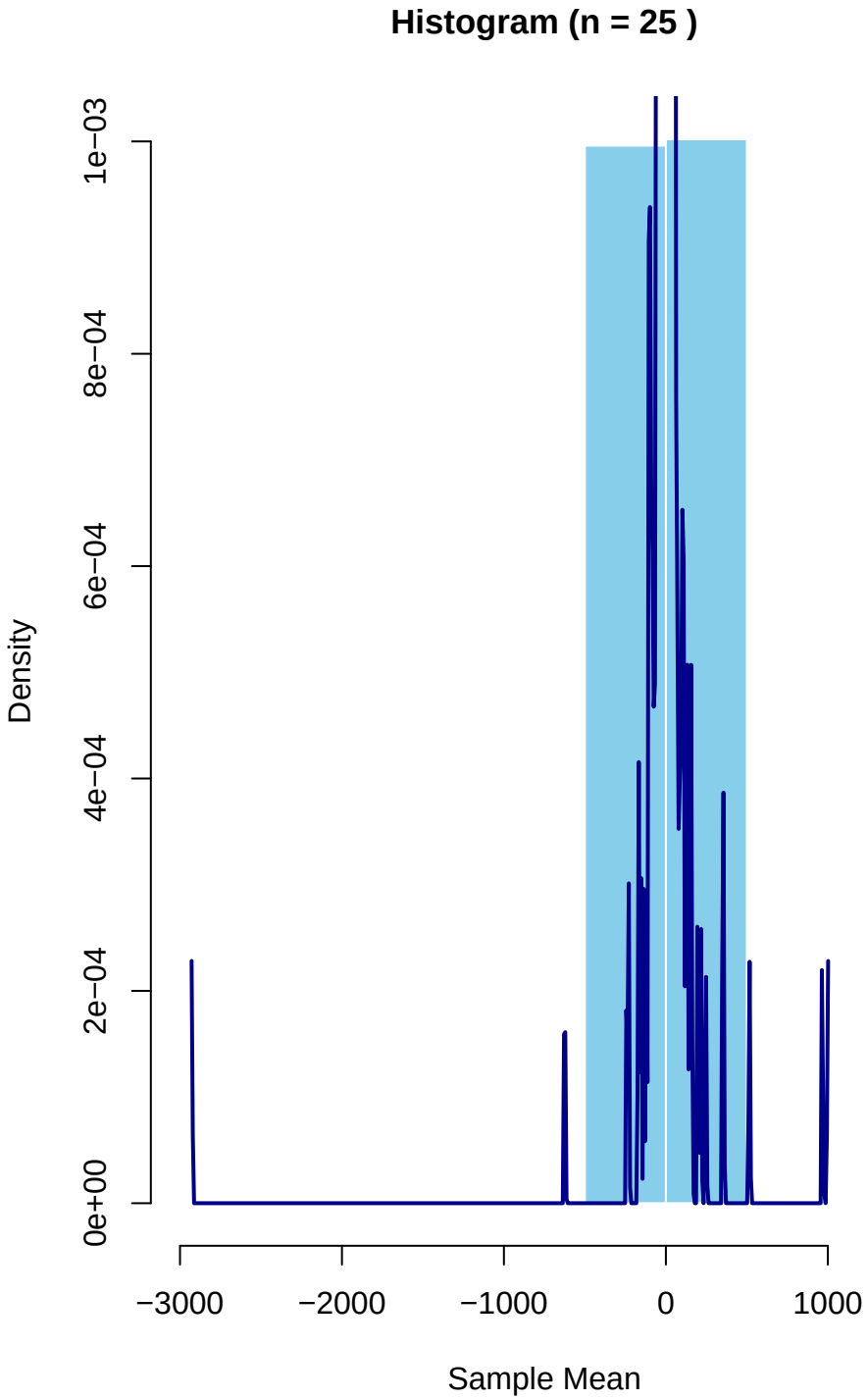
Cauchy (Location=0, Scale=1) — n = 24

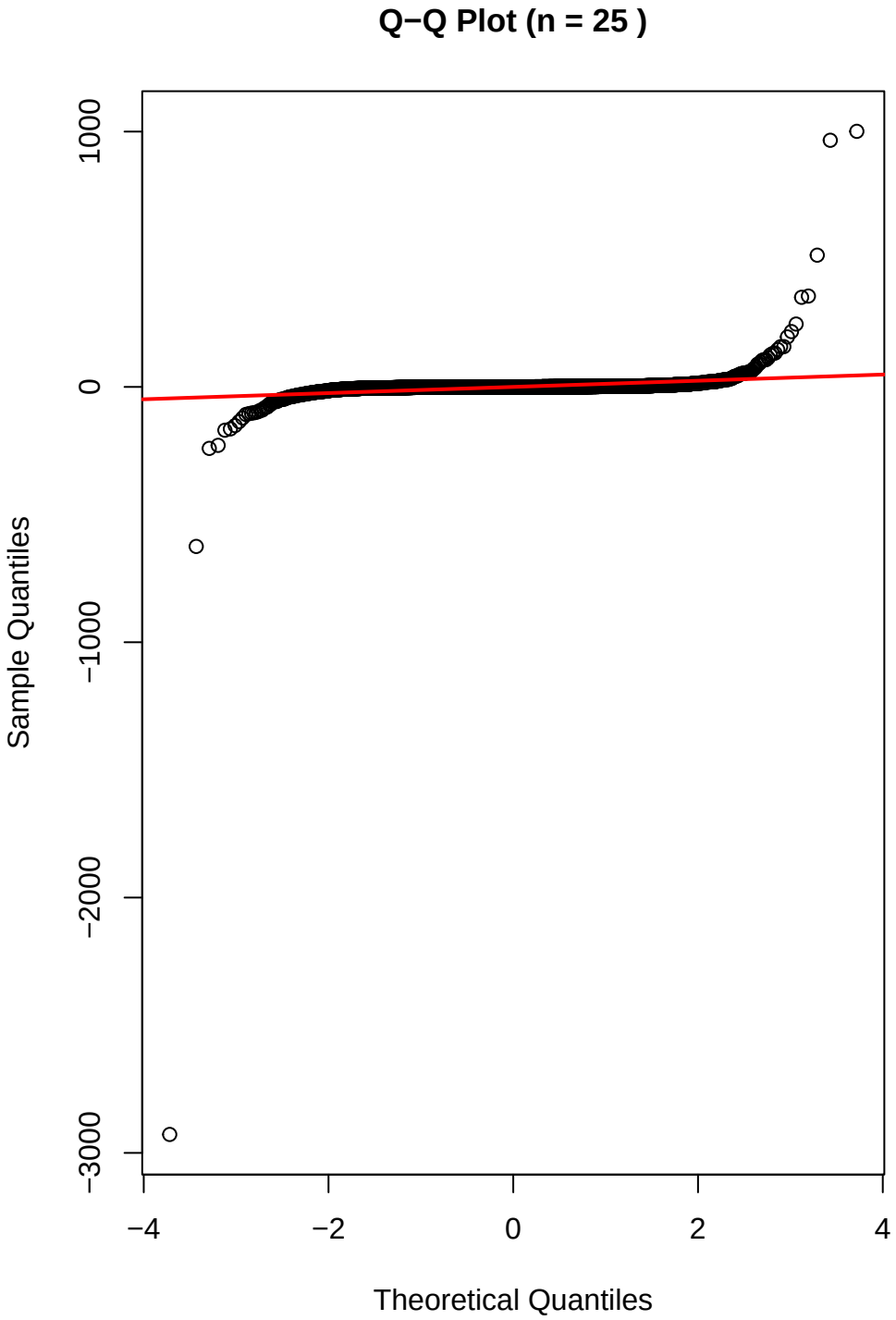




Excess Kurtosis = 4750.4915

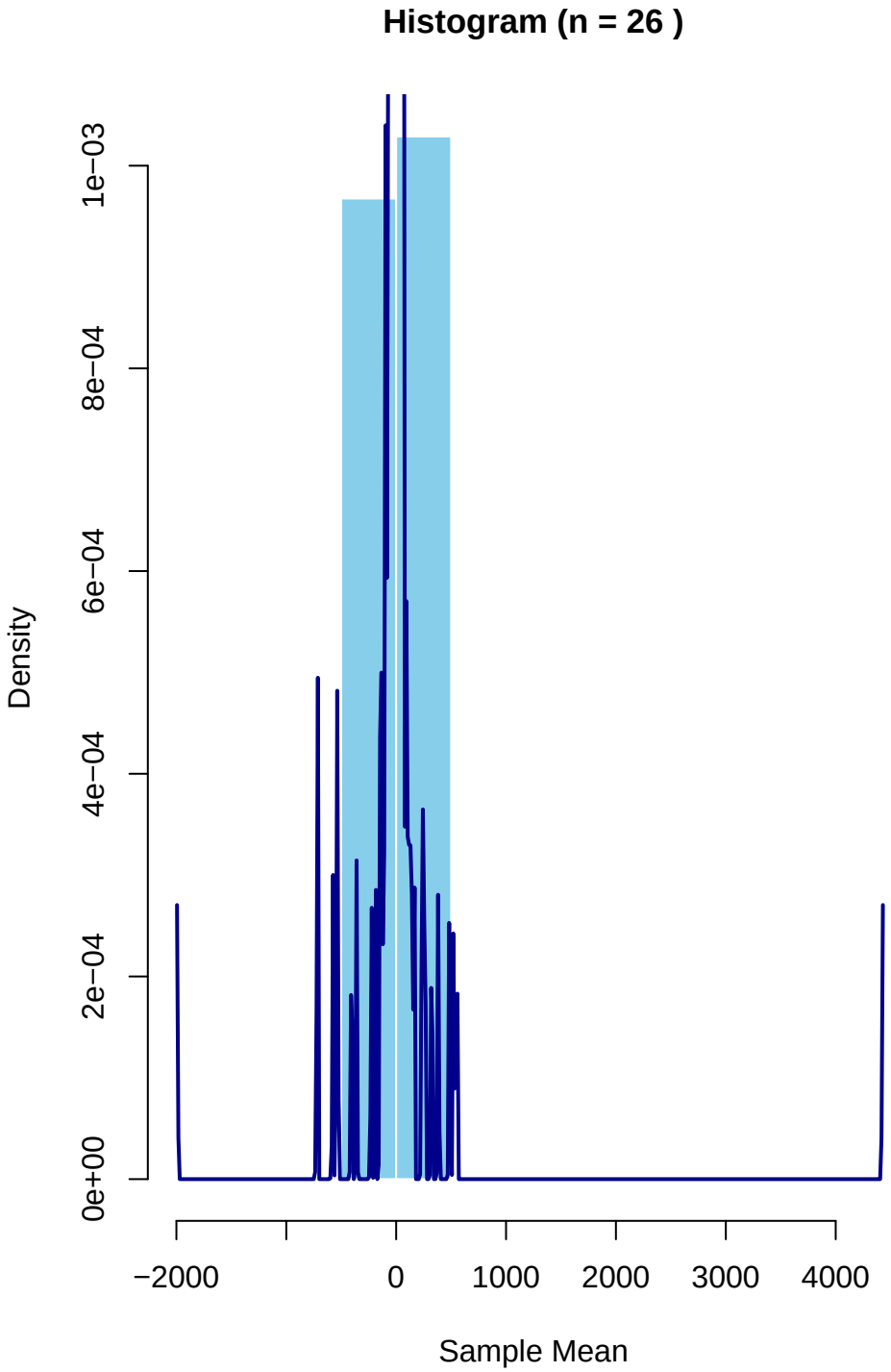
Cauchy (Location=0, Scale=1) — n = 25

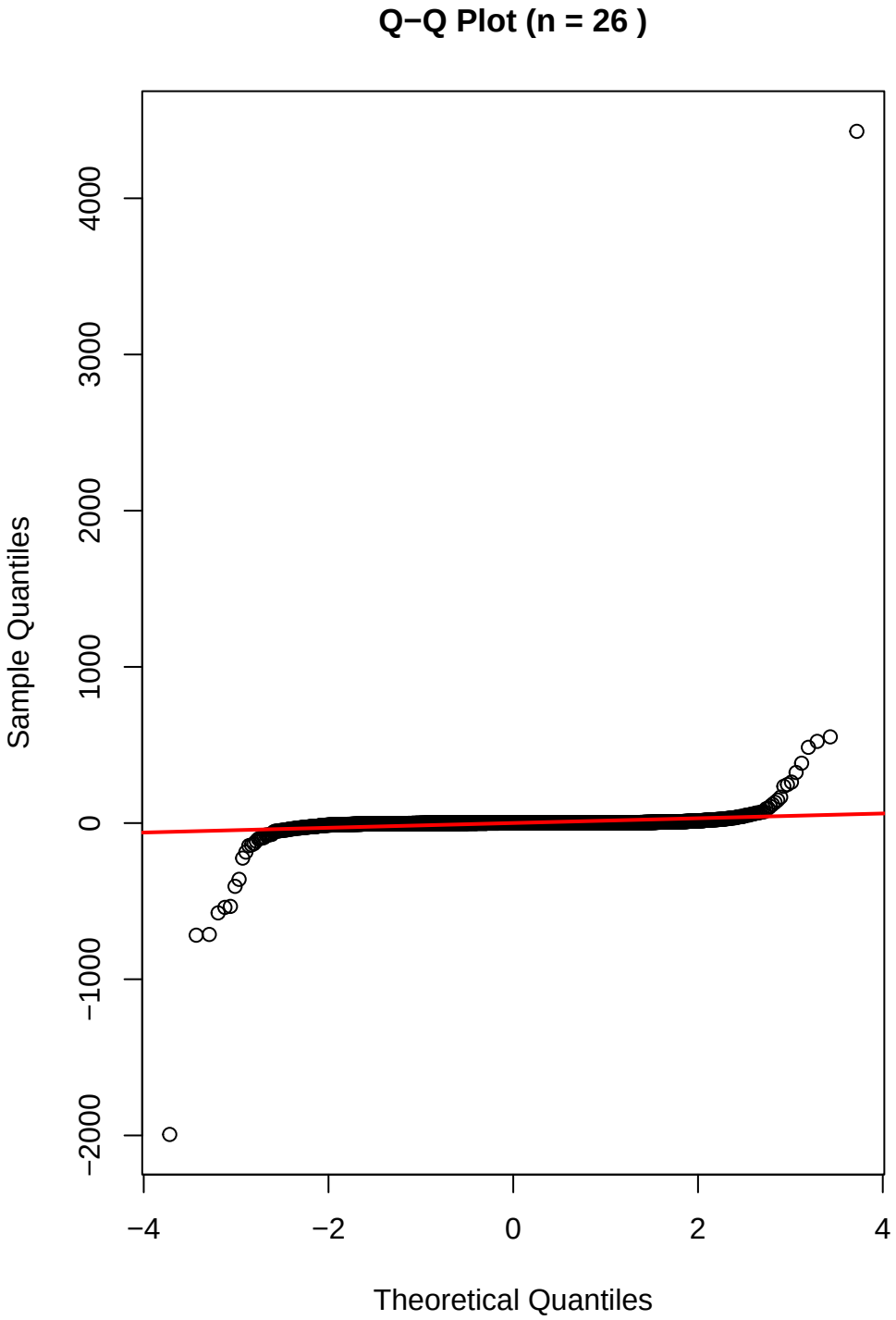




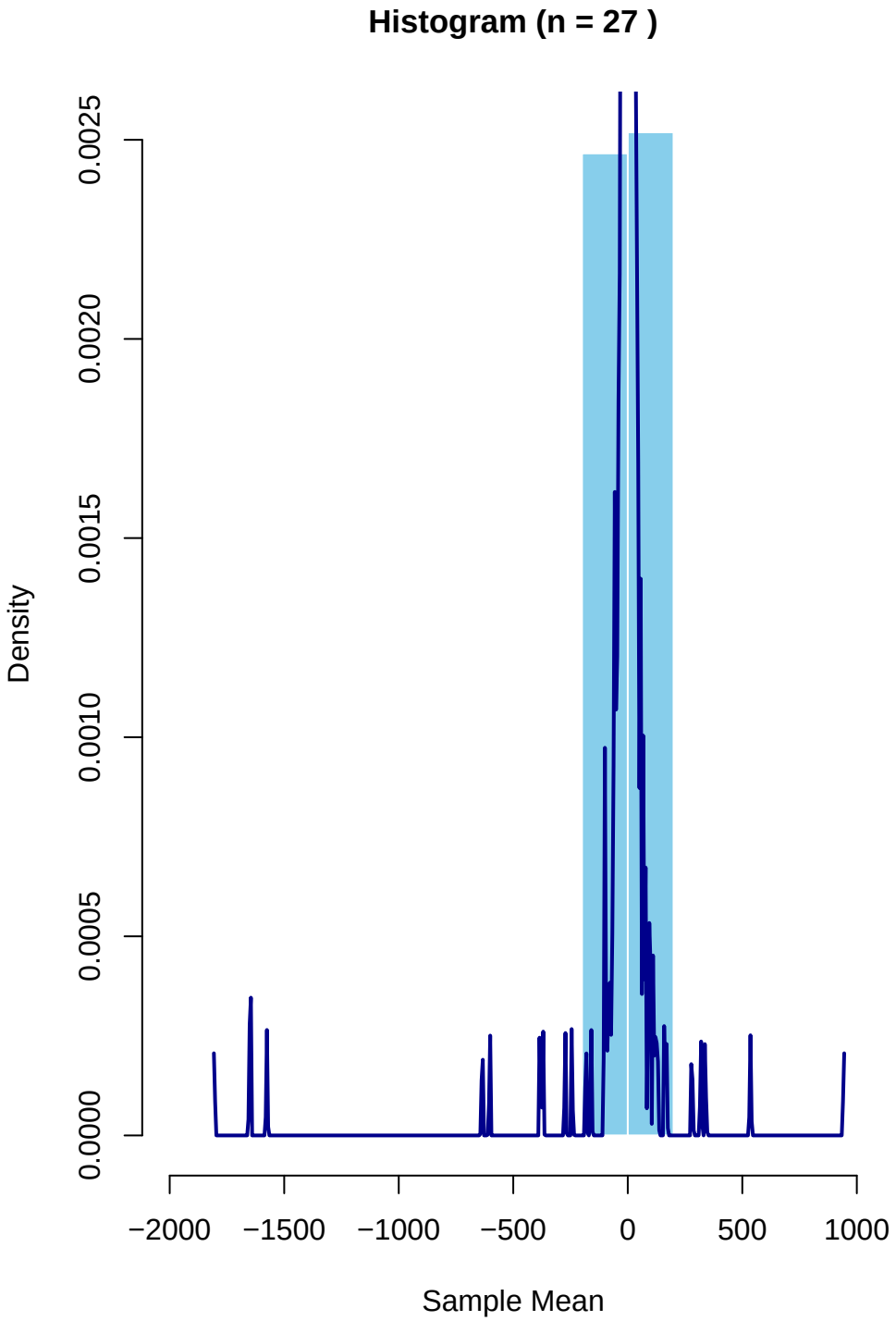
Excess Kurtosis = 2490.4264

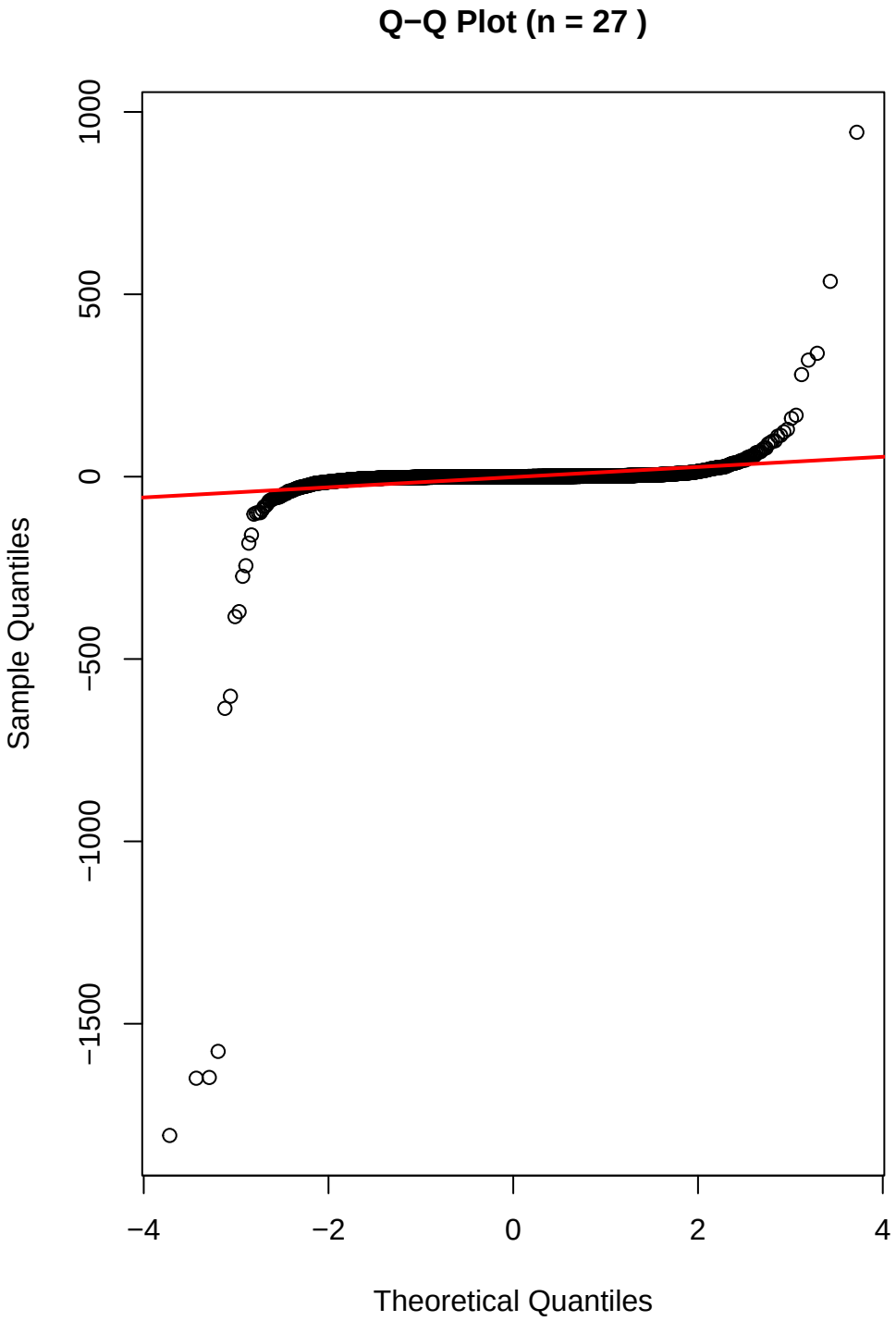
Cauchy (Location=0, Scale=1) — n = 26





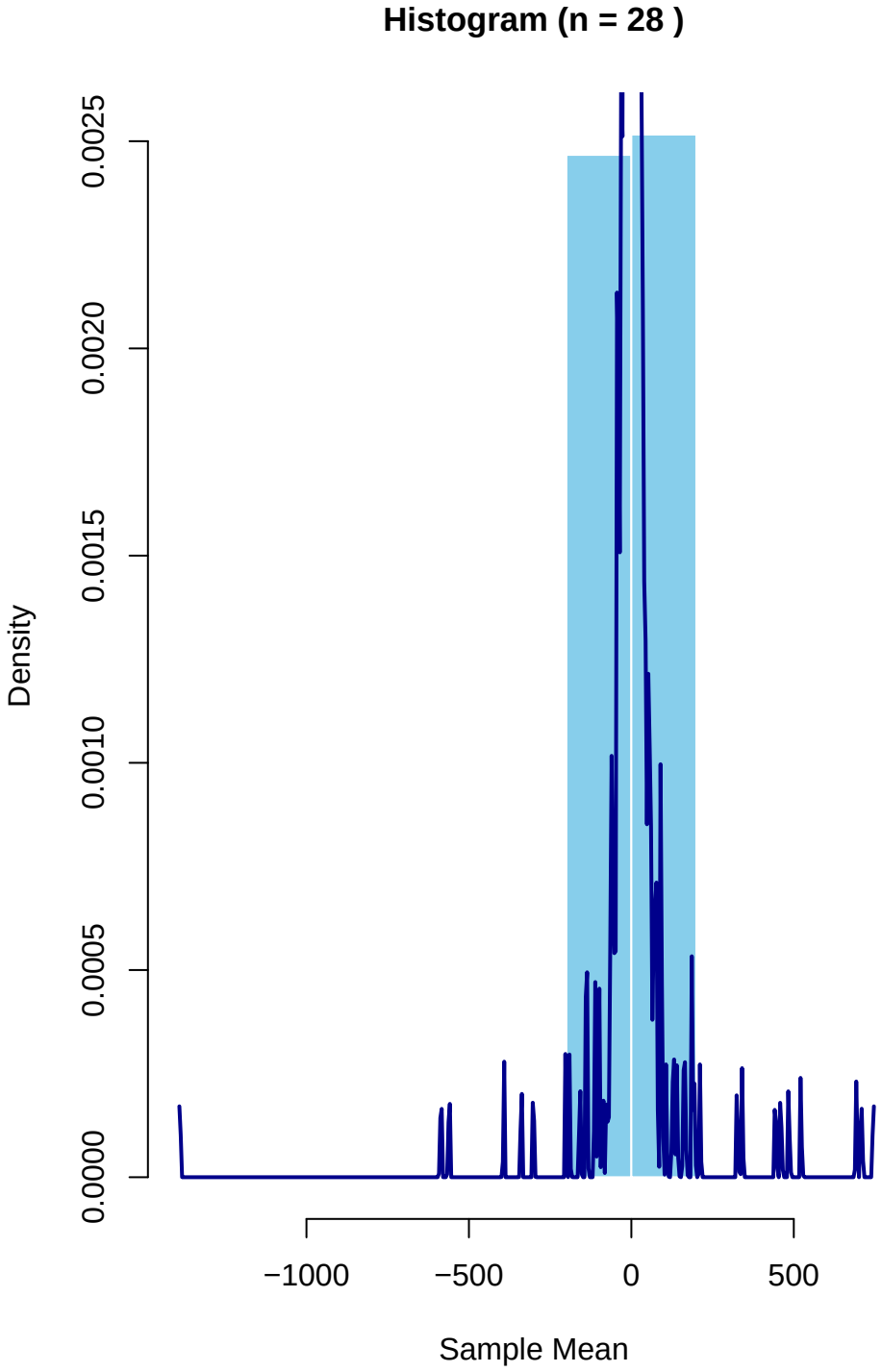
Cauchy (Location=0, Scale=1) — n = 27

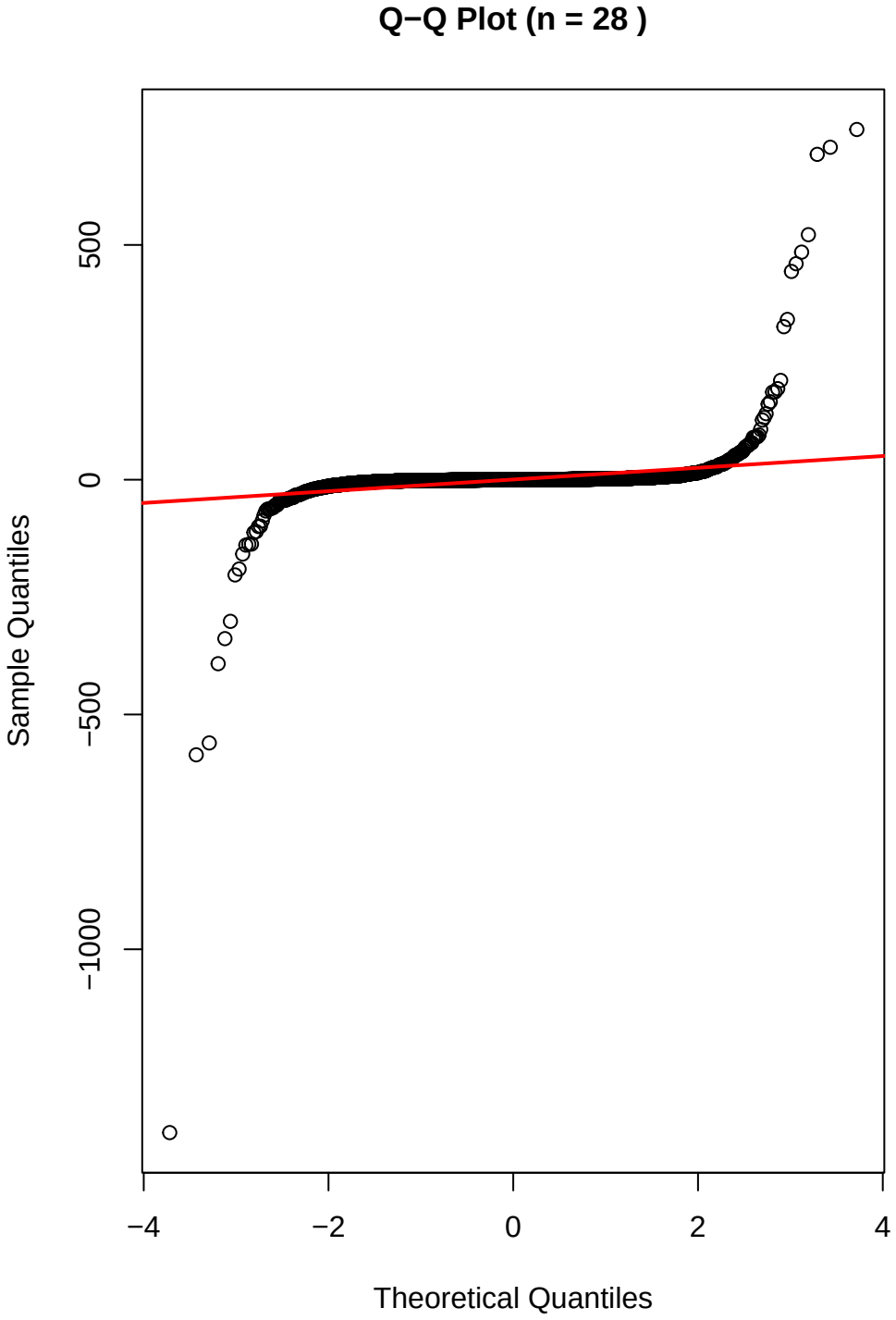




Excess Kurtosis = 790.815

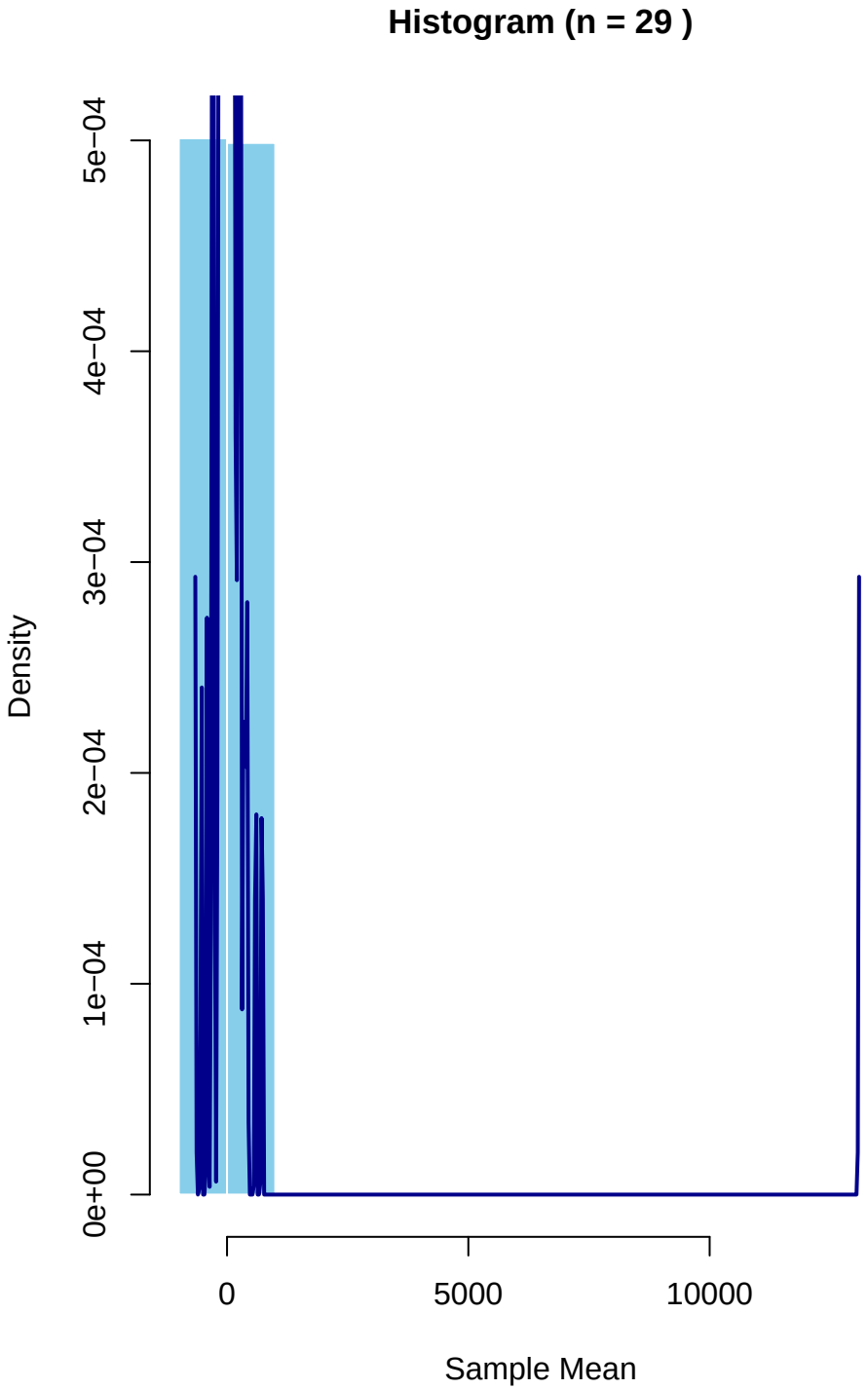
Cauchy (Location=0, Scale=1) — n = 28

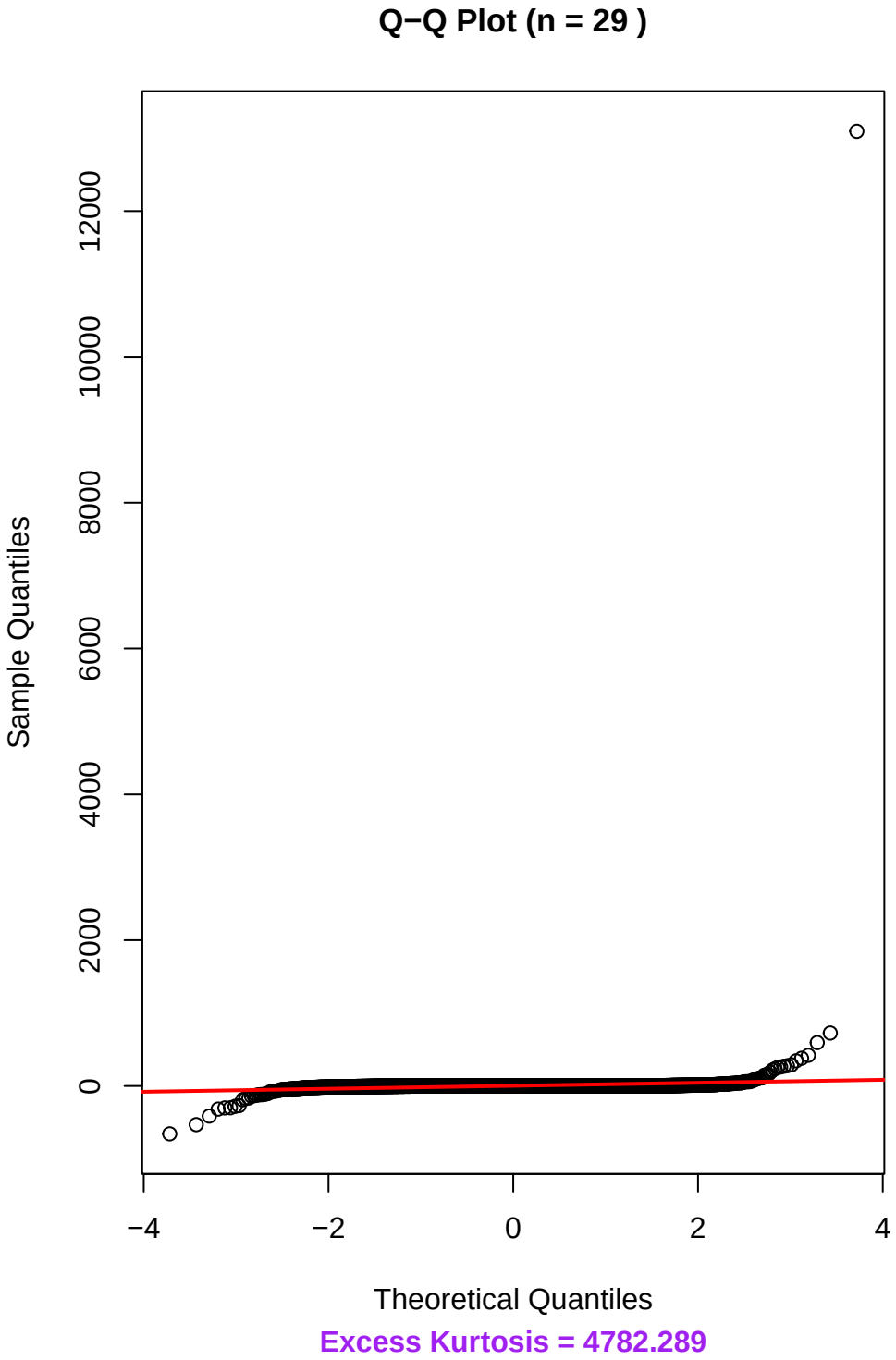




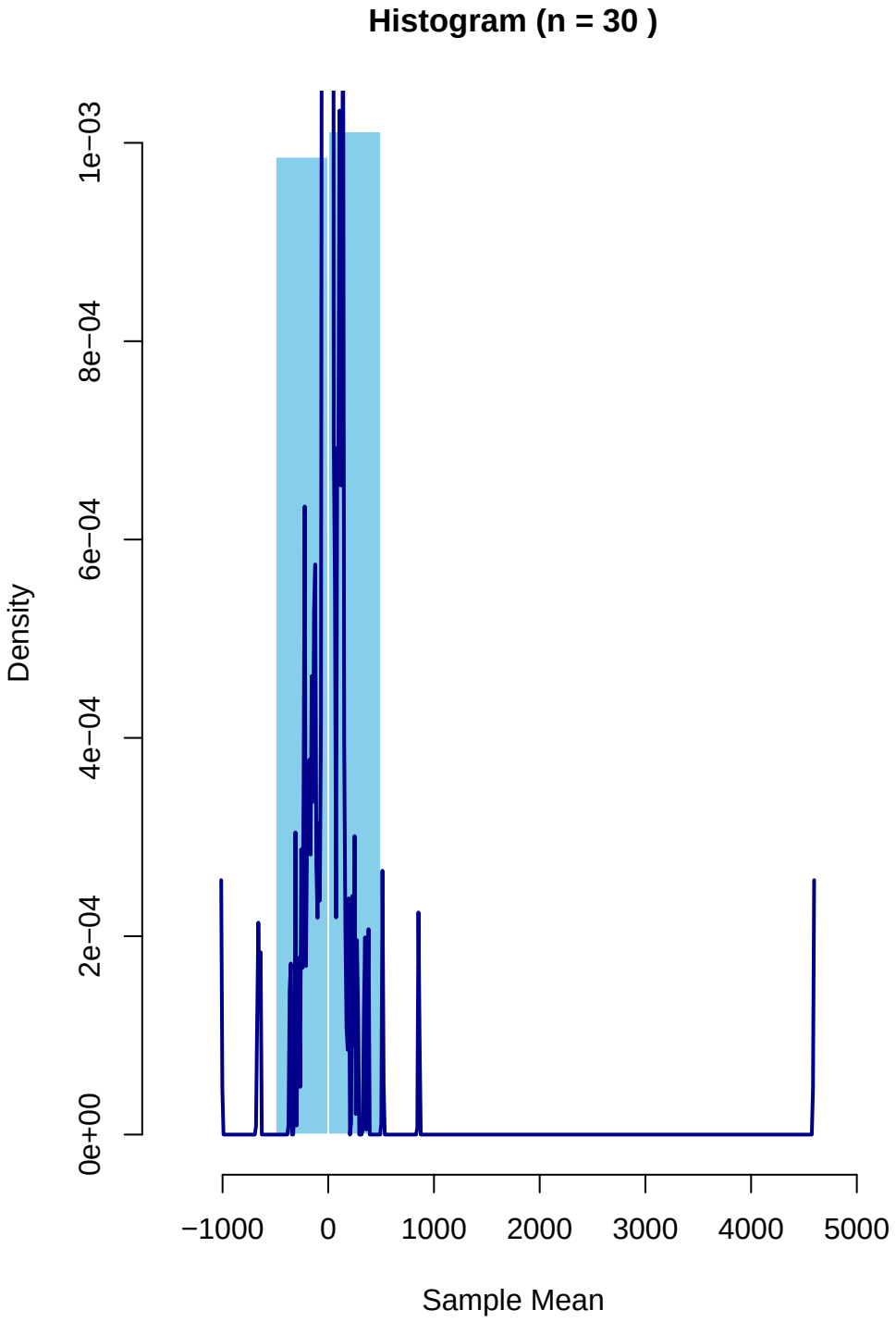
Excess Kurtosis = 607.9999

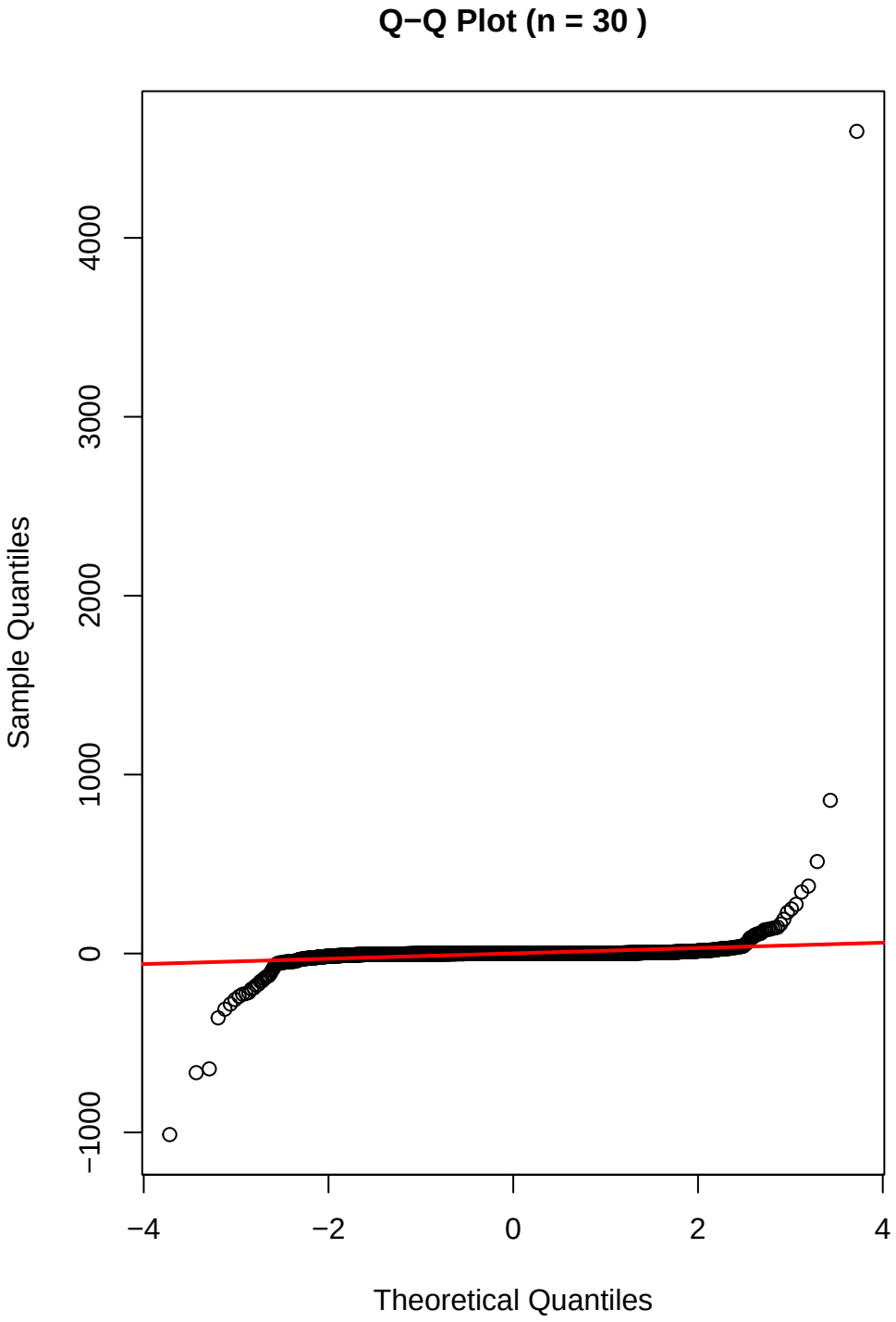
Cauchy (Location=0, Scale=1) — n = 29



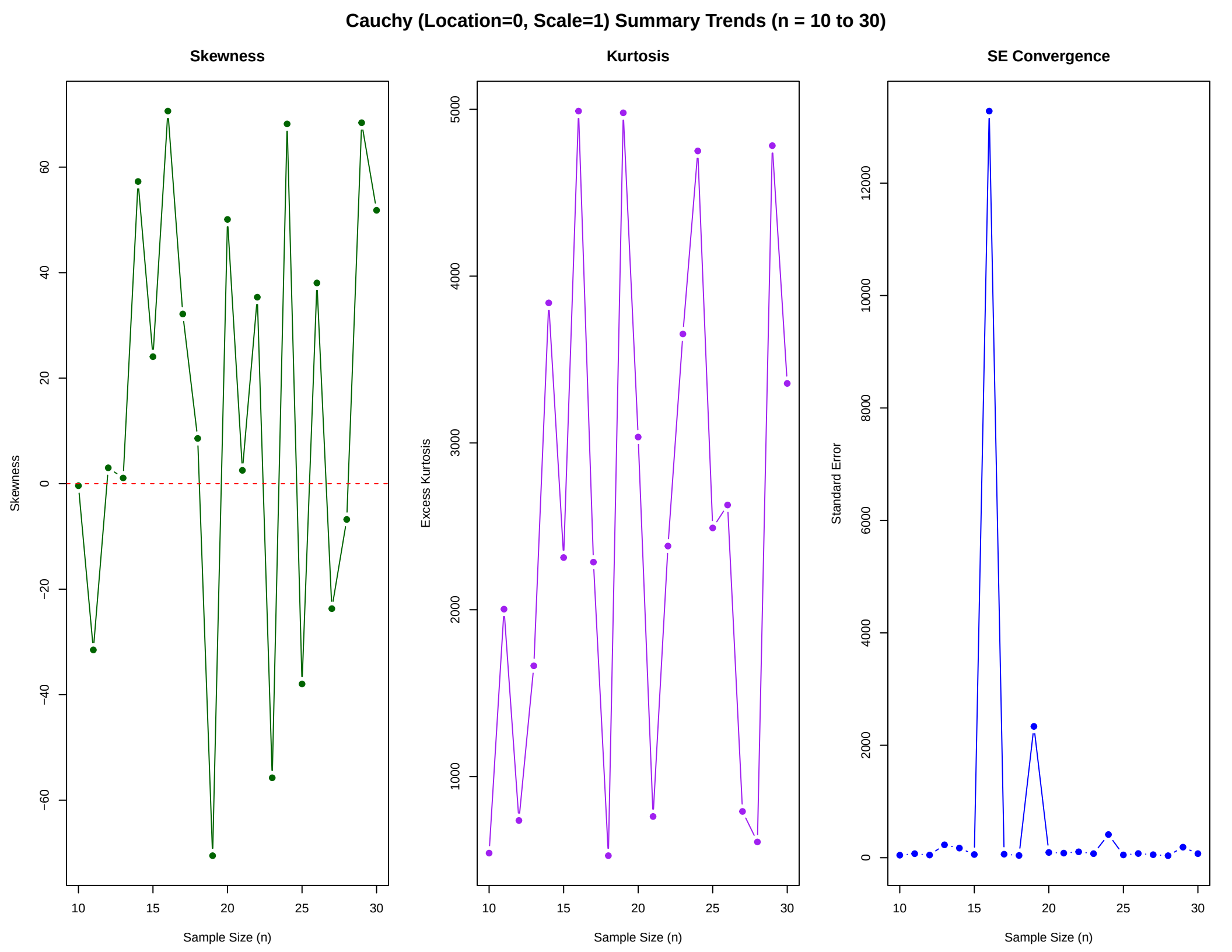


Cauchy (Location=0, Scale=1) — n = 30





Excess Kurtosis = 3356.7779


Cauchy (Location=0, Scale=1) Summary Trends (n = 10 to 30)
Skewness
Skewness
Sample Size (n)
Kurtosis
Excess Kurtosis
Sample Size (n)
SE Convergence
Standard Error
Sample Size (n)